SPINE — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations · 69 of 69 topics –

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Spine

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer

– – Active space – –

ADOLESCENT IDIOPATHIC SCOLIOSIS

1 Basics

Epidemiology

Definition	coronal deformity, **Cobb > 10°**, age **10–18 yrs**; **commonest type of scoliosis**
Incidence	**3%** for 10–20° curves · **0.3%** for curves > 30°
Sex ratio	**1:1** small curves · **10:1 female** for curves > 30°
Commonest pattern	**right thoracic** curve + **thoracic hypokyphosis** (T5–T12 ≈ 8°; normal 20–50°) ✗ Wrong options: right thoracic + hyperkyphosis · left thoracic
Family history	most patients **positive**

Aetiology

Pathophysiology	**unknown / multifactorial**: hormonal (**melatonin**), brain stem, proprioception, platelet, **calmodulin**
Neurocentral synchondrosis (NCS)	cartilaginous plate between **centrum + posterior neural arches** ✗ Wrong options: anterior arches

NCS closure order	Age
Cervical	5–6 yrs
Lumbar	11–12 yrs (~100% closed at age 10)
Thoracic	**14–17 yrs** (~50% closed at age 10)

NCS order trap	**cervical → lumbar → thoracic** ✗ Wrong options: cervical → thoracic → lumbar

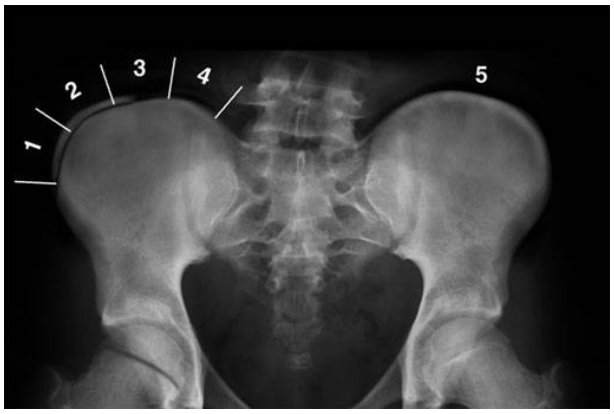
Skeletal maturity & progression risk

Best predictor of progression	curve magnitude at **peak height velocity (PHV)**; **> 30° at PHV → > 70%** reach surgical range ✗ Wrong options: > 20° at menarche · > 30° at Risser 2 · skeletal age 12
PHV timing	**before menarche** and **before Risser 1** (occurs during **Risser 0**) · menarche ≈ 1 yr after PHV · girls mature **~1.5 yrs (18 months) after menarche**
Skeletal method best matching curve acceleration / PHV	**Tanner-Whitehouse III RUS** (radius, ulna, metacarpals + phalanges of digits 1, 3, 5) (✗ Wrong options: Greulich-Pyle · Oxford · Risser · Lenke) · TW **stage III** (✗ Wrong options: stage 2)
PHV correlates closest with (TO)	**olecranon apophysis fusion**; olecranon **stage 5 = PHV** (stage 1 = 3.0 yrs before, stage 7 = 3.4 yrs after) ✗ Wrong options: triradiate closure · Risser 1 · menarche

Other PHV surrogates (TO) modified Oxford score **19–21** · calcaneal apophysis **stage 3** · humeral head ossification between **stages 2–3** · Sauvegrain elbow score · TW2 = most accurate/reproducible, GP = simpler, less reliable in Asians

Risk factor for progression	Detail
Curve magnitude (immature)	**> 25°** before maturity → continues to progress
Curve magnitude (mature)	Thoracic **> 50°** · lumbar **> 40°** → progress **1–2°/yr**
Remaining growth	Age **< 12 yrs** · Tanner **< 3** (F) · **Risser 0–1** · **open triradiate cartilage** · PHV
Curve type	Thoracic > lumbar · double > single

Risser	Iliac apophysis	Growth
0	No ossification	Covers **first 2/3 of pubertal spurt → greatest velocity of linear growth**; highest progression risk with open triradiate
I	25%	Velocity descending
II	50%	GT apophysis unites; **~3 cm sitting growth** left, no more LL growth
III	75%	~2 cm sitting growth
IV	100% ossified, not fused	~1 cm sitting growth; = **skeletal maturity** for bracing
V	Fused	Complete maturity



Risser staging of the iliac apophysis

Open triradiate + < 18 h brace wear factor **most correlated with progression** (Karol) → Risser 0 needs **≥ 18 h/day**
✗ Wrong options: curves < 25° · Risser 1 · Sanders 6

Karol – progression to ≥ 50° in brace Risser 0 **44.2%** · Risser 1 **6.9%** · Risser 2 **0%**

Natural history into adulthood

Untreated AIS (Weinstein 50-yr) ↑ **chronic back pain** (✓)
✗ Wrong options: difficult vaginal delivery / ↑ caesarean · pulmonary dysfunction at 38°

Cobb > 60° at maturity ↑ **acute + chronic back pain** vs controls
✗ Wrong options: premature death · disability · depression · ADL limitation

Curves < 30° at maturity rarely progress · thoracic **> 50°** most likely to progress

– – Active space – –

Curves > 90° cardiopulmonary dysfunction, early death, pain, ↓ self-image

3 Presentation

School screening **7° on scoliometer** (Adams forward bend) abnormal → correlates with **~20° Cobb**

Adams forward bending axial rotation (rib prominence) = **structural** curve · **sitting** forward bend eliminates **leg-length inequality**

Back pain 25–30% of all adolescents; mild, intermittent, non-limiting + typical right curve → **stretching + lighter backpack** · atypical pain → MRI / bone scan
✗ Wrong options: MRI · bone scan · TLSO for 20° · pain clinic

Atypical features requiring MRI

Exam finding (non-idiopathic)	Suggests
Hairy patch	Spina bifida / **diastematomyelia**
Subcutaneous tumours · pigmented lesions · **café-au-lait**	**Neurofibromatosis**
Excessive flexibility	**Ehlers-Danlos / Marfan**
Joint contractures	CP / arthrogryposis
Craniofacial anomalies	Syndromic scoliosis
Limb weakness	Neuromuscular scoliosis
Cavovarus foot	Neural axis anomaly → MRI
Asymmetric abdominal reflexes	**Syringomyelia** (curve with little rotation)

Truncal shift (TO) loss of coronal balance in **large idiopathic** curves → NOT a sign of non-idiopathic scoliosis (EXCEPT answer)

4 Diagnosis & Classification

Key radiology findings

First films (TO) **standing PA + lateral whole spine**; PA = ↓ breast radiation (AP gives better quality)
✗ Wrong options: standing AP · supine · segment only

Cobb angle lines along the **most tilted** end vertebrae · **> 10° = scoliosis** · error **3–5°**

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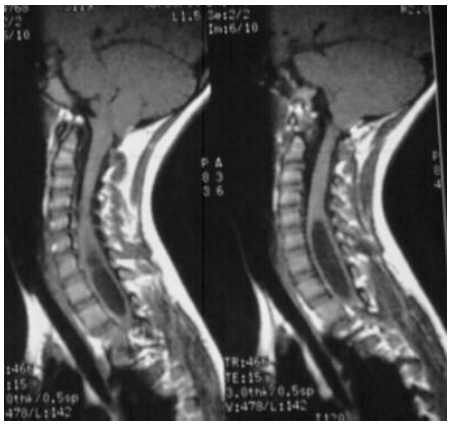
Radiographic term	Definition
Coronal balance	**C7 plumb line** vs **central sacral vertical line (CSVL)**
Sagittal balance	C7 plumb vs posterosuperior corner of S1
Stable zone	Between vertical lines from lumbosacral facets
Stable vertebra	Most proximal vertebra most closely **bisected by CSVL**
Neutral vertebra	**Rotationally neutral** (spinous process equidistant from pedicles)
End vertebra	**Most tilted** from horizontal apical vertebra
Apical vertebra	Disc / vertebra **deviated farthest** from centre
Touched vertebra (TV)	**Last cephalad vertebra touched by CSVL**
Clavicle angle	**Best predictor of post-op shoulder balance**

MRI **posterior fossa → conus**; rules out Chiari / syrinx / cord cyst / tether

MRI indicated	NOT an MRI indication
Left thoracic curve	**Left thoracolumbar / lumbar curve** (TO – common pattern)
Short angular curve · **apical kyphosis** / excessive kyphosis	Age 10–18
Rapid progression	Risser stage
Neurologic signs, pain (atypical)	Curve magnitude alone (without rapid progression)
Asymmetric abdominal reflexes · foot deformity (cavovarus)	Mild activity-related back pain
Male, structural abnormality	Routine initial work-up

Left thoracic curve in a neurologically normal child

total spine MRI = next step
✗ Wrong options: bracing · PT · bending films · CT



Syringomyelia on sagittal MRI — reason for MRI in a left thoracic curve

– – Active space – –

Lenke classification

Components (TO) **curve type 1–6** + **lumbar modifier A/B/C** + **sagittal (thoracic) modifier –/N/+**
✗ Wrong options: "thoracic modifier A/B/C"

Regions proximal thoracic: apex **T3–T5** · main thoracic: apex **T6 → T11–12 disc** · thoracolumbar: apex **T12–L1** · lumbar: apex **L1–2 disc → L4**

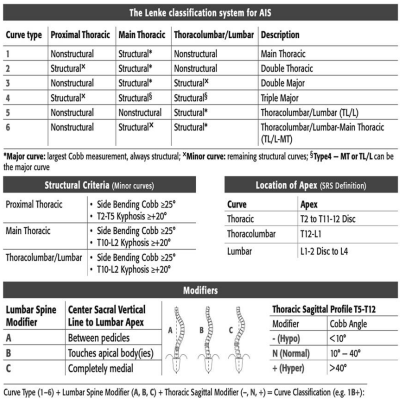
Major curve largest Cobb, always structural

Structural minor curve residual **≥ 25° on supine side-bending** OR kyphosis **≥ 20°** (T2–T5 for PT; **T10–L2** for TL/L)

Lenke type	Name	Structural curves
1	Main thoracic	MT only
2	**Double thoracic** (TO)	**MT (major) + PT** structural; TL/L non-structural
3	Double major	MT (major) + TL/L structural; PT non-structural
4	Triple major	All three structural
5	TL/L	TL/L only (major)
6	TL/L–MT	TL/L major (≥ 5° > MT) + MT structural

Lumbar modifier	CSVL at lumbar apex
A	Between pedicles
B	Touches apical pedicle
C	Completely medial to apical body/pedicle

Sagittal modifier (T5–T12)	Kyphosis
–	**< 10°** (hypokyphosis)
N	**10–40°**
+	**> 40°** (hyperkyphosis)



The Lenke classification system for AIS

Curve type	Proximal Thoracic	Main Thoracic	Thoracolumbar/Lumbar	Description
1	Nonstructural	Structural*	Nonstructural	Main Thoracic
2	Structural[X]	Structural*	Nonstructural	Double Thoracic
3	Nonstructural	Structural*	Structural[X]	Double Major
4	Structural[X]	Structural§	Structural§	Triple Major
5	Nonstructural	Nonstructural	Structural*	Thoracolumbar/Lumbar (TL/L)
6	Nonstructural	Structural[X]	Structural*	Thoracolumbar/Lumbar-Main Thoracic (TL/L-MT)

*Major curve: largest Cobb measurement, always structural; *Minor curve: remaining structural curves; §Type4 – MT or TL/L can be the major curve

Structural Criteria (Minor curves)

Proximal Thoracic	• Side Bending Cobb ≥25° • T2-T5 Kyphosis ≥+20°
Main Thoracic	• Side Bending Cobb ≥25° • T10-L2 Kyphosis ≥+20°
Thoracolumbar/Lumbar	• Side Bending Cobb ≥25° • T10-L2 Kyphosis ≥+20°

Location of Apex (SRS Definition)

Curve	Apex
Thoracic	T2 to T11–12 Disc
Thoracolumbar	T12-L1
Lumbar	L1-2 Disc to L4

Modifiers

Lumbar Spine Modifier	Center Sacral Vertical Line to Lumbar Apex
A	Between pedicles
B	Touches apical body(ies)
C	Completely medial

Thoracic Sagittal Profile T5-T12

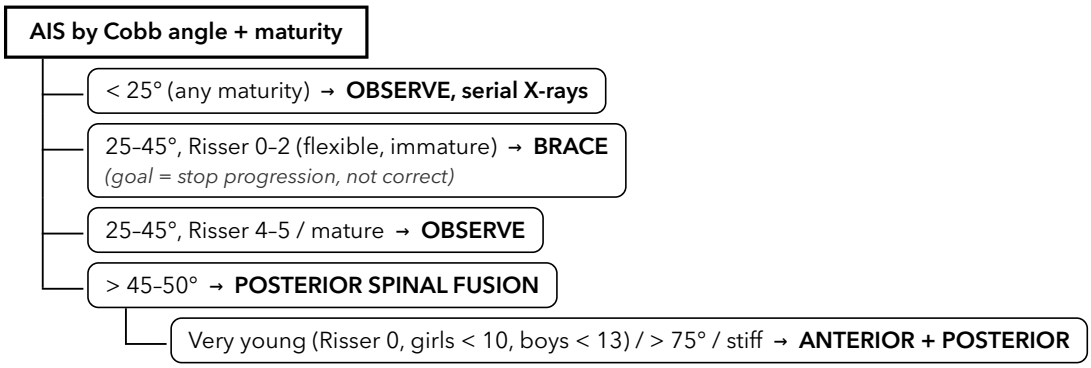
Modifier	Cobb Angle
- (Hypo)	<10°
N (Normal)	10° – 40°
+ (Hyper)	>40°

Curve Type (1–6) + Lumbar Spine Modifier (A, B, C) + Thoracic Sagittal Modifier (–, N, +) = Curve Classification (e.g. 1B+):

Lenke classification summary table

King-Moe 5 thoracic curve types for **Harrington** instrumentation; coronal only, no sagittal plane → replaced by Lenke

5 Management



Bracing – indications, wear time & efficacy

Parameter	Value
Cobb range	**25–45°** (Orthobullets) · **20–40°** (OB) · **20–25° with documented progression** → brace – read the options
Maturity	**Risser 0, 1, 2** · Sanders 1–3 · age 10–15
Prescribed	**16–23 h/day** until maturity or surgery
Minimum effective (actual)	**> 12 h/day** (12.9 h → **90–93% success**, BrAIST) (✗ Wrong options: 6–8 h · 9–10 h)
Dose-response	**17.7 h → 93%**; > 12 h → 82% no progression; < 7 h → 69% progressed; prescribed 16 vs 23 h had no effect
Effect	**50% ↓ need for surgery** · **NNT = 4** (compliant)
Best predictor of brace success	**Time in brace** (✗ Wrong options: in-brace correction · PT first)

Brace success / failure (TO) success = **< 5°** progression · failure = **≥ 6°** progression at discontinuation, **> 45°** before/at maturity, or switch to surgery
✗ Wrong options: "success = correction > 5°"

Skeletal maturity = stop brace **Risser 4** · **< 1 cm height change over 2 visits 6 months apart** · **2 yrs post-menarche**

Poor brace prognosis poor in-brace correction · **hypokyphosis** (relative CI) · **male** · **obese** · non-compliant

Brace (TO)	Curve apex
Milwaukee (CTLSO)	**Above T7** (TO, Orthobullets) · at/proximal to **T6** (OB) · double thoracic / apex above T8 do not respond to TLSO
TLSO / Boston (underarm)	**T7 or below** (e.g. apex T12)
Charleston bending	**Night-time** only

– – Active space – –

Sanders staging — **AP hand X-ray** + curve magnitude → risk of progression **> 50° despite bracing** in **Lenke 1 + 3** curves · stage 3 = **complete capping of phalangeal epiphyses**

25° curve + Sanders stage	Risk > 50° despite brace
1	**99%**
3	**29%** → brace most likely to change natural history (✓)
4–8	**0%**
45° curve	~100%

TABLE III Logistic Projection of the Probability of Lenke Type-1 and Type-3 Curves Progressing to Surgery Assuming a >50° Threshold*†

Curve	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	Stage 7, 8
10°	2% (0% to 40%)	0% (0% to 15%)	0% (0% to 0%)	0% (0% to 0%)	0% (0% to 0%)	0% (0% to 0%)	0% (0% to 1%)
15°	23% (4% to 69%)	11% (1% to 58%)	0% (0% to 2%)	0% (0% to 0%)	0% (0% to 0%)	0% (0% to 0%)	0% (0% to 7%)
20°	84% (40% to 98%)	92% (56% to 99%)	0% (0% to 14%)	0% (0% to 1%)	0% (0% to 1%)	0% (0% to 1%)	0% (0% to 26%)
25°	99% (68% to 100%)	100% (92% to 100%)	29% (3% to 84%)	0% (0% to 5%)	0% (0% to 5%)	0% (0% to 2%)	0% (0% to 64%)
30°	100% (83% to 100%)	100% (98% to 100%)	100% (47% to 100%)	0% (0% to 27%)	0% (0% to 22%)	0% (0% to 11%)	0% (0% to 91%)
35°	100% (91% to 100%)	100% (100% to 100%)	100% (89% to 100%)	0% (0% to 79%)	0% (0% to 65%)	0% (0% to 41%)	0% (0% to 98%)
40°	100% (95% to 100%)	100% (100% to 100%)	100% (98% to 100%)	15% (0% to 99%)	0% (0% to 94%)	0% (0% to 83%)	0% (0% to 100%)
45°	100% (98% to 100%)	100% (100% to 100%)	100% (100% to 100%)	88% (2% to 100%)	1% (0% to 99%)	0% (0% to 98%)	0% (0% to 100%)

*Unshaded cells correspond with combinations of curve size and maturity stage for which surgery would be a plausible treatment if >50° at maturity is accepted as the threshold for surgical treatment. Shaded cells correspond with combinations for which surgery would not be a plausible treatment. †Cells with wide 95% confidence intervals (shown in parentheses) correspond with groups that had too few patients for accurate estimates (or groups that had no patients) and should be interpreted with caution.

Sanders skeletal maturity staging (hand radiograph)

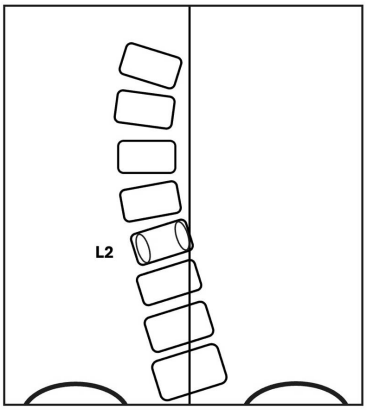
Bracing traps — **scoliosis-specific exercises / PT** not effective · spinal manipulation unsupported · 12° curve with 6° progression → observe · Risser 4 / Sanders 7 + 30° → observe · boy with 50° → surgery · bracing large (> 45°) curves can still **delay** surgery to maturity · congenital curve → no bracing

Fusion level selection

PSF indications — Cobb **> 45°** (Orthobullets) / **> 50°** (OB) · **gold standard** for thoracic + double major curves · usable for all idiopathic types

ASF — **thoracolumbar / lumbar** curves with **normal sagittal profile**; saves lumbar levels; fuse **end → end vertebra**; not if **thoracic hyperkyphosis > 40°** (↑ pseudarthrosis)

Technique	Fusion levels
Typical	Proximal end vertebra → 1–2 levels cephalad to **stable** vertebra; double/triple major → **distal end vertebra**
Harrington	1 above + 2 below end vertebrae if within **stable zone**
Moe	To **neutral** vertebra
Lenke	All major curves + minor curves that are **stiff or kyphotic**
Touched vertebra	Fuse **to or caudad to TV** → best correction; cephalad to TV → **truncal shift**, poor long-term outcome



Touched vertebra: last cephalad vertebra touched by CSVL

Avoid L4 / L5	Cochran: ↑ **low back pain** with fusion to **L5** (less with L4) · pelvis almost **never** needed in AIS
Lenke 1B (65° MT, flexible PT + lumbar)	**PSF of thoracic curve only** ✗ Wrong options: T2–L4 · brace · stapling · A/P
Selective thoracic fusion	**AVT ratio** = thoracic AVT (C7 plumb → thoracic apex) ÷ TL/L AVT (CSVL → lumbar apex); **> 1.2** → selective thoracic OK (e.g. 45/15 = **3.0** ✓)
LIV to prevent DJK / adding-on	include **stable sagittal vertebra** (≥ 50% of body anterior to **posterior sacral vertical line**) · **Lenke 1A-R (L4 tilted right)** → ↑ adding-on · e.g. TL kyphosis T8–L2 with L3 the first SSV → **LIV = L3** ✗ Wrong options: T12 · L2 · L5 · pelvis
Isolated long PSF indicated	**13-yr Risser 3 with 55°** ✗ Wrong options: JIS → needs anterior too · IIS → Mehta cast · hemivertebra + bar → bar excision · 27° → brace
Vertebral body stapling	small thoracic curves **< 35°**, immature

Surgical technique

Pedicle screws	insertional torque ∝ **pullout resistance**; ↑ by **undertapping 1 mm** + **straightforward trajectory**; not tapping may be stronger still · segmental screws → ↑ coronal correction, ↓ need for anterior release ✗ Wrong options: anatomic trajectory ↓ torque
Pedicle subtraction osteotomy (PSO)	posterior closing wedge hinged on **anterior column**, usually **L2 / L3** · commonest complication **pseudarthrosis 25–30%** · 1 PSO ≈ **3 Smith-Petersen** osteotomies, more blood loss ✗ Wrong options: cord ischaemia · aorta < 1–2% · neuro deficit < 10%
Opioid-sparing	**short-course post-op steroids** ↓ opioid use ✗ Wrong options: bracing · avoiding drain (drains ↓ opioids) · delaying PT · pre-op opioids ↑ use
ED visits after PSF	~10%, mostly minor → **older age + more fusion levels** ✗ Wrong options: BMI · Lenke type · ERAS ↓ visits · pelvic fixation/Ponte

Neuromonitoring

Standard of care	**SSEP + MEP**; MEPs warn earliest of cord dysfunction, change faster than SSEPs
Alert thresholds	amplitude drop **> 50%** (Orthobullets) · critical: **MEP 80%**, **SSEP 50% / latency 10%** (OB)

– – Active space – –

Neuromonitoring signal drop

- Technical / anaesthetic cause? → **CHECK leads + IV agents**
- Not yet critical (e.g. MEP 60%, SSEP 30%) → **REPOSITION, raise BP, check Hb**
  (do not deepen anaesthesia — suppresses MEPs)
- Persistent / critical → **REVERSE CORRECTION**
  - Still lost → STAGNARA WAKE-UP TEST → **remove implants if spine stable**

6 Complications

Complication	Key facts
Neurologic injury	Paraplegia **1:1000**; ↑ with **kyphosis**, excessive correction, **sublaminar wires**
Pseudarthrosis (1–2%)	Late pain, progression, **rod breakage**; asymptomatic + no loss of correction → observe
Infection (1–2%)	**Late pain**, clean wound; **P. acnes** commonest delayed organism (**2-week** culture); I&D + retain hardware if not loose and < 6 months
Flat back	Loss of lumbar lordosis → fatigue + back pain; rare with sagittal rod contouring; Rx **posterior closing wedge osteotomies** (± anterior release)
Crankshaft	Continued **anterior growth** after **PSF alone** in very young → rotation; prevent with **anterior discectomy + fusion**
SMA syndrome	**3rd part of duodenum** compressed between SMA (arises **L1**) and aorta → bilious vomiting in **1st post-op week**, electrolyte disturbance; risk: height **< 50th**, weight **< 25th** centile, kyphosis; Rx **NG tube + IV fluids**

X vs Y

	Typical AIS	Needs MRI
Curve	**Right thoracic**, left TL/L	**Left thoracic**, short angular
Sagittal	**Hypokyphosis**	**Apical kyphosis** / hyperkyphosis
Progression	Slow	**Rapid**
Exam	Normal neuro, symmetric reflexes	Asymmetric abdominal reflexes, cavovarus, skin signs

	Milwaukee (CTLSO)	Boston TLSO	Charleston
Apex	Above T7	T7 or below	Any (night)
Wear	Full time	Full time	**Night only**

	Risser	Sanders	TW-III RUS
Site	**Iliac apophysis**	**Hand**	Radius, ulna, MC/phalanges 1, 3, 5
Use	Common; PHV during Risser 0	Predicts > 50° despite brace	**Best correlate of curve acceleration**

How it's asked

– – Active space – –

"Which is NOT an MRI indication in AIS?"	→ **left thoracolumbar / lumbar curve** (TO) ✗ Wrong options: left thoracic · rapid progression · neuro signs
"Exam finding NOT suggesting non-idiopathic scoliosis?"	→ **truncal shift** (TO) ✗ Wrong options: hairy patch · pigmented lesions · hyperflexibility
"12-yr girl, left thoracic curve, absent left abdominal reflexes — next step?"	→ **MRI** ✗ Wrong options: observe · TLSO · PSF
"First-visit evaluation EXCEPT?"	→ **MRI** (✓ leg lengths, abdominal reflexes, waist asymmetry, café-au-lait)
"Curve magnitude at which time point best predicts progression?"	→ **peak height velocity** ✗ Wrong options: menarche · Risser 3 · TW stage 2
"PHV correlates closest with?"	→ **olecranon apophysis fusion** (TO) ✗ Wrong options: triradiate closure · Risser 1 · menarche
"Which Risser grade = greatest velocity of linear growth?"	→ **Risser 0**
"Which X-ray shows highest progression risk?"	→ **Risser 0 with open triradiate cartilage** ✗ Wrong options: late Risser 0 with fusing triradiate · Risser 3–5
"Which statement on curve progression is WRONG?"	→ **rib phase 1 worse than rib phase 2** (TO — phase 2 progresses rapidly; RVAD > 20° → progression)
"NCS lies between and closes in which order?"	→ **centrum + posterior arches; cervical, lumbar, thoracic**
"Brace indicated?"	→ **premenarchal girl, 30° curve · 28°, Risser 1 · 30° MT, Risser 1** ✗ Wrong options: 38° Risser 4 · 52° · 19°
"Minimum wear for 90–93% success?"	→ **12–13 h actual wear** ✗ Wrong options: 6–8 · 15–16 · 19–20
"Correct statement on bracing (TO)?"	→ **Milwaukee for apex above T7** ✗ Wrong options: Boston only below T12 · 8 h minimum · success = > 5° correction
"Expected effect of bracing a 32° immature curve?"	→ **rigid TLSO 13 h/day → 50% ↓ need for surgery**
"Bracing true statement (Sanders 2, 32°)?"	→ **NNT = 4**
"Full-time TLSO most likely to improve natural history?"	→ **26° MT, Sanders 3**

– – Active space – –

"Braced 28° → 11° curve, now Risser 4 — next?"	→ **discontinue brace** ✗ Wrong options: continue full-time/ night · PSF · MRI
"Which curve suits a TLSO (Risser 0)?"	→ **30°, apex T12** ✗ Wrong options: apex T6 → Milwaukee · 15° · 53°
"Lenke classification includes all EXCEPT?"	→ **thoracic modifier (A, B, C)** (TO)
"Double thoracic curve = ?"	→ **Lenke type 2**: structural MT (major) + PT, non-structural TL (TO)
"Lenke 1B, 65° — management?"	→ **PSF of thoracic curve only**
"AVT 45 mm / 15 mm?"	→ **ratio 3.0 → selective thoracic fusion**
"LIV relative to touched vertebra?"	→ **to or caudad to TV** ✗ Wrong options: cephalad → truncal shift
"Anterior single-rod + cages — pseudarthrosis risk?"	→ **thoracic hyperkyphosis > 40°**
"Commonest complication of PSO?"	→ **pseudarthrosis** ✗ Wrong options: neuro deficit · aortic injury
"Undertapping 1 mm + straightforward trajectory?"	→ **↑ maximal insertional torque**
"MEP ↓ 60%, SSEP ↓ 30%, anaesthesia unchanged — immediate step?"	→ **reposition patient** ✗ Wrong options: deepen anaesthesia · remove screw · wake-up test · abort
"Reduces post-op opioid use?"	→ **short-course adjunctive steroids**
"Untreated AIS, 38° at maturity — counsel?"	→ **↑ lifetime chronic back pain** ✗ Wrong options: caesarean · pulmonary decline
"Bilious vomiting day 5 after PSF in a thin girl?"	→ **SMA syndrome → NG tube + IV fluids**
"Late pain, clean wound, months after PSF?"	→ **P. acnes infection** (hold cultures 2 weeks)

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ADULT ISTHMIC SPONDYLOLISTHESIS

– Spine · 2 / 69 –

1 Basics

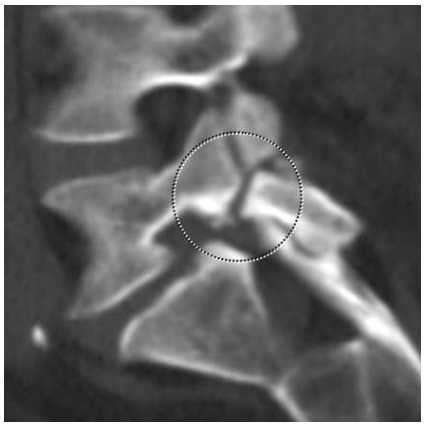
Pathogenesis, prevalence & classification

Definition	forward slip of a vertebra on the one below due to a **pars interarticularis defect** (spondylolysis)
Pars	thin bicortical bridge where **lamina + inferior articular process** meet **pedicle + superior articular process**; defect = fibrocartilaginous cleft
Prevalence	spondylolysis **4–6%** of North American adults · 4.4% at age 6 → 6% adult (Beutler) ✗ Wrong options: < 2% · 8–10%
Sex	**male** (2/3 of defects); **slip progression** commoner in **females**
Level	**L5-S1 82%** (Orthobullets) · 86–90% (OB) · L4-5 11% ✗ Wrong options: thoracolumbar junction
Why L5-S1	highest lumbar forces + facet orientation: "**more sagittal**" (Orthobullets) · "**more coronal**" (OB) – read the options
Sports	**repetitive hyperextension**: gymnasts, weightlifters, football linemen · high incidence in **Inuits** (genetic) ✗ Wrong options: hyperflexion
Natural history	only **5%** of spondylolysis → spondylolisthesis · progression mostly in **adolescence**, rare after maturity ✗ Wrong options: adults progress more

Wiltse-Newman classification

Type	Name	Lesion
I	**Dysplastic**	Congenital L5 arch / upper sacrum anomaly
IIA	**Isthmic – lytic**	**Pars fatigue (stress) fracture** – commonest adult
IIB	Isthmic	**Elongated intact pars** (healed stress fractures)
IIC	Isthmic	**Acute pars fracture**
III	**Degenerative**	Facet instability, **pars intact** (L4-5)
IV	**Traumatic**	Acute fracture of arch **other than pars** (e.g. pedicle)
V	**Pathologic**	Local / generalised bone disease

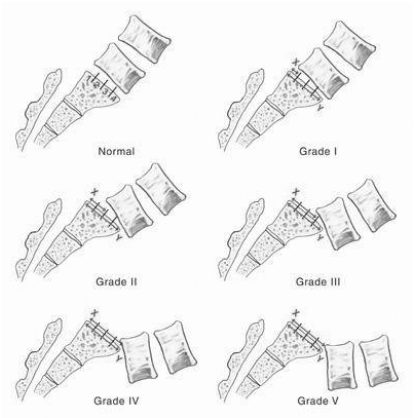
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Isthmic spondylolisthesis: pars defect (Wiltse II)

Meyerding grade

Grade	Slip	Group
I	< 25%	Low grade
II	25–50%	Low grade
III	50–75%	**High grade**
IV	75–100%	High grade
V	> 100%	**Spondyloptosis**



Meyerding grading

2 Mechanism

Foraminal stenosis (main cause of leg pain)	**exiting L5 root** in L5-S1 foramen compressed by **hypertrophic fibrocartilage of pars defect** · **uncinate spur** of posteroinferior L5 body · bulging L5-S1 disc · facet hypertrophy
Lateral recess stenosis	facet arthrosis + hypertrophic ligamentum flavum
Central stenosis = rare	posterior arch stays behind as body slips → canal **enlarges**; only in spondyloptosis ✗ Wrong options: "central stenosis compresses traversing root"
Disc	bilateral pars defect → ↑ shear across disc → early degeneration → pain
Nutcracker vs shear (TO)	**low PI + low SS** → L5 posterior elements **clamped between L4 and S1** in extension = **nutcracker** (low progression risk) · **high PI + high SS** → ↑ **shear** on L5 pars

3 Presentation

Axial back pain	**commonest**; long history, episodic
Leg pain	**L5 radiculopathy** (foraminal) → **EHL + ankle dorsiflexion weakness**, dorsum-of-foot dysaesthesia, hip abductor weakness ✗ Wrong options: FHL · quadriceps · gastrocsoleus · iliopsoas
Tight hamstrings	common accompanying sign
Neurogenic claudication / cauda equina	**rare** (slips rarely exceed grade II)
Root logic traps	central L3-4 / far-lateral L4-5 HNP → **L4** · central L5-S1 HNP → **S1** (plantar flexion, lateral foot) · isthmic L3-4 → not L5

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Lateral X-ray	Slip + pars defect (diagnosis)
Oblique X-ray	Pars defect ("Scottie dog collar")
Flexion-extension	**Instability = 4 mm translation or 10° angulation** vs adjacent segment
AP X-ray	Coronal deformity · **inverted Napoleon hat sign** = **marked anterolisthesis L5 on S1** (TO) (✗ Wrong options: retrolisthesis · facet dislocation · sacralisation)
CT	Best bony pars defect
MRI – T2 parasagittal	**Best for foraminal stenosis** / neural compression

Pelvic incidence & spinopelvic parameters

Pelvic incidence (PI)	angle between **perpendicular to S1 endplate at its midpoint** and line from **S1 endplate midpoint → femoral head centre**
PI = PT + SS	PT = PI − SS · SS = PI − PT

Parameter	Measured	Type
Pelvic incidence	Perpendicular to S1 endplate vs line to hip axis	**Morphologic – fixed**, same standing/sitting/lying (TO)
Pelvic tilt	**Vertical** through hip axis vs line to S1 midpoint	Positional
Sacral slope	S1 endplate vs **horizontal**	Positional
Slip angle	L5 **inferior** endplate vs perpendicular to posterior sacral cortex	Local L5-S1 kyphosis; **> 30°** predicts progression
Lumbosacral angle	L5 **upper** endplate vs perpendicular to upper sacrum	**> 10°** predicts progression
Lumbar index	Anterior ÷ posterior L5 height	L5 wedging

– – Active space – –

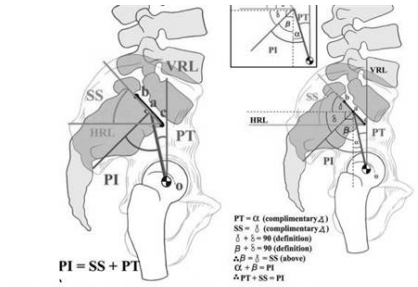


Figure 4. **A**, the association between PI, SS, and PT. PI, which describes the shape of the pelvic unit, is linked to the orientation (SS and PT) of the pelvis in space since it can be demonstrated that PI = SS + PT. Depending on the shape and orientation of the sacropelvic foundation, the lumbosacral junction will experience a different combination of normal and shear forces. **B**, the "geometric proof" demonstrates the mathematical association of PI, PT, and SS, where a change in one parameter affects the other measurements and the overall alignment of the sacropelvic foundation.

PI = PT + SS

PI values	normal **55° ± 10°** (TO) · ~47° childhood → ↑ in adolescence → ~57° adult, then constant ✗ Wrong options: "PI greater sitting than standing"
PI correlates with	**severity / Meyerding grade** of slip (direct linear) · determines SS, PT, **lumbar lordosis** · correlates with **sacral slope** (TO)
Pelvic tilt (TO)	varies with position · opens at the back · mean **13° ± 6°** standing · ↑ PT → sacral plateau becomes more **horizontal** ✗ Wrong options: "more vertical"
Balanced pelvis (TO)	**high SS + low PT** · unbalanced = **low SS + high PT** (retroverted pelvis, vertical sacrum) · PT is the better indicator (SS changes secondary to PT)
Slip progression predictor	**slip angle** (Orthobullets, OB – most predictive of progression + surgical outcome) · **pelvic incidence** (TO answer) – read the options
Global balance	SVA (C7 plumb vs posterosuperior S1); **SVA > 50 mm** → worse outcome after decompression-fusion

SDSG classification (L5-S1)

Grade	Type	Defining feature
Low (< 50%)	**1 – Nutcracker**	**PI < 45°**
Low	2	PI 45–60°
Low	**3 – Shear**	**PI > 60°** → prone to progress
High (> 50%)	4	**Balanced pelvis** (high SS, low PT)
High	5	Retroverted pelvis, **balanced spine** (C7 plumb on/behind hips)
High	6	Retroverted pelvis, **unbalanced spine**

High-grade slips mean **PI > 60°**

5 Management

First line activity restriction, **NSAIDs**, PT; bracing in acute phase; injections unclear · acute pars defect after fall in adult → **observe + mobilise**
✗ Wrong options: cast · TLSO 6–8 wks · ORIF · fusion

Operative criteria

Indication for decompression + fusion
Incapacitating pain failing 6 months non-op (commonest)
Progressive neurologic deficit
Slip progression
Cauda equina syndrome

Isthmic spondylolisthesis failing 6 months non-op

- Low grade (I–II) → **L5-S1 DECOMPRESSION + INSTRUMENTED FUSION**
  (Gill laminectomy + foraminal decompression must be fused)
- High grade (III–V) → **L4-S1 DECOMPRESSION + FUSION + ANTERIOR COLUMN SUPPORT**
  - Reduction? → **PARTIAL (correct slip angle)**

Surgery – decompression & fusion; ALIF vs TLIF

Gill laminectomy — removes loose lamina + scarred pars → decompresses root; **destabilising** → always fuse
✗ Wrong options: laminectomy alone → further slip · decompression alone in high grade = poor

Pars repair — not if > grade I slip

Contraindicated — **stand-alone ALIF / stand-alone PLIF** (high failure) · **total disc replacement** (not for isthmic slip)

Recurrent L5 radiculopathy after L5-S1 discectomy in isthmic slip — **posterior foraminal decompression + instrumented fusion + graft**
✗ Wrong options: revision hemilaminotomy · TDA · Gill alone

Best technique — **surgeon preference; outcomes similar**
✗ Wrong options: PSF highest fusion · reduction lowest L5 injury · TLIF lowest dural tear (TLIF ↑ root injury + dural tears)

	ALIF (+ posterior fixation)	TLIF
Approach	Retroperitoneal; ALL released	Posterior; facetectomy, lateral to dura
Decompression	**Indirect** (disc height → foraminal distraction)	**Direct** visualisation of roots
Disc height / segmental lordosis	**Greater** (✓)	Less
Fusion rate	Similar (some ↑)	Similar
1-yr pain / overall complications	Similar	Similar
Specific risks	**Retrograde ejaculation** (superior hypogastric plexus), retroperitoneal haematoma, persistent radiculopathy, pars-defect pain fibres left behind	Dural tear, root injury
Use	**Low-grade** slips, even with radicular pain (87–94% good at 2 yrs) · not high grade	Any

– – Active space – –

PLIF vs TLIF PLIF cage **medial to facets** · TLIF = facetectomy, more lateral · interbody ↑ time, blood loss, stay

Reduction – L5 nerve root injury

Root at risk with reduction **L5** (exits under L5 pedicle; S1 traverses) → post-op **EHL / great-toe extension weakness**
✗ Wrong options: S1 plantar flexion · L2–3 hip flexion · L4 patellar reflex · genitofemoral (anterior approach) · cauda equina

Strain biomechanics **71% of L5 strain in the second half** of complete reduction (non-linear after > 50%) → **partial reduction** · ↑ disc height and ↑ lordosis protect L5

Reduction goal correct **slip angle / lumbosacral kyphosis** first; slip % second

High-grade slip (SDSG)

- Balanced pelvis (type 4) → **IN-SITU FUSION**
- Retroverted pelvis, spine clinically balanced (type 5) → **PARTIAL REDUCTION + FUSION**
- Retroverted pelvis + sagittal imbalance / stooping (type 6) → **REDUCTION MANDATORY (360°)**

6 Complications

Complication	Note
L5 root stretch	Reduction
Pseudarthrosis	–
Dural tear	↑ with TLIF
Sacral insufficiency fracture	Constructs **ending at S1** without pelvic fixation
Retrograde ejaculation	ALIF

X vs Y

	Isthmic	Degenerative
Level	**L5-S1**	**L4-5**
Pars	**Defect**	Intact (facet arthrosis)
Age / sex	Young, **male**	Older, **female**
Root	**Exiting L5** (foraminal)	Traversing L5 (lateral recess / central)
Grade	Can be high	Rarely > grade I
Canal	**Enlarged**	Stenotic

	Nutcracker	Shear
PI / SS	**Low / low**	**High / high**
SDSG	Type 1	Type 3
Progression	Low	High

How it's asked

"Inverted Napoleon hat sign on AP indicates?"	→ **anterolisthesis of L5 on S1** (TO)
"Cause of this spondylolisthesis (pars fracture on CT)?"	→ **fatigue fracture of pars** ✗ Wrong options: congenital arch · degenerative · traumatic non-pars
"Commonest level + sex for pars defect?"	→ **L5-S1, male**
"Prevalence of spondylolysis in adults?"	→ **4–6%**
"Lacrosse player, bilateral pars defect — true?"	→ **foraminal stenosis from fibrocartilage** ✗ Wrong options: central stenosis · TL junction · adults progress more · fuse first
"Grade I isthmic L5-S1 — weakness of?"	→ **EHL**
"Great-toe extension weakness + tight hamstrings in a 33-yr basketballer?"	→ **grade 1 isthmic L5-S1**
"True about isthmic spondylolisthesis?"	→ **caused by pars defect** ✗ Wrong options: hyperflexion · L4-5 · no PI correlation · adults progress more
"Balanced pelvis correlates with?"	→ **low pelvic tilt** (TO)
"Pelvic incidence correlates with?"	→ **sacral slope** (TO)
"Progression in spondylolisthesis correlates with?"	→ **pelvic incidence** (TO) · **slip angle** (OB image question) — read the options
"Pelvic tilt — all true EXCEPT?"	→ **"↑ PT makes sacral plateau more vertical"** (TO)
"Pelvic incidence — NOT true?"	→ **"greater sitting than standing"** (TO)
"Nutcracker type = ?"	→ **low PI + low SS** (TO)
"Parameter correlating with severity?"	→ **PT + SS (= PI)**
"Which angle predicts slip progression and surgical outcome?"	→ **slip angle** (L5 inferior endplate vs perpendicular to posterior sacrum)
"Grade 3, retroverted pelvis, clinically balanced spine — treatment?"	→ **partial reduction + fusion** (TO) ✗ Wrong options: complete reduction · in-situ fusion · conservative
"Failed non-op high-grade slip — surgery?"	→ **decompression + L4-S1 fusion + anterior column support**
"Grade 2 isthmic with 3/5 dorsiflexion — surgery?"	→ **L5 laminectomy + fusion** ✗ Wrong options: pars repair · laminectomy alone · stand-alone ALIF · TDR
"Best technique for high-grade isthmic slip in gymnast?"	→ **surgeon preference; similar outcomes**

– – Active space – –

"Advantage of ALIF + PF over TLIF?"	**→ greater sagittal (segmental lordosis) correction · greater disc-height restoration** ✗ Wrong options: lower sexual dysfunction · better fusion · direct root view
"Deficit after reduction + fusion of L5-S1 slip?"	**→ L5 root → great-toe extension weakness**
"Most L5 strain during reduction occurs?"	**→ second half of complete reduction (71%)**

ADULT PYOGENIC VERTEBRAL OSTEOMYELITIS

– Spine · 3 / 69 –

1 Basics

Epidemiology – lumbar spine most common

Spectrum (spondylodiscitis) discitis · vertebral osteomyelitis · epidural abscess

Age median **50–60 yrs**

Region	Share
Lumbar	**50–60%**
Thoracic	30–40%
Cervical	**~10%** (Orthobullets) · 20% (OB)
Sacrum	Rare

Order **lumbar > thoracic > cervical > sacrum**
✗ Wrong options: lumbar = thoracic · cervical first

Risk factors **IV drug use** · **diabetes** · **recent systemic infection (UTI, pneumonia)** · malignancy · immunosuppression · obesity · **malnutrition (albumin < 3 g/dL)** · trauma · smoking

Organisms

Organism	Setting
Staph aureus	**Commonest (50–65%)**
Staph epidermidis	2nd commonest
Gram-negatives	↑ last decade; GU + respiratory sources
Pseudomonas	**IV drug users**
Salmonella	**Sickle cell disease**
Fungal · TB	Immunocompromised

Commonest fungal infection of spine (TO) **Aspergillus**; Candida + Aspergillus = commensals entering via IV lines/implants/surgery · ↑ during COVID (steroid overuse) · Coccidioides + Blastomyces = geographic
✗ Wrong options: Candida · Cryptococcus · Blastomyces

2 Mechanism

Haematogenous seeding (commonest) arterial or venous → **endplate** (low-flow vascular anastomosis) → **disc** → adjacent endplate

Direct inoculation penetrating trauma, open fracture, surgery

– – Active space – –

Contiguous spread	retropharyngeal + retroperitoneal abscess
Neurologic deficit (10–20%)	direct neural infection · **epidural abscess** compression · **instability**
Epidural abscess	pus/granulation between dura and epidural fat · in **~18%** of spondylodiscitis · **50%** of abscess patients have neuro symptoms
Psoas abscess	associated

3 Presentation

Pain	severe, **insidious**, unrelenting, worse with activity; **night pain** → infection / malignancy
Fever	**only 33%**
History	UTI, pneumonia, skin infection, transplant
Neuro	radiculopathy / myelopathy in 10–20% → careful neuro exam

4 Diagnosis & Classification

Genes & lab markers

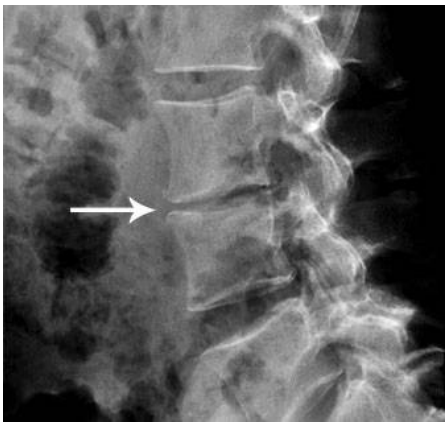
Test	Fact
WBC	↑ in only **~50%** → insensitive
ESR	↑ in **90%**; serial monitoring, less reliable than CRP
CRP	↑ in **90%**; **best for tracking treatment response**
Blood cultures	**All patients before antibiotics**; least invasive; positive **~33% (25–66%)**; **85% accurate** when positive; ↑ yield off antibiotics + during fever
CT-guided biopsy	If blood cultures negative + no urgent surgery; diagnostic **68–86%** (Orthobullets) · needle ~70%, open > 80% (OB); send **aerobic, anaerobic, fungal, AFB**
Open biopsy	Non-invasive fails; anterior, costotransversectomy, transpedicular

Post-discectomy: severe pain + spasm, no fever, normal TLC + ESR, CRP very high (TO)

infective discitis
✗ Wrong options: instability · pseudomeningocele · residual disc

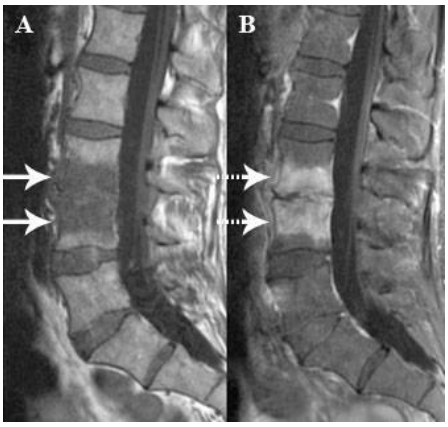
Key radiology findings

Modality	Findings
X-ray	Delayed by weeks · **loss of psoas shadow** (first days) · **disc space narrowing at 7–10 days** · **endplate erosion/sclerosis at 10–21 days** · local osteopenia
CT	Bony destruction, abscess extent
MRI with gadolinium	**Gold standard**; **most sensitive (96%)** + specific (93%); **best to separate infection from tumour**; early scan may mimic degeneration → repeat
Tc-99m bone scan	If MRI impossible; **90% sensitive, not specific**; + gallium-67 → ↑ sensitivity + specificity
Indium-111	Poor sensitivity (**17%**)



Disc space narrowing with endplate erosion on lateral X-ray

Classic discitis MRI triad — **T2 bright disc** (fluid/pus) · **T1 dark**, disc–body margin lost · **T1 fat-sat + gad: rim enhancement** of abscess + adjacent bodies → pathognomonic
✗ Wrong options: lymphoma · primary epidural abscess · DDD · early AS



Spondylodiscitis on T2 and T1 MRI

MRI limit — less useful than **labs** for treatment response

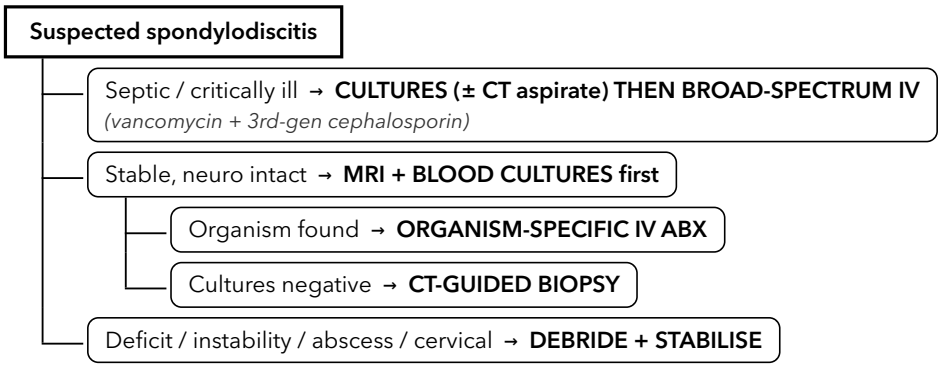
Infection vs tumour — infection = **disc involved**, endplate erosion, marked inflammation · tumour spares disc

Discitis vs Modic I — Modic I: T1 dark, T2 bright marrow but **disc desiccated (normal / dark T2)**, no endplate erosion, no soft-tissue mass, normal ESR/CRP

– – Active space – –

Pyogenic vs TB on MRI (TO)	collection with **relatively preserved disc → pyogenic** (acute, presents early) · **collapsed disc**, subligamentous spread → TB (late presentation) · end result of spinal TB = **bony ankylosis** (weight-bearing + cancellous surfaces) ✗ Wrong options: fibrous ankylosis as in limb joints

5 Management



Diagnosis – MRI, blood cultures & CT-guided biopsy

Organism ID is mandatory before targeted therapy · stable patient → **hold antibiotics** until cultures/biopsy
✗ Wrong options: empirical IV antibiotics · INH-RIF-PZA before biopsy in suspected TB

IVDU with UTI, back pain, ↑ ESR	**admit + blood cultures + MRI with/without gad** ✗ Wrong options: oral abx · bone scan · renal US
Blood cultures negative, neuro intact	**CT-guided closed biopsy** ✗ Wrong options: IV abx · open surgery · repeat MRI
MSSA in blood cultures	**organism-specific IV antibiotics** ✗ Wrong options: CT biopsy · debridement

Treatment – organism-specific antibiotics

Course	**6–12 weeks** ± bracing · IV until improvement (~**4–6 wks**) → oral · success **80%**
Bracing	pain + deformity prevention · **cervical → rigid cervicothoracic orthosis or halo**
Resistant organisms	MRSA, VRSA, VRE → **linezolid, daptomycin**

Infection	Agent
Gram-positive	Cefazolin, vancomycin
Gram-negative	Fluoroquinolone (ciprofloxacin), 3rd-gen cephalosporin
Fungal	**Amphotericin B** first line (± azoles)
TB	**INH + rifampicin + PZA + ethambutol**

Operative criteria

Debridement + stabilisation indicated
Cervical osteomyelitis / cervical epidural abscess
Progressive neurologic deficit
Progressive deformity / gross instability
Refractory to medical therapy
Large abscess

Open biopsy alone cultures + CT biopsy fail; lumbar disease without canal abscess

Surgical technique

Gold standard **anterior debridement + strut graft ± posterior instrumentation** · posterior debridement alone usually inadequate

Graft autograft (tricortical iliac crest, rib, fibula) safe in acute infection · allograft OK · **titanium mesh cage + autograft + posterior instrumentation** → best deformity correction

Instrumentation in active infection controversial (staged vs single stage) · **titanium > stainless steel**

Posterior instrumentation severe kyphosis or multilevel anterior construct

Two-level cervical osteomyelitis + rapidly progressive gait loss + ↑ WBC/ CRP/ESR **anterior corpectomy + posterior instrumentation (A + P)** · cervical abscess → early surgery (little space for cord) · lumbar abscess → often antibiotics alone

X vs Y

	Pyogenic	Tuberculous
Onset	Acute, painful, early	Insidious, late
Disc (TO)	**Relatively preserved** at presentation	**Collapsed / destroyed**
Spread	Endplate → disc → endplate	Subligamentous, skip lesions
End result	–	**Bony ankylosis**

	Infection	Tumour
Disc	**Involved**	Spared
Endplate	Eroded	Preserved
Best test	**MRI + gad**	MRI

How it's asked

"Spondylodiscitis with collection but preserved disc?" → **pyogenic** (TO)
✗ Wrong options: TB

– – Active space – –

"End result of spinal TB?"	→ **bony ankylosis** (TO)
"IVDU, back pain, negative blood cultures, neuro intact – initial management?"	→ **CT-guided closed biopsy**
"IVDU, UTI, focal back pain, ↑ ESR – next step?"	→ **admission + blood cultures + MRI with/without gadolinium**
"Spondylodiscitis on MRI in a diabetic, no deficit – plan?"	→ **needle biopsy + IV antibiotics + TLSO** ✗ Wrong options: Modic I · antibiotics before biopsy · anterior surgery
"HIV-positive man, suspected TB spondylitis, cultures negative?"	→ **CT-guided biopsy + cultures** ✗ Wrong options: start anti-TB drugs · broad antibiotics
"IVDU, thoracic discitis, no sepsis or deficit – first step?"	→ **blood cultures** ✗ Wrong options: start IV antibiotics · corpectomy
"T2 bright disc, T1 loss of margins, rim enhancement on gad – pathognomonic for?"	→ **discitis**
"Stable 17-yr with discitis after pyelonephritis – next step?"	→ **MRI with contrast + blood cultures** ✗ Wrong options: aspiration first · start broad abx
"MSSA in 2 blood cultures, no deficit – next?"	→ **organism-specific IV antibiotics**
"Commonest fungal infection of spine?"	→ **Aspergillosis** (TO)
"Amphotericin B used for?"	→ **fungal osteomyelitis**
"Order of spinal involvement?"	→ **lumbar > thoracic > cervical > sacrum**
"Which scenario needs anterior + posterior fusion?"	→ **diabetic with 2-level cervical osteomyelitis + rapid gait loss**
"Post-discectomy pain, normal ESR + TLC, high CRP, afebrile?"	→ **infective discitis** (TO)
"Best lab to follow treatment response?"	→ **CRP**
"Most sensitive and specific imaging?"	→ **MRI with gadolinium** ✗ Wrong options: indium scan (17% sensitive)
"Organism in sickle cell / IVDU?"	→ **Salmonella / Pseudomonas**

– – Active space – –

ADULT SPINAL DEFORMITY

– Spine · 4 / 69 –

1 Basics

Epidemiology & types

Demographics	mean age **60**; M = F
Coronal imbalance	lateral deviation of vertical line **> 10°**
Sagittal imbalance	radiographic **> 5 cm** (C7 plumb vs posterosuperior S1)
Degenerative pathoanatomy	**asymmetric degeneration** of discs ± facets
Loss of sagittal balance	osteoporosis · pre-existing scoliosis · iatrogenic instability · DDD

	Idiopathic (residual)	**Degenerative (de novo)**
Origin	Untreated AIS	Degeneration, iatrogenic, paralysis
Curve pattern	**Classic** patterns	Lacks classic pattern
Segments	**More**	Fewer
Location	**Thoracic**	**Lumbar**
Magnitude	Larger	Smaller

Natural history & progression

Progression by curve	thoracic > lumbar > thoracolumbar > double major · right thoracic **1°/yr** · right lumbar **0.5°/yr** · TL **0.25°/yr**
By magnitude	**< 30° rarely progress · > 50° commonly progress**
Other risks	**intercrestal line below L4-5** · pre-existing rotation
Symptoms progress to	side of curve **convexity**

3 Presentation

Low back pain	**commonest (40–90%)**; spondylosis, instability, discogenic; more severe/ recurrent than general population
Neurogenic claudication	stenosis on **concave side**; no relief with sitting / flexion (unlike classic claudication)
Radicular pain	foraminal + lateral recess stenosis, worse in **concavity** (rotation + translation)
Exam	thoracic prominence on forward bend, compensatory posture, weakness

4 Diagnosis & Classification

Key radiology findings

Study	Use
36-inch standing AP + lateral + R/L bending	Cobb, coronal (C7PL vs CSVL) + sagittal balance, flexibility
CT	Bony deformity, facet arthrosis
CT myelogram	**Best for stenosis + bony anatomy** when rotation makes MRI hard
MRI	When leg pain present: stenosis, facet hypertrophy, foraminal encroachment
DEXA	Bone density for planning

Sagittal balance – SVA, PI-LL mismatch, pelvic parameters

Sagittal vertical axis (SVA)
horizontal distance **C7 plumb line → posterosuperior corner of S1**; neutral = through it · positive = anterior · negative = posterior

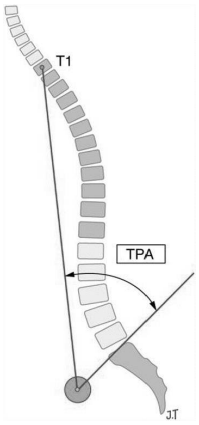
Strongest predictor of symptoms / HRQOL / PROMs / poor surgical result
sagittal balance (C7 SVA)
✗ Wrong options: coronal imbalance · Cobb · levels · apex level · residual scoliosis > 25° · L5-S1 disc left out · truncal shift · PI

Radiographic parameter correlating with pain + disability (VAS, ODI)
thoracolumbar kyphosis (✗ Wrong options: SS · TK · PT · PI) · TL/lumbar curves also correlate (✗ Wrong options: thoracic scoliosis · thoracic hypokyphosis · lumbar hyperlordosis · disc degeneration)

Pelvic incidence
only permanent anatomic parameter; **PI = PT + SS**; not posture dependent (TO); stabilises after adolescence; determines lordosis needed

Compensation for positive SVA
pelvic retroversion (↑ PT, ↓ SS) · thoracic hypokyphosis · cervical hyperlordosis · knee flexion → SVA may look normal (< 5 cm) → look for regional imbalance

T1 pelvic angle (T1PA)
angle T1 centre → femoral head → S1 endplate centre = **T1 spinal inclination + PT**; accounts for **spinal inclination + pelvic retroversion**; no calibration needed; usable intra-op · goal **< 14°** · **> 20° ≈ ODI 40** (severe) · ↑ T1PA = ↑ ODI



T1 pelvic angle

Diagnostic criteria – surgical alignment targets

Target	Value
SVA	**< 5 cm** (best reliable improvement correcting to **+3 cm**)
PI – LL	**LL = PI ± 9°** (PI – LL < 10°) (✗ Wrong options: within 15° · LL = PI + 25° · LL = TK + 15°)
Pelvic tilt	**< 20°** (OB) · < 25° (alternate goal)
T1PA	**< 14°**
TK + LL + PI	**≤ 45°** – most predictive of maintaining correction
GAP score	**< 3** → fewer mechanical failures / revisions

Normal lumbar lordosis 40–60° (~70% from L4-S1; apex L3-4 / L4)

5 Management

Non-op coronal curves **< 30°** rarely progress · NSAIDs · **TCAs for sleep** · PT core strengthening · injections / root blocks (diagnostic + therapeutic) · bracing (comfort)

Operative criteria

Indication for instrumented fusion
Curve **> 50°**
Sagittal imbalance
Curve progression
Intractable back / radicular pain failing non-op
Cardiopulmonary decline: thoracic **> 60° → PFTs ↓; > 90° → mortality ↑**
Cosmesis (controversial)

ASD surgical approach

- Thoracic / double structural > 50° → **POSTERIOR-ONLY FUSION**
- Isolated TL / lumbar, > 70°, rigid, L5-S1 interbody → **ANTERIOR RELEASE + POSTERIOR**

Goals restore **spinal balance** · relieve pain · solid fusion

Worse outcomes **baseline depression** · **obesity** · ≥ 4 comorbidities · insufficient sagittal correction · pseudarthrosis
✗ Wrong options: ethnicity · pre-op ODI · pre-op SVA if corrected

– – Active space – –

Fusion level selection

Proximal extend to a **neutral + horizontal vertebra above main curve**

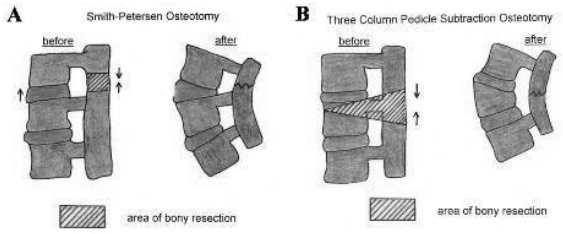
Distal level	Indication / outcome
L5	Only if **no L5-S1 pathology**; lowest future L5-S1 degeneration with **normal C7 plumb + normal sacral inclination**
S1	**Any L5-S1 pathology**: spondylolisthesis, spondylolysis, facet arthrosis, prior laminectomy · ✓ **better correction + maintenance of sagittal balance** · (✗ Wrong options: ↑ pseudarthrosis, OR time, reoperation, **sacral insufficiency fracture**, altered gait)
Ilium (sacropelvic)	Sacrum in fusion of **> 3 levels**; iliac screws/bolts → **↑ fusion success** at L5-S1 + **↓ sacral fractures** in osteopenic; prominent hardware

Maximise fusion rate **ilium + sacrum** both in distal fixation
✗ Wrong options: ilium alone · sacrum alone · coronal balance

Pubic ramus fractures occur with sacral **and** iliac fixation

Osteotomies – correction per level

Osteotomy	Correction	Indication / prerequisite
Smith-Petersen (SPO) / Ponte	**~10° per level** (Orthobullets) · posterior column osteotomy **3–5°** (OB)	Mild–moderate imbalance; needs **mobile disc + disc height**, **no anterior fusion**; more in lumbar
Pedicle subtraction (PSO)	**30–35° lumbar**, 25° thoracic (Orthobullets) · **25–35°** (OB)	**SVA > 12 cm** (Orthobullets) · **> 10 cm** (TO); **anterior fusion present** (correction through body)
Vertebral column resection (VCR)	**Up to / > 45°** (avg 62° coronal, 45° sagittal)	**Rigid sharp angular** kyphosis (tumour, fracture, infection), rigid scoliosis, congenital kyphosis, hemivertebra
TLIF	~8°	–
TLIF + ALL release + hyperlordotic cage	10–15° (22° with PCOs)	–



Osteotomy types for sagittal correction

3 SPO ≈ 1 PSO PSO more blood loss · neuromonitoring preferred

PSO level **L3 / L4** (apex of lordosis L3-4; 70% of LL from L4-S1)
✗ Wrong options: L1 → non-anatomic · SPOs above apex · thoracic SPOs ↓ TK

Flat back after prior lateral interbody / L3-S1 fusion **L3 PSO + extension to pelvis**
✗ Wrong options: SPOs through fused discs · T12 PSO · VCR

Bone health optimisation

Osteoporosis (T < −2.5) or ≥ 2 vertebral compression fractures (= osteoporosis without DEXA)	**teriparatide** before surgery (≥ **6 months** → ↓ pseudarthrosis + reoperation) ✗ Wrong options: continue/start alendronate (impairs fusion) · vit D/Ca alone · denosumab switch · repeat DEXA
Osteopenia (T −1.7)	no anabolic referral needed
Cement (PMMA) augmentation of end vertebrae	↑ **fusion rate** + ↓ **junctional failure** + ↑ pullout strength, no extra complications · osteoporotic spine: **≥ 3 fixation points above + below** ✗ Wrong options: fewer levels fused
Active smoker	**smoking cessation clinic + nicotine testing**, postpone elective fusion ✗ Wrong options: schedule now
Smoking	↑ pseudarthrosis, infection, implant loosening, fracture risk; **cessation programmes improve outcomes** (EXCEPT answer: "cessation doesn't help" is FALSE)

Perioperative considerations

Neuromonitoring (SSEP ↓ 60% at correction)	**stop instrumenting + raise MAP ≥ 85 mmHg**, recheck, then reverse correction ✗ Wrong options: wake-up test first · MAP 60 · steroids first · inhalational only (reduce inhalational agents)
Neuro deficit on monitoring after correction	remove instrumentation, consider wake-up test
Jehovah's Witness	**closed-circuit cell saver** ✗ Wrong options: whole blood, PRBC, plasma, platelets
Open thoracotomy for anterior release (adult)	**pulmonary function ↓ and often stays ↓ long term**
Pedicle anatomy	cortical:cancellous ratio **thoracic pedicles > lumbar pedicles > vertebral bodies > sacrum** (implant failure likeliest in sacrum) · lumbosacral pedicle **diameter + medial angulation ↑ caudally** (L3 more medial than L1; S1 > L1) · **L5 root on sacral ala** (at risk with S-I screws) · S1 screw start **inferior + lateral** to L5 inferior articular process tip
Posterior instrumentation system (TO)	**Cotrel-Dubousset** (rods + hooks/screws, posterior) ✗ Wrong options: Dwyer · Zielke (anterior) · Boston brace

– – Active space – –

6 Complications

Complication	Rate / fact
Overall	**~13.5%**; ↑ with osteotomies, revisions, A/P
Major	**~10% (10–20%)**; irreversible; **VTE** → worst outcomes
Strongest predictor of major complication	**Age > 60** (✗ Wrong options: smoking · BMI > 30 · PSO · comorbidities)
Pseudarthrosis	**5–25%**; posterior-only 15%; **L5-S1 + TL junction**; **commonest reason for revision within 3 yrs**
Implant failure / rod fracture	**Commonest cause of reoperation**; rod breakage L4-S1 from **large PI-LL mismatch** (✗ Wrong options: titanium rods · coronal balance)
Proximal junctional failure	2nd commonest revision; within 6–9 months
Dural tear	2.9%
Infection	Deep 1.5%, superficial 0.9%; **smoking** strongest SSI factor (✗ Wrong options: drains · drain output · prolonged abx)
Neuro deficit	Acute ~1%, delayed 0.5% · **after PSO 18%** (Orthobullets) · 11%, 2.8% permanent (OB) · ~20%
PSO pseudarthrosis	**29% at 5 yrs**, at TL junction above osteotomy
Epidural haematoma / PE / DVT	~0.2% each · death 0.3%

Pseudarthrosis risks — **age > 55** · kyphosis > 20° · **positive SVA > 5 cm** · hip OA · smoking · **thoracoabdominal approach** · incomplete lumbopelvic fixation · osteoporosis · TL Cobb > 20°
✗ Wrong options: posterior midline · Cobb 60°

Post-PSO quadriceps weakness — screw malposition (30% of deficits → CT), large PI-LL correction, prone positioning (femoral nerve), stroke · **least likely = apical derotation** (coronal manoeuvre)

Post-op neuro decline + fluid collection at 3 months — **I&D, retain hardware**
✗ Wrong options: observation · antibiotics only · remove hardware

X vs Y

	SPO / Ponte	PSO	VCR
Columns	Posterior	3-column wedge (hinge anterior)	All 3 removed
Correction	~10°/level	25–35°	> 45°
Needs mobile disc	**Yes**	No	No
Prior anterior fusion	No	Yes	Yes
Commonest complication	–	**Pseudarthrosis**	High blood loss, instability

	Fusion to L5	Fusion to S1 + ilium
L5-S1 healthy	✓	–
Sagittal correction	Less	**Better + maintained**
Pseudarthrosis	–	↑ (sacrum alone) · ↓ with iliac fixation

How it's asked

"PI – all true EXCEPT?"	→ **posture dependent** (TO)
"Angle x on lateral (perpendicular to S1 endplate vs line to femoral head)?"	→ **pelvic incidence** (TO)
"75° TL curve, SVA 12 cm – what most reliably ↓ disability?"	→ **correct SVA to +3 cm** ✗ Wrong options: LL within 15° of PI · Cobb < 25° · PT > 20° · stop at L5
"Symptoms in adult scoliosis correlate with?"	→ **sagittal imbalance**
"Which figure correctly measures sagittal imbalance?"	→ **C7 plumb → posterosuperior S1**
"Parameter correlating with VAS + disability?"	→ **thoracolumbar kyphosis**
"Correct which parameter to improve HRQOL?"	→ **C7 SVA**
"Formula for lumbar lordosis correction?"	→ **LL = PI ± 9°**
"Most predictive of poor result after multilevel fusion?"	→ **sagittal imbalance**
"Back pain in adult idiopathic scoliosis correlates with?"	→ **thoracolumbar kyphosis**
"Permanent pelvic parameter guiding lordosis?"	→ **pelvic incidence**
"PI = which two added?"	→ **PT + SS**
"Greatest association with worse PROMs?"	→ **C7 plumb line deviation**
"Best post-op equation (PI 70, LL 45)?"	→ **PI – LL = 7**
"T1 pelvic angle – true?"	→ **accounts for spinal inclination + pelvic retroversion**
"Advantage of fusing to sacrum vs L5?"	→ **better correction + maintenance of sagittal balance**
"Maximise fusion rate?"	→ **include ilium + sacrum**
"Primary reason for iliac fixation?"	→ **improved fusion success**
"Benefit of pelvic fixation in osteopenic long fusion?"	→ **↓ sacral fractures**
"Osteotomy for SVA > 10 cm?"	→ **pedicle subtraction** (TO) ✗ Wrong options: Ponte · SPO
"Commonest complication of lumbar PSO?"	→ **pseudarthrosis** (29% at 5 yrs)
"Correction paired correctly?"	→ **PSO 25–35°** ✗ Wrong options: PCO 15–20° · VCR 35–40° · TLIF 15–20°
"Post lateral interbody fusion flat back?"	→ **L3 PSO + fusion to pelvis**

– – Active space – –

"Prior L3-S1 fusion, LL 22, PI 55?"	→ **PSO at L3**
"Factor ↑ nonunion in adult idiopathic scoliosis surgery?"	→ **thoracoabdominal approach** ✗ Wrong options: age > 35 · SVA < 5 cm
"FALSE about smoking?"	→ **"cessation doesn't change outcomes"**
"Incidence of major complications?"	→ **10–20%**
"Most predictive of major complication?"	→ **age > 60**
"Neuro decline + collection 3 months after TLIF?"	→ **I&D (keep hardware)**
"Least likely cause of quad weakness after PSO?"	→ **apical derotation**
"Factor linked to worse PROMs after ASD surgery?"	→ **obesity**
"Strongest SSI risk factor (smoker, DM, BMI 37, drains)?"	→ **smoking**
"Commonest complication needing revision within 3 yrs?"	→ **pseudarthrosis**
"Broken rod at L5 after lumbopelvic fixation?"	→ **large PI-LL mismatch**
"Most common cause of reoperation — expected finding?"	→ **implant fracture**
"Purpose of PMMA at end vertebrae?"	→ **↑ fusion rate + ↓ junctional failure**
"T-score −3.2 on alendronate, planning T11-pelvis fusion?"	→ **refer for teriparatide**
"Smoker with apixaban + oxycodone, T −1.7?"	→ **smoking cessation + nicotine tests in 6 weeks**
"Two compression fractures, low BMD, pre-op?"	→ **teriparatide before surgery**
"Lowest pseudarthrosis risk?"	→ **teriparatide started 9 months ago**
"Fusion level selection — true?"	→ **proximal to neutral + horizontal vertebra**
"Pedicle anatomy — true?"	→ **thoracic pedicle cortical:cancellous ratio > sacrum**
"SSEP ↓ 60% at correction?"	→ **stop instrumenting, MAP ≥ 85**
"Jehovah's Witness may accept?"	→ **closed-circuit cell saver**
"Thoracotomy effect on adult PFTs?"	→ **↓ and often stays ↓**
"Posterior scoliosis system?"	→ **Cotrel-Dubousset** (TO)

ANKYLOSING SPONDYLITIS

– Spine · 5 / 69 –

1 Basics

Epidemiology

Definition	chronic **seronegative** autoimmune spondyloarthropathy: bridging syndesmophytes, **enthesitis**, sacroiliitis, uveitis
Incidence	~**0.2%** of Caucasians
Sex / age	**4:1 male** · presents **3rd decade** · juvenile form < 16 yrs ✗ Wrong options: more common in females
HLA-B27 carriers	**< 10%** of B27-positive people develop AS · B27 in **~6%** of white population

Pathoanatomy

Enthesitis	inflammation at tendon/ligament insertion → erosion → soft-tissue ossification → **ankylosis**; targets **SI joints**, spinal apophyseal joints, pubic symphysis ✗ Wrong options: synovial process = RA
Disc	annulus inflammation → **syndesmophytes**
Theories	autoimmune reaction to environmental pathogen in a genetically susceptible host; B27 peptide aggregation; cytotoxic T-cells against B27
Spread	SI fusion first → cephalad spinal progression → often **entire spine** ✗ Wrong options: "only pelvis + spine": large joints too

Genes & lab markers

Marker	Fact
HLA-B27	**Class I** MHC antigen, **B locus, chromosome 6**; positive **~90%** (Orthobullets, TO) · ~95% (OB)
B27 prevalence (TO)	**AS 90% · juvenile SpA 70% · RA 6%**
RF	**Negative** (seronegative)
ESR / CRP	Non-specific mild ↑ in active disease (✗ Wrong options: normal)
✗ Traps	COL1A1 = OI · **CD20** = lymphoma · **EXT1** = MHE · **HLA-B8** = DISH + diabetes · HLA-DR3 = SLE · urine phosphoethanolamine = hypophosphatasia

Inheritance	genetic predisposition, mode **unknown**

– – Active space – –

Systemic manifestations

System	Manifestation
Eye	**Acute anterior uveitis / iritis** — commonest extra-articular; confirms diagnosis with B27 + sacroiliitis
Heart	**Conduction abnormalities**
Aorta	Aortitis, stenosis, **regurgitation**
Lung	**Pulmonary fibrosis** (↓ FEV), costovertebral fusion → ↓ chest expansion
Kidney	Amyloidosis
Synovitis	**Klebsiella pneumoniae** (gram-negative bacillus) — B27-positive more susceptible (✗ Wrong options: Strep pneumoniae · gonococcus · C. acnes · Staph aureus)
Other	IBD, psoriasis

Orthopaedic **bilateral sacroiliitis** · progressive **kyphosis** · **cervical fractures** · large-joint arthritis (**hip + shoulder**, asymmetric)

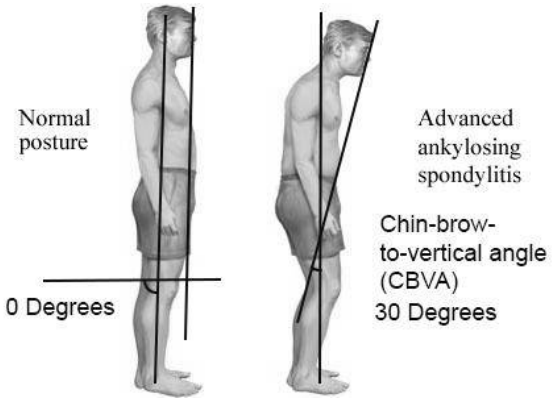
3 Presentation

Lumbosacral pain + stiffness insidious, 3rd decade, **worse in morning**, improves with exercise

Neck / upper thoracic pain later; **acute neck pain = fracture until proven otherwise**

Other sciatica (piriformis), **loss of horizontal gaze**, dyspnoea (costovertebral)

Exam test	Detects
Chest expansion < 2 cm	**More specific than HLA-B27**
Schober test	Lumbar stiffness
Wall (Fleche) test (TO)	**Cervical spine involvement in AS** — occiput can't touch wall
FABER (Patrick)	SI joint pain (✗ Wrong options: FADDIR = FAI)
Chin-brow to vertical angle (CBVA)	Chin-on-chest deformity; **standing lateral**; osteotomy planning — correction ∝ outcome
Supine + sitting exam	Separates **hip flexion contracture** from spinal kyphosis



Chin-brow to vertical angle

Juvenile AS sacroiliitis, spondylitis, **enthesitis** (more than adult), kyphosis, stiffness, peripheral LL arthritis, uveitis · not **urethritis** (= reactive arthritis triad)

– – Active space – –

4 Diagnosis & Classification

Diagnostic criteria

Orthobullets **bilateral sacroiliitis** ± uveitis + **HLA-B27**

Modified New York criteria (TO) HLA-B27 is **NOT** a criterion

Clinical criteria	Radiological criterion
LBP **> 3 months**, better with exercise, not relieved by rest	**Bilateral sacroiliitis grade 2–4** or **unilateral grade 3–4**
↓ lumbar motion in **sagittal + frontal** planes	–
↓ chest expansion for age/sex	–

Definite AS radiological + **≥ 1 clinical** · **Probable** – 3 clinical alone, or radiological alone

SI joint injection often the **most sensitive diagnostic test**

Key radiology findings

Modality	Finding
X-ray spine	**Squaring** of vertebrae · **marginal (vertical) syndesmophytes** · **bamboo spine** (late) · ALL/PLL calcification · osteoporosis · **dagger sign** (AP: central line = ossified supra/interspinous ligaments)
X-ray pelvis – Ferguson view	Beam **10–15° cephalad** → anterior SI joint
SI joint	**Bilateral symmetric erosions**; **earliest = iliac side** erosion → narrowing → ankylosis
CT	Bony change, not inflammation; **most sensitive for cervical fracture**
MRI (STIR / gad)	Detects inflammation → **earliest diagnosis**; **Romanus lesion** = "shiny corner" erosions at vertebral corners (**TO: MRI of LS spine**) · **Andersson lesion** = spondylodiscitis / pseudarthrosis · active sacroiliitis → dynamic fat-sat MRI
Bone scan	SI uptake, not specific



Bilateral sacroiliitis

– – Active space – –



Bamboo spine with marginal syndesmophytes

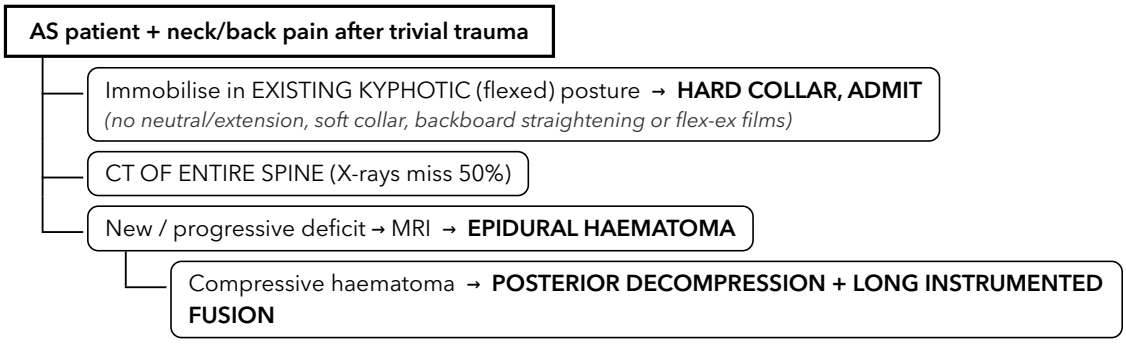
Fractures on X-ray	**negative in 50%** → CT

5 Management

First line	**NSAIDs / COX-2** + PT (flexibility) · no **oral steroids** (oral or IV – no effect on progression)
Second line	**TNF-α inhibitors**: infliximab, etanercept, **adalimumab** (also approved for RA + PsA → "husband AS + wife RA on same drug" = adalimumab) ✗ Wrong options: denosumab · tofacitinib (RA) · imatinib · rituximab (RA)
Early AS with fused SIJs	**NSAIDs + PT + HLA-B27 test** ✗ Wrong options: SI injections · SI fusion · aspiration

Spine fractures – image whole spine, immobilise in existing posture

Epidemiology	risk **4×** general population · **mid-cervical + cervicothoracic** (C5–7) commonest (> 50%), then TL junction ✗ Wrong options: lumbar commonest
Pattern	**extension**, **3-column**, behaves like a **long-bone fracture** (long lever arm + osteoporosis)
Outcomes	neuro involvement **~75%** (67% AS vs 40% DISH) · occult multi-level fractures **13%** · diagnostic delay up to **40%** · mortality up to **32–34%** · complications up to 84% (pneumonia, respiratory failure, pseudarthrosis) · fusion rates **high** once fixed



Neuro deficits often delayed	admit + serial exams even if CT negative

Epidural haematoma & neurological deterioration

Incidence	up to **15%**; hypervascular epidural tissue + torn epidural veins in a rigid canal; high mortality
Delayed weakness hours after fall (CT negative or nondisplaced)	**epidural haemorrhage** ✗ Wrong options: further displacement once immobilised · degenerative stenosis · vertebral artery · myelomalacia · cord infarct
Dorsal epidural haematoma + progressive deficit	**posterior laminectomy + instrumented fusion** ✗ Wrong options: steroids · repeat MRI · anterior fusion · laminectomy alone · halo (↑ mortality)

Fracture fixation – long-segment posterior instrumentation

Operative indications	**progressive deficit** · **epidural haematoma with compromise** · **unstable fracture** (virtually all, even minimally displaced, neuro intact)
Construct	**long posterior**, **≥ 3 levels above + 3 below**, multiple fixation points (osteoporosis + lever arm) · **do NOT correct deformity** · 360° if very osteoporotic / incomplete SCI with unstable C6 · (✗ Wrong options: short-segment · uninstrumented fusion · halo (septic + pulmonary issues in elderly)) ✗ Wrong options: stand-alone anterior → high failure
Non-op	stable, no deficit, or unfit: immobilise in existing kyphosis, admit; low-weight traction may reduce
Complications	progressive deformity, nonunion, hardware failure, infection

Kyphotic deformity correction

Exclude hip flexion contracture first · goal = sagittal balance + **horizontal gaze**

Osteotomy	Key facts
Smith-Petersen (SPO)	Posterior column (pars + facets), pivot = **posterior annulus**; needs **mobile anterior disc**; **10° per 10 mm** resected (TO); **simplest** (TO); C7 plumb 6–8 cm positive; ↓ foraminal height
Ponte	Modification of SPO; **wider** facet + lamina resection; originally **thoracic** (SPO originally **lumbar**, AS flat back) – Schwab grade 2 vs SPO grade 1 (TO)
Pedicle subtraction (PSO)	**Procedure of choice for TL kyphosis**; closing wedge, hinge **anterior cortex**; **30–40° per level**; single level, no disc resection; direct bony apposition → better fusion; works with fused discs / 360° fusion; SVA > 4 cm (> 12 cm ideal); transient deficits up to **20%**
Opening wedge (single)	Hinges on **posterior** body edge; splits ALL
Multi-segment opening	Less bone loss, preserves ALL
Bone-disc-bone	35–60°; hard in AS (ossified annulus)
Vertebral column resection	Most correction; requires disc resection
C7-T1 cervicothoracic extension osteotomy	**Chin-on-chest** deformity; **posterior extension osteotomy + posterior instrumented fusion** (✗ Wrong options: anterior osteotomy (no access to C7 pedicle) · halo traction · in-situ fusion)

– – Active space – –

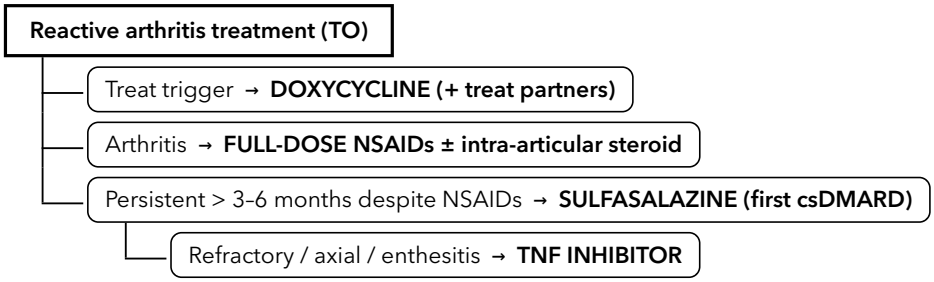
Why lumbar osteotomy	below conus, wide canal, avoids thoracic cage, long lever arm → more correction; upper lumbar level · **spinal motion not increased** (TO)
Why C7-T1	**vertebral artery outside transverse foramen at C7**, wide canal · remove C7 lateral mass + parts of C7/T1 pedicles · aim **slight undercorrection (CBVA ~10°)** · halo post-op if poor bone · **sitting position → venous air embolus**; paradoxical embolus risk ↑ with **patent foramen ovale** ✗ Wrong options: N_2O · hypotension · obesity · emphysema
THA before spinal osteotomy (TO)	THA carries less risk; operate both hips together if significant flexion contractures ✗ Wrong options: "cervicothoracic osteotomy before THA"

Hip arthroplasty in AS

Acetabula	more **vertical + anteverted**; altered spinopelvic relationship → ↑ **dislocation**, especially **anterior** (posterior approach still → anterior dislocation risk) ✗ Wrong options: infection · LLD · sciatic palsy (dysplasia) · abductor lurch (lateral approach)
Bilateral THA	kyphosis due to hip flexion contractures

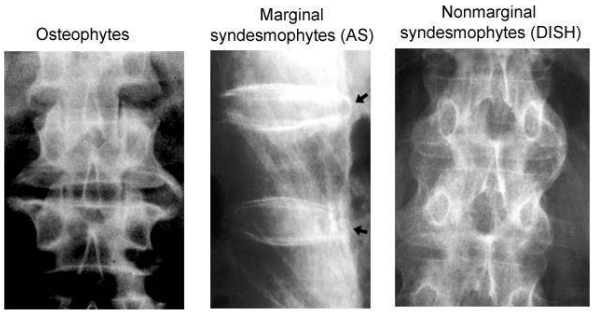
Reactive arthritis – pathogenesis & treatment

Definition (TO)	HLA-B27-associated seronegative SpA after **GU (Chlamydia trachomatis 40–60%)** or enteric (Salmonella, Shigella, Yersinia, Campylobacter) infection; onset **1–6 wks (2–4)** later; M:F 5–10:1
Features	**asymmetric lower-limb oligoarthritis**, **dactylitis**, enthesitis (Achilles, plantar fascia), **asymmetric sacroiliitis**, conjunctivitis, anterior uveitis, **circinate balanitis**, **keratoderma blennorrhagicum** (soles), urethritis; full Reiter triad in only 1/3 · HLA-B27 **60–80%**
Pathogenesis (TO)	local immune response to **persistent, metabolically active, non-culturable bacteria / antigens in synovium**; synovial fluid WBC 10–50k, culture negative ✗ Wrong options: "completely sterile"



X vs Y

	AS	DISH
Syndesmophytes	**Marginal**, vertical	**Non-marginal**, "flowing candle wax" (≥ 4 levels)
X-ray	**Bamboo spine**, squaring, **Romanus shiny corners**	Disc space **preserved**
Disc	Involved / ossified (cervical)	Preserved
Bone density	**Osteopenia**	Normal / ↑
HLA	**B27**	**B8** (with diabetes)
Age	Young	Older
SI joint	**Bilateral sacroiliitis**	Spared
Diabetes	No	**Yes**
Fracture neuro deficit	67%	40%



Syndesmophytes: AS (marginal) vs DISH (non-marginal)

	AS	Reactive arthritis
Sacroiliitis.....249	**Bilateral symmetric**	**Asymmetric**
Peripheral	Hip / shoulder	Asymmetric LL oligoarthritis, dactylitis
Urethritis	No	Yes
B27	90%	60–80%

	Smith-Petersen	Ponte	PSO
Pivot	Posterior annulus	–	**Anterior cortex**
Correction	10°/10 mm	Slightly > 10°	30–40°
Needs mobile disc	Yes	Yes	No

How it's asked

"Romanus lesion seen in?"	→ **MRI of LS spine in AS** (TO) ✗ Wrong options: X-ray
"Wall test useful in?"	→ **cervical spine involvement in AS** (TO)

– – Active space – –

"HLA-B27 prevalence in AS, juvenile SpA, RA?"	→ **90%, 70%, 6%** (TO)
"AS diagnostic criteria — all EXCEPT?"	→ **HLA-B27 positivity** (TO, modified New York)
"Bilateral sacroiliitis + B27 — what confirms diagnosis?"	→ **uveitis** ✗ Wrong options: erythema marginatum · HLA-DR3 · RF · urine phosphoethanolamine
"Juvenile AS — all EXCEPT?"	→ **urethritis**
"AS patient expresses?"	→ **class I antigen, B locus, MHC, chromosome 6**
"Bicycle fall, AS on X-ray — true?"	→ **HLA-B27 positive** ✗ Wrong options: Klippel-Feil · Japanese OPLL · DISH 4-level · RF positive
"26-yr with hip flexion contractures + SI sclerosis?"	→ **HLA-B27 positive** ✗ Wrong options: FADDIR · normal ESR/CRP
"Fused SIJs in 35-yr — next?"	→ **NSAIDs + PT + HLA-B27 testing**
"RF-negative bilateral sacroiliitis — additional finding?"	→ **eye inflammation** ✗ Wrong options: ↑ FEV · non-marginal syndesmophytes · champagne-glass pelvis · diabetes
"AS — true?"	→ **often involves entire spine** ✗ Wrong options: female · diabetes · SI rarely involved · oral steroids
"B27-positive AS — ↑ synovitis risk from?"	→ **gram-negative bacillus (Klebsiella)**
"Husband AS, wife RA, same drug?"	→ **adalimumab**
"AS after fall on ice, neck pain, neuro intact — next?"	→ **CT cervical spine** ✗ Wrong options: flex-ex · bone scan · soft collar
"Chin-on-chest + bamboo spine after fall — management?"	→ **hard collar in existing kyphosis, admit, advanced imaging**
"Immobilise cervicothoracic kyphosis during work-up?"	→ **maintain flexed position**
"Marginal syndesmophytes, posterior neck pain after fall?"	→ **immobilise + CT C-spine + X-ray T/L spine**
"AS on etanercept, rear-ended, X-rays negative?"	→ **CT cervical, thoracic + lumbar**
"AS fracture — true statement?"	→ **CT entire spinal axis** ✗ Wrong options: occult injuries rare · lumbar commonest · delay uncommon · low mortality
"Deficit 7 h after nondisplaced extension fracture, dorsal haematoma on MRI?"	→ **posterior laminectomy + instrumented fusion**

"C7-T1 fracture + posterior haematoma + progressive weakness?"	→ **posterior decompression + instrumented fusion** ✗ Wrong options: halo (↑ mortality)
"Principal concern with AS fracture-dislocation?"	→ **epidural haematoma**
"Normal CT, severe weakness 3 h later — MRI shows?"	→ **epidural haemorrhage**
"HLA-B27 man, TL fracture after 2-stair fall, intact?"	→ **posterior fusion with long-segment pedicle screws**
"AS shear fracture C5-6 with hand weakness?"	→ **posterior decompression + fusion C3-T2**
"Minimally displaced 3-column lumbar AS fracture, intact?"	→ **multilevel posterior instrumentation + fusion**
"C6 fracture, incomplete SCI, absent BCR — definitive?"	→ **open reduction, decompression, A + P stabilisation**
"AS cervical fracture — true?"	→ **long posterior instrumentation ± decompression** ✗ Wrong options: low fusion rate
"Which fracture suits posterior cervicothoracic fusion?"	→ **3-column AS fracture** ✗ Wrong options: flexion teardrop → anterior · Jefferson → halo · type 3 hangman → C2-3 ACDF · type 3 odontoid → collar
"Osteotomy in AS — true?"	→ **SPO gives 10° per 10 mm** (TO) ✗ Wrong options: ↑ motion · cervical before THA · thoracic for lungs
"SPO vs Ponte — incorrect statement?"	→ **"SPO described for thoracic, Ponte for lumbar"** (TO — reverse)
"True about osteotomies in AS?"	→ **SPO simpler than PSO** (TO) ✗ Wrong options: PSO pivots on posterior body
"Most single-level correction without disc resection?"	→ **PSO**
"Chin-on-chest deformity — management?"	→ **posterior extension osteotomy + posterior instrumented fusion**
"Greatest risk for paradoxical VAE in beach-chair?"	→ **patent foramen ovale**
"THA in AS (posterior approach) — risk?"	→ **anterior dislocation**
"Bilateral THA in AS — counsel about?"	→ **periprosthetic dislocation**
"Chlamydia reactive arthritis — pathogenesis?"	→ **persistent non-culturable bacteria/ antigens in synovium** (TO)
"SARA failing 14 weeks of NSAIDs?"	→ **sulfasalazine** (TO) ✗ Wrong options: TNF first · high-dose steroids · methotrexate superior

– – Active space – –

ATLANTOAXIAL INSTABILITY

– Spine · 6 / 69 –

1 Basics

Aetiology

C1-2 "transitional zone" → instability from degenerative + traumatic causes

	Degenerative / congenital	Traumatic / infective
Adult	**Down syndrome** · **rheumatoid arthritis** · **os odontoideum**	Type I odontoid fracture (very rare) · atlas fractures · transverse ligament injury
Paediatric	**JRA** · **Morquio** (lysosomal storage)	**Rotatory atlantoaxial subluxation** · Grisel syndrome

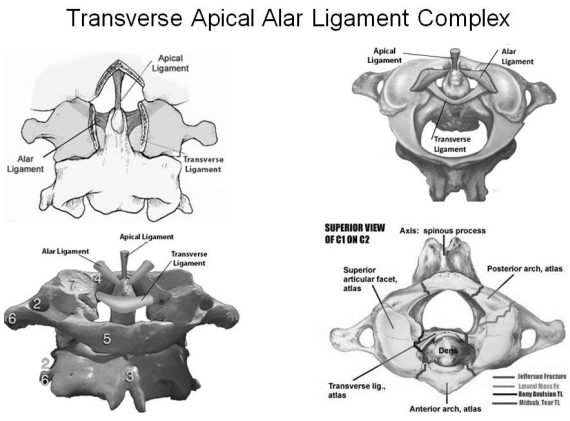
Os odontoideum = ADULT cause (TO) not paediatric (EXCEPT answer) · aetiology debated, most support **post-traumatic**

Other causes (TO) skeletal dysplasias (**pseudoachondroplasia – COMP**), neurofibromatosis, Grisel (infection)

Pseudoachondroplasia vs achondroplasia pseudoachondroplasia → **atlantoaxial instability** · achondroplasia → **lumbar stenosis**

Anatomy & ligaments

Ligament	Connects / role
Transverse atlantal ligament (TAL)	Horizontal limb of cruciate; medial tubercles of C1; straps dens to anterior arch; **most important stabiliser – strongest ligament of the spine**
Alar ligaments	**Paired parasagittal**; dens → occipital condyles (+ C1 lateral mass); "check" ligaments → **limit rotation**
Apical ligament	**Single midline**; dens tip → foramen magnum; little strength
Tectorial membrane	Cephalad continuation of **PLL**; dens/C2 → foramen magnum
Anterior / posterior atlanto-occipital membranes	ALL continuation / posterior FM → posterior C1 arch



Transverse, alar and apical ligament complex

Biomechanics	C1-2 provides **~50%** (Orthobullets) · **60%** (TO) of cervical rotation; peg-in-ring; near-horizontal facets

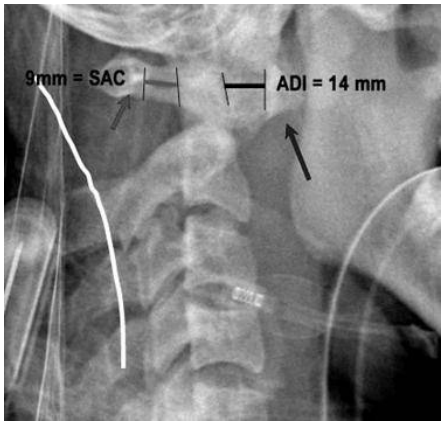
3 Presentation

Symptoms	neck pain, neuro symptoms
Deficits appear late	capacious canal at C1 (Steel's rule of thirds)
Myelopathy	hyperreflexia, weakness, **broad-based gait**, ↓ hand dexterity, loss of motor milestones, bladder dysfunction
Child, trivial fall → transient quadriparesis, recovered (TO)	**congenital atlantoaxial instability** ✗ Wrong options: ossified LF (age) · syrinx (wouldn't recover) · bifid atlas

4 Diagnosis & Classification

Key radiology findings

Measure	Definition	Threshold
Atlantodens interval (ADI)	Dens → **posterior border of anterior C1 arch**, flexion-extension	Adult **> 3.5 mm** unstable (Orthobullets) · **> 3 mm** (TO) · child **> 5 mm** · **> 10 mm → surgery in RA**
Space available for cord (SAC / PADI)	Posterior dens → anterior posterior arch	Normal 17–29 mm · **< 14 mm** → ↑ neuro risk, surgery in RA
Sum of lateral mass displacement (open mouth)	C1 masses over C2	**> 8.1 mm** → TAL rupture, unstable (Orthobullets) · **> 6.9 mm** Rule of Spence (TO)



ADI and SAC on lateral flexion view

Lateral masses can displace only if	C1 ring fractured **or** TAL ruptured
Flexion-extension views	mandatory pre-op in **all high-risk patients** (Down, RA, dysplasia) before any surgery
Pseudoachondroplasia + myelopathy	**flexion-extension C-spine; ADI difference 6 mm** ✗ Wrong options: lateral mass 9 mm = atlas fracture · hip OA · scoliosis · lumbar stenosis
CT	essential for rotated C1 on C2 · MRI for TAL

– – Active space – –

Fielding & Hawkins classification (TO)

Type	Displacement	TAL	Treatment
I	Simple **rotatory**, no anterior shift	Intact	Collar 6–8 wks
II	Anterior **3–5 mm**, one lateral mass pivots	Deficient	Collar / surgery
III	Anterior **> 5 mm**	Disrupted	**C1-2 fusion** (unstable)
IV	**Posterior** displacement (dens erosion)	–	**C1-2 fusion** (unstable)

5 Management

Determined by specific condition (RA, Down, os odontoideum, fracture, TAL injury, JRA, Morquio, AARS)

Operative criteria (TO)

Adults	Children (≥ 1 of)
ADI > 5 mm	**Neurologic deficit**
Moderate displacement / instability on dynamic films	**Persistent ADI > 4 mm**
RA: ADI > 10 mm or SAC < 14 mm	Deformity **> 3 months**
–	Recurrence despite **6 weeks** immobilisation

C1-2 instability – os odontoideum

Os odontoideum

- Asymptomatic, stable → **STOP CONTACT SPORTS**
- Neuro signs or widened ADI → **POSTERIOR C1-2 FUSION**
  (not anterior C1-2 fusion – osseous anatomy)

Grisel syndrome

Definition (TO) inflammatory **atlantoaxial rotatory subluxation** after **upper respiratory / head-neck infection** (or **tonsillectomy, adenoid** inflammation); hyperaemia damages facet capsules + TAL

Presentation (TO) child, **painful torticollis** ("cock-robin": chin rotated one way, neck tilted the other, slight flexion), **facial asymmetry**, **change in voice / nasal resonance**, painful attempted correction
✗ Wrong options: muscular torticollis (painless, infant) · Sandifer (reflux)

Grisel type	Treatment (TO)
I	**Soft collar**
II	**Hard collar**
III	**Halo**
IV	**Open fixation** (neuro complications)

X vs Y

	Grisel	Muscular torticollis
Age	Child (~4–12)	Infant
Trigger	URTI / tonsillectomy	Birth (intrauterine positioning)
Pain	**Painful**	Painless
Mass	No	SCM tumour
Head	**Cock-robin**: tilt one side, chin to the other, slight flexion	Tilt toward, chin away from tight SCM
Lesion	C1-2 rotatory subluxation	SCM fibrosis

	ADI	SAC (PADI)
Measures	Anterior instability	Cord space
Abnormal	> 3.5 mm adult, > 5 mm child	**< 14 mm**
RA surgery	> 10 mm	< 14 mm

How it's asked

"NOT a cause of paediatric atlantoaxial instability?"	→ **os odontoideum** (TO)
"10-yr, trivial fall, transient quadriparesis?"	→ **congenital atlantoaxial instability** (TO)
"11-yr os odontoideum with myelopathic signs — treatment?"	→ **posterior C1-2 fusion** ✗ Wrong options: PT · soft collar · stop sports only · anterior fusion
"Pseudoachondroplasia with myelopathy — study + finding?"	→ **flexion-extension C-spine, ADI difference 6 mm**
"4-yr girl, painful torticollis, facial asymmetry, voice change after URTI?"	→ **Grisel syndrome** (TO)
"Most important C1-2 stabiliser?"	→ **transverse ligament**
"Single midline CVJ ligament?"	→ **apical ligament**
"Sum of lateral mass overhang implying TAL rupture?"	→ **> 8.1 mm** (Orthobullets) · **> 6.9 mm** (TO, Spence) — read the options
"SAC predicting neuro injury in RA?"	→ **< 14 mm**
"ADI needing surgery in RA?"	→ **> 10 mm**

– – Active space – –

ATLANTOAXIAL ROTATORY DISPLACEMENT (AARD)

– Spine · 7 / 69 –

1 Basics

Aetiology

Definition paediatric C1-2 rotatory instability: spectrum from mild subluxation / muscle spasm → **fixed facet dislocation** (mechanical block)

Cause	Share
Infection (pharyngitis, otitis media, URTI, retropharyngeal abscess)	**~35%** – **Grisel's disease**; lymphatic oedema around C-spine
Trauma (even trivial – fall from bed)	**~24%**
Recent head / neck surgery (tonsillectomy)	**~20%**
Idiopathic	–
Associated	**Down syndrome**, **RA**, tumours, congenital anomalies

Mechanism ligamentous laxity

TAL intact canal stenosis only with **severe rotation + facet dislocation**

TAL ruptured + anterolisthesis > 5 mm stenosis with **less rotation (45°)**; vertebral arteries at risk

Anatomy

C2 ossification **5 centres** · **subdental (basilar) synchondrosis** between dens + body fuses **~6 yrs** · secondary (ossiculum terminale) centre appears **~3 yrs**, fuses **~12 yrs**

Ligaments **transverse** limits anterior translation of atlas · **apical + alar** limit rotation

3 Presentation

Symptoms head tilt, neck pain, headache, ↓ rotation

Posture **"cock-robin"**: chin rotated **away from** the subluxed facet, head tilted toward it (≈ 20° tilt, 20° rotation, 10° flexion)

SCM spasm on the **SAME side as the chin** (protective, secondary) – e.g. right facet subluxation → chin left → **left SCM spastic**

5-yr after tonsillectomy right neck bend, left head turn, SCM spasm → AARS

4 Diagnosis & Classification

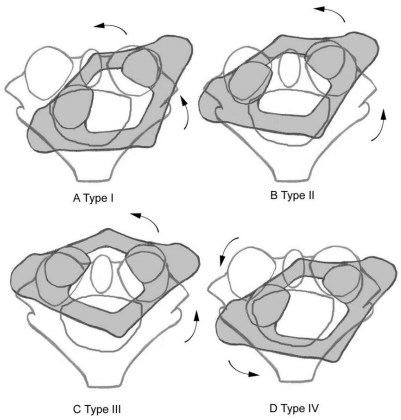
Key radiology findings

Study	Finding
AP open-mouth	Asymmetric **size + distance from midline** of lateral masses
Lateral	Facet appears **anterior + wedge-shaped** (not oval)
Flexion-extension	Exclude instability (hard to obtain)
Dynamic CT	**Gold standard** – neutral + maximal right + left rotation → **fixed C1-on-C2 rotation unchanged** with turning
MRI	Little value unless neuro signs

Diagnosis is primarily clinical (OB) · CT/MRI for persistent cases · labs (CBC, ESR, CRP) for infection

Fielding (& Hawkins) classification

Type	Lesion	TAL	Note
I	**Unilateral** facet subluxation; **odontoid = pivot** (one facet forward, one back)	**Intact**	**Commonest, benign**
II	Unilateral, **3–5 mm** anterior; one facet = pivot	Injured	–
III	**Bilateral** anterior **> 5 mm**	Disrupted	Rare; neuro risk / sudden death
IV	**Posterior** C1 displacement (odontoid fracture / hypoplastic dens)	–	Rare; neuro risk / sudden death



Fielding type I: rotation about the dens

5 Management

Chronicity is the key factor earlier treatment → better conservative success; many reduce spontaneously before presentation

– – Active space – –

AARD by duration (Orthobullets)

- < 1 week → **SOFT COLLAR + NSAIDs + exercises**
- > 2 weeks / failed collar × 2 wks → HALTER TRACTION (~5 lb) + BENZODIAZEPINES → **hard collar 3 months**
- > 1 month / failed halter × 2 wks → HALO TRACTION → **halo vest 3 months**
- > 3 months / neuro deficit / failed halo × 2 wks / recurrent → **POSTERIOR C1-2 FUSION**

Timing variants (OB)	soft collar < 7 days · halter 1 wk–1 month · halo 1–3 months · or sequential **2-week trials** (collar → halter → skeletal traction → arthrodesis) — at least **2 weeks** of collar before traction

Early management – collar, NSAIDs & traction

Acute (3–5 days, post-URTI / tonsillectomy)	**soft collar** + NSAIDs + antispasmodic ✗ Wrong options: halter · halo · fusion · observation alone
1 week of collar without resolution	**continue collar ≥ 1 more week** then reassess ✗ Wrong options: immediate halter · stretching
Fixed torticollis 4 weeks after fall	**cervical (halter) traction** · halo needed when child's skin can't tolerate halter weight ✗ Wrong options: soft collar · SCM release · fusion

Chronic / irreducible – C1-2 arthrodesis

Failed halter + halo traction, still subluxed on dynamic CT	**posterior atlantoaxial (C1-2) fusion** ✗ Wrong options: no treatment · closed reduction under sedation/GA · occipitocervical fusion (loses motion) · more sedation · more weight · halo vest/C-collar unless reduced
Halo vest	only to **maintain** a reduction achieved by halo traction

6 Complications

Missed / delayed diagnosis	commonest problem
Type III/IV	neuro injury, vertebral artery compromise, death

X vs Y

	AARD	**Congenital muscular torticollis**
Primary lesion	C1-2 subluxation	SCM fibrosis
SCM spasm vs chin	**Same side** as chin	**Opposite** side to chin
Age	Child, acute after URTI / trauma / surgery	Infant
Pain	Painful	Painless
Rx	Collar → traction → C1-2 fusion	Stretching → SCM release

How it's asked

"5-yr, fixed torticollis 4 weeks after fall, CT rotatory fixation — next?"	→ **cervical traction** ✗ Wrong options: soft collar · SCM release
"6-yr, head turned 3 days after URTI, no deficit?"	→ **soft collar** ✗ Wrong options: halter · halo · arthrodesis
"7-yr, 1 week of collar, still tilted?"	→ **continue soft collar ≥ 1 week, re-evaluate**
"Painful head tilt 5 days after tonsillectomy?"	→ **cervical collar + NSAIDs + antispasmodic**
"3-yr, 8 months of torticollis, failed halter + halo, still subluxed?"	→ **posterior atlantoaxial fusion** ✗ Wrong options: occipitocervical fusion
"12-yr, 10 weeks, failed 2 wks halter + 3 wks halo?"	→ **C1-2 arthrodesis**
"Gold-standard diagnosis?"	→ **dynamic CT**
"Commonest Fielding type?"	→ **type I — unilateral, TAL intact, dens pivot**
"Right facet subluxation — chin + SCM spasm?"	→ **chin left, left SCM spastic**

– – Active space – –

ATLAS FRACTURES & TRANSVERSE LIGAMENT INJURIES

– Spine · 8 / 69 –

1 Basics

Epidemiology

Incidence	**~7% of cervical fractures** · 25% of CVJ injuries · 1–3% of all spinal injuries · **commonly missed**
Bimodal	**20–30 yrs** high-energy axial load (~85% MVC) · **elderly** low-energy ground-level fall (OA, osteoporosis, gait imbalance)
Associated injuries	**50%** other spine injury · **40% axis (C2) fracture** · head injury
Neuro injury rare	capacious canal: cord occupies only **~35%** at C1 (vs 50% lower) · burst **increases** space at C1

Mechanism

Force	Injury
Axial compression (blow to vertex, diving)	**Jefferson burst**
Hyperextension (blow to forehead)	**Posterior arch** fracture; low-energy falls; ↑ with **odontoid fractures**; 30% less energy needed in extension
Lateral compression	Anterior arch fracture
Lateral distraction	Comminuted lateral mass

Anatomy

Atlas	ring with 2 lateral masses, **no body, no spinous process**; **3 ossification centres**
Incomplete posterior arch	common **congenital variant**, not trauma
Vertebral artery	C1 transverse foramen → bends posteromedially across **sulcal groove** (common site of posterior arch fracture)

Ligament	Role
Transverse (TAL)	**Main stabiliser of C1-2** (TO); posterior dens → anterior arch, inserts on **lateral mass tubercles**; prevents dens migrating into canal
Alar (paired)	Dens → occipital condyles; strong (5–6 mm); occipitocervical stability
Apical	Weak midline, dens → foramen magnum
Tectorial membrane	Posterior C2 body → anterior foramen magnum; **cephalad PLL**

Layers anterior to dura (dorsal → ventral)	tectorial membrane → cruciate ligament → odontoid (alar + apical) ligaments

Motion	**occiput-C1 ~50% of flexion-extension · C1-2 ~50% of rotation**

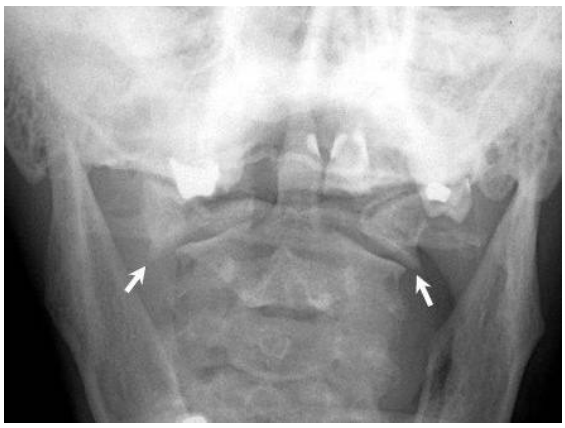
3 Presentation

Symptoms	neck pain, spasm, ↓ motion, suboccipital pain, sense of instability
C2 / greater occipital nerve	**occipital neuralgia** (TO: commonest nerve injury – emerges below posterior arch), occipital numbness, rarely alopecia, neck flexion/ extension weakness
Vertebral artery dissection	LOC, diplopia, vertigo, blindness, ataxia, dysphagia, nausea
Neuro deficit	uncommon in isolated C1; ↑ with associated C2 fracture

4 Diagnosis & Classification

Key radiology findings

Measure	Normal / abnormal
ADI	Adult **< 3 mm** normal (child < 5) · **3–5 mm** = **TAL torn, alar + apical intact** · **> 5 mm** = TAL + alar + tectorial membrane (OB: **> 7 mm**)
Sum of lateral mass displacement (open mouth / coronal CT)	**> 6.9 mm (rule of Spence)** or **> 8.1 mm with magnification (rule of Heller)** → TAL rupture, unstable (OB: ≥ 7 mm / 8.2 mm)
SAC (PADI)	Normal ≥ 13–14 mm; doesn't predict TAL injury
Retropharyngeal soft tissue	**> 9.5 mm** → anterior arch injury
Basion-dens interval	< 12 mm normal (occipitocervical)
Power's ratio	Occipitocervical instability
C2 pars displacement	Hangman fracture



Jefferson fracture: lateral mass overhang on open-mouth view

X-rays	lateral, **60° obliques** (posterior arch), **open-mouth** · flexion-extension only for **late instability** after non-op (not in ED)
CT	every suspected injury; **sagittal** recon = occult anterior arch fracture · **axial** = Dickman II avulsion · **coronal** = LMD · CTA for vertebral artery
Paediatric pseudospread	asymmetric atlas vs axis growth → lateral mass overhang without fracture (CT)

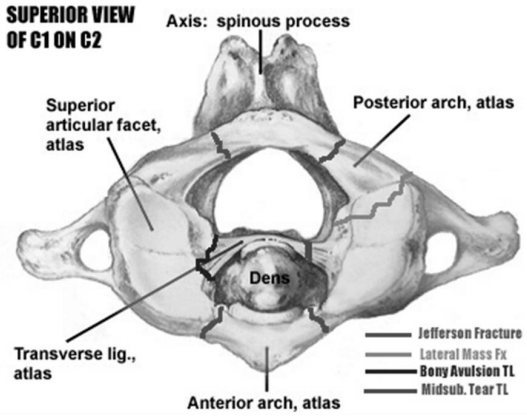
– – Active space – –

MRI every confirmed C1 fracture; **↑ T2 signal in TAL = intrasubstance tear**; **more sensitive than CT for TAL**

LMD caveat measurements may underestimate due to **rebound** after injury

Landells classification

Type	Pattern	Treatment
1	**Isolated anterior or posterior arch** – **commonest**; "plough" = isolated anterior arch from dens driven through it	**Hard collar**
2	**Jefferson burst** – bilateral anterior + posterior arch, axial load	TAL intact → **hard collar** · intrasubstance tear → **C1-2 fusion** · bony avulsion → **halo**
3	**Unilateral lateral mass**	TAL intact → hard collar · disrupted → halo



Landells atlas fracture patterns

Dickman classification (TAL injury)

Type	Lesion	Treatment
I	**Intrasubstance tear** (won't heal)	**C1-2 fusion**
II	**Bony avulsion** of C1 lateral mass tubercle	**Halo** (75% success)

5 Management

Atlas fracture

- TAL intact (LMD < 7 mm, normal ADI) → **RIGID COLLAR 6–12 WKS** *(not soft collar or observation)*
- TAL bony avulsion (Dickman II) → **HALO 3 months**
- TAL intrasubstance tear (Dickman I) → **POSTERIOR C1-2 FUSION**
  - Comminuted C1 / no C1 purchase → **OCCIPUT-C2 FUSION**

Elderly **never halo** (↑ mortality) → hard collar / surgery · clear C-spine expeditiously

Non-op reduce with halo traction first if displaced · **post-treatment flexion-extension films** for late instability

Operative indications unstable type 2/3 (controversial) · **Dickman I** · **combined C1 + C2** fractures (type II odontoid, hangman) · some prefer occiput-C2 (lower revision)

C1 internal fixation lateral mass split; preserves C1-2 motion (small series)

Fusion technique

Technique	Fact
Harms / Goel (C1 lateral mass + C2 pedicle screws)	**10° medial trajectory** protects **internal carotid** (bicortical) (✗ Wrong options: vertebral artery · cord (at risk if too medial))
C1-2 transarticular screw (Magerl)	**Vertebral artery** at risk in C2 groove (~10% on CT planning) (TO)
Gallie sublaminar wiring	Needs intact posterior arch → valid for **TAL disruption** (✗ Wrong options: OC dissociation · comminuted C1 · type I/III odontoid)
Occiput-C2	Comminuted C1; occipital plate

Motion loss C1-2 fusion → **~50% rotation** · occiput-C2 → **50% rotation + 50% flexion-extension** (TO)

6 Complications

Complication	Fact
Vertebral artery	Displaced posterior ring through **sulcal groove**
Cock-robin deformity	Displaced **unilateral sagittal split lateral mass** → occipital condyle settles on C2 → **occipitocervical fusion ± osteotomy**
Nonunion	**~20%** non-op
Neck pain / stiffness	14–80% pain, 8–20% stiffness, 34% activity limits
Infection	↑ with posterior approach
Prognosis	Worse PROMs with **> 7 mm** displacement

X vs Y

ADI	Ligaments torn
< 3 mm	None
3–5 mm	TAL only
> 5 mm (> 7 mm OB)	TAL + alar + tectorial membrane

	Dickman I	Dickman II
Lesion	Midsubstance	Bony avulsion
Heals?	No	Yes (75%)
Rx	C1-2 fusion	Halo

– – Active space – –

How it's asked

"Main stabiliser of C1-C2?"	→ **transverse ligament** (TO)
"ADI > 7 mm implies?"	→ **TAL + alar ruptured + tectorial membrane ruptured**
"Jefferson fracture – which indicates TAL disruption?"	→ **combined LMD 8.2 mm** ✗ Wrong options: ADI 2 · PADI 16 · pars shift · Power's ratio
"Diving injury, LMD 8.3 mm – true?"	→ **TAL disrupted, unstable → rigid orthosis / halo / surgery**
"Comminuted C1, which suggests TAL rupture?"	→ **LMD 9 mm** ✗ Wrong options: SAC 15 · ADI 2 · BDI 11 · PVS 5 mm
"Occiput-C2 fusion – motion loss?"	→ **50% rotation + 50% flexion-extension** (TO)
"Gallie C1-2 wiring valid for?"	→ **TAL disruption**
"Jefferson with intact TAL?"	→ **rigid collar 6–12 weeks**
"MRI shows intrasubstance TAL rupture 2 weeks after injury?"	→ **surgical stabilisation** ✗ Wrong options: collar · halo
"Elderly, stable C1 fracture after slip?"	→ **hard collar** ✗ Wrong options: halo · flex-ex films · fusion
"10° medial C1 lateral mass screw protects?"	→ **internal carotid artery**
"Transarticular C1-2 screw – structure at risk?"	→ **vertebral artery** (TO)
"True about Jefferson fracture?"	→ **greater occipital neuralgia** (TO) ✗ Wrong options: high neuro injury · type 1 → halo · intact TAL → fusion

CAUDA EQUINA SYNDROME

– Spine · 9 / 69 –

1 Basics

Epidemiology

Incidence **rare**: 1.5–3.4 per million/yr; prevalence ~1/65,000 · in **~3%** of lumbar disc herniations (TO: 1–10%)

Demographics **males**, **4th decade (30s)**

Level **L4-5** commonest

Causes

Cause	Note
Disc herniation	**Commonest** (large central HNP) (✗ Wrong options: burst fracture commonest)
Spinal epidural haematoma	Spontaneous on high-dose DVT prophylaxis, **epidural anaesthesia**, early post-op prophylaxis
Tumours	Myxopapillary ependymoma, schwannoma, meningioma
Synovial facet cyst · epidural abscess	Ask about **IVDU**
Trauma	**Lumbar burst** retropulsion, dislocation
High-grade developmental spondylolisthesis	Intact lamina

Pathobiology

Compression → ↓ nutrient delivery (↓ blood flow, ↓ CSF diffusion) → venous congestion + **intraneural oedema** → "intraneural compartment syndrome" → **root ischaemia**

Vulnerability roots covered only by **endoneurium** (no Schwann cell covering per TO) + hypovascular central zone

Anatomy

Conus medullaris terminal cord, ends **T12–L1** · **filum terminale** → coccyx

Cauda equina **L1–S5 roots** in lumbar canal → **lower motor neuron** lesion

Bladder innervation	Effect
Parasympathetic (pelvic splanchnic, S2–4)	**Voiding** – detrusor contracts, internal sphincter relaxes
Sympathetic (hypogastric plexus)	**Retention** – detrusor relaxes, internal sphincter contracts
Pudendal nerve	**External sphincter** (voluntary)

CES bladder LMN interrupts reflex arc → can't sense filling → **retention → overflow incontinence**

– – Active space – –

3 Presentation

History heavy lifting with flexed spine, coughing spell · ask **anticoagulants**, recent procedures (haematoma), **IVDU** (infection)

Symptom	Frequency
Back pain	**Commonest** (Orthobullets, OB) – may be the only initial symptom · 83%
Leg pain (radicular)	**2nd commonest** (Orthobullets) · 90% – "commonest presenting symptom" in some OB stems – read the options
Urinary retention	60%
Urinary incontinence	55%
Leg weakness / sensory deficit / absent ankle jerk	55% / 62% / 62%
Perianal sensory deficit	76%
↓ rectal tone	50% (relatively **late**)
Sexual / bowel dysfunction	Rare

Order of frequency **leg pain > urinary retention > urinary incontinence**

Saddle anaesthesia less common but **most specific** → trigger emergency surgery protocol

Saddle anaesthesia alone, no other LL deficit (TO) lesion **below S2** (S3–S5)

Exam palpate bladder · ↓ voluntary rectal tone · **pinprick** loss S2–S4 perianal/perineum/posterior thigh · ↓ / absent LL reflexes · **↓/absent bulbocavernosus + anal wink**
✗ Wrong options: **Babinski** (UMN sign – EXCEPT answer)

Classic triad (TO) **asymmetric areflexic paralysis** + **bladder/bowel (S2–4)** + **saddle anaesthesia** (S3–4, **perianal S5 spared initially**)

32-yr, unilateral radiculopathy + saddle anaesthesia (TO) **CES + urinary retention + absent ankle jerk**
✗ Wrong options: perianal anaesthesia + areflexia · conus

4 Diagnosis & Classification

Diagnosis is clinical (history + exam); MRI confirms cause + plans surgery

Key radiology findings

Study	Use
MRI	**Study of choice**, emergent (**within 1–2 h** of ED arrival); large **central disc** obliterating canal, haematoma, abscess
CT myelography	**Can't have MRI (pacemaker)** (TO: myelo-CT; ✗ Wrong options: plain CT · angiography · no imaging)
X-ray	Usually normal; don't delay MRI; burst fracture / high-grade slip

Genes & lab markers

CBC, ESR, CRP — if abscess suspected (fever, IVDU) – not routine
✗ Wrong options: labs before MRI without infection signs

Post-void residual (PVR) — pre-op if it doesn't delay surgery · normal **< 50–100 mL** · **PVR < 200 mL → 97% NPV** · **> 200 mL highly suggestive**
✗ Wrong options: "> 200 mL excludes"

Classification – bladder function

Grade	Definition
Incomplete (CES-I)	↓ urgency / sensation, **no** retention or incontinence
Complete (CES-R)	Clear **retention** and/or incontinence

5 Management

Suspected CES (saddle anaesthesia / bladder change + back-leg pain)

- EMERGENT MRI (CT myelogram if MRI contraindicated)
- Compression confirmed → **DECOMPRESSION ASAP**
  (< 24 h preferable · < 48 h standard of care)

Timing of decompression

< 48 h vs > 48 h — significant improvement in **sensory, motor, urinary + rectal** function when < 48 h · **no added benefit < 24 h vs 24–48 h** (Ahn meta-analysis)
✗ Wrong options: < 24 h better than < 48 h · < 60 h · < 1 week

Not indicated — epidural steroid injection · methylprednisolone · outpatient MRI · reassurance/bed rest

Surgical technique

Procedure	Indication
Microdiscectomy (unilateral laminotomy + medial facetectomy + discectomy)	Massive soft disc, **young**, minimal degeneration
Wide laminectomy (bilateral laminectomy, LF + medial facetectomy, discectomy)	**Older**, hypertrophic LF + lateral recess stenosis; ↓ dural retraction, better visualisation
Laminectomy + fusion	High-grade spondylolisthesis; insidious CES with degenerative spondylolisthesis

Microdiscectomy vs wide laminectomy — **similar outcomes**; goal is adequate thecal sac decompression
✗ Wrong options: laminectomy → greater recovery

– – Active space – –

6 Complications

Prognosis – bladder dysfunction recovery

Worse prognosis — **saddle anaesthesia + bladder dysfunction** at presentation · **decompression > 48 h** · pre-op chronic back pain (urinary/rectal) · pre-op rectal dysfunction · older age (sexual function, neuro recovery)
✗ Wrong options: age per se in some OB stems

Least likely to recover — **bladder dysfunction** · **long-term urinary dysfunction common (~50%)** · bladder improves up to **16 months**; motor up to **1 yr**; sexual over years
✗ Wrong options: EHL weakness · perianal sensation · foot numbness · leg pain

Expected course — **pain + sensory improve; residual bladder dysfunction possible**
✗ Wrong options: complete recovery · no change

Complication	Fact
Dural tear	1–4%; primary repair ± fat graft, bed rest 4–7 d, ± lumbar drain; outcomes unaffected if treated
Iatrogenic instability	Aggressive medial facetectomy
Epidural fibrosis	Back/leg pain at **~3 months**; **gad MRI** separates from recurrent HNP; poor response to re-exploration
Wound infection	~1%; diabetics
Vascular injury	Curette through **ALL**; mortality up to **50%**
DVT	Prophylaxis held for haematoma fear; antiplatelets safe at **48–72 h** post-op

X vs Y

Feature (TO)	Cauda equina	Conus medullaris
Level of lesion	Roots, **L1 vertebra and below**	Sacral cord, **T12 (D12)–L1** (T11–L2)
Presentation	**Asymmetric**	**Symmetric**
Radicular pain	**Severe**	Usually absent
Sensory	**Saddle (S2–S4)**, perianal S5 spared initially	**Perianal (S5)** anaesthesia; sensory dissociation possible
Motor	Asymmetric **flaccid** (LMN)	Symmetric, usually flaccid, sometimes hyperreflexic
Reflexes	**Areflexia** (ankle; knee lost if L3–4 involved)	**Knee reflex preserved**
Bladder / bowel	**Late** (TO table) — retention first → overflow	**Early**
BCR / anal wink	Absent	**Persistently absent** (irreversible sacral cord injury)
Babinski	No	–

Diabetic, bilateral saddle anaesthesia + bladder/bowel + absent BCR (TO) → **conus medullaris**
✗ Wrong options: central cord · Brown-Séquard · mixed

	Hemi-cauda equina	Complete CES
Legs	**Unilateral**	Bilateral
Saddle	Unilateral S1–S5	Bilateral
Motor	Often absent	Present
Sphincter	**Retention / incontinence present**	Present
Cause	**Disc herniation (same — EXCEPT answer, TO)**	Disc herniation

Neurology rather than spine referral → non-compressive cause (e.g. MS / demyelination)
✗ Wrong options: burst fracture · large disc · sacral metastasis · epidural lymphoma

How it's asked

"Young man, unilateral radiculopathy + saddle anaesthesia — diagnosis + signs?"	→ **CES, urinary retention, absent ankle jerk** (TO)
"Order of findings in CES, commonest → least?"	→ **leg pain > urinary retention > urinary incontinence**
"Back pain + perianal numbness, bilateral weakness — next?"	→ **immediate MRI lumbar spine** ✗ Wrong options: CT myelogram · labs · X-ray · discharge
"Cough → back pain, retention, faecal incontinence, saddle paraesthesia?"	→ **immediate MRI**
"Commonest presenting problem?"	→ **back + leg pain**
"Bladder + bowel loss, saddle numbness — management?"	→ **immediate MRI + acute surgery**
"Decompression window for bladder + motor recovery?"	→ **< 48 hours**
"After decompression, least likely to improve?"	→ **bladder dysfunction**
"Prognosis after urgent surgery with dense perineal anaesthesia?"	→ **pain + sensation improve, bladder may remain**
"12 h of retention + buttock numbness — true?"	→ **saddle anaesthesia + bladder dysfunction → poor prognosis** ✗ Wrong options: CT myelogram · ESI · steroids
"CES treatment — true?"	→ **long-term urinary dysfunction common** ✗ Wrong options: < 24 h better than < 48 h · laminectomy better · sexual dysfunction commonest · burst fracture commonest

– – Active space – –

"L4-5 HNP occluding sac – true?"	→ **urinary function can have a prolonged recovery** ✗ Wrong options: PVR > 200 mL excludes · elderly recover better
"Not true about CES?"	→ **positive Babinski** (TO)
"Diabetic, saddle anaesthesia, bladder/bowel, absent BCR?"	→ **conus medullaris** (TO)
"Hemi-CES differs from CES EXCEPT?"	→ **cause is mostly herniated disc** (TO)
"Saddle anaesthesia only – lesion level?"	→ **below S2** (TO)
"Pacemaker + conus-cauda picture – pre-op study?"	→ **myelo-CT** (TO)

– – Active space – –

CEREBRAL PALSY - SPINAL DISORDERS

– Spine · 10 / 69 –

1 Basics

Epidemiology

Scoliosis in CP overall **20%**; ↑ with severity · **spastic quadriplegia** highest (esp. non-sitters) · **bedridden → ~100%** · rare in ambulators

Spondylolisthesis **4–21%** in **spastic diplegia** → check lateral film

Pathophysiology

Cause muscle weakness + truncal imbalance (little evidence) · **pelvic obliquity** drives deformity

CP scoliosis vs idiopathic	
Progression	**More likely**; **1–2° per month** from age 8–10
Onset	Earlier
Shape	**Long, stiff C-shaped**; left curves common
Sagittal	Greater kyphosis / lordosis
Pelvis	**Pelvic obliquity**
Maturity	Delayed
Bracing	**Less effective**
Fusion	Longer, often **to pelvis**
Host	Medically fragile → multidisciplinary

4 Diagnosis & Classification

Weinstein classification

Group	**Curve**	**Pelvic obliquity**
I	**Double** thoracic + lumbar	Minimal
II	Large **lumbar / thoracolumbar**	**Marked**

Key radiology findings

X-rays same technique over time; erect films when possible · AP + lateral (rib deformity, wedging, rotation, spondylolisthesis) · **bending films**: push-pull / fulcrum if uncooperative

MRI **not routine** pre-op; only for **rapid progression** or **neuro change**

– – Active space – –

Exam — **hamstring contracture → ↓ lumbar lordosis** · **hip contracture → ↑ lordosis** · curve flexibility, balance, pelvic obliquity

5 Management

Non-op — **non-progressive curves < 50°** or age **< 10** (delay surgery) · custom seat (sitting only) · **TLSO improves sitting balance**, doesn't change natural history (may slow progression only if immature) · **Botox** short-term (2–3 months)

Operative criteria

Procedure	Indication
PSF ± pelvis	**Group I, 50–90°**, ambulators, progressive / interfering with sitting; **age > 10**; adequate hip ROM; stable nutrition
PSF ± ASF ± pelvis	**Group I > 90°**, **non-ambulators**, **Group II**; immature (**crankshaft**); anterior release for **> 100–120°** rigid curves (don't bend < 60°)
Growing rods	Young child
Pre-op traction	Severe rigid curves

Surgery threshold (OB) — progressive **> 40–50°** interfering with sitting in **> 10 yrs**

Fusion levels

Proximal — **T1 or T2** (avoid proximal junctional kyphosis)

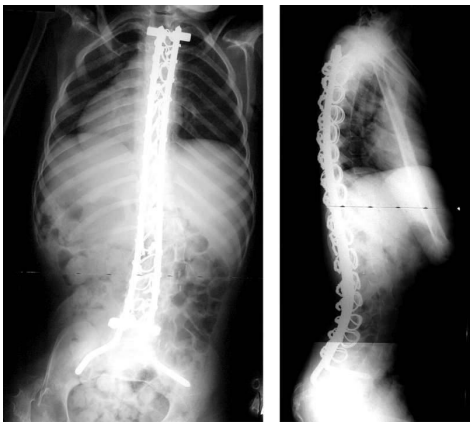
Distal — L4/L5 for long curves · **pelvis if obliquity > 15°** (else loss of correction + pseudarthrosis)

CP scoliosis needing fusion

- Pelvic obliquity > 15° / non-sitter → POSTERIOR T2 → **PELVIS**
  (do not fuse to L4 – late loss of correction)
- Immature or > 100–120° rigid → **ADD ANTERIOR RELEASE (same day)**

Fixation techniques

Technique	Note
Luque rod + sublaminar wires	Classic
Unit rod + sublaminar wires	Pelvic limb may penetrate pelvis
Pedicle screws	Better correction, may avoid anterior surgery
Galveston	Rod bent from **S1 lamina → PSIS → between iliac tables** just above **sciatic notch**
Others	Bilateral sacral screws, iliosacral screws, spinopelvic transiliac, **Dunn-McCarthy** (S-rod over sacral ala)



Unit rod with Galveston pelvic fixation

A/P same day ↓ complications vs staged (nutrition, blood loss, stay) · post-op brace usually unnecessary

Pre-operative assessment

Domain	Action
Nutrition	Weight < 5th centile → ↑ infection, intubation, stay; **albumin > 3.5 g/dL**; consider **G-tube**
Respiratory	History, exam, CXR (formal PFTs hard)
GI	Treat **GERD** (aspiration)
Seizures	Control; **valproic acid → check bleeding time**

Goal painless solid fusion, balanced spine, **level pelvis**; family goals + risk-benefit

6 Complications

Complication	Fact
Pulmonary	**Insufficiency = commonest**; aspiration, pneumonia
Implant failure	Often asymptomatic
Wound infection	**3–5%**, > idiopathic; local debridement usually enough
GI	GERD, malnutrition
Neuro	Seizures
Death	**0–7%**

Natural history larger curves progress more; pelvic obliquity; decubitus (conflicting)

X vs Y

	Weinstein I	Weinstein II
Curve	Double	Large lumbar/TL
Obliquity	Minimal	Marked
Fusion	PSF (50–90°, ambulator)	PSF ± ASF to pelvis

– – Active space – –

How it's asked

"GMFCS V, 75° C-curve with pelvic obliquity, mature?"	→ **posterior fusion T2 to pelvis** ✗ Wrong options: brace · anterior-only · A/P · T2–L4
"When add anterior release in CP?"	→ **> 100–120° rigid curves** or **immature (crankshaft)**
"Extend fusion to pelvis when?"	→ **pelvic obliquity > 15°**
"Commonest post-op complication?"	→ **pulmonary insufficiency**
"Pre-op lab for patient on valproate?"	→ **bleeding time**

CERVICAL ADJACENT SEGMENT DISEASE

– Spine · 11 / 69 –

1 Basics

Definitions

Adjacent segment degeneration	**radiographic** degeneration next to a fusion, ± symptoms
Adjacent segment disease (ASD)	**clinical** radiculopathy / myelopathy from the adjacent motion segment
Adjacent level ossification development (ALOD)	large anterior marginal osteophyte / ALL heterotopic bone next to a plated level; disc height preserved, no herniation; clinical significance unclear

Epidemiology

Incidence	clinically relevant ASD **1.6–2.4% / yr** (reoperation ~2.4%/yr) · **25.6% at 10 yrs** after ACDF
Levels	C4-5, **C5-6 (highest)**, C6-7 · **C2-3 lowest**

Risk factor for ASD	
Smoking	**Strongest**
Female	Yes (✗ Wrong options: male)
Age < 60	Yes
< 3 levels fused	Yes (more remaining motion segments)
Fusion ending next to C5-6 / C6-7	Most mobile levels
Pre-op dural indentation / disc protrusion at C5-7	Yes
Fusion vs non-fusion (foraminotomy, discectomy)	**No difference**

Pathophysiology

ASD	↑ compensatory motion → ↑ intradiscal pressure, stress, shear → herniation, height loss, osteophytes → stenosis · also ALL / longus colli damage · **needle marker in wrong disc**
Counter-evidence	**CDA (motion-preserving) does not reduce ASD** (Orthobullets ASD note) – partly natural history of spondylosis
ALOD	**anterior plate within 5 mm of adjacent disc**; variable-angle screws allow subsidence toward the disc ✗ Wrong options: diabetes · static screws (protective) · male · age > 60

– – Active space – –

Park classification of adjacent level ossification

Grade	Ossification
0	None
1	< 50% of disc space
2	> 50% of disc space
3	**Complete bridging**



Grade 3 adjacent-level ossification

3 Presentation

Radiculopathy dermatomal pain / weakness at adjacent root

Myelopathy gait instability, hand clumsiness, urinary change

Exam inspect wound · painful ROM arc · rule out **cubital / carpal tunnel** for sensory change · **Hoffmann** (myelopathy) · **Spurling** (radiculopathy) · **shoulder abduction relief** (radiculopathy) · **Lhermitte** (myelopathy)

4 Diagnosis & Classification

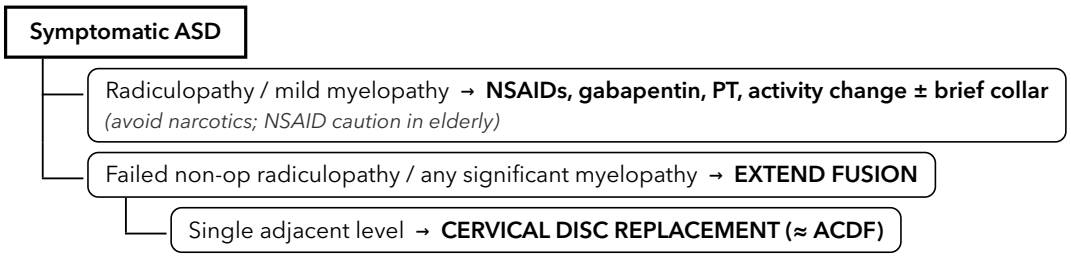
Key radiology findings

Study	Use
X-ray AP, lateral, **flexion-extension**	Disc narrowing, foraminal stenosis, osteophytes, facet arthropathy; **poor correlation with symptoms**
CT	**Pseudarthrosis**, OPLL
MRI	Foraminal / central stenosis at adjacent level

Differential (top 5) **pseudarthrosis** · progressive DDD · **indolent infection** · myelopathy / radiculopathy at a non-adjacent level

Prevalence context radiographic degeneration in **> 80%** of asymptomatic unfused people > 60 · in **25–91%** of fused patients

5 Management



Extension of fusion — approach by pathology location · **re-operate from the prior side** (assume ipsilateral RLN palsy unless ENT scope is normal → protects contralateral nerve) · include C5-6/C6-7 · fusing **> 3 levels** ↓ further ASD · plate **> 5 mm** from adjacent disc

CDR for ASD — single level; **equivalent to ACDF**; ↓ adjacent ossification vs plating; align centre of rotation · as index procedure **no clear ASD protection** (meta-analyses: possibly ↓ reoperation)

6 Complications

Progression — new adjacent segment degenerates after extension → further revision

Reoperation — overall **6.1–25.6%**; **0.7–3.7%/yr**

X vs Y

	Adjacent segment degeneration	Adjacent segment disease	ALOD
Definition	Radiographic	Clinical symptoms	Anterior ossification
Cause	Motion / natural history	Same	**Plate < 5 mm from disc**
Rx	Observe	Non-op → extend fusion / CDR	Observe

How it's asked

"Asymptomatic anterior ossification 2 yrs after ACDF – associated factor?"	→ **plate placed within 5 mm of adjacent disc** ✗ Wrong options: diabetes · static screws · male · age > 60
"Strongest risk factor for cervical ASD?"	→ **smoking**
"Level with highest ASD risk?"	→ **C5-6**
"Revision anterior approach – which side?"	→ **same side as prior surgery** (unless ENT shows normal cords)

– – Active space – –

CERVICAL DISC ARTHROPLASTY

– Spine · 12 / 69 –

1 Basics

History

1966 Fernström	stainless-steel **ball bearing**; device complications → abandoned as ACDF rose
2002	modern CDA; premise = maintain motion → ↓ adjacent segment disease
2010 →	RCTs: **superior to fusion for reoperation rates + quicker return to work**

Benefits & risks

Benefit	Risk
Preserves motion	**Hardware failure** → retropulsion into canal, **paralysis**
Pseudarthrosis not a concern	**Persistent facet-origin neck pain**
Quicker return to activities	–

4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP + lateral	Alignment, disc height
CT	Implant **positioning + sizing**; facet arthrosis
MRI	Central + foraminal stenosis

5 Management

Indications & contraindications

Indications	Contraindication
1- and 2-level radiculopathy	**Significant facet degeneration (advanced facet arthrosis)** → persistent axial neck pain → fuse instead
1- and 2-level myelopathy (thorough decompression)	–
Revision CDA **within 2 weeks** = primary results	–

Not contraindications two-level disease · **loss of disc height** · neutral lordosis/kyphosis · bilateral Hoffmann (myelopathy)

Evidence

	vs ACDF
Neurologic improvement	**Equivalent** (1 + 2 level)
PROMs	**Equivalent**
Reoperation	**Superior (lower)**; CDA reoperations mostly **index-level, implant-related**
Adjacent segment disease	"Mounting evidence of ↓" (Orthobullets CDA note) · "no protective effect, possibly ↓ reoperation" (Orthobullets ASD note) – read the options

Single-level myelopathy, no fixed kyphosis (smoker, inverted brachioradialis)

CDA at C5-6 gives best function
✗ Wrong options: NSAIDs/PT · laminoplasty C4-7 · laminectomy-fusion C4-7 (multilevel, ↑ NDI) · C6 corpectomy

Technique

Anterior approach · **align centre of rotation in coronal + sagittal planes** (critical in 2-level)



Single-level cervical disc arthroplasty (lateral)

6 Complications

Hardware failure catastrophic retropulsion

Facet pain if arthritic facets ignored

X vs Y

	CDA	ACDF
Motion	Preserved	Lost
Pseudarthrosis	N/A	Possible
Reoperation	**Lower**	Higher
Facet arthrosis	contraindicated	✓
Levels	1–2	Any

– – Active space – –

How it's asked

"51-yr single-level myelopathy, no kyphosis — best functional outcome?"	→ **C5-6 disc arthroplasty**
"C7 radiculopathy — contraindication to CDA?"	→ **advanced facet arthrosis** ✗ Wrong options: two-level · disc height loss · neutral alignment · Hoffmann
"Where is CDA superior to ACDF?"	→ **reoperation rate**

– – Active space – –

CERVICAL FACET DISLOCATIONS & FRACTURES

– Spine · 13 / 69 –

1 Basics

Epidemiology

Young	high-energy (MVC, motorcycle, contact sport) · **elderly** – low-energy
C7 / C7-T1	**17%** of injuries → always visualise cervicothoracic junction
Associated	head injury, **non-contiguous spine injuries** (TL, CT, OC junctions), limb injuries

Pathoanatomy

Injury	Key facts
Facet fracture	**Superior facet** more often; ↓ threshold for dislocation (loses interlock); may be a **spontaneously reduced dislocation**
Unilateral facet dislocation	**Most frequently missed** C-spine injury on X-ray · **~25%** subluxation · **monoradiculopathy** that improves with traction (inferior facet of cephalad vertebra in foramen) · rotation component
Bilateral facet dislocation	**~50%** subluxation · **SCI in ~80%**

Mechanism	**flexion-distraction** ± rotation → posterior structures torn (ISL, capsule, LF, posterior annulus), **ALL usually preserved** ✗ Wrong options: flexion-compression · vertical compression · extension
PLL / posterior annulus disruption	**40%** unilateral · **80%** bilateral

– – Active space – –

2 Mechanism

Allen & Ferguson classification (research)

Group	
1 Flexion-compression	–
2 Vertical compression	–
3 Flexion-distraction	**Stage 1** facet sprain, < 10° kyphosis · **Stage 2 unilateral dislocation** · **Stage 3 bilateral, 50% (perched)** · **Stage 4 complete (100%)**
4 Extension-compression	–
5 Extension-distraction	–
6 Lateral flexion	–

Clinical (descriptive) compression, burst, flexion-distraction, unilateral/bilateral dislocation, facet fracture

3 Presentation

Unilateral single-arm paraesthesia: **C5-6 → C6** (wrist extension weakness, thumb numbness) · **C6-7 → C7** (triceps + wrist flexion weakness, index + middle finger numbness) · angular deformity

Bilateral bilateral UL + LL weakness, sensory change → **ASIA** documentation

History AS / DISH, prior fusion · scalp lacerations → head injury

4 Diagnosis & Classification

Key radiology findings

Study	Findings / indication
X-ray (AP, lateral, oblique, open mouth)	Subluxation (25% unilateral, 50% bilateral), disc height loss (retropulsed disc), widened interspinous distance, focal hypolordosis, soft-tissue swelling
Flexion-extension	**Whenever a facet fracture is seen** (spontaneous reduction / occult instability)
CT	Most cases; subtle facet subluxation, facet/pedicle/lamina fractures
MRI	**Before any open reduction / surgery**; **obtunded** patient (emergent); **failed closed reduction**; **neuro worsening during reduction** – looks for **disc herniation**, PLC injury, cord compression; **cord haematoma = poor motor prognosis**

MRI timing (TO) best sensitivity for haemorrhage/oedema **within 48 h**

– – Active space – –



Bilateral facet dislocation: 50% subluxation on lateral X-ray

Differential — **lateral mass fracture-separation** → needs **2-level** fusion (dislocation = 1 level)

5 Management

Operative criteria – facet fractures

Stable (non-op 6–12 wks, collar / halo)	Unstable (single-level instrumented fusion)
Reduced unilateral fracture, **no instability on flex-ex**	**Bilateral** facet fractures
< 40% of lateral mass or **< 1 cm** height	**> 40% of lateral mass** or **> 1 cm** height

Non-op risk — **> 30% re-subluxation / redislocation**, persistent pain

Halo — suboptimal in lower C-spine; **most likely to fail with facet fracture-dislocation** · hard orthosis often satisfactory
✗ Wrong options: C1 lateral mass · C2 pars · burst

Closed reduction vs MRI first

Facet dislocation

- Awake, alert, cooperative → **EMERGENT CLOSED REDUCTION (Gardner-Wells)**
  → *MRI after reduction → URGENT STABILISATION (< 24 h)*
- Obtunded / intoxicated / intubated / GCS ↓ → **EMERGENT MRI FIRST**
  → *OPEN REDUCTION + STABILISATION*
- Failed closed reduction or neuro worsening → STOP → MRI → **OPEN REDUCTION**

Awake patient – source conflict — **immediate closed traction** (Orthobullets, OB – especially with deteriorating neuro, ASIA E → D) · **hard collar + MRI first** (TO answer for awake, intact patient 30 min post-injury) – read the options

Pre-reduction MRI — not routine in awake patients (delays reduction, early reduction ↑ recovery)

– – Active space – –

MRI finding that changes plan — **herniated disc at dislocated level → anterior discectomy before/with reduction**
✗ Wrong options: cord oedema · PLC disruption · lamina fracture · ALL signal

Reduction fact	Value
Unilateral	**Harder** to reduce, **more stable** after; closed success ~25%
Bilateral	**Easier** (PLL torn), **less stable** after
Fail closed reduction	**26%**
Never	Closed reduction with **altered mental status**

Closed reduction technique

Gardner-Wells tongs — **1 cm above pinna**, in line with **EAM**, **below skull equator** · **posterior pin placement** → flexion moment

Weights — 5–10 lb increments · up to **140 lb or 70% body weight** · ~**9.4–9.8 lb per level above injury** · carbon-fibre tongs for MRI (~80 lb limit)

After reduction — drop to **10–15 lb**, add **extension** (pulley / thoracic pad)

Abort — neuro worsening (→ immediate MRI) · **overdistraction > 1.5× adjacent disc space**

Requirements — sedation, respiratory supervision, serial cross-table laterals + neuro exams

Surgical options

Procedure	Indication / technique
ACDF (single level, plate)	Disc herniation compressing cord after reduction; anterior open reduction of **unilateral** dislocation — Caspar pin distraction + **rotate cephalad pin toward dislocated side**
Anterior reduction of bilateral	Converging Caspar pins (10–20°) compressed to unlock; curette pushes rostral body back · **lamina spreaders not effective for bilateral**
Posterior reduction + lateral mass screws	No disc herniation; irreducible from front/closed; Penfield 4 lever, remove tip of caudal superior facet; often **2 levels** (poor purchase)
Anterior → posterior	Disc needs removal but can't reduce anteriorly: discectomy, plate fixed to **superior body only** (stops graft kick-out, allows rotation)

Biomechanics — posterior fixation > anterior plating for flexion-distraction (Sutterlin, Coe) · anterior overdistraction risks cord + hardware

6 Complications

Complication	Fact
SSI	↑ posterior approach
Redislocation	Unilateral treated in orthosis → ACDF
Respiratory	ARDS, pneumonia, tracheostomy
Vertebral artery	Up to **11%**; ↑ with lateral mass / transverse process involvement; often unrecognised
Oesophageal	Anterior reduction → primary repair
Pin-tract	Halo; meningitis if inner table breached

Prognosis worse with severe deficit, older age, **cord haematoma** · **bilateral dislocation = worst outcome** among subaxial patterns

X vs Y

	Unilateral	Bilateral
Subluxation	**25%**	**50%**
Neuro	**Radiculopathy**	**SCI ~80%**
PLL torn	40%	80%
Reduce	Harder, stable after	Easier, unstable after
X-ray	Most missed	Obvious

How it's asked

"Characteristic mechanism of facet dislocation?"	→ **flexion-distraction**
"Right C6-7 facet dislocation — which CT image?"	→ **unilateral right perched/jumped facet**
"Awake, alert, 30 min post-injury, C4-5 jumped facets — next (TO)?"	→ **hard collar + MRI** (TO) · OB awake-patient stems → **immediate closed reduction** — read the options
"Awake, cooperative, ASIA E → D after CT?"	→ **immediate closed reduction with traction** ✗ Wrong options: MRI first · steroids
"GCS 3, intubated, bilateral C5-6 dislocation?"	→ **MRI**
"Intubated, moving limbs, bilateral C6-7?"	→ **cervical MRI then reduction**
"Intoxicated, obtunded, bilateral C4-5?"	→ **immediate MRI then surgery**
"MRI finding mandating surgery instead of closed reduction?"	→ **herniated disc at dislocated level**
"GCS 10, C7-T1 fracture-dislocation?"	→ **immediate MRI**
"Altered mental status — MRI finding that changes approach?"	→ **herniated disc (→ anterior)**

– – Active space – –

"Nondisplaced C6 facet fracture, stable on flex-ex?"	→ **hard collar**
"Halo most likely to fail?"	→ **C6-7 facet fracture-dislocation**
"Worst outcome with incomplete SCI?"	→ **bilateral facet dislocation**

CERVICAL LATERAL MASS FRACTURE SEPARATION

– Spine · 14 / 69 –

1 Basics

Epidemiology

Demographics **M:F 2:1**, mean age **35** (20–70)

Level **C6 > C5 > C7 > C4 > C3**

Pathoanatomy

Lesion fracture lines through **lamina + pedicle** → lateral mass (with its superior + inferior facets) floats free

Two-level instability both the facet joint above and below are compromised → **fuse 2 levels** (vs facet dislocation = 1 level)

Associated finding	Rate
Anterior listhesis of fractured vertebra	**77%**
Listhesis of vertebra above / below	24% / 10%
Coronal translation	33%
Body collapse	33% (less in type A)

2 Mechanism

Mechanism traffic accident, fall, heavy object on head → **hyperextension + lateral compression + rotation**

Allen-Ferguson groups extension-compression · **lateral flexion → type B comminution** · flexion-distraction

3 Presentation

Neuro symptoms common (**up to 66%**): radiculopathy, SCI / myelopathy (Frankel / ASIA)

Exam **torticollis**, paravertebral spasm

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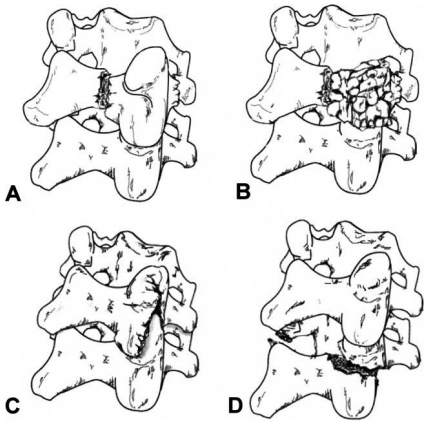
4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray (AP, lateral, oblique)	**Low sensitivity – 38%**; disc narrowing; instability = **> 3.5 mm** displacement, **> 10° kyphosis**, **> 10° rotation** vs adjacent level
CT	**Diagnosis**; fracture extends into adjacent facets, foramen transversarium, transverse process, pedicle, lamina; translation, uncovertebral subluxation
MRI	**ALL rupture 50–75%** · PLL 30–35% · ISL/SSL 10–75% · disc disruption, bone bruise

Kotani classification

Type	Description	Same-level / adjacent translation
A – Separation	**2 fracture lines: unilateral lamina + pedicle**	91% / 20%
B – Comminution	Multiple lines, lateral wedging (lateral flexion)	– / 50%
C – Split	Vertical coronal line; superior articular process of caudal vertebra invaginates	80% / 0%
D – Traumatic spondylolysis	Bilateral horizontal pars fractures → anterior/ posterior elements separate	100% / 50%



Kotani classification of lateral mass fractures

5 Management

Non-op (Miami J / halo) stable, no deficit; **halo poorly controls hyperextension/rotation**; spontaneous fusion only **20%**; late instability + pain → surveillance

Procedure	Indication
Posterior decompression + 2-level instrumented fusion	**Most cases** (injured structures posterior); lateral mass / pedicle screws or lateral mass plating; late instability after non-op
Two-level ACDF	**Minimally displaced**, no posterior reduction needed; **controls anterior collapse + rotation**
Single pedicle screw	**Type A** separation **without instability**
A + P	Extra anterior support; failed anterior reduction / late presentation

Posterior caveats	anterior disc collapse + late kyphosis; **midline (wiring) fusion doesn't control rotation**

C5 lateral mass fracture-separation + ASIA C

- Posterior → **INSTRUMENTED FUSION C4-6 (both levels)**
  (not C5-6 only or C5 corpectomy – body intact)
- Anterior alternative → **2-LEVEL ACDF C4-6**

Lateral mass screw safety

Technique (TO)	start **1 mm medial to pillar midpoint**, aim **30° lateral + 15° cephalad**
Structure most at risk (TO)	**nerve root** (exits anterolateral to facet; long screw) ✗ Wrong options: vertebral artery (too medial) · cord (safe) · thoracic duct

6 Complications

Vertebral artery injury (pedicle screws) · **late kyphosis** · **late instability** · chronic neck pain · radiculopathy

X vs Y

	Lateral mass fracture-separation	Facet dislocation
Mechanism	**Hyperextension + lateral compression + rotation**	**Flexion-distraction**
Unstable levels	**2**	1
Fusion	**2 levels** (above + below)	1 level
X-ray detection	38%	Unilateral often missed

How it's asked

"ASIA C, C5 lateral mass fracture-separation – management?"	→ **posterior instrumented fusion C4-6** ✗ Wrong options: C5-6 · corpectomy · 1-level ACDF
"Structure most at risk with lateral mass screw?"	→ **nerve root** (TO)
"Why fuse two levels?"	→ **superior + inferior facets of the floating mass both incompetent**

– – Active space – –

CERVICAL MYELOPATHY

– Spine · 15 / 69 –

1 Basics

Aetiology

Commonest cause	**degenerative cervical spondylotic myelopathy (CSM)**: osteophytes, disc-osteophyte complex, degenerative spondylolisthesis, **LF hypertrophy**
Others	**congenital stenosis** (symptoms when spondylosis adds on) · **OPLL** · tumour · epidural abscess · trauma · kyphosis
Cord injury mechanism	direct compression + **ischaemia (anterior spinal artery compression)**
Tandem stenosis (cervical + lumbar)	**~20%** (Orthobullets) · **5–25%** (OB) → myelopathy **takes precedence**

Epidemiology & canal diameter

Commonest level of spondylosis (TO)	**C5-6**, then C6-7 (most subaxial motion) ✗ Wrong options: C3-4 · C1-2

Canal (AP)	Value (TO)
Normal	~17 mm (C1 23, C2 20, C3–6 17, C7 15)
Relative stenosis	**10–13 mm**
Absolute stenosis	**< 10 mm** (area < 100 mm²)
Cord compression	Canal **< 13 mm** (Orthobullets)
Pavlov (Torg) ratio	**< 0.8** → congenitally narrow canal

X-ray degeneration	**70% by age 70** · MRI false positives: **28%** of > 40 yrs have HNP / foraminal stenosis

Natural history

Pattern	**slow, stepwise deterioration with plateaus** · rarely improves non-op ✗ Wrong options: stable · rapid continuous decline · improves with steroids/rest
Worse non-op outcome	segmental kyphosis, circumferential compression

3 Presentation

Symptoms	neck pain/stiffness (often absent; **~15%** of severe cases painless), occipital headache · **diffuse non-dermatomal** bilateral paraesthesia · **hand clumsiness** (buttons, handwriting, dropping things) · **gait instability** (most important clinical predictor; stairs) · urinary retention (**rare, late**)
Severe CSM	proximal LL muscles weaker than distal (opposite in lumbar stenosis)

Clinical diagnosis – UMN signs

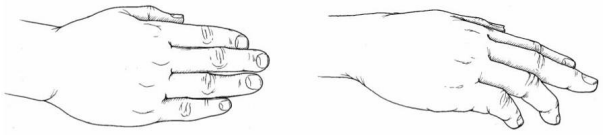
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Sign	Fact
Gait ataxia	**Commonest physical finding (91%)** (OB)
Hoffmann	Flick middle finger DIP → thumb/index flexion; "most common exam finding" (Orthobullets) · **80%** (OB) / 59% in another series; ↑ with severity
Hyperreflexia	LL 81%, UL 67%; **absent if concomitant peripheral neuropathy / root compression / diabetes**
Inverted radial reflex	Tap distal brachioradialis → finger flexion
Sustained clonus (> 3 beats)	**Sensitivity ~13%, specificity ~100%**
Babinski	**Least common** (≤ 1/3)
Finger escape (Wartenberg)	Small finger abducts/flexes (intrinsic weakness)
Grip-and-release	Normal **20 in 10 s**
Lhermitte	Neck **flexion → electric shock** down spine/limbs; specific, insensitive; also MS
Romberg / tandem gait	Posterior column dysfunction
Proprioception loss	Advanced, **poor prognosis**
Pinprick ↓	Lateral spinothalamic tract

UMN vs LMN — **fasciculations, atrophy, tongue fasciculations, areflexia = LMN** (EXCEPT answers) · weakness = both

Myelopathy hand — intrinsic/webspace atrophy, finger escape, **dysdiadochokinesia**, Hoffmann
✗ Wrong options: **intrinsic plus posture**

Other tests — Spurling (radiculopathy) · Jackson (hyperextension) · Lasègue (lumbar SLR)



Finger escape sign

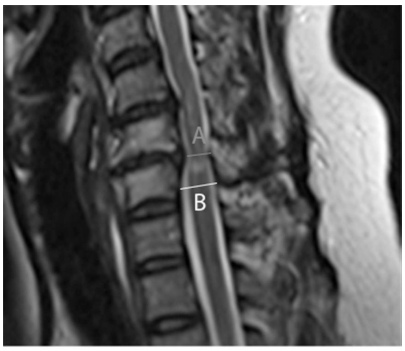
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4 Diagnosis & Classification

Key radiology findings

Study	Findings
X-ray (AP, lateral, oblique, flex-ext)	Uncovertebral + facet degeneration, osteophytes, disc narrowing; **Pavlov < 0.8**; **C2-C7 alignment** + **local kyphosis angle**; obliques → foraminal stenosis; flex-ext → instability / compensatory subluxation
MRI	**Study of choice**; CSF effacement = functional stenosis; **T2 bright = myelomalacia**; **T1 signal change → poorer prognosis**; **compression ratio < 0.4 → poor prognosis**
CT	OPLL, osteophytes
CT myelogram	Pacemaker / hardware artefact (C1-2 or lumbar puncture)
NCS	High false negatives; separates peripheral vs central (ALS)

Compression ratio smallest **AP** cord diameter ÷ largest **transverse** diameter



Compression ratio of the cord

Prognostic MRI **focal** T2 signal better than **multisegmental** T2 signal · isolated **low T1** = gliosis/cavitation, little improvement · cord shape boomerang → teardrop → **triangular** (worst)

Non-op success predictor **transverse cord area > 70 mm²**
✗ Wrong options: ↑ CMCT · low T1 signal · canal < 13 mm · younger age

Differential normal ageing, stroke, movement disorders, **B12 deficiency**, **ALS**, **MS**

Tandem stenosis

Lumbar claudication + UMN signs (hyperreflexia, clonus, broad gait, clumsy hands) **MRI cervical (± thoracic) spine before lumbar surgery** · pacemaker → **CT myelogram of C-spine** · gait problem with 5/5 strength + mild lumbar stenosis → **MRI thoracic + cervical** (brain MRI second-line)
✗ Wrong options: lumbar decompression ± fusion · ESI · PT · discogram · ABI

Classifications

Nurick	
0	Root symptoms only
1	Cord signs, **normal gait**
2	Gait difficulty, **fully employed**
3	Gait prevents employment, walks unassisted
4	**Walks with assistance**
5	Chair / bedbound

Ranawat	
I	Pain, no deficit
II	Subjective weakness, hyperreflexia, dysaesthesia
IIIA	Objective weakness + long-tract signs, **ambulatory**
IIIB	Same, **non-ambulatory**

JOA / mJOA (17 points) UL motor, LL motor, sensory, bladder · **severe = mJOA < 12** · excellent outcome ≥ 16

5 Management

Non-op mild disease **without functional impairment** or poor surgical candidate (function > exam findings) · NSAIDs, gabapentin, collar in slight flexion, PT · no traction/chiropractic · **cord compression without myelopathy signs → PT**
✗ Wrong options: surgery

Surgery significant functional impairment / progression; **mild myelopathy by mJOA** also indicated; **operate as soon as medically stable**
✗ Wrong options: wait for Nurick IV, multisegmental T2 signal, intolerable pain, incontinence

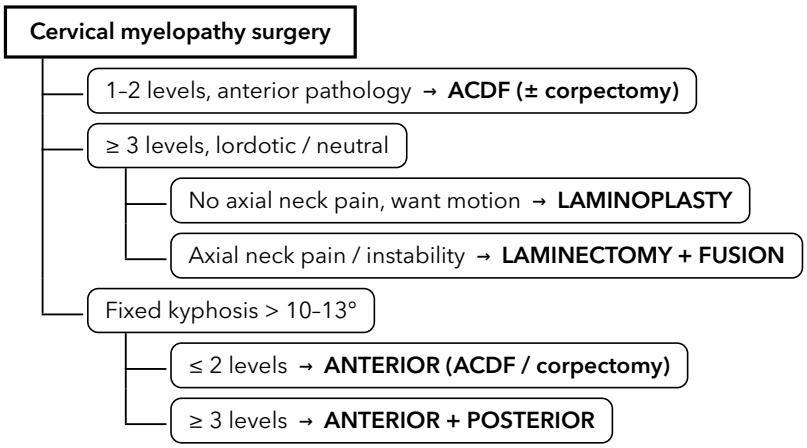
Goals & outcomes of surgery

Goal **halt stepwise neurologic decline**
✗ Wrong options: complete recovery · mJOA > 16 · resolve myelomalacia · restore normal lordosis (aim neutral)

Expected **significant mJOA improvement at 1 yr even in severe (mJOA 10) CSM**; gait / sensory improvement variable

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Approach selection



Operative criteria – alignment thresholds

Threshold	Meaning
Fixed kyphosis > 10°	Anterior procedure (corrects kyphosis); posterior alone not enough
Local kyphosis > 13°	Contraindication to posterior decompression / **strongest predictor of poor laminoplasty outcome** (✗ Wrong options: multilevel stenosis · osteoporosis · CSF effacement)
Kyphosis > 12° + > 2 levels	**Combined A/P** (OB)
≥ 3-level corpectomy	Anterior alone → **> 70% catastrophic failure** → add posterior

Bowstringing — in kyphosis cord drapes over anterior osteophytes; posterior decompression can't let it drift back

Anterior decompression – ACDF/corpectomy

ACDF — **mainstay for 1–2 levels**, anterior pathology (soft disc, disc-osteophyte, focal OPLL), fixed kyphosis > 10° · single-level CSM gold standard (TO) · Smith-Robinson approach · avoid with poor swallowing

Anterior vs posterior — anterior: **↓ infection, ↓ blood loss, ↓ post-op pain** · no difference in C5 palsy, C7 radiculopathy, overall complications

Corpectomy — retrovertebral disease / kyphosis · 2-level corpectomies vulnerable (prefer **1 corpectomy + adjacent discectomy**) · **graft dislodgement 7–20%** (can be fatal) · **2-level → add posterior fixation**; **3-level → A + P mandatory**

3-level corpectomy, anterior plate failure — prevented by **combined anterior + posterior fixation**
✗ Wrong options: discectomies (can't decompress OPLL) · mesh cage · longer collar/halo · screw mix

Cervical disc arthroplasty

Ideal (TO) — skeletally mature, 20–70 yrs, **1–2 levels C3–C7**, soft disc, minimal spondylosis, preserved facets, disc height ≥ 3 mm, ≥ 6 wks failed non-op · single-level myelopathy without kyphosis → CDA gives best function (OB)

Contraindications (TO) — **≥ 3 levels** · **instability (> 3 mm translation / > 11° rotation)** · **facet degeneration** · axial neck pain alone · severe spondylosis (bridging, > 50% height loss, < 2° motion) · osteoporosis / metabolic bone disease · adjacent fusion / prior surgery at level · **morbid obesity (BMI > 40)** · **RA / spondyloarthropathy** · infection, malignancy, steroids, pregnancy · kyphotic deformity
✗ Wrong options: **2-level disease** (NOT a contraindication)

Complication — hardware failure → retropulsion

Posterior decompression – laminectomy/fusion vs laminoplasty

	Laminectomy + fusion	Laminoplasty
Indication	**Multilevel**, kyphosis < 10°, **axial neck pain** (facet)	**Multilevel**, **preserved lordosis**, **congenital stenosis**, **OPLL**, motion desired, pseudarthrosis risk (diabetes, steroids)
Contraindication	Fixed kyphosis > 10–13°	**Kyphosis > 13°** (e.g. 15°, 20°); severe axial neck pain (relative)
Motion	Lost	**Preserved** (some loss)
Wound complications	Higher	**Lower**
Cost (2 yr)	**Higher**	Lower
Revision	Comparable	Comparable
ASD	Higher	Lower
C5 palsy	Higher (some series)	Similar / lower

Laminoplasty mechanism — ↑ **AP** canal dimension + cord drifts dorsally · **open door** = unilateral hinge, opening held by bone/anchors/plate · **French door** = bilateral hinges, midline opening · C3–7 · facets not violated · bilateral exposure still needed
✗ Wrong options: medial-lateral

Not contraindications to laminoplasty — OPLL · multilevel stenosis · 30° flex-ext motion · age > 55 · SAC < 8 mm · prior posterior surgery · radiculopathy (add keyhole foraminotomy) · C7 SVA +5 cm · compression ratio 0.3

Laminoplasty + fusion after ACDF — preserves **paraspinal muscle attachments / ↓ atrophy**
✗ Wrong options: infection · neuro · stay benefit

Laminectomy alone — rarely; **post-laminectomy kyphosis 11–47%**

Posterior-alone ideal image — **multilevel stenosis with preserved lordosis**
✗ Wrong options: teardrop fracture · kyphotic multilevel → A/P · skip-level lordotic → skip ACDF · anterior abscess

Central cord syndrome on background CSM + axial neck pain, multilevel — **posterior decompression + fusion**
✗ Wrong options: 2-level corpectomy without posterior · lumbar MRI (proximal weaker)

Combined anterior-posterior

Indications — **multilevel stenosis in rigid kyphotic spine** · multilevel corpectomies · **post-laminectomy kyphosis**

– – Active space – –

Occipitocervical fusion	periodontoid pannus (posterior-only resolves pannus; transoral ↑ morbidity > 4 h)

6 Complications

C5 palsy

Definition	post-op **deltoid ± biceps** weakness ≥ 1 grade without myelopathy worsening
Incidence	**~4.6%** (Orthobullets) · 5–12% after laminoplasty · 0–30% range
Timing	immediate to **weeks** (most within 1 week; up to 20 days); **92% unilateral**; half with C5 pain/sensory change
Risk factors	**male** (strongest) · **laminectomy + fusion** (higher in some studies) ✗ Wrong options: age · myelopathy severity · OPLL · level
Anterior vs posterior	**no significant difference** (Orthobullets, OB, TO); laminoplasty lower in some studies
Mechanism	**dorsal cord drift tethers C5 root** (C5 at apex of lordosis) vs segmental cord ischaemia/reperfusion · prevention: prophylactic **C4-5 keyhole foraminotomy**, avoid excessive cord shift/lordosis, deltoid/biceps MEP monitoring
Prognosis	**good**; observe · ~80% recover by 6–12 months · **prolonged recovery: multilevel paresis, motor grade ≤ 2, sensory involvement + intractable pain** ✗ Wrong options: revision · extension · steroids

Anterior approach complications

Complication	Fact
RLN palsy (hoarseness)	Commonest nerve injury (~1%); **prolonged tracheo-oesophageal retraction / ET cuff**; **right = left** in studies; **↑ in revision (up to 9.5%)**; observe → ENT at **6 wks** (Teflon); revision on **same side** as prior palsy · left RLN loops **aortic arch**, right **subclavian**
Dysphagia	30–50%; **local retropharyngeal steroid** before closure ↓ it (✗ Wrong options: TXA · incision type)
Airway compromise	**> 5 h surgery, > 300 mL blood loss, multilevel at/above C3-4 (C2–C4 exposure) → keep intubated 24–48 h**
Retropharyngeal haematoma (TO)	**6–24 h**, tense incision, dyspnoea/dysphagia/dysphonia → **immediate evacuation**
Angioedema (TO)	**6–12 h**, face/tongue swelling → IV steroids, antihistamine, adrenaline
Pharyngolaryngeal oedema (TO)	**12–72 h**, voice change, worse supine → may need **cricothyrotomy**
Vertebral artery	**Burr through lateral wall** of body in corpectomy
Horner	Lateral retraction on **longus colli** (sympathetic chain)
Superior laryngeal nerve	Rare; C4 and above
Hypoglossal	Rare; tongue deviation
Oesophageal	Rare; trauma, revision, diverticula

Posterior approach & general complications

Complication	Fact
Infection	**Higher posterior** than anterior
Pseudarthrosis	**12% single, 30% multilevel** (3-level ACDF + plate fusion ~80%)
Post-laminectomy kyphosis	Loss of posterior tension band + muscle denervation; ↑ with pre-op kyphosis, **> 30–50% facet resection**, **children (~100%) → A/P revision**
Epidural haematoma	**1/1000**; new weakness → **emergent MRI + evacuation**
Hardware failure	7–20% after 2-level corpectomy → stabilise posteriorly

Epidural haematoma age	T1	T2
Hyperacute (< 24 h)	**Hypo**	**Hyper** (STIR bright)
Acute (1–3 d)	Iso	Hypo
Subacute (3 d–2 wk)	Hyper	Hypo
Chronic (> 2 wk)	Hypo	Hypo

Root-level localisation

Disc → root	Motor	Sensory	Reflex
C4-5 → C5	**Deltoid** (± biceps)	**Lateral arm**	Biceps
C5-6 → C6	**Biceps (elbow flexion) + wrist extensors**	**Radial forearm, thumb**	**Supinator (brachioradialis)**
C6-7 → C7	**Triceps + wrist flexors**	Index, middle (ring)	**Triceps**
C7-T1 → C8	**Finger flexors (FDP)**	**Little finger, ulnar hand/ forearm**	–
T1	Finger abduction	Medial forearm	–

Cervical rule disc compresses the **exiting lower root** (C5-6 → C6)

Shoulder abduction relief sign radiculopathy (C7 here; elbow flexion weakness = C6)

X vs Y

	Myelopathy	Radiculopathy
Lesion	Cord (UMN)	Root (LMN)
Signs	Hoffmann, clonus, broad gait	Dermatomal pain, weak reflex
Tests	Lhermitte	Spurling, shoulder abduction relief

– – Active space – –

Early airway problem after ACDF (TO)	Time	Treatment
Angioedema	6–12 h	Steroids / antihistamine / adrenaline
Retropharyngeal haematoma	**6–24 h**	**Immediate evacuation**
Pharyngolaryngeal oedema	12–72 h	Reintubation / cricothyrotomy

How it's asked

"UMN sign EXCEPT?"	→ **fasciculations**
"Neck flexion → electric shocks?"	→ **Lhermitte sign**
"Clumsy hands, Hoffmann, finger escape – next?"	→ **cervical MRI** ✗ Wrong options: splint · carpal tunnel injection · EMG
"Least common finding in myelopathy?"	→ **Babinski**
"Myelopathy hand EXCEPT?"	→ **intrinsic plus posturing**
"Finding supporting UMN condition?"	→ **sustained clonus** ✗ Wrong options: absent reflexes · atrophy · tongue fasciculations
"Lumbar stenosis + clumsy hands + 3+ patellar reflexes – next?"	→ **MRI cervical spine**
"Lumbar DS + myelopathy, pacemaker?"	→ **CT myelogram of cervical spine**
"Wide-based gait, 5/5 strength, mild lumbar stenosis?"	→ **MRI thoracic + cervical spine**
"Non-op success predictor?"	→ **transverse cord area > 70 mm²**
"Natural history?"	→ **stepwise deterioration after stable periods**
"Cord compression, no myelopathic signs?"	→ **physical therapy**
"Most likely outcome after surgery?"	→ **halt stepwise decline**
"When to operate?"	→ **when medically stable**
"mJOA 10, post C3-7 fusion – expected?"	→ **significant mJOA improvement at 1 yr**
"Goal of surgery in severe CSM?"	→ **prevent further decline**
"Myelomalacia, lordotic, C4-7 – best pairing?"	→ **posterior decompression + fusion C4-7 to prevent deterioration**
"Single-level CSM – best approach?"	→ **ACDF** (TO)
"Myelopathy + > 10° kyphosis?"	→ **anterior decompression + fusion**
"Kyphotic, myelomalacia, broad gait?"	→ **multilevel anterior decompression + fusion**
"C4/5 + C5/6 compression, progressive?"	→ **ACDF both levels**
"Kyphosis + 3-level compression, needs walker?"	→ **multilevel anterior strut + plate, then posterior fusion**

– – Active space – –

"Anterior vs posterior — anterior has?"	→ **lower blood loss**
"Poor laminoplasty outcome predictor?"	→ **local kyphosis > 13°**
"Laminoplasty contraindicated in which image?"	→ **rigid kyphosis > 13°**
"Posterior-only most appropriate?"	→ **multilevel stenosis, preserved lordosis**
"Laminoplasty — true?"	→ **preserves more motion than laminectomy + fusion**
"Strongest contraindication to open-door laminoplasty?"	→ **C2-7 rigid kyphosis 15°**
"Laminoplasty decompresses by?"	→ **↑ AP canal dimension**
"Relative contraindication to laminoplasty?"	→ **kyphosis 20°** ✗ Wrong options: OPLL · age
"Contraindication to laminoplasty?"	→ **fixed cervical kyphosis** ✗ Wrong options: SAC < 8 mm · OPLL · prior posterior surgery · radiculopathy
"Long-standing multilevel spondylotic compression?"	→ **multilevel laminectomy + fusion**
"Contraindication to isolated posterior approach?"	→ **rigid kyphotic alignment**
"Best indication for laminoplasty alone?"	→ **multilevel congenital stenosis with lordosis**
"Laminoplasty vs laminectomy after ACDF — benefit?"	→ **preserved muscle attachments**
"80-yr fall, CSM, axial neck pain, multilevel?"	→ **posterior decompression + fusion**
"Laminoplasty vs laminectomy-fusion — true?"	→ **local kyphosis > 13° = poor laminoplasty outcome**
"3-level corpectomy failed — prevention?"	→ **combined anterior + posterior fixation**
"Fixed kyphosis + multilevel compression?"	→ **combined anteroposterior decompression + fusion**
"Favours combined A/P?"	→ **axial pain, kyphosis 17°, 3 levels**
"Bilateral deltoid weakness day 2 after C5-6 surgery?"	→ **C5 palsy** (TO)
"Commonest neuro complication after laminoplasty?"	→ **biceps weakness (C5)**
"Radiculopathy pattern from cord drift?"	→ **motor-dominant, deltoid**
"Root affected after laminoplasty?"	→ **C5**
"New 3/5 weakness after C4-6 ACDF?"	→ **C5, biceps**
"Deltoid 3/5 two weeks after C2-4 fusion — treatment?"	→ **reassurance + observation**
"C5 palsy risk factor?"	→ **male sex**
"Deficit common to post-ACDF and post-laminectomy palsy?"	→ **can't abduct arm against gravity**
"C5 palsy — true?"	→ **no difference anterior vs posterior**

– – Active space – –

"C5 palsy prolonged recovery?"	→ **motor grade ≤ 2**
"Shoulder abduction weakness + lateral shoulder numbness after C4-7 fusion?"	→ **C5 root**
"Post-CDA presentation needing only reassurance?"	→ **bilateral deltoid/biceps 3/5 + lateral arm numbness from day 5**
"12 h after ACDF, dyspnoea, dysphagia, tense incision, SpO_2 normal?"	→ **retropharyngeal haematoma → immediate evacuation** (TO)
"Burr through C5 lateral wall?"	→ **vertebral artery**
"4-level ACDF, 6 h, 800 mL – post-op?"	→ **keep intubated 24–48 h**
"Hoarseness after ACDF – true?"	→ **↑ with revision surgery** ✗ Wrong options: right > left · immediate re-exploration
"Reduces dysphagia?"	→ **local retropharyngeal steroid**
"Prolonged retraction at tracheo-oesophageal junction?"	→ **vocal cord paresis**
"Most concerning after 6-h C3-6 ACDF?"	→ **airway complications**
"Neck pain, can't hold horizon after prior laminectomy?"	→ **post-laminectomy kyphosis**
"True about CSM surgery?"	→ **posterior has higher infection rate**
"Hyperacute epidural haematoma on MRI?"	→ **T1 hypo, T2 hyper**
"Little finger dermatome?"	→ **C8** (TO)
"Deltoid 3/5 + lateral arm numbness?"	→ **C4-5 disc** (TO)
"C6-7 disc, shoulder abduction relief – EXCEPT?"	→ **elbow flexion weakness** (TO)
"Wrist extensor weakness – sensory change?"	→ **radial forearm** (TO)
"Weak wrist dorsiflexion + sluggish supinator reflex?"	→ **C6** (TO)
"Can't flex fingers, ulnar hand numb?"	→ **C8** (TO)
"NOT a contraindication to cervical disc replacement?"	→ **2-level involvement** (TO) ✗ Wrong options: morbid obesity · RA · instability
"Image shows anterior discectomy + ?"	→ **disc replacement** (TO)
"Canal diameter suggesting stenosis?"	→ **< 10 mm** (TO)
"Commonest site of cervical spondylosis?"	→ **C5-6** (TO)

CERVICAL RADICULOPATHY

– Spine · 16 / 69 –

1 Basics

Epidemiology

Incidence	**107.3 / 100,000 men**, 63.5 / 100,000 women per year · peak **50–54 yrs**
Risk factors	white race, **smoking**, prior lumbar radiculopathy
Degeneration prevalence	X-ray changes in **70% by 70** · MRI: HNP/foraminal stenosis in **28%** of asymptomatic > 40 · **≥ 1 degenerated disc in ~57% (50–60%)** of asymptomatic > 40 (Boden) · up to 19% asymptomatic root compression → **treat only if clinical findings match MRI**

Aetiology

Cause	Note
Cervical spondylosis	Disc-osteophyte, height loss, **uncovertebral + facet spurs**
Soft disc	**Posterolateral = commonest** (between uncinate and lateral PLL) → mostly **motor** · intraforaminal → **radicular pain** · midline → **myelopathy**
Double crush	Root + distal entrapment (↓ axoplasmic flow)
Rare	Tumour, root avulsion, synovial / meningeal cyst, dural AVF, tortuous vertebral artery

Chemical mediators	**IL-1, IL-6, substance P, neuropeptide Y, CGRP, bradykinin, TNF-α, prostaglandins**

Anatomy

Pedicle-root mismatch	cervical root exits **above** its pedicle (C6 above C6 pedicle) · lumbar root exits **below** (L5 under L5) · extra **C8** root makes the switch
Disc → root	cervical disc compresses the **lower (exiting) root**: **C6-7 → C7**
Horizontal cervical root	**central/paracentral and foraminal discs hit the SAME root** (vs lumbar: paracentral → traversing, foraminal → exiting) · far-lateral discs not typically described in C-spine
Compression sources	facet hypertrophy (posterior) · **uncovertebral osteophyte (anterior)** · disc height loss (↓ foraminal volume) · ventrolateral root course → ventral compression
Disc	annulus **type I collagen** (thicker ventrally) · nucleus **type II collagen + GAGs**; water **90% (< 30 yrs) → 70%** (8th decade)

3 Presentation

Symptoms	occipital headache, **trapezial / interscapular pain**, neck pain, unilateral arm pain (often non-dermatomal), dermatomal paraesthesia, weakness (overhead = C7, grip = C8)

– – Active space – –

Root-level localisation

Root (disc)	Motor	Reflex	Sensory
C4 (C3-4)	**Scapular winging**	–	Base of neck; Tinel at levator scapulae
C5 (C4-5)	**Deltoid + biceps** (shoulder abduction, ER, elbow flexion)	**Biceps**	**Superior shoulder, lateral arm**
C6 (C5-6)	**Brachioradialis (elbow flexion in mid-pronation) + wrist extension (ECRL)**, biceps	**Brachioradialis (supinator)**	**Thumb, (index), dorsoradial forearm**
C7 (C6-7)	**Triceps, wrist flexion**, finger extension, pronation	**Triceps**	**Middle finger** (index/ring), dorsal forearm, posterior arm
C8 (C7-T1)	**Finger flexion – DIP of middle + index (FDP)**, grip, **thumb extension**	–	**Ring + little finger**, ulnar hand/forearm
T1 (T1-2)	**Hand intrinsics** (finger abduction)	–	**Axilla / medial arm**, ipsilateral **Horner**

Commonest root **C7** (several studies)

C8 vs ulnar neuropathy C8 weakens **index/middle FDP** (AIN-innervated) – ulnar entrapment does not

C6 vs tennis elbow vs PIN C6: thumb paraesthesia + relief with shoulder abduction · lateral epicondylitis: pain with resisted wrist/long-finger extension · PIN: painless wrist extension weakness **with radial deviation**

Facet fracture localisation **C6 superior facet fracture → C5-6 instability → C6 root** (radial forearm + thumb numbness)

Provocative tests

Test	Fact
Shoulder abduction relief (SAR)	Hand on head relieves pain (↓ root tension) – **virtually pathognomonic**, best separates cervical vs shoulder/peripheral causes; analogue of SLR
Spurling	**Extension + rotation + lateral bend to affected side + axial load →** ipsilateral arm pain; **specific, not sensitive** (✗ Wrong options: contralateral rotation)
Upper limb tension, Valsalva, neck distraction	Supportive
Hoffmann, inverted radial reflex, clonus, Babinski	**Myelopathy** (✗ Wrong options: radiculopathy)

Peripheral mimics Froment (ulnar), **"OK" sign** failure (AIN), Guyon (no dorsal wrist sensory change), cubital tunnel (worse with elbow **flexion**), diabetic neuropathy (glove-stocking)

Differential carpal / cubital tunnel, **Parsonage-Turner**, thoracic outlet, myelopathy, brachial plexus injury, rotator cuff (history must match MRI)

4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP, lateral, **oblique**	**Oblique = best for foraminal stenosis** from osteophytes; flex-ext for instability
MRI (T2 axial)	Red flags (fever, weight loss, night pain), **> 6 weeks** of failed care, motor weakness; loss of perineural fat
CT	Bony anatomy, **OPLL**, instrumentation planning; **post-op pseudarthrosis study of choice**
CT myelogram	Pacemaker, hardware artefact
Discography	Rarely; oesophageal puncture, discitis

Genes & lab markers – electrodiagnostics

EMG / NCS finding	Fact
Fibrillations + positive sharp waves	In affected root muscles (✓) – appear **~3 weeks**; **paraspinals first**
SNAPs normal	Lesion **proximal to DRG**
CMAP	↓ in proportion to atrophy
F-wave	↓ / absent in severe cases (✗ Wrong options: ↑)
H-reflex	S1 only (✗ Wrong options: cervical)
Prolonged distal latency	Peripheral compression (✗ Wrong options: radiculopathy)
Sensitivity	**40–70%** – not a screening test; not needed with concordant exam + MRI

Selective nerve root block localises level in multilevel disease

5 Management

Natural history & non-op care

Natural history favourable: **75% (75–90%) improve non-op**
✗ Wrong options: stepwise decline = myelopathy · needs PT/surgery to resolve

Medications **NSAIDs** = most appropriate for chronic radicular pain (TO) · brief **oral steroids** · gabapentin · **nortriptyline** if depressed (↓ narcotic need)
✗ Wrong options: narcotics · muscle relaxants (habituation, depression) · anticonvulsants

Other short immobilisation **< 1–2 weeks** (avoid prolonged → atrophy) · isometrics, heat, traction (not in myelopathy)

Selective nerve root steroid injection long-term relief **40–70% (50–70%)**; steroid + LA = LA alone long term; ↑ risks vs lumbar (dural puncture, meningitis, abscess, root injury)

– – Active space – –

Axial neck pain only (no root/cord signs, solid old fusion)	**education + PT** ✗ Wrong options: hardware removal · revision · PSF · ESI
Return to play	after symptom resolution + **repeat MRI without cord compression**; **oral methylprednisolone** speeds return; **no ↑ catastrophic SCI risk** after C4-5 HNP · single-level ACDF not a contraindication · collegiate C5 radiculopathy → **NSAIDs + PT, RTP after resolution + repeat imaging** ✗ Wrong options: immediate RTP on steroids · EMG · surgery

Operative criteria

Indication
Disabling pain after **3 months** of non-op care
Progressive / significant neurologic deficit
Static deficit with significant radicular pain

Surgical options

Procedure	**Indication / outcome**
ACDF	**Gold standard**; ventral compression; e.g. 6 months C7 radiculopathy + ventral cord compression (✗ Wrong options: posterior foraminotomy (cord risk) · anterior foraminotomy (vertebral artery) · corpectomy · laminoplasty)
Posterior foraminotomy	**Foraminal soft disc / osteophyte, single level**, prior anterior surgery; **> 91% success**, ~3% complications, = ACDF outcomes, faster return to work, lower cost · not for central disc, myelopathy, instability, OPLL, kyphosis
Anterior foraminotomy	Unilateral root, avoid fusion (smokers, diabetics); uncovertebral resection; **vertebral artery + sympathetic chain risk**; not widely accepted
CDA	Single level, minimal facet arthrosis; **= ACDF** (arm pain, NDI, SF-36, neuro; **PROMs equivalent at 2–5 yrs**); ASD + reoperation + dysphagia similar; less neck pain; **heterotopic bone 60%** (no motion effect)

Best posterior foraminotomy candidate	**young, foraminal soft disc at C6-7, C7 radiculopathy** ✗ Wrong options: myelopathy · SCI · midline disc-osteophyte · MS

ACDF technique

Approach	Smith-Robinson · **transverse incision 1–2 levels**, longitudinal for multilevel / short thick neck
Landmarks	mandible **C1-2** · hyoid **C3-4** · thyroid cartilage **C4-6** · **cricoid C5-6**
Side	left-sided **lower RLN risk** theoretically; **C7-T1 left → thoracic duct** risk · **deflate ET cuff after retractor placement** (RLN compressed against tube)
Plates	↑ fusion in **multilevel** (↓ graft complications, reoperation, earlier return to work); **single-level: no difference** · dynamic (controlled settling) vs static (fixed angle) · old Caspar bicortical → neuro risk · screw length > 80% AP → ↓ pseudarthrosis
Graft	**ICBG autograft gold standard** (donor pain) · allograft ↑ pseudarthrosis (**41% vs 27%**) + subsidence (**28% vs 16%**) (Orthobullets) · **no difference single level** (OB) – read the options

Post-op walk day 0, soft collar briefly (hard collar if no plate), ROM at 6 wks, full activity 3 months

6 Complications

Pseudarthrosis

Rates single level **5–10%** (Orthobullets complications) · 20% (Orthobullets outcomes) · **2.6–4.8%** (OB) · multilevel **30% / > 50%** · doesn't always correlate with outcome

Risk factors **smoking + diabetes** (strongest, modifiable) · multilevel · revision (✗ Wrong options: graft type · plate · level · alignment single level) · **smoking cessation ≥ 4 weeks pre-op** = greatest fusion impact (✗ Wrong options: plate · autograft · collar)

Treatment asymptomatic → observe · symptomatic → **posterior cervical fusion** (↑ fusion, ↓ revision; but ↑ blood loss, stay, complications) or anterior revision if radiculopathy (EXCEPT: posterior does NOT ↓ fusion rate)

ACDF complications

Complication	Fact
RLN palsy	**Commonest nerve injury** (1–11%; 2.7%); vagus branch; **left hooks aortic arch, right hooks subclavian** (shorter, crosses lateral → medial higher, can be aberrant) → right **theoretically** riskier (TO: left approach preferred) · **no proven clinical difference** (Orthobullets, OB); posterior cricoarytenoid paralysis; observe → ENT at 6 wks (Teflon)
Revision after RLN palsy	**Same side** as prior approach; **pre-op laryngoscopy** for revision / prior thyroidectomy (bilateral palsy → stridor) · not contralateral approach
Superior laryngeal nerve	Runs with **superior thyroid artery** (upper C-spine); cricothyroid → pitch / voice fatigue
Hypoglossal	Upper C-spine (C2-3); **tongue deviates toward injured side**
Horner	Sympathetic chain on **lateral longus colli at C6**; retractors not under longus colli → ptosis, miosis, anhidrosis, enophthalmos
Vertebral artery	**Lateral discectomy / uncovertebral** bleeding; aberrant medial course (C6)
Dysphagia	↑ at C3-4; **zero-profile cages** ↓ prominence; local steroid helps
Oesophageal perforation	Early: sharp instruments → repair now · late: plate loosening → NG + TPN
Airway	**> 5 h surgery**, exposure **above C4**, **> 4 levels / ≥ 4 bodies** (✗ Wrong options: smoking · lung disease · myelopathy · drains)
CSF leak (TO)	~1% (↑ OPLL 12.5%, revision, corpectomy, men); intra-op → watertight repair; post-op → bed rest → **lumbar subarachnoid drain** → open repair
Adjacent-level ossification	**Plate < 5 mm from supra-adjacent disc** (✗ Wrong options: needle · stripping · longus colli); **start screws near inferior endplate** of top vertebra ↓ it
ASD	~3%/yr, 30% at 10 yrs

– – Active space – –

X vs Y

	Cervical	Lumbar
Root vs pedicle	Exits **above**	Exits **below**
Root course	Horizontal	Vertical
Paracentral disc	**Same root as foraminal**	Traversing root
Example	C5-6 → C6	L4-5 paracentral → L5, foraminal → L4

	Radiculopathy	Myelopathy
Natural history	**75% improve**	Stepwise decline
Signs	Dermatomal, SAR, Spurling	Hoffmann, clonus, gait
First line	Non-op	Surgery if impaired

How it's asked

"Left foraminal C4/5 disc – exam?"	**→ shoulder abduction + elbow flexion weakness**
"Triceps weakness, ↓ triceps reflex, middle finger numbness – level?"	**→ C6-7**
"Posterolateral C5/6 disc?"	**→ C6: brachioradialis + wrist extension weakness**
"C5-6 disc – sensory loss?"	**→ dorsoradial forearm + thumb**
"Thumb numbness, weak wrist extension + supination?"	**→ C5/6, C6 root**
"Small finger numbness, weak index/middle DIP flexion + thumb extension?"	**→ C7/T1**
"Index-long-ring numbness, triceps weak, SAR positive?"	**→ foraminal C6-7 disc**
"Lateral forearm pain relieved by abduction, thumb numb, wrist extension 4/5?"	**→ paracentral C5/6 disc** ✗ Wrong options: ECRB tendinosis · PIN
"C7 distribution pain – motor + MRI?"	**→ wrist flexion weakness + C6-7 uncovertebral foraminal stenosis**
"Biceps, brachialis, wrist extensors weak, thumb/index numb – posterolateral HNP?"	**→ C5-6**
"Middle finger numb, finger extensors 4/5, triceps 1+?"	**→ C6-7**
"C6-7 HNP – distribution?"	**→ forearm/hand pain, middle finger numbness, triceps + finger extensor weakness**
"Anterior arm → dorsoradial forearm → index, biceps weak, BR reflex lost?"	**→ C5-C6**
"Long finger extension weak, SAR positive, normal obliques?"	**→ C7 radiculopathy, paracentral C6/7 disc**

"EMG finding with C6 radiculopathy?"	→ **fibrillations + positive sharp waves in C6 muscles**
"Foraminal C7/T1 disc?"	→ **small finger numbness + weakness of all finger flexion**
"C5-6 axial disc, negative Hoffmann?"	→ **brachioradialis weakness**
"Unstable superior facet fracture + radial forearm/thumb numbness?"	→ **C6 superior facet, C5-6 instability**
"C6-7 paracentral disc – exam?"	→ **weak elbow extension, ↓ triceps reflex, dorsal middle finger numbness**
"Supports radiculopathy over peripheral neuropathy?"	→ **relief with arm above head**
"Diabetic, ulnar hand tingling, better with hand above head?"	→ **cervical radiculopathy**
"Exam supporting radiculopathy?"	→ **shoulder abduction test · Spurling sign** ✗ Wrong options: Hoffmann · inverted BR reflex · intrinsic wasting
"Hypothenar pain – finding most suggestive of cervical cause?"	→ **relief with shoulder abduction**
"Lateral shoulder pain but C7 compression on MRI?"	→ **further history + exam**
"Chronic radicular pain – drug?"	→ **NSAIDs** (TO)
"Axial neck pain 12 yrs after solid C6-7 ACDF, no deficit?"	→ **education + PT**
"C7 radiculopathy, wants injection – level + 1-yr success?"	→ **C6/7, 50–70%**
"Expected outcome with NSAIDs + rest?"	→ **> 75% chance of resolution**
"NFL player after C4-5 HNP, now normal – added SCI risk?"	→ **none**
"Triceps weakness, sleeps with arm overhead, Spurling positive – true?"	→ **will improve with NSAIDs + PT** ✗ Wrong options: surgery now · EMG next · C7/T1
"Asymptomatic > 40 – degenerated disc prevalence?"	→ **50–60%**
"Collegiate linebacker, C5 radiculopathy, wants fast return?"	→ **NSAIDs + PT; RTP after resolution + repeat imaging**
"C6 radiculopathy with 3+ patellar reflexes – true?"	→ **likely resolves non-op** ✗ Wrong options: interferon · EMG required · root exits below pedicle · Hoffmann
"Best posterior foraminotomy candidate?"	→ **foraminal soft disc C6/7, young male**
"6 months C7 radiculopathy, ventral compression?"	→ **ACDF**
"CDA vs ACDF at 2–5 yrs?"	→ **equivalent PROMs**
"Anterior approach CSF leak – correct?"	→ **may need lumbar CSF drain** (TO)

– – Active space – –

"Structure at more risk on RIGHT-sided approach?"	→ **recurrent laryngeal nerve** (TO)
"Ptosis + miosis after left ACDF?"	→ **retractor placement (sympathetic chain)** (TO)
"True about anterior approach (TO)?"	→ **left preferred – right RLN more vulnerable**
"Hoarseness + unilateral cord palsy after ACDF?"	→ **RLN palsy** (TO)
"Left hypoglossal injury at C2-3?"	→ **tongue deviates left**
"Prior left ACDF + left cord palsy, now C4/5 myelopathy?"	→ **hardware removal + C4/5 ACDF via left approach**
"Left cord paralysis – CONTRAINDICATED revision?"	→ **right-sided ACDF**
"True about RLN?"	→ **commonest nerve injury in ACDF**
"Airway complication risk factor?"	→ **surgery > 5 h**
"Brisk bleeding during lateral discectomy?"	→ **vertebral artery**
"Posterior cricoarytenoid paralysis?"	→ **RLN**
"Prolonged longus colli retraction?"	→ **Horner syndrome**
"Bilateral cord palsy after left ACDF with prior thyroidectomy – prevention?"	→ **pre-op laryngoscopy**
"Posterior fusion for pseudarthrosis – EXCEPT?"	→ **decreased fusion rate**
"Greatest impact on fusion in a smoker?"	→ **smoking cessation 4 weeks pre-op**
"Pseudarthrosis after single-level ACDF – factor?"	→ **diabetes + tobacco**
"Osteophyte at disc above C6-7 fusion?"	→ **plate close to adjacent disc**
"Factor ↑ adjacent segment disease?"	→ **plate distance from adjacent disc**
"Adjacent segment ossification?"	→ **plate < 5 mm from supra-adjacent disc**

CERVICAL SPINE ANATOMY

– Spine · 17 / 69 –

1 Basics

Embryology

Hox (homeobox) genes	direct craniocaudal differentiation + segmentation
Atlas (C1)	**3 ossification centres**: 2 lateral masses (join posteriorly → posterior arch) + anterior arch (appears **~1 yr**); lateral masses fuse to body at **7 yrs**
Axis (C2)	**5 centres**: 2 neural arches/lateral masses, 2 halves of dens (usually fused at birth; persistence = dens bicornis), body · **subdental (basilar / dentocentral) synchondrosis** open in **all children to 3 yrs**, half at 4–5, fused **~6 yrs** · secondary tip centre (ossiculum terminale) appears **~3 yrs**, fuses **~12 yrs** (persistence = ossiculum terminale persistens, insignificant)
3-yr-old with synchondrosis between dens and body	**normal physiologic finding** ✗ Wrong options: developmental anomaly · type 1/3 odontoid fracture

Osteology

Feature	Levels
Transverse foramen	**C1–C7** (all subaxial vertebrae in most people)
Vertebral artery passes through	**C6 → C1**; **C7 foramen usually empty**
Bifid spinous process	**C2–C6** (C7 not bifid)
Carotid (Chassaignac) tubercle	**C6** – anterior approach landmark
C8	No vertebra; **C8 root** exists
Atlas	**No body, no spinous process**

Canal	normal **17 mm**; **< 13 mm** → possible cord compression
Alignment	lordosis measured C2–C7

Facets & kinematics

Subaxial superior facets	face **posteromedial at C3 → posterolateral at C7** ✗ Wrong options: reverse
Axial view	**superior articular process is anterior** to the inferior articular process

– – Active space – –

Motion greatest at **cephalad** subaxial levels (facet orientation)
✗ Wrong options: caudal

Joint	Contribution
Occiput-C1	**~50% of flexion-extension**
C1-C2 (diarthrodial)	**50° of 100° rotation**, 10° of 110° flex-ext, 0° lateral bend
C2–C7	50° flex-ext, 50° rotation, **60° lateral bend**

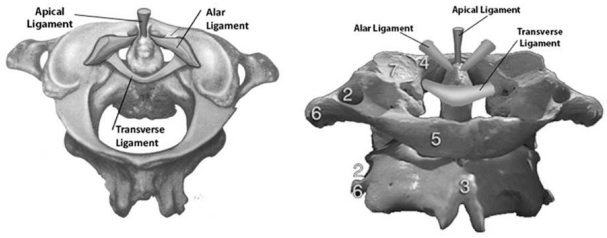
Odontoid blood supply

Watershed between **apex (internal carotid branches)** and **base (vertebral artery branches)** → poor healing of **type II odontoid** fractures

Ligaments

Transverse limits anterior translation of atlas (strongest ligament of spine) · **apical + alar** – limit rotation

Connects	Ligaments (TO)
Atlas → occiput	**Anterior atlanto-occipital membrane** (ALL extension) · **posterior atlanto-occipital membrane** · **ascending band of cruciate**
Axis (dens) → occiput	**Tectorial membrane** (PLL extension; C2–3 bodies → FM) · **alar** (dens → condyles) · **apical** (dens tip → FM, weak)
Axis → atlas	**Transverse atlantal ligament** · atlanto-alar part of alar · descending band of cruciate



Transverse, alar and apical ligaments

Vertebral artery segments (TO)

Segment	Course
V1 **Pre-foraminal**	Subclavian → front of C7 transverse process (between longus colli + scalenus anterior)
V2 **Foraminal**	Transverse foramina **C6 → C2**
V3 **Extradural (atlantic)**	Above C2 → **through C1 transverse foramen** → horizontal in occipital triangle → canal
V4 **Intradural**	Pierces dura → anterior medulla → joins opposite → **basilar** at lower pons

C1 transverse foramen contains **extradural segment of vertebral artery** + venous plexus + sympathetic plexus (TO)

Medial / enlarged transverse foramen on CT before ACDF ↑ **vertebral artery injury** risk
✗ Wrong options: IJV · ICA · vagus

Pedicle & body dimensions (TO)

Measure	Level
Smallest transverse pedicle diameter	**C3** (5.1 mm); cervical pedicles C2 largest, C3 smallest, ↑ C3 → C7
Narrowest pedicle overall (average)	**T4**
Largest pedicle diameter / longest pedicle	**L5**
Shortest pedicle	T10
Largest canal cross-section	**C2** (374 mm²)
Smallest canal	**T3–T8**
Smallest body	C3 + C5 · narrowest body width C4
Tallest / shortest disc	L2-3 / T1-2
Lumbar pedicles	Angle more medially caudally; L1 smallest lumbar

C1 lateral mass screw

Trajectory — **10° medial + 22° cephalad** · safe even with **anomalous vertebral artery** that precludes transarticular screws · medial angle protects **internal carotid**

Muscles

Sternocleidomastoid (TO) — one side: **rotates head to OPPOSITE side** + tilts to same side · both: **flex neck + extend head**, lift head off pillow, accessory inspiration
✗ Wrong options: rotate to same side (EXCEPT answer)

X vs Y

	C1 (atlas)	C2 (axis)	C7
Body	No	✓ + dens	Yes
Spinous process	No	Bifid	Non-bifid (prominens)
Ossification centres	3	5	–
Vertebral artery in foramen	Yes	Yes	Usually no

How it's asked

"Structure in C1 transverse foramen?"	→ **extradural segment of vertebral artery** (TO)
"3-yr-old, synchondrosis between dens and body?"	→ **normal**
"Subaxial anatomy – true?"	→ **all subaxial vertebrae have a foramen transversarium** ✗ Wrong options: artery through all · rotation greatest caudally · facets posterolateral → posteromedial caudally · inferior process anterior

– – Active space – –

"Medially placed, enlarged transverse foramen before ACDF — risk?"	→ **vertebral artery injury**
"C1 lateral mass screw trajectory?"	→ **10° medial, 22° cephalad**
"SCM functions EXCEPT?"	→ **rotate neck to same side** (TO)
"Smallest pedicle diameter (C2–C5)?"	→ **C3** (TO)
"Ligament connecting atlas to occiput?"	→ **ascending band of cruciate** (TO) ✗ Wrong options: tectorial membrane · alar · TAL

CERVICAL SPINE TRAUMA EVALUATION

– Spine · 18 / 69 –

1 Basics

Rule	every trauma patient has a C-spine injury **until proven otherwise**
Clearance	confirming **absence** of injury by exam ± imaging; do it **efficiently** (delay ↑ complications)
Missed injuries	permanent disability; causes: **inadequate imaging of the injured level**, LOC, multisystem trauma
Noncontiguous spinal injuries	**10–15%** (Orthobullets) · up to **16%** (OB) → **image the whole spine** once one fracture is found
High-energy markers	MVA **> 35 mph**, fall **> 10 ft**, closed head injury, neuro deficit, pelvis/limb fractures
Mechanism clues	elderly fall onto forehead / high-speed rear-end → **hyperextension**
Comorbidities to identify	**AS, DISH**, prior fusion, ligamentous laxity disorders

3 Presentation

Primary survey	ABC; **manual in-line traction** whenever collar off for airway; inspect whole spine; **seat-belt sign** → TL flexion-distraction
Secondary survey	remove collar, inspect face/scalp, head posture (rotation → **unilateral facet dislocation**), **palpate posterior midline** (no tenderness in alert patient → low probability of injury), **log roll**, full neuro exam

4 Diagnosis & Classification

Diagnostic criteria – clearance

Clinical clearance (no imaging) – ALL of	Radiographic clearance – ANY of
Awake, alert, not intoxicated	Intoxicated
No neck pain, tenderness or neuro deficit	Altered mental status / obtunded
No distracting injury	Neck pain or tenderness
–	Distracting injury

Paediatric	**NEXUS-based clinical guidelines → fewer CTs** = most effective radiation reduction · < 3 yrs can't cooperate; X-ray screen first, CT if abnormal ✗ Wrong options: fluoroscopy · low-dose CT · avoiding radiographs · MRI for all

– – Active space – –

Key radiology findings

Study	Requirement
Lateral X-ray	Must show **inferior endplate of C7** (Orthobullets) · **top of T1** (OB)
AP + **open-mouth** odontoid	Standard trio
Swimmer's view (TO)	**Cervicothoracic (C7-T1) junction** (✗ Wrong options: Hill-Sachs · Bankart · AO junction)
CT	Higher sensitivity, faster; must include **superior endplate of T1** (Orthobullets) · **inferior endplate of T1** (OB answer) – read the options
MRI	Ligament / cord / disc
Flexion-extension	Not in acute / known instability or deficit; only to exclude occult instability **after** clearance

Lateral X-ray lines anterior vertebral line · posterior vertebral body line · spinolaminar line



Lateral C-spine alignment lines

Prevertebral / retropharyngeal soft tissue (TO)	Upper limit
C2	**6–7 mm** ("above cricoid" → **< 7 mm**)
C3	7 mm
C6 (cricoid)	**18–22 mm** (retropharyngeal **< 22 mm**)
Child at C6	14 mm

Obtunded / intubated after 3 days, X-rays negative, occipital skin erythema
definitive clearance with CT ± MRI
✗ Wrong options: keep collar until surgeries done · halo · wait to examine

GCS 9, multitrauma
CT C-spine
✗ Wrong options: remove collar · wait without imaging · X-rays to C7 only · flex-ext

Burst fracture found (C5 / C7)
CT of entire spine before surgery
✗ Wrong options: steroids · flex-ext · MRA · EMG · lumbar MRI

5 Management

Collar from scene until directed exam

Whiplash	rear-end low-speed extension injury, normal neuro + X-rays → **soft collar + early ROM / PT** (✗ Wrong options: Philadelphia collar · halo · C2 screws) · persistent pain at 4 wks → **NSAIDs + activity modification + PT** (✗ Wrong options: observation · ESI · facet blocks · MRI)
Injury → procedure (OB)	**flexion teardrop → anterior corpectomy + strut + plate** · AAI with cord compression → posterior C1-2 fusion · bilateral facet dislocation → anterior or posterior reduction + fusion · central cord + spondylosis → ACDF or posterior decompression-fusion · C7 Chance (bony) → halo or posterior fusion

Transient quadriparesis – return to play

Recurrent transient quadriparesis + cervical stenosis on CT/MRI	**no further contact sport** · stenosis **< 13 mm**, absolute **< 10 mm** · **Torg ratio has low predictive value** (Cantu) ✗ Wrong options: neck roll · EMG · ESI · steroid pack

6 Complications

Delayed clearance	**aspiration**, ↓ respiratory function, **occipital / submandibular pressure ulcers**, ↑ ICP

Prognosis

Elderly (75 vs 60 yrs) incomplete cervical SCI	**↑ mortality** · elderly: lower-energy falls with head strike, fewer deficits ✗ Wrong options: more severe deficits · more paralysis · epidural haematoma · chronic pain
Preadolescent SCI (e.g. 3-yr-old T2/3 paraplegia)	**≥ 90% develop paralytic scoliosis**; bracing doesn't prevent it · uncommon post-adolescence

X vs Y

	Clinical clearance	Imaging clearance
Mental state	Alert, sober	Obtunded / intoxicated
Symptoms	No pain, tenderness, deficit	Pain, tenderness, deficit
Distracting injury	None	Present

How it's asked

"C5 burst on X-ray, MRI shows retropulsion – before surgery?"	→ **CT of entire spine**
"Intubated 3 days, X-rays negative, occipital erythema?"	→ **definitive CT and/or MRI clearance**
"C7 burst on MRI with deficits – next study?"	→ **CT cervical, thoracic + lumbar**
"Most effective radiation reduction in paediatric trauma?"	→ **clinical guidelines → fewer CTs**
"GCS 9, collar on, multiple fractures?"	→ **CT C-spine to T1**

– – Active space – –

"Injury suited to anterior corpectomy + plate?"	→ **flexion teardrop fracture**
"< 5 mph rear-end, neck stiffness, normal X-rays?"	→ **soft orthosis + early PT**
"4 weeks of whiplash pain?"	→ **NSAIDs, activity modification, PT**
"75- vs 60-yr incomplete SCI — 75-yr has ↑?"	→ **mortality**
"3-yr-old with complete paraplegia — later deformity?"	→ **≥ 90%**
"Recurrent transient quadriparesis with stenosis?"	→ **stop football**
"Normal prevertebral space above cricoid?"	→ **< 7 mm** (TO)
"Normal retropharyngeal space at cricoid?"	→ **< 22 mm** (TO)
"Swimmer's view shows?"	→ **cervicothoracic junction** (TO)

– – Active space – –

CERVICAL SPONDYLOSIS

– Spine · 19 / 69 –

1 Basics

Prevalence & high-risk sports

Onset	40–50 yrs · **85%** show spondylosis regardless of symptoms: **> 65 yrs** (Orthobullets) · **> 60 yrs** (OB) · men > women
Levels	**C5-6 > C6-7** (most subaxial flexion-extension) · risk: excessive driving, smoking, lifting, professional athletes
Highest spinal degeneration in sport	**jockeys (horse racing)** · also wrestling, rugby, ski-jumping, gymnastics ✗ Wrong options: badminton · baseball · volleyball · golf

Pathoanatomy

Motion segment disc + **2 facet joints + 2 uncovertebral joints (Luschka)**

Degenerative cascade

- Disc → **desiccation, height loss, bulge / herniation**
- Joints → **uncinate spurs + facet arthrosis**
- Ligaments → **LF thickening + infolding (height loss)**
- Deformity → kyphosis → load shifts to facets/uncovertebral → **more spurs**

Clinical syndromes axial / discogenic neck pain · radiculopathy · myelopathy

2 Mechanism

Root compression (radiculopathy)	foraminal spurs (facet + uncovertebral) · posterolateral disc / disc-osteophyte between uncinate and lateral PLL · foraminal soft disc → **exiting root** (C6-7 → C7)
Central stenosis (myelopathy)	canal **< 13 mm** (normal 17) · **worse in extension**: cord pinched between disc anteriorly and facets + infolded LF posteriorly

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Findings
X-ray	Uncovertebral + facet degeneration, osteophytes, disc narrowing, **endplate sclerosis**, ↓ sagittal diameter; **poor symptom correlation** · lateral: alignment, canal size
Oblique	**Foraminal stenosis** (uncovertebral arthrosis)
Flexion-extension	Angular / translational instability; compensatory subluxation next to stiff segment
MRI (axial)	Modality of choice; disc degeneration, myelomalacia; **28%** false positives > 40 yrs
CT myelogram	Bony detail + neural status; pacemaker / hardware
Discography	Rarely; oesophageal puncture, discitis



Degenerative cervical spondylosis on lateral X-ray

5 Management

Treatment	by pain severity/chronicity, instability, neuro deficit: observation → medical → surgery
Axial neck pain only, normal neuro (70-yr, 2 months)	counsel that **85% of > 60s show these X-ray findings** ✗ Wrong options: surgery to prevent decline · MRI for radiculopathy (clinical diagnosis) · "< 10% improve non-op"

How it's asked

"Sport with highest cervical + lumbar degeneration?"	**→ jockeys**
"70-yr axial neck pain, spondylotic X-ray, no deficits – counsel?"	**→ 85% of > 60-yr-olds have these findings**
"Commonest level?"	**→ C5-6**

CHANCE FRACTURE (FLEXION-DISTRACTION INJURY)

– Spine · 20 / 69 –

1 Basics

Definition	thoracolumbar **flexion-distraction (seat-belt / jack-knife)** injury; described by **GQ Chance 1948** (purely osseous)
Level	**T12–L2 / T12–L3** (transitional zone; TO: D12–L2)
Types	**bony** (fracture through spinous process → lamina → pedicles → body; heals well) · **ligamentous / discoligamentous** (through disc + ligaments; heals poorly) · mixed
Neuro deficit	relatively **low** incidence (TO)

Associated intra-abdominal injuries (lap belt)

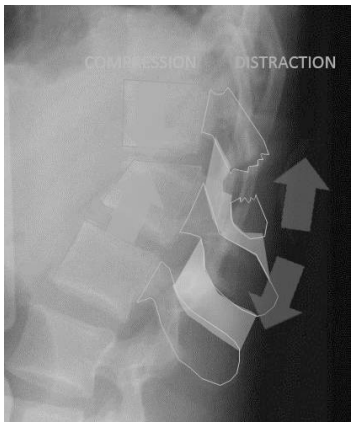
GI injury ~50%	**commonest associated injury** (✗ Wrong options: head injury · pulmonary contusion · pelvis · GU) · colon rupture, small bowel / **duodenal** tear (✗ Wrong options: tamponade · bladder rupture · cholecystitis)

Seat-belt sign (abdominal ecchymosis) → suspect **flexion-distraction injury** + bowel injury · lap-only belts (old cars, rear seat, children)
✗ Wrong options: L5 TP fracture (pelvic ring) · burst · pars · HNP

2 Mechanism

Axis of rotation	**anterior to the spine** (abdominal wall / **abdominal viscera**) – lap belt acts as fulcrum ✗ Wrong options: ISL · PLL · ALL
Column failure	**posterior + middle columns fail in tension (distraction)** · **anterior column**: fails in **compression** (Orthobullets, OB) · "no crushing of the body" (TO) – TO answer: **middle + posterior columns fail in distraction**
Posterior column failure mode	**tension** ✗ Wrong options: torsion · shear · compression
Pure ligamentous variant (X-ray)	**splayed spinous processes + ↑ posterior disc height**, no body fracture

– – Active space – –



Chance fracture: posterior + middle tension, anterior compression

4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP + lateral	**Interspinous widening**, horizontal fracture line, empty pedicle sign on AP
Flexion-extension	Instability
CT	Bony extent; retropulsion
MRI	**Posterior ligamentous complex (PLC)** – ↑ T2 in ISL / PLL = ligamentous injury



Lateral X-ray: Chance fracture through posterior elements

AO thoracolumbar classification (TO)

Type	Pattern
A0	Spinous / transverse process fracture
A1	Wedge, one endplate, posterior wall intact
A2	Split, both endplates, posterior wall intact
A3	Incomplete burst – one endplate + posterior wall
A4	Complete burst – both endplates + posterior wall
B1	**Monosegmental osseous posterior tension band failure = bony Chance**
B2	**Osseoligamentous posterior tension band** disruption (± bone; ISL T2 signal; heals slower)
B3	**Hyperextension** – anterior tension band failure (ankylosed spines)
C	Translation / dislocation (no subtypes; all operative)

Neuro modifier N0 intact · N1 transient · N2 radicular · N3 incomplete SCI / cauda equina · N4 complete · NX unknown · "+" ongoing compression

Case modifiers **M1** indeterminate tension band injury · **M2** comorbidity (AS, skin, rheumatologic)

B1 vs B2 B1: CT line through spinous process, MRI posterior elements may look normal · B2: ISL hyperintense, body comminution favours B2

Operative criteria (TO – TLICS)

TLICS component	Points
Distraction mechanism	**4**
PLC injury suspected / confirmed	2 / 3
Neuro intact	0
Total ≥ 5	**Surgery** (≤ 3 non-op)

5 Management

Chance / flexion-distraction injury

- Purely bony, PLC intact, neuro intact, kyphosis < 15–20° → **HYPEREXTENSION CAST / TLSO (12–16 wks)**
  (serial X-rays for nonunion + kyphosis)
- Ligamentous / mixed (PLC torn) → **POSTERIOR COMPRESSION INSTRUMENTATION + FUSION**
  (1 level above + 1 below with pedicle screws)
- Neuro deficit → **DECOMPRESSION + STABILISATION**

Non-op union > 90% in selected bony injuries (cancellous bone heals)

Surgical construct **compression** construct for Chance · historically 3 above + 2 below · modern pedicle screws **1 above + 1 below** (short segment ± index screws)
✗ Wrong options: distraction (that's for burst)

Anterior vertebrectomy + strut + instrumentation (decompression)

– – Active space – –

Laminectomy	only for compressive deficit; further destabilises posterior column
Bowel perforation	laparotomy first; spine staged or same sitting
Paediatric soft-tissue Chance (8-yr, L3-4, CT negative for bone)	**L3-4 posterior instrumented fusion** ✗ Wrong options: percutaneous screws (bony only) · TLSO · L1-5 fusion + laminectomy · PLIF
Bony-ligamentous L2 (PLC + PLL signal, through superior endplate), ASIA E (TO)	**posterior short-segment instrumentation** ✗ Wrong options: cast (not purely osseous) · surgery justified by seat-belt sign · TLSO because neuro intact

6 Complications

Complication	Note
Pain	Commonest
Progressive kyphosis	**Unrecognised PLL/PLC injury**
Scoliosis, flat back	Forward-flexed posture, fatigue
Post-traumatic syringomyelia	–
Nonunion	Ligamentous / non-op

X vs Y

	Chance (flexion-distraction)	Burst
Mechanism	Flexion + distraction, axis **anterior**	**Axial load** ± flexion, axis middle column
Posterior column	**Tension** failure	Usually intact
Middle column	Tension	**Compression** (retropulsion)
Associated	**Bowel injury 50%**	Calcaneus, other spine
Construct	**Compression**	Distraction

	Bony Chance (AO B1)	Ligamentous (AO B2)
Healing	Good	Poor
Rx	**Hyperextension brace / cast**	**Posterior fusion**

How it's asked

"Lap-belt bruising + Chance fracture – commonest associated injury?"	→ **colonic / GI injury**
"Seat-belt sign, GCS 8 – suspect?"	→ **flexion-distraction injury**
"Lap belt + small bowel tear – spinal injury?"	→ **flexion-distraction**
"Lap belt, posterior column involved – likely associated injury?"	→ **gastrointestinal**
"Mechanism of Chance fracture?"	→ **flexion-distraction** (TO)

Chance Fracture (Flexion-Distraction Injury)	
"Columns failing in Chance fracture?"	→ **middle + posterior in distraction** (TO)
"Posterior column failure mode?"	→ **tension**
"Centre of rotation, ligamentous Chance?"	→ **abdominal viscera (anterior to spine)**
"8-yr soft-tissue Chance L3-4, no deficit?"	→ **L3-4 posterior instrumented fusion**
"Bony Chance — AO type?"	→ **B1** (TO)
"L2 bony-ligamentous Chance with PLC signal, ASIA E?"	→ **posterior short-segment instrumentation** (TO)

– – Active space – –

– – Active space – –

CONGENITAL MUSCULAR TORTICOLLIS

– Spine · 21 / 69 –

1 Basics

Epidemiology

Commonest cause of infantile torticollis	**0.3–2.0%** ✗ Wrong options: Klippel-Feil · Chiari · AARD · paroxysmal torticollis
Sex	**M:F 3:2** · mostly **right** side (TO)
Risk factors	**first pregnancy**, **breech**, traumatic delivery, oligohydramnios (packaging)

Pathophysiology

Lesion	**sternocleidomastoid (SCM) contracture / fibromatosis** ✗ Wrong options: platysma · omohyoid · longus colli · trapezius
Theories	compression/stretch injury · **venous outflow obstruction → compartment syndrome** of SCM
Mass	palpable at birth or **within 4 weeks**, middle–lower third of sternal head; regresses over months → fibrous cord

Associated packaging disorders

Condition	Note
DDH	**5–15%** → examine hips
Metatarsus adductus	✓
Calcaneovalgus feet	✓
Plagiocephaly	**Contralateral** side
Congenital atlanto-occipital anomalies	✓

Anatomy – SCM

Origin	sternal head (manubrium) + clavicular head (medial third of clavicle)
Insertion	mastoid process + lateral occipital bone
Nerve	**spinal accessory (CN XI)** – at risk in release
Action	ipsilateral tilt/flexion + **contralateral rotation** of the chin

3 Presentation

Posture	**head tilt TOWARD** affected SCM + **chin rotated AWAY** ("cock-robin")
Painless passive motion · older child: tight band, ↓ rotation + lateral flexion	
Exam	vision, hearing, neuro, **hips**, feet, spine

4 Diagnosis & Classification

Key radiology findings

Study	Indication / finding
Ultrasound	**Modality of choice with palpable mass** + ↓ ROM; enlarged **hyperechoic** (fibrotic) SCM; separates CMT from serious causes; follows fibrosis over time
X-ray C-spine	Tilt/rotation **without a mass**; exclude bony causes
Dynamic CT C1-2	Exclude AARD
MRI brain + C-spine	Non-muscular / central causes

5 Management

Congenital muscular torticollis

- < 1 yr, < 30° ROM loss → **PASSIVE STRETCHING (90–95% respond in year 1)**
  (tilt away + rotate chin toward affected side)
- Failed ≥ 1 yr stretching → **BIPOLAR RELEASE / Z-LENGTHENING of SCM**

Surgical timing after **≥ 1 yr failed stretching**; 92% success even in older children; facial asymmetry improves if done **before 10 yrs** (Orthobullets) · **after 3–4 yrs** – **release before 3 yrs → deep painful scarring** (TO)

Technique short proximal incision behind ear (mastoid) ± distal sternal/clavicular incision · unipolar (clavicular) or **bipolar (preferred)**; Z-plasty clavicular head + mastoid release · CN XI at risk · botox for refractory (OB)

6 Complications

Untreated permanent rotational deformity, **positional plagiocephaly**, craniofacial asymmetry / **hemihypoplasia**, compensatory scoliosis

X vs Y

	CMT	AARD / Grisel	Klippel-Feil	Paroxysmal torticollis of infancy
Pain	**Painless**	**Painful**	Painless	Episodic
Mass	**SCM mass**	No	No	No
SCM spasm vs chin	**Opposite** side (SCM primary)	Same side (secondary)	–	–
Key	Breech, first-born, DDH	URTI / trauma	Short webbed neck, low hairline, ↓ ROM	Vestibular; pallor, ataxia; resolves by 2–3 yrs
Rx	Stretching	Collar → traction → fusion	–	Self-limiting

– – Active space – –



Congenital muscular torticollis: tilt toward, chin away

How it's asked

"True about SCM contracture?"	→ **surgery before 3 yrs → deep painful scarring** (TO) ✗ Wrong options: disappears after infancy · best age 1–2 · unipolar sternal release suffices
"Treatment of choice in infancy?"	→ **passive stretching** (TO)
"Muscle implicated?"	→ **sternocleidomastoid**
"6-week infant, head tilt since birth + neck mass — commonest cause?"	→ **congenital muscular torticollis**
"Imaging with palpable mass?"	→ **ultrasound**
"Associated condition to screen?"	→ **DDH**

CONGENITAL SCOLIOSIS

– Spine · 22 / 69 –

1 Basics

Epidemiology & aetiology

Definition failure of normal vertebral development at **4–6 weeks of gestation** (mesenchymal anlage) – same window as **heart + kidneys**

Prevalence **1–4%**

Causes mostly **sporadic**; maternal **diabetes, alcohol, valproic acid, hyperthermia** · environmental / vascular disruption

Genes & inheritance

Familial inheritance **not** associated → "deformity NOT familial" = congenital scoliosis (e.g. child with myelomeningocele) · genetics uncertain
✗ Wrong options: unlike AIS

Jarcho-Levin (spondylocostal dysostosis) most commonly **autosomal recessive**

Associated anomalies

System	Rate
Any systemic anomaly	**Up to 60–61%**
Cardiac	**10%** (Orthobullets) · **25–26%** (OB) – **VSD** commonest, then ASD, PDA, ToF, TGA
Genitourinary	**25%** (Orthobullets) · 20% / up to one-third (OB) – unilateral absent kidney, horseshoe, obstructive uropathy, duplicate ureters, hypospadias
Intraspinal (neural axis)	**20–40%** (Orthobullets) · 37% (OB) · 18% pre-MRI era – diastematomyelia, **tethered cord**, syrinx, **Chiari**, intradural lipoma, dural bands; ↑ with segmentation + mixed defects
Orthopaedic	Clubfoot, DDH, limb hypoplasia, **Sprengel**, Klippel-Feil, vertical talus, cavus, **thumb hypoplasia / radial club hand**

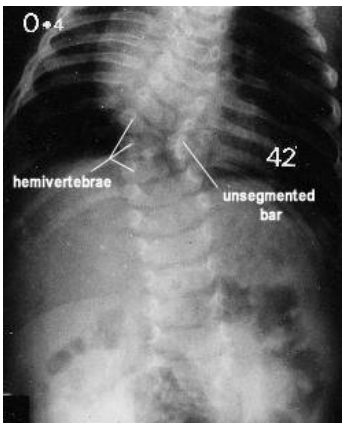
Syndrome	Features
VACTERL (38–55%)	Vertebral, anal atresia, cardiac, TE fistula, renal, radial / limb
Goldenhar (oculo-auriculo-vertebral)	**Hemifacial microsomia + epibulbar dermoids**
Jarcho-Levin	**Short-trunk dwarfism, multiple vertebral + rib fusions → thoracic insufficiency syndrome**
Klippel-Feil	Short neck, low hairline, cervical fusion
Alagille	Peripheral pulmonic stenosis, cholestasis, facial dysmorphism

Not associated Sever's disease (calcaneal apophysitis)

4 Diagnosis & Classification

Classification

Category	Type	Note
Failure of formation	**Fully segmented hemivertebra**	Disc + growth plate above and below → progresses; **commonest anomaly (38%)** (TO)
	Semisegmented	Fused one side
	Unsegmented	Fused both sides
	Incarcerated	Within margins of adjacent vertebrae → little progression
	Unincarcerated	Laterally placed
	Wedge / butterfly	–
Failure of segmentation	**Block vertebra**	**Least progression (< 2°/yr)**
	Unilateral unsegmented bar	Common, likely to progress
Mixed	**Unilateral bar + contralateral hemivertebra**	**Most rapid progression**



Hemivertebra with contralateral unsegmented bar

Progression risk

Anomaly (fastest → slowest)	Rate
1 Unilateral bar + contralateral (convex) hemivertebra	**5–10°/yr** (Orthobullets) · **6–7°/yr**, most > 50° by age 2 (TO)
2 Unilateral unsegmented bar	~5°/yr; > 50° by 10, > 70° at maturity
(Double convex hemivertebra)	3–4°/yr (TO)
3 Fully segmented hemivertebra	1–2°/yr
4 Unincarcerated hemivertebra	–
5 Incarcerated hemivertebra	–
6 Unsegmented hemivertebra	–
7 Block vertebra	**< 2°/yr**

Timing fastest in **first 3 years** · **fused ribs** ↑ progression

Anterior failure of formation (kyphosis) **progressive sagittal deformity + paralysis** · anterior failure of segmentation → progressive but **rarely paralysis**

Key radiology findings & work-up

Study	Purpose
AP + lateral X-ray	Usually diagnostic
3D CT	Posterior anatomy, surgical planning (judicious radiation)
MRI whole spine	**All before surgery**; immediately if **hairy patch / neuro signs** (sedation in infants → may defer if no surgery planned)
Renal US (or MRI)	**All patients** – "renal US indicated in?" → **congenital scoliosis**
Echocardiogram	Cardiac evaluation

New congenital scoliosis in infant **echo + renal US (+ spinal MRI)** first (✗ Wrong options: HLA-B27 · bone scan · skeletal survey · EMG · brace · surgery yet) · MRI already normal → **renal US** next (✗ Wrong options: blood cultures · flex-ext C-spine · platelets)

5 Management

Observation non-progressive: incarcerated, non-segmented, some partially segmented hemivertebrae; serial X-rays every 4–6 months; **document progression before surgery**

Bracing **NOT effective** for congenital curves; may control supple compensatory curves · derotational casting may delay growing constructs
✗ Wrong options: primary treatment

Congenital scoliosis

- Unilateral bar + contralateral hemivertebra → **SURGERY AT ANY AGE**
  (hemivertebra excision + short-segment posterior fusion)
- Progressive fully segmented hemivertebra, young → **HEMIVERTEBRECTOMY + SHORT FUSION**
- Unilateral bar, minimal deformity → **IN-SITU FUSION**
- Multiple fused ribs (> 4) / thoracic insufficiency → **VEPTR / rib osteotomies / growing rods**

– – Active space – –

Operative criteria & procedures

Procedure	Indication
Hemivertebra excision + short posterior fusion	Progressive hemivertebra (esp. **lumbosacral** → oblique takeoff, truncal imbalance); best **< 5–6 yrs, flexible < 40°** (Orthobullets) · ages 3–8 (posterior; younger = poor anchors) · indication = **documented progression** (✗ Wrong options: failed bracing · Cobb > 45° · Risser 4 · maturity)
In-situ arthrodesis (A/P or posterior)	Unilateral bars with minimal deformity
Posterior fusion ± osteotomy	Hemivertebra opposite bar not needing vertebrectomy; older patients with progression/deficit/respiratory decline; many pedicle screws ↓ crankshaft
A/P fusion ± vertebrectomy	Young (**girls < 10, boys < 12**) with progression, deficit, respiratory decline
Convex hemiepiphysiodesis	Concave growth intact, **< 5 yrs, < 40–50°**; mixed results
Distraction growing rods	Control deformity during growth, delay fusion; lengthen ~6-monthly
VEPTR / rib osteotomies	**> 4 fused ribs**, **thoracic insufficiency syndrome** (may stiffen chest)
Spinal column shortening resection	Late, severe, rigid deformity; fixed pelvic obliquity

22-month / 14-month with bar + contralateral hemivertebra (~40°), normal MRI	**hemivertebrectomy + short-segment posterior instrumented fusion** ✗ Wrong options: observe · brace · Risser cast · VEPTR · growing rods
7-yr, hemivertebra curve ↑ 10°	**hemivertebra excision + limited fusion**
18-month, lumbar hemivertebra	echo + renal US first; no long TL fusion (crankshaft, limits growth)

Congenital kyphosis

Level (TO) — **T7–T9** (gibbus) · associations: MPS, SED congenita, Morquio, pseudoachondroplasia

Type	Prognosis
I – failure of formation (anterior)	**Worst**; relentless progression → **paraplegia**
II – failure of segmentation	Rounded, slower, paraplegia uncommon

Treatment (TO)	**no bracing** · **< 5 yrs + < 50–55° → posterior in-situ fusion** (anterior growth continues → some correction) · **> 5 yrs, > 55°, or neuro deficit → anterior release/decompression + strut + posterior instrumented fusion** (neuromonitoring)
Supports early in-situ posterior fusion (8-month)	50° curve, failure of formation, bar, neuro weakness ✗ Wrong options: **block vertebra** (EXCEPT – least progression)

6 Complications

– – Active space – –

Complication	Note
Crankshaft	Posterior fusion alone in young
Short stature	Fusion limits trunk growth (younger worse)
Neurologic injury	Overdistraction / shortening, overcorrection, segmental vessel ligation → SSEP + MEP
Soft tissue	Optimise nutrition pre-op

X vs Y

	Congenital	Idiopathic (AIS)
Cause	Vertebral anomaly (4–6 wks)	Unknown, familial
Inheritance	No	Yes
Brace	Ineffective	Effective 25–45°
Work-up	Echo + renal US + MRI	MRI only if atypical

	Anterior failure of formation	Anterior failure of segmentation
Deformity	Sharp kyphosis	Rounded kyphosis
Paralysis	**Often**	Rare

How it's asked

"2-yr-old with hemivertebrae – additional work-up?"	→ **echocardiography**
"Renal ultrasound indicated in?"	→ **congenital scoliosis**
"LEAST likely association?"	→ **Sever's disease**
"Infant with congenital scoliosis + hairy patch, one kidney?"	→ **MRI of entire spine**
"3-month-old, MRI normal – next?"	→ **cardiac + renal evaluation**
"Screening tests with congenital vertebral anomalies?"	→ **renal US + echo + spinal MRI**
"2-yr, progressed 15°, MRI normal – additional study?"	→ **renal ultrasound**
"18-month, new congenital scoliosis – next step?"	→ **echocardiogram + renal US**
"Most progressive anomaly?"	→ **unilateral bar with convex hemivertebra** (TO)
"Worst prognosis?"	→ **hemivertebra with contralateral bar** (TO)
"Pattern with worst prognosis, surgical indication?"	→ **hemivertebra + contralateral bar**
"Anterior failure of formation leads to?"	→ **progressive sagittal deformity + paralysis**
"When to resect a hemivertebra (4-yr)?"	→ **on deformity progression**

– – Active space – –

"8-month congenital kyphosis — EXCEPT for early fusion?"	**→ block vertebra**
"14-month, 40°, convex hemivertebra + concave bar?"	**→ hemivertebra excision + short posterior fusion**
"7-yr hemivertebra curve ↑ 10°?"	**→ hemivertebra excision + limited fusion**
"22-month, bar + hemivertebra, 41°?"	**→ hemivertebrectomy + short-segment fusion**
"Deformity NOT familial?"	**→ congenital scoliosis**
"Commonest level of congenital kyphosis?"	**→ T7–9** (TO)

DEGENERATIVE SPONDYLOLISTHESIS

– Spine · 23 / 69 –

1 Basics

Epidemiology

Definition anterior slip with an **intact pars** (Wiltse **type III**) (TO)

Prevalence ~5% men, ~9% women (5–9%)

Demographics **women > 40** (**~8×**; 5–10× in OB), **African Americans**, **diabetics**; hormonal ligamentous laxity

Level **L4-5** (5× other levels)
✗ Wrong options: L5-S1 = isthmic

Risk factors **sacralisation of L5** (transitional L5) · **sagittally oriented facets**

Pathoanatomy

Cascade disc degeneration → facet capsule degeneration → microinstability → macroinstability → anterolisthesis (worse with congenitally sagittal facets)

Grade rarely > Meyerding II (posterior arch intact → canal narrows as body slides)

Wiltse classification (TO)

Type	Name
I	Dysplastic (no pars defect)
II	Isthmic (lytic, elongated, acute pars)
III	**Degenerative**
IV	Traumatic (non-pars arch fracture)
V	Pathologic (e.g. OI)
VI	Iatrogenic (added later)

2 Mechanism

Stenosis	L4-5 slip compresses	Cause
Central + lateral recess	**Traversing L5**	Slip, **LF hypertrophy**, facet osteophytes
Foraminal	**Exiting L4**	Vertical: disc height loss + posterolateral body osteophyte pushes root against pedicle · AP: superior facet + posterior body changes

3 Presentation

Back pain **commonest**; mechanical, relieved by rest/sitting · **start-up pain** (sit → stand) suggests instability

– – Active space – –

Neurogenic claudication	2nd commonest: buttock/leg pain with walking upright, **relieved by sitting / flexing** (shopping cart), **not by standing still** · stationary bike tolerated
L4	quadriceps (sit-to-stand), ↓ patellar reflex, dorsiflexion (heel walk)
L5	dorsiflexion (heel walk), **EHL**, gluteus medius (hip abduction) · EHL weakness = commonest documented deficit (L5)
Hamstring tightness	common; separate from neurogenic pain · cauda equina very rare

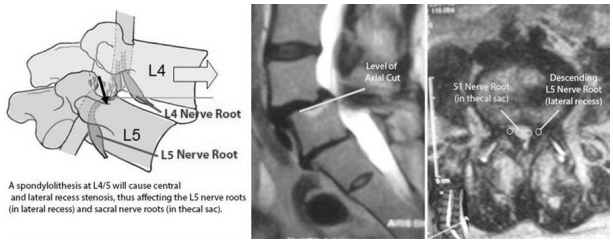
Neurogenic vs vascular claudication

	Neurogenic	Vascular
Relief	**Sitting / flexion**	**Standing still**
Stationary bike	**Tolerated**	Painful
Pulses / ABI	Normal (**ABI > 0.9**)	↓
Walking test	Stop + stand → no relief; sit → relief	Stop + stand → relief

4 Diagnosis & Classification

Key radiology findings

Study	Use
Standing lateral + flexion-extension	Slip; **instability = 4 mm translation or 10° angulation** vs adjacent segment
MRI (T2 sagittal + axial)	**Persistent leg pain after failed non-op**; neural compression
CT	Bony pathology; **most reliable for pseudarthrosis** after fusion
CT myelogram	MRI contraindicated (**pacemaker**)



Degenerative L4-5 spondylolisthesis with stenosis

5 Management

Non-op (most)	activity restriction, NSAIDs, PT · **ESI second-line**

Operative criteria

Indication for surgery
Persistent incapacitating pain after 6 months non-op **incl. ESIs** (commonest)
Progressive motor deficit
Cauda equina syndrome
Active smoker → smoking cessation first (continue PT)

Symptomatic L4-5 DS failing 6 months non-op

- Standard → **DECOMPRESSION + INSTRUMENTED POSTEROLATERAL FUSION (± TLIF/PLIF)**
- Medically frail, can't tolerate fusion → **DECOMPRESSION ALONE**
  (~31% progressive instability)
- Revision pseudarthrosis → **ALIF**

SPORT	surgery → **greater pain relief + better function at 4 yrs** vs non-op
Herkowitz & Kurz	decompression + fusion **90% good/excellent** vs **44%** decompression alone · satisfaction ~79% with fusion vs ~69% decompression alone
Not appropriate	decompression alone (iatrogenic instability, ~30% progress) · microdiscectomy · TDR (doesn't address facet pain generator) · stand-alone ALIF (suboptimal fusion, cage extrusion; doesn't decompress) · fusion alone (must decompress lateral recess + foramen) · uninstrumented fusion (↑ nonunion)

Fusion technique & outcomes

Pedicle screws	↓ **pseudarthrosis** (↑ fusion rate); outcome benefit inferred, not directly proven · slip reduction: limited role in adults, not correlated with outcome
Solid fusion	**most important for long-term relief** of back + leg pain (short-term relief similar with pseudarthrosis) ✗ Wrong options: interbody device · rhBMP-2 · preserving facet capsules
Interbody cage (TLIF/PLIF)	↑ **hospital cost** without ↑ fusion, ODI, SF-36 or LOS (OB) · "may ↑ fusion + ↓ slip progression" (Orthobullets) – read the options
Cortical bone trajectory (CBT) screws	start **more medial + caudal**, aim **cephalad + lateral**, smaller screws · ↓ blood loss, smaller incision, ↓ 1-wk pain; fusion + function = pedicle screws; usually with PLIF/TLIF · e.g. single-level PLIF with **local autograft + CBT screws**
Fusion after laminectomy (no DS/scoliosis)	if destabilised: **pars fractured** or whole facet removed ✗ Wrong options: dural tear repair · 2 levels · smoker · < half facets removed

Smoking

Smoking	↑ pseudarthrosis, infection, reoperation; ↓ **functional outcome scores** ✗ Wrong options: LOS · airway after corpectomy · haematoma

Post-op cessation ≥ 6 months → ↑ fusion rate vs continued smoking, but risk **stays above non-smokers** whether quitting pre- or post-op · pre-op cessation ≥ 4 wks ↓ wound complications · multilevel fusion ↑ nonunion risk in smokers

– – Active space – –

6 Complications

Complication	Fact
Pseudarthrosis	**5–30%**; **CT** best (✗ Wrong options: MRI · flex-ext (miss 60–80%) · bone scan · ESR/CRP)
Adjacent segment disease	30–40% radiographic (Orthobullets) · **20–29% need surgery at 10 yrs** (18% at 8 yrs SPORT) · ↑ with **laminectomy at cranial adjacent level** (✗ Wrong options: 1-level fusion · DS itself · obesity)
SSI	**0.1–2%**; **I&D + retain hardware** (stability aids eradication) (✗ Wrong options: aspiration · antibiotics alone · remove hardware)
SSI risk factors	**Age > 60–65**, **diabetes**, **smoking**, ↑ BMI, alcohol, blood loss, operative time, levels, **staged A/P**, prior SSI (✗ Wrong options: **previous spine surgery** (EXCEPT))
LFCN palsy	**Prone iliac bolster** → lateral thigh numbness (meralgia paraesthetica)
Ulnar nerve / brachial plexopathy	Arm positioning
Dural tear	–
In-hospital complications	**Age** (strongest), ≥ 3 comorbidities, blood loss, operative time, levels fused (✗ Wrong options: gender, race, hospital size)

Adjacent-level symptoms not matching MRI (vague radicular pain, L5/S1 weakness, L3-4 degeneration)

EMG before surgery

X vs Y

	Degenerative	Isthmic
Level	**L4-5**	L5-S1
Pars	**Intact**	Defect
Sex	**Female**	Male
Root	**Traversing L5** (+ exiting L4 foraminal)	Exiting L5
Canal	**Narrowed**	Widened
Grade	≤ II	Can be high

How it's asked

"L4/5 DS, failed 6 months incl. ESIs – next?"	→ **decompression + fusion** ✗ Wrong options: TDR · microdiscectomy · decompression alone · fusion alone
"Failed non-op, flexion-extension + MRI DS?"	→ **laminectomy + partial facetectomy + foraminotomy + instrumented PSF**

"DS with EHL weakness, failed ESIs?"	→ **decompressive laminectomy + posterior instrumented fusion**
"Pacemaker, CT myelogram DS, ABI 0.95, bikes painlessly?"	→ **decompression + instrumented fusion**
"70-yr, trouble heel-walking, failed non-op?"	→ **laminectomy + instrumented fusion**
"SPORT — surgery vs non-op at 4 yrs?"	→ **greater pain relief + improved function**
"Greatest effect on pain, function, ODI (4 mm translation)?"	→ **decompression + instrumented fusion**
"Laminectomy for stenosis — fuse if?"	→ **pars fractured**
"Best long-term improvement?"	→ **open laminectomy + instrumented fusion**
"Pedicle screws in posterolateral fusion?"	→ **↓ pseudarthrosis**
"Most important for long-term relief?"	→ **solid L4/5 fusion**
"Fusion vs pseudarthrosis outcome?"	→ **better long-term back + leg pain relief**
"Interbody spacer adds?"	→ **increased hospital cost**
"↓ blood loss + incision, high fusion?"	→ **PLIF with local autograft + cortical trajectory screws**
"Smoker quits post-op — nonunion risk?"	→ **still higher than non-smokers**
"Smoking associated with?"	→ **lower functional scores**
"Post-op cessation needed to ↑ fusion?"	→ **≥ 6 months**
"Heavy smoker, failed 8 months non-op, no deficit?"	→ **continue PT + smoking cessation**
"Six-week wound erythema, ↑ ESR, dorsal collection?"	→ **debridement, delayed closure, retain hardware**
"SSI risk factors EXCEPT?"	→ **previous spine surgery**
"Recurrent pain 1 yr after fusion, normal labs — confirm diagnosis?"	→ **CT lumbar spine**
"Complication risk factors?"	→ **age + blood loss + levels fused (all)**
"Lateral thigh numbness after prone fusion?"	→ **positioning (LFCN)**
"Highest in-hospital complication risk?"	→ **age**
"Highest risk of ASD needing surgery?"	→ **laminectomy at cranial adjacent level**
"Need for adjacent-level surgery at 10 yrs?"	→ **20–29%**
"Expected weakness (L4-5 DS)?"	→ **EHL**
"Vague radicular pain, symptoms don't match adjacent-level MRI?"	→ **EMG**
"Wiltse type 3?"	→ **degenerative** (TO)

– – Active space – –

DISC SPACE INFECTION - PEDIATRIC

– Spine · 24 / 69 –

1 Basics

Epidemiology

Who	children > adults · **males** · usually **< 5 yrs**
Where	**lumbar spine (50–60%)**

Pathophysiology

Why children	vessels run from **cartilaginous endplate into nucleus pulposus** → direct haematogenous seeding of disc → spreads to endplate/body · **adults**: vessels reach **annulus only** → isolated discitis rare

Organism	Note
Staph aureus	**> 80%**
TB	**Always consider** – esp. not improving on first-line antibiotics
Salmonella	**Sickle cell**

3 Presentation

Toddler	**limp / refuses to sit or walk**, **loss of appetite**, abdominal pain, fever only **25%**
Older child	back pain with **point tenderness**, ↓ ROM

4 Diagnosis & Classification

Genes & lab markers

Test	Fact
ESR, CRP, WBC	**High-normal or mildly ↑**; WBC normal in ~45%
CRP	**Most sensitive for monitoring treatment** (half-life **19 h**) (**✗** Wrong options: ESR (3 wks to normalise) · WBC · blood cultures · procalcitonin)
Blood cultures	Identify organism (often negative)

Key radiology findings

X-ray unreliable; earliest changes at **~1 week**

Order	Finding	Timing
1 (earliest)	**Loss of lumbar lordosis**	Before bony change
2	**Disc space narrowing**	**10–21 days** (Orthobullets) · from 1 wk (OB)
3	**Endplate erosion**	10–21 days / 1–3 wks
Later	"Sawtooth" erosion 3–4 wks · endplate scalloping	Long-standing
Healed	**Vertebra magna**, permanent height loss, **block vertebra** (spontaneous fusion)	Resolved

MRI **diagnostic test of choice**



Paediatric discitis on T2 MRI

5 Management

Paediatric discitis

- Early, no abscess / thecal sac displacement → **ADMIT: BED REST + IV ANTI-STAPH ABX**
  (7–10 d IV → total 4–6 wks; ± immobilisation; follow CRP)
- No response at 7–10 d → **CT-GUIDED BIOPSY (rule out TB)**
- Late / abscess + deficit / failed non-op → **SURGICAL DEBRIDEMENT + ABX + BRACE**

Bacteraemic 5-yr, no collection **admit, bed rest + IV antibiotics**
✗ Wrong options: oral antibiotics at home · surgery · cast alone (slow, recurrent)

IV antibiotics speed resolution + ↓ recurrence

6 Complications

Long-term disc space narrowing, **interbody fusion**, back pain

– – Active space – –

X vs Y

	Paediatric discitis	Spondylolysis
Labs	↑ ESR/CRP	Normal
Exam	Pain flexion + extension, refuses to walk	Pain with **extension** (gymnast)
Rx	**IV antibiotics**	Activity restriction ± brace

How it's asked

"Disc space infection in children — all true EXCEPT?"	→ **loss of disc height is the earliest sign** (TO) — earliest = loss of lordosis
"Order of radiographic findings?"	→ **loss of lordosis → disc narrowing → endplate erosion**
"6-yr gymnast, ↑ ESR/CRP, discitis on MRI — true?"	→ **IV antibiotics usually resolve it** ✗ Wrong options: urgent debridement · bracing first line (spondylolysis)
"Best marker to monitor antibiotic response?"	→ **CRP**
"5-yr, gram-positive bacteraemia, no collection?"	→ **admit for bed rest + IV antibiotics**

DISCOGENIC BACK PAIN

– Spine · 25 / 69 –

1 Basics

Definition	**axial** low back pain from disc degeneration, **no radicular symptoms**
Biology	disc characteristics modified by **bone morphogenetic proteins**

3 Presentation

Pain ↑ with	**bending, sitting, axial loading** · flexion aggravates, extension relieves
Exam	**SLR / nerve tension negative**, neuro normal
Acute lumbago (3 wks), no red flags	**PT + NSAIDs**, no imaging · red flags: diabetes / IVDU (infection), cancer, smoking, **night pain** ✗ Wrong options: X-ray · MRI · discography · steroid taper

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray	**Best initial study** – degeneration, **vacuum phenomenon**; excludes other causes
MRI	Degenerated discs **without** significant stenosis / herniation
Provocative discography	Primary confirmatory test but **accelerates degeneration**: ↑ herniation, ↓ disc height, **reactive endplate changes** (✗ Wrong options: myelogram · TFESI · spinal anaesthesia)

Diagnostic criteria – positive discogram

Criterion
Concordant pain on injection
Abnormal disc morphology on fluoroscopy / post-discography CT
Negative (painless) control level(s)

5 Management

First line (majority)	NSAIDs, PT, **cognitive therapy**, lifestyle change
Fusion vs cognitive-exercise therapy	**equivalent ODI improvement** at **1 + 10 yrs** ✗ Wrong options: fusion superior · therapy superior
Fusion	controversial; **poor results when based on positive discography**

– – Active space – –

Lumbar total disc replacement (TDR)

Indication **single-level DDD with disease-free facets** (controversial)

TDR vs fusion / ALIF	Finding
Safety + efficacy at 2 yrs	**Comparable / better**
Hospital stay	**Shorter**
Reoperation	**Lower**
Disc height restoration	**Greater**
Subsidence	**Less**
Adjacent segment disease	**Lower**

Contraindications pars defect, fracture, **facet arthrosis**, **spondylolisthesis**, posterior element insufficiency, prior laminectomy, scoliosis, instability

Metal-on-metal TDR **↑ serum cobalt + chromium, like MoM THA**; persist; renally excreted
✗ Wrong options: transient · only cobalt · hepatic neutralisation · negligible

Failed lumbar TDR

- Asymptomatic subsidence, no extrusion → **OBSERVE**
- Facet pain / micro-instability, implant well placed → **POSTERIOR STABILISATION**
- Poly dislocation / malposition → **REVISION ARTHROPLASTY or ARTHRODESIS VIA RETROPERITONEAL APPROACH**
  (not far-lateral – iliac crest blocks L5-S1)

X vs Y

	Discogenic pain	Radiculopathy
Pain	Axial	Leg > back
SLR	Negative	Positive
Worse with	Sitting, bending, loading	Sitting/cough (HNP)

How it's asked

"3 weeks of lumbago, no red flags?"	**→ PT + NSAIDs**
"Best initial study for discogenic pain?"	**→ radiography**
"MoM vs poly TDR – labs?"	**→ ↑ cobalt + chromium like MoM THA**
"TDR vs fusion for single-level DDD?"	**→ comparable safety + efficacy at 2 yrs**
"Anterior poly dislocation 6 wks after TDR?"	**→ revision to arthrodesis via retroperitoneal approach**
"Fusion vs cognitive/exercise therapy?"	**→ equivalent improvement**
"Procedure accelerating disc degeneration?"	**→ provocative discography**

DISH (DIFFUSE IDIOPATHIC SKELETAL HYPEROSTOSIS)

– Spine · 26 / 69 –

1 Basics

Epidemiology

Also — **Forestier disease**; **enthesopathy** of spine + extremities; ossification of the **anterior longitudinal ligament** (TO: non-inflammatory, "degenerative")

Incidence — **6–12%** overall · uncommon < 50 · > 50: 25% M / 15% F · > 80: 28% M / 26% F · non-diabetics 1.6–13% vs **diabetics 13–49%**

Race — commonest in **Caucasians**; less in Black, Native American, Asian
✗ Wrong options: "most common in African Americans" (EXCEPT answer)

Location — **thoracic (right side) > cervical > lumbar** · thoracic **T7–T11**, often isolated · **right-sided thoracic** because the **pulsatile aorta protects the left** · cervical + lumbar **symmetric**

Genes & lab markers

Marker	Fact
HLA-B8	Associated (esp. DISH + diabetes)
HLA-B27	**Not** associated (✗ Wrong options: "90% B27 positive")
↑ HbA1c / diabetes	Image showing DISH → elevated HbA1c
↑ Growth hormone + IGF-1	✓
ESR	Normal

Risk factors — **gout, hyperlipidaemia, diabetes**, coronary artery disease

Associated conditions

Lumbar — **spinal stenosis**

Cervical (large anterior osteophytes) — **dysphagia**, stridor, hoarseness, **sleep apnoea**, difficult intubation, **myelopathy**

Fracture — ankylosed segments → **long lever arms**; **hyperextension** injuries; low-energy falls → unstable fractures

Extra-spinal — elbow and other entheses · **heterotopic ossification after THA 30–50%** (vs < 20% without DISH)

3 Presentation

Often asymptomatic (incidental)

Thoracolumbar — **mild chronic back pain** (ankylosis stabilises) + **stiffness worse in morning + cold weather**

– – Active space – –

Cervical	pain, stiffness, **dysphagia** (food "stuck"), stridor, hoarseness, sleep apnoea
Exam	↓ spinal ROM; myelopathy / stenosis signs

4 Diagnosis & Classification

Diagnostic criteria

Criterion (Resnick)
Flowing ossification along anterolateral aspect of ≥ 4 contiguous vertebrae (= **non-marginal syndesmophytes at 3 successive levels**)
Preserved disc height; little degenerative change (no marginal sclerosis, no vacuum)
No facet ankylosis; **no SI joint erosion / sclerosis / fusion**

Key radiology findings

Study	Finding
X-ray	**"Flowing candle wax"** non-marginal ossification; thoracic **right-sided**; **lateral C-spine**: anterior bone with **preserved disc space** (best view to separate from AS)
Bone scan	↑ uptake – may **mimic metastases**
CT	**Any neck pain after trauma** → rule out occult fracture
MRI	Ligament / cord injury



DISH: flowing anterior ossification with preserved discs

Dysphagia + neck stiffness in a diabetic	**X-rays of cervical, thoracic, lumbar spine** first ✗ Wrong options: HLA-B27 · MRI first · ESI · inflammatory labs

Osteophyte vs syndesmophyte vs DISH (TO)

Structure	Origin / orientation
Traction osteophyte	Small, horizontal, parallel to endplate, few mm from margin (Sharpey fibres)
Claw osteophyte	Larger, curved, may bridge – progression of traction osteophyte in degenerated disc (TO image answer)
Syndesmophyte (AS)	Ossified **annulus**, **vertical**, at the **endplate margin** (marginal)
DISH	Ossified **ALL**, undulating, away from endplate margin (non-marginal)

Osteophytes 80% men / 60% women by 50, 95% by 70; most at **L3-4 + T9-10**

5 Management

Non-op (most) activity modification, PT, brace, NSAIDs, **bisphosphonates**

Cervical traction (fracture) caution: **over-distraction** (compromised ligaments)

Surgery for sequelae: **lumbar stenosis, cervical myelopathy, adult spinal deformity**

Fractures – CT & long posterior fusion

DISH + trauma (even standing-height fall)

- Neck pain → **CT C-SPINE (not soft/hard collar + discharge or flex-ext)**
- Fracture into posterior elements → **LONG POSTERIOR FUSION**
  (≥ 3 levels above + 3 below, e.g. C6 → C3-T2)
  (not C5-7 short, anterior alone – failure, or halo / collar – ↑ mortality)

6 Complications

Cervical trauma mortality **15% operative vs 67% non-operative** · ↑ morbidity + mortality vs non-ankylosed spines · vs AS: "higher" / "similar" (Orthobullets) · **equivalent** (OB) – read the options

Neurologic injury with trauma **AS > DISH** (67% vs 40%); AS also more surgery + epidural haematoma

HO after THA **30–50%**

– – Active space – –

X vs Y

	DISH	Ankylosing spondylitis
Syndesmophytes	**Non-marginal**, flowing candle wax	**Marginal**, bamboo spine, Romanus shiny corners
Disc space	**Preserved**	Ossified (cervical)
Bone density	Normal / ↑	**Osteopenia**
HLA	**B8**	**B27**
Age	> 50	Young (15–40)
SI joints	**Spared** (abnormality excludes DISH)	**Bilateral sacroiliitis**
Facets	Spared	Ankylosed
Diabetes	**Yes**	No
ESR	Normal	↑
Thoracic	**Right-sided**	Whole spine
Trauma neuro injury	Lower	**Higher**

How it's asked

"50-yr, neck stiffness, vertigo – lateral C-spine flowing ossification?"	→ **DISH** (TO)
"Marked structures on X-ray/CT (bridging curved spurs)?"	→ **claw osteophytes** (TO)
"67-yr neck + back pain, anterior flowing ossification?"	→ **DISH** ✗ Wrong options: OPLL · RA · AS · OA
"Neck stiffness + dysphagia, lateral X-ray?"	→ **DISH**
"DISH – NOT true?"	→ **most common in African Americans**
"Which image → elevated HbA1c?"	→ **DISH**
"AS vs DISH – true?"	→ **AS has higher neuro injury rate with trauma**
"Diabetic, dysphagia, stiff neck – next?"	→ **X-rays of cervical, thoracic + lumbar spine**
"Fracture in DISH – true?"	→ **↑ morbidity + mortality**
"80-yr, fall, DISH, mild neck pain?"	→ **CT cervical spine**
"C6 fracture into posterior elements with DISH?"	→ **posterior fusion C3-T2**

HALO ORTHOSIS IMMOBILIZATION

– Spine · 27 / 69 –

1 Basics

Biomechanics

Principle fixes **skull to torso → most rigid external cervical immobilisation** (TO)

Best controls **upper cervical** motion: **rotation at C1-2** / occipitocervical junction
✗ Wrong options: subaxial flex-ext, rotation, lateral bend, CT junction

Weakness **intercalated paradoxical motion** in subaxial spine; **lateral bending least controlled → "snaking"**: recumbent lateral shows mid-cervical **kyphosis**, upright shows lordosis → not ideal for **lower cervical** injuries (incl. facet fracture-dislocation – fails)

5 Management

Indications

Adult	Paediatric
Occipital condyle fracture	**Atlanto-occipital dissociation**
Occipitocervical dislocation	**Jefferson / atlas fractures**
Stable Jefferson (type II atlas)	**Unstable odontoid fractures**
Type II odontoid in young	**Persistent AARS**
Type II / IIA hangman	C1-2 dissociation; subaxial trauma
Post-op adjunct	**Pre-op traction** for deformity (halo-gravity)

Contraindications

Absolute	Relative
Cranial fracture	Polytrauma, severe chest trauma, barrel chest, obesity
Infection	**Advanced age** – mortality **21% if > 79 yrs** (elderly → avoid halo)
Severe soft-tissue injury at pin sites	–

Pre-application CT suspected **cranial fracture** · **children < 10 yrs** (bone thickness, avoid **sutures**)
✗ Wrong options: skull X-ray

– – Active space – –

Pin technique

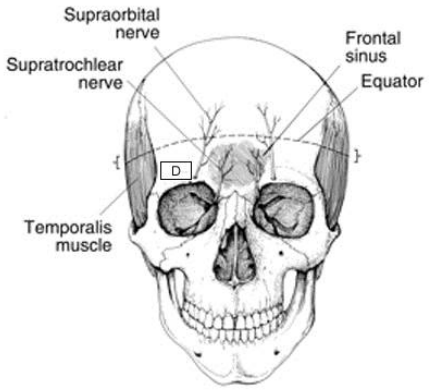
	Adult	Child
Pins	**4** (2 anterior, 2 posterior diagonally opposite)	**6–8** (Orthobullets) · **8–12** (OB) – **more pins**
Torque	**8 in-lb** (Orthobullets, OB); 6–8 in-lb range	**2–4 in-lb** ("finger-tight") · **1–5 in-lb** (OB) – **lower torque** (✗ Wrong options: ft-lb · maximal hand tightening)
Vest	Standard	Custom vest > 2 yrs · **Minerva cast < 2 yrs**
Retighten	Day 2	–

Anterior safe zone **1 cm above the orbital rim**, over the **lateral third** of the orbit (Orthobullets) · "lateral two-thirds", ≥ 4.5 cm from midline (OB) – **at/below skull equator** · **lateral to supraorbital nerve**, anterior + medial to **temporalis**

Ring below widest part of skull, ~1 cm above ear helix · pins perpendicular to cortex

Anterior pin position

- Too MEDIAL → **SUPRAORBITAL + SUPRATROCHLEAR NERVE injury, frontal sinus** *(forehead / upper eyelid numbness)*
- Correct (1 cm above lateral third, below equator) → **NO NEURO COMPLICATION**
- Too LATERAL (temporal fossa) → THIN BONE + TEMPORALIS → **jaw pain, chewing pain, infection**



Halo anterior pin safe zone and nerves

6 Complications

Overall **children ~70% (68%)** vs **adults ~35% (36%)**

Complication	Rate / management
Pin loosening	**36% — commonest in adults**; retighten; persistent → exchange
Pin-site infection	**12–20%**; esp. posterior pin in **temporalis fossa**; **not loose → keep pin + oral antibiotics + local care** (✗ Wrong options: remove · tighten) · loose / deep → remove + new site, IV abx, drain abscess
Discomfort	18%; release skin around pin
Dural puncture	1%; meningitis if inner table breached
Abducens (CN VI) palsy	**Commonest cranial nerve** (✗ Wrong options: oculomotor · trochlear) — **traction injury**, not pin; diplopia + **loss of lateral gaze**; ↓ traction + **observe** (resolves)
Supraorbital / supratrochlear palsy	**Medial anterior pins**
Medical	Pneumonia, ARDS, arrhythmia; ↑ mortality elderly



Abducens palsy: loss of lateral gaze

X vs Y

Nerve	Cause	Deficit
Supraorbital	Medial anterior pin	Upper eyelid / forehead / scalp numbness
Supratrochlear	More medial pin	Medial forehead + medial eyelid numbness
Abducens (VI)	**Traction**	Diplopia, ↓ lateral gaze

How it's asked

"Anterior pins too medial — nerve at risk?"	**→ supraorbital nerve**
"Region for anterior pin?"	**→ 1 cm above lateral orbit, below equator**
"Red (medial) zone — structures at risk?"	**→ supraorbital + supratrochlear nerves**

– – Active space – –

"Anterior pin position?"	→ **1 cm above, within lateral third of superior orbital rim** ✗ Wrong options: temporalis · medial third
"6-yr, 6 pins at 3 in-lb, correctly placed anterior pins — complication?"	→ **none**
"2-yr halo — pin/torque?"	→ **8 pins at 2–4 in-lb**
"7-yr, 10 pins at 6 in-lb, pin complication — change?"	→ **lower insertional torque**
"Preadolescent halo?"	→ **8–12 pins at 1–5 in-lb**
"2-yr Down syndrome, O-C2 fusion + halo — recommended?"	→ **pre-op skull CT**
"Halo — all true EXCEPT?"	→ **oculomotor is commonest injured CN** (TO) — it's abducens
"Commonest neurologic complication of halo traction?"	→ **deficit in lateral eye movement (CN VI)**
"Lateral gaze loss after 2 wks halo traction?"	→ **CN VI injury** ✗ Wrong options: surgery needed
"Erythematous, non-loose pin, afebrile?"	→ **leave pin + oral antibiotics**
"Halo best controls?"	→ **atlantoaxial rotation**
"Occipital condyle fracture in halo — true?"	→ **pins tightened to 8 in-lb** ✗ Wrong options: lower C-spine · lateral bend best controlled · middle third · trochlear nerve

INCOMPLETE SPINAL CORD INJURIES

– Spine · 28 / 69 –

1 Basics

Definition

Incomplete SCI — some motor or sensory function preserved below the level: **sacral sparing** (voluntary anal contraction, perianal sensation) or palpable/visible muscle contraction below level · **sacral sparing separates complete from incomplete**

Epidemiology

US — ~11,000 new SCI/yr · **34% incomplete tetraplegia** (central cord commonest) · 17% incomplete paraplegia · 47% complete

Anatomy – tracts

Tract	Function	Crosses
Lateral corticospinal (LCT)	**Motor** (voluntary) – **upper limb fibres central/medial, lower limb lateral**	Brainstem (medulla) → ipsilateral in cord
Ventral corticospinal	Motor (smaller)	–
Dorsal columns (gracilis = legs/ lower trunk, cuneatus = arms)	**Fine touch, vibration, proprioception**	Brainstem
Lateral spinothalamic (LST)	**Pain, temperature**, crude sensation	**In cord, 1–2 levels above entry**
Ventral spinothalamic	**Light (crude) touch**	In cord

Blood supply — **anterior spinal artery → anterior 2/3 of cord** (fed by segmental arteries incl. **artery of Adamkiewicz**: largest, usually **left, T8–L1**)

4 Diagnosis & Classification

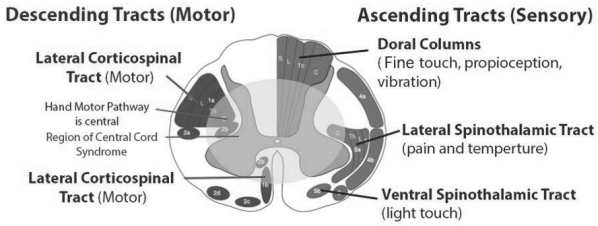
ASIA impairment scale

Grade	Definition
A	**Complete** – no motor or sensory in **S4-S5**
B	Sensory incomplete – sensation preserved below level incl. S4-S5, **no motor**
C	Motor incomplete – **> half key muscles below level grade < 3**
D	Motor incomplete – **≥ half key muscles grade ≥ 3**
E	Normal

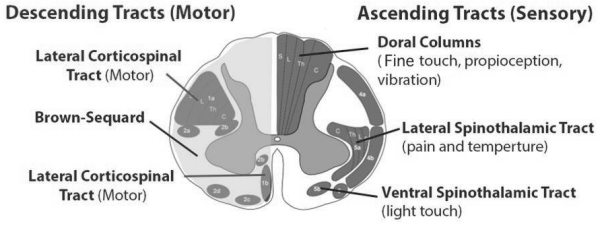
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Incomplete cord syndromes

Syndrome	Mechanism	Lost	Preserved	Prognosis
Central cord	**Elderly, hyperextension**, spondylotic stenosis (anterior osteophytes + infolded LF)	Motor **UE > LE**, **hands > arms**; burning dysaesthesia (hyperpathia) in arms	**Sacral sparing**; LE motor	**Good** (ambulation > 80%); **clumsy hands**
Anterior cord	**Flexion-compression** (bone) or **anterior spinal artery** injury (aortic surgery, T12 corpectomy)	**Motor (LCT) + pain/ temperature (LST)** ± light touch; **LE > UE**	**Dorsal columns** (proprioception, vibration)	**Worst**; 10–20% motor recovery; mimics complete
Brown-Séquard	**Hemitransection** – **penetrating** (stab)	**Ipsilateral motor + proprioception/ vibration**; **contralateral pain + temperature**	–	**Best – 99% (98%) ambulatory**
Posterior cord	Very rare; posterior spinal artery / tumour / trauma	**Proprioception**	Motor, pain, temperature, light touch	–



Central cord syndrome: central (upper limb) fibres of the lateral corticospinal tract



Brown-Séquard: ipsilateral motor, contralateral pain/temperature

5 Management

Central cord syndrome

Pathophysiology transient compression → **central cord oedema** → selective **lateral corticospinal white matter** injury
✗ Wrong options: central grey haemorrhage · anterior/posterior spinal artery · chronic compression = myelopathy · disc herniation · OPLL

Late picture **UE LMN signs** (clumsy) + **LE UMN signs** (spastic)

Treatment **controversial** · no active compression → non-op · active compression / instability → decompression · ASIA C may benefit from **early (< 24 h) decompression** (OB)
✗ Wrong options: high-dose steroids · routine non-op for all

Recovery order	**LE first → bowel/bladder → proximal UE → hands last · age < 50** → best recovery · usually walk + regain bladder; **hands rarely normal** ✗ Wrong options: stepwise decline = myelopathy · rapid decline = abscess/tumour
Better neurologic outcome	**ASIA C/D** (mild) · T2 cord signal → less early decline (OB)

Prognosis & surgical indications

Most important prognostic variable	**completeness / severity of neurologic deficit** ✗ Wrong options: mechanism · steroids < 8 h · gender · early surgery
Gunshot wounds	surgery only for **retained canal fragment + incomplete SCI with progressive deficit** (or persistent CSF leak) · trans-colonic GSW → **broad antibiotics, no surgery** · stable burst without deficit → non-op (Wood RCT) · central cord without compression → non-op · type III odontoid neuro intact → non-op

X vs Y

Finding	Syndrome
Arms + hands weak > legs, elderly fall on forehead	**Central cord**
Legs paralysed, pain/temp lost, proprioception kept	**Anterior cord**
Ipsilateral motor + proprioception, contralateral pain/temp	**Brown-Séquard**
Proprioception lost only	**Posterior cord**
After thoracoabdominal aneurysm repair / left-sided T12 corpectomy	**Anterior cord (Adamkiewicz)**

	ASIA C	ASIA D
Key muscles below level	> half grade **< 3**	≥ half grade **≥ 3**

How it's asked

"78-yr fall, hands weakest, urinary dysfunction — prognosis non-op?"	→ **likely to walk independently** ✗ Wrong options: full hand recovery · stepwise · < 10% bladder recovery
"79-yr forehead strike, UE weak, legs normal — pathophysiology?"	→ **selective injury to central lateral corticospinal tract**
"Tract causing motor deficit (UE 3/5, LE 4/5)?"	→ **lateral corticospinal**
"Wrist/finger flexion 3/5, legs 4+/5?"	→ **lateral corticospinal tract**
"NOT associated with central cord?"	→ **LE dysaesthesias**
"Who gets central cord syndrome?"	→ **67-yr after hyperextension** ✗ Wrong options: penetrating → BSS · aortic repair → anterior cord
"73-yr, arms 3/5, legs 4+/5 after forehead strike — pathogenesis?"	→ **transient compression + central cord oedema**
"75-yr falls on head, stenotic MRI — symptoms?"	→ **UE weakness primarily**

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"Central cord – true?"	**→ age < 50 → better ambulation + motor recovery** ✗ Wrong options: steroids · routine non-op
"88-yr, arms > legs weakness – pattern?"	**→ central cord diagram**
"C5-6 dislocation with Brown-Séquard – findings?"	**→ ipsilateral motor loss, contralateral pain/temp loss**
"Stab with hemitransection?"	**→ ipsilateral motor + proprioception, contralateral pain/temp**
"Right neck stab, left pain loss, right proprioception loss – motor + temp?"	**→ right motor loss, left temperature loss**
"Most consistent with anterior cord?"	**→ paraplegia after thoracoabdominal aneurysm repair**
"T12 corpectomy (left approach), new leg weakness, sensation intact, canal patent?"	**→ occlusion of arterial supply (Adamkiewicz)**
"MVC, legs weak, ↓ sensation below T9, position sense kept – spared region?"	**→ dorsal columns**
"Posterior cord syndrome loses?"	**→ proprioception**
"Most important prognostic variable?"	**→ severity of neurologic deficit**
"Surgical indication?"	**→ GSW, retained fragment compressing cauda equina, progressive motor deficit**

INFANTILE IDIOPATHIC SCOLIOSIS

– Spine · 29 / 69 –

1 Basics

Epidemiology

Definition	idiopathic scoliosis presenting at **≤ 3 years**
Incidence	**4%** of idiopathic scoliosis
Sex	**males > females** (**3:1**, OB)
Curve	usually **left thoracic**
Risk factor	family history ✗ Wrong options: predicts progression

Genes & associations

Inheritance	**autosomal dominant, variable penetrance**
Associations	**plagiocephaly**, congenital defects (DDH, congenital heart disease), **neural axis anomalies** (**22% of curves > 20°**; 80% of these need neurosurgery; **syrinx ~20%**), **thoracic insufficiency syndrome**

Early-onset scoliosis (EOS)

Definition (TO)	curve **> 10° with onset before age 10**, any aetiology (GSSG / CSSG) · categories (Vitale): **idiopathic, neuromuscular, congenital, syndromic** (Marfan, Down) ✗ Wrong options: "< 5 yrs only" · better prognosis than older children
Growth	**T1-L5 grows fastest in first 5 yrs**; thoracic height doubles birth → maturity · **thoracic spine height: 12 cm at birth, 18 cm at 5 yrs, 27 cm at maturity** · thoracic volume **6% at birth, 30% at 5, 50% at 10** · target **≥ 22 cm T1–T12** before fusion
Lungs (TO)	**alveolar number ↑ until ~age 8, THEN volumetric growth** · earlier onset + larger curve → pulmonary failure · infantile/juvenile or severely lordotic curves cause pulmonary problems; typical AIS does not ✗ Wrong options: reverse (EXCEPT answer)
Thoracic insufficiency syndrome	↓ thoracic growth + lung volume → pulmonary hypertension, **cor pulmonale** · PFT impairment > 60°, cardiopulmonary issues > 90°

3 Presentation

History	age noticed, progression, perinatal, milestones; **drooling** → neuro cause
Inspection	**café-au-lait** (NF), hair patch, **dimple above gluteal fold** (below = benign), nevi, **plagiocephaly**
Neuro	milestones, **cavovarus**, **asymmetric abdominal reflexes (syrinx)**, clonus, Hoffmann, Babinski, gait

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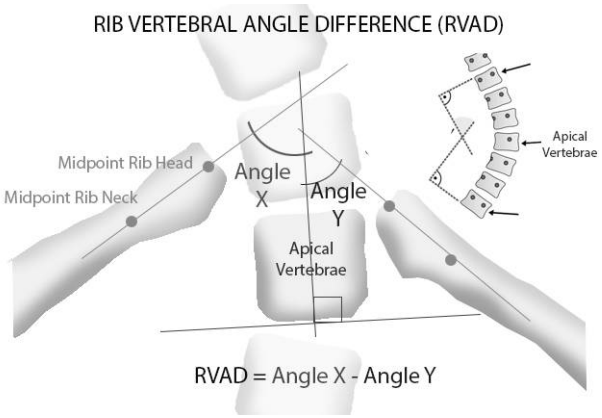
4 Diagnosis & Classification

Types **resolving** (most) vs **progressive**

Key radiology findings

X-ray **standing PA + lateral**; **supine if can't stand (underestimates curve)**; exclude congenital anomalies

Measure	Technique	Progression risk
Cobb	Standard	**> 20°** → progression
Rib phase	Convex apical rib head vs apical body on AP	**Phase 1** = no overlap · **Phase 2 = overlap → high risk** (✗ Wrong options: "phase 1 worse" (TO))
RVAD (Mehta)	Perpendicular to apical endplate; rib-vertebral angles via rib head–neck line; **concave − convex**	**> 20° → progression** (83%); **< 20° → resolves** (83%) (✗ Wrong options: > 10°)



Rib-vertebral angle difference (Mehta)

Mehta predictors **Cobb > 20° · RVAD > 20° · phase 2 rib**
✗ Wrong options: age · sex · family history

MRI (whole spine) **first**, before treatment: tether, cyst, tumour, **syrinx** · abnormal abdominal reflexes or < 10 yrs with significant curve → **MRI spinal axis**
✗ Wrong options: PT · brace · observe · fusion

5 Management

Infantile idiopathic scoliosis (MRI normal)

- Cobb < 30° (< 20–25°) + RVAD < 20° + phase 1 → **OBSERVE (~90% resolve)**
- Flexible, Cobb > 30° (> 20–25°), RVAD > 20°, phase 2 → **SERIAL MEHTA (EDF) DEROTATIONAL CASTING**
  (or TLSO); brace after casting / late flexible
- Cobb > 50–60° or failed casting → **GROWING RODS (dual / magnetic) or VEPTR**
  (fusion delayed to ~10 yrs)

Mehta casting

Elongation-derotation-flexion (EDF) casting	corrects in 3D while allowing growth · **start early**: before 18 months–4 yrs → better chance of curve < 15° (worse if started > 4 yrs) · smaller initial Cobb + good first in-cast correction predict success · may cure idiopathic infantile curves; in **syndromic EOS → delays surgery, curve may still progress** but spine keeps growing · complications: skin necrosis, infection
1-yr-old, 35°, RVAD > 20°, phase 2	**serial derotational casting** ✗ Wrong options: TLSO · observe · fusion · VBT

22-month, RVAD 26° → casting indication
✗ Wrong options: Cobb 18°/28° · RVAD 16°/5°

Growing constructs

Growing rods	anchors proximal + distal; **lengthen every 6–8 months**; permit up to **5 cm** growth; ↑ **FVC** · traditional or **magnetic** · 5-yr failing casting with ↓ FEV1/FVC → **growing rods** ✗ Wrong options: more casts · TLSO · fusion
Fusion before age 10	**pulmonary compromise** → delay
VEPTR (TO)	rib-based distraction (rib hooks proximally; spine/pelvic anchors distally) → **EOS with rib anomalies / TIS**, also neuromuscular (SMA) · **facilitates** infant use · pelvic anchors in ambulators → **crouched gait** (limit to non-ambulators) ✗ Wrong options: "pelvic anchors make infant use difficult" (EXCEPT answer)
Growth-friendly classes (Skaggs)	**distraction**: TGR, VEPTR, MCGR · **compression**: vertebral body stapling, VBT · **guided growth**: Luque trolley, Shilla

6 Complications

Surgery	high complication rate · progressive untreated curves can be **fatal** · **progressive by age 5 → > 50% exceed 70°**

X vs Y

	Infantile	Juvenile	Adolescent
Age	**0–3**	4–10	10–18
Sex	**Male**	F (older)	**Female**
Curve	**Left thoracic**	Right thoracic	Right thoracic
Natural history	Most **resolve**	Most progress	Variable
Key Rx	**Mehta cast**	Brace / growing rods	Brace / PSF

How it's asked

"Suggests progression in infantile scoliosis?"	→ **rib overlap of apical vertebra (phase 2)** ✗ Wrong options: RVAD > 10° · male · family history

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"Indication for Mehta casting / TLSO?"	**→ RVAD 26°**
"RVAD calculation?"	**→ concave angle − convex angle (X − Y)**
"VEPTR – incorrect statement?"	**→ "pelvic anchors make use in infants with TIS difficult"** (TO)
"5-yr failing Mehta casts, ↓ FEV1/FVC?"	**→ traditional or magnetic growing rods**
"Mehta casting for syndromic EOS – expected?"	**→ continued progression but further spine growth**
"1-yr-old, 35°, RVAD > 20°, phase 2 rib?"	**→ serial derotational casting**
"22-month, 48°, RVAD 24°, MRI normal – true?"	**→ earlier EDF casting → better chance of curve < 15°**
"6-yr painless curve, asymmetric abdominal reflexes?"	**→ MRI spinal axis**
"Lungs and spinal deformity – EXCEPT?"	**→ "volumetric growth until 8 then alveolar increase"** (TO – reverse)
"EOS definition?"	**→ scoliosis developing before age 10** (TO)

INTERVERTEBRAL DISC

– Spine · 30 / 69 –

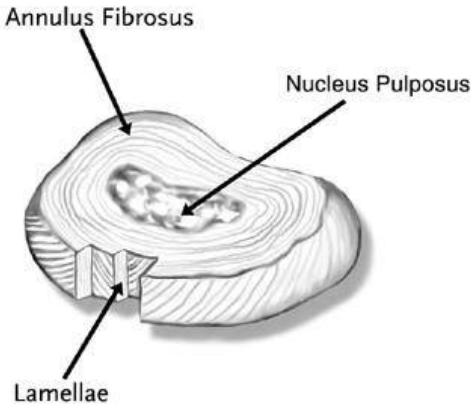
1 Basics

Composition & structure

Function motion + stability; links bodies; **25% of spinal column height**

Endplates **hyaline cartilage** attaches disc to bodies

	Annulus fibrosus	Nucleus pulposus
Collagen	**Type I** (obliquely oriented, 30° alternating "criss-cross" lamellae); inner annulus also type II; minor type V	**Type II** (minor type XI)
Collagen : proteoglycan	**High collagen / low PG**	**Low collagen / high PG**
Water	Lower (~65% outer annulus)	**~88%** (80–90%) (✗ Wrong options: 60–70%)
Cells	**Fibroblast-like** (annulus only)	**Chondrocyte-like** (survive hypoxia); notochordal cells gone by ~10 yrs
Origin	**Sclerotome**	**Notochord**
Property	**Highest tensile stress / tensile modulus** – resists torsion, axial + tensile loads; prevents distraction	**Highest compressive stress**; compressibility; distributes load



Disc: annulus fibrosus and nucleus pulposus

Aggrecan **most abundant proteoglycan**; hydrophilic → **maintains water content** (compressive strength) · GAGs: chondroitin + keratan sulphate · other PGs: decorin (fibrillogenesis), versican (adhesion), biglycan, lumican, fibromodulin (more in annulus)
✗ Wrong options: tensile strength (collagen) · sensory transmission · cross-linking

Aggrecan : collagen **nucleus > 1** · annulus < 1 · endplate < 1

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Nutrition & metabolism

Largest avascular structure — capillaries end at endplates; **nutrition by diffusion through pores in the (central) endplates** · outer third of annulus from epidural vessels (TO) · loading "pumps" diffusion
✗ Wrong options: CSF · annular vessels · diffusion across annulus (not porous enough)

Nucleus cells — energy by **anaerobic glycolysis** → need **glucose**; die with ↓ glucose or acidic pH, not hypoxia
✗ Wrong options: Krebs

Endplate damage (smoking, calcification, sclerosis) → degeneration

Innervation

Sinuvertebral nerve — from **dorsal root ganglion** (Orthobullets, OB) · "branch of ventral ramus" (TO) → **superficial annulus only** (nucleus + inner annulus aneural) · painful degenerate discs: nerves grow to inner third (~50%) with granulation tissue (neo-innervation) · central endplate also innervated

Pain neuropeptides — **substance P**, calcitonin (CGRP), **VIP**, CPON (neuropeptide Y flank)

Biomechanics

Viscoelastic — **creep** (deformation over time) + **hysteresis** (energy absorption with repetitive loading; ↓ with age)

Intradiscal pressure — **lowest supine** < standing < **highest sitting flexed forward holding weights**; load held closer to body → lower pressure
✗ Wrong options: highest standing (EXCEPT answer)

Diurnal — ~25% of water lost by day, regained at night
✗ Wrong options: "water ↓ at night"

After subtotal discectomy — **extension** = most stable loading mode

2 Mechanism

Degeneration biology

↓ Decreases with age / degeneration	↑ Increases
Water	**Keratan : chondroitin sulphate ratio**
Proteoglycans / aggrecan (large aggregating PG degradation = **pathognomonic**)	**Lactate**
pH	**Degradative enzymes – MMPs**
Viable cell number, matrix synthesis, disc height	Fibroblast-like cell density (**annulus only** – not in nucleus)
Nutritional transport	Type X collagen
Elasticity, GAG : collagen ratio, sulphation	Inner annulus thickness
No change: absolute collagen quantity	Nucleus stays **type II**

Conversion to fibrocartilage — annulus–nucleus distinction lost, endplate hyalinisation, fissures, **Schmorl nodes**, vascularisation of posterior annulus

Annular tears commonest **posterolateral**
✗ Wrong options: central

Herniated disc spontaneous ↑ **IL-1β, OPG, RANKL, PTH**; **IL-1β** stimulates disc cells to ↑ **MMPs, nitric oxide, IL-6, PGE2** · **MMPs = most abundant product** of herniated disc cells → **ECM proteolysis**
✗ Wrong options: caspases · decorin

Kirkaldy-Willis stages (TO)

Stage	Age	Features
Dysfunction	15–45	Circumferential + radial annular tears, facet synovitis
Instability	35–70	Internal disruption, disc resorption, facet laxity/subluxation → **HNP**
Stabilisation	> 60	Hypertrophic bone, segmental stiffening → **stenosis**

4 Diagnosis & Classification

Key radiology findings

Vacuum phenomenon (TO) **nitrogen** gas in disc/vertebra: **DDD** (commonest), osteoporotic collapse, myeloma, Schmorl node, osteonecrosis (Kümmell), rarely infection
✗ Wrong options: **endplate avulsion** (EXCEPT)

Pfirrmann classification (TO – T2 MRI)

Grade	Signal	Nucleus vs annulus	Height
I	Homogeneous, bright white	Clear	Normal
II	Inhomogeneous, white ± grey band	Clear	Normal
III	Intermittent grey	Unclear	Normal / slightly ↓
IV	Dark grey	Lost	Slightly–moderately ↓
V	Black	Lost	**Collapsed**

5 Management

Biological therapy **BMP-2, BMP-7, TGF-β = morphogens**: ↑ **chondrogenic phenotype + matrix synthesis** without proliferation · **IGF-1, FGF = mitogens** (↑ proliferation)

HNP natural history (TO) mostly favourable; surgical benefit mainly first year; non-progressive deficits (except CES) treated non-op; surgery can wait 6–12 wks

X vs Y

	Morphogen	Mitogen
Examples	BMP-2, BMP-7, TGF-β	IGF-1, FGF
Action	↑ Chondrogenic phenotype, matrix	↑ Cell number

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How it's asked

"Discs — all true EXCEPT?"	→ **intradiscal pressure highest standing** (TO)
"True about IVD?"	→ **nucleus type II, annulus type I** (TO) ✗ Wrong options: nucleus water < annulus · water ↓ at night
"Annular tear — collagen type?"	→ **type I**
"Aggrecan : collagen in nucleus, annulus, endplate?"	→ **> 1, < 1, < 1**
"Aggrecan function?"	→ **maintains water content**
"Highest tensile modulus?"	→ **annulus fibrosus**
"Main nutrition in adults?"	→ **diffusion through porous central endplates** (TO)
"Main blood supply?"	→ **diffusion through endplate pores** (TO)
"Nucleus cells generate energy by?"	→ **anaerobic glycolysis**
"Nucleus pulposus — true?"	→ **nutrition by endplate diffusion** ✗ Wrong options: type I · 60–70% water · low PG ratio · sclerotome · innervated · fibroblast-like cells
"Ageing disc?"	→ **water ↓ with ↓ proteoglycans** (TO)
"IL-1β stimulates?"	→ **MMPs, NO, IL-6, PGE2**
"Age-related degeneration process?"	→ **↑ degradative enzyme activity**
"Ageing at endplate?"	→ **conversion to fibrocartilage**
"Progressive ageing change?"	→ **↑ keratan : chondroitin ratio**
"Pathognomonic of degeneration?"	→ **degradation of large proteoglycans in nucleus**
"Nucleus birth → adulthood — EXCEPT?"	→ **↑ fibroblast-like cells in nucleus**
"L4-5 degeneration pathology?"	→ **↓ proteoglycan · ↓ aggrecan · ↑ MMP activity**
"Herniated disc cells produce abundantly — function?"	→ **ECM proteolysis (MMPs)**
"BMP-2/7, TGF-β mechanism?"	→ **↑ chondrogenic phenotype + matrix synthesis**
"Vacuum phenomenon — all EXCEPT?"	→ **endplate avulsion** (TO)
"Pfirrmann uses?"	→ **T2 MRI** (TO)

JUVENILE IDIOPATHIC SCOLIOSIS

– Spine · 31 / 69 –

1 Basics

Epidemiology

Definition idiopathic scoliosis presenting at **4–10 yrs** (3–10 in OB)

Incidence **15%** of idiopathic scoliosis

Sex / curve **females > males** · **right main thoracic** commonest

Natural history **high risk of progression**: **70% need treatment** (half brace, half surgery); spontaneous resolution rare; can be fatal untreated

Associated neural axis abnormalities

Prevalence **18–25%** (~20% with curves > 20°)

Anomaly	Clue
Syringomyelia	Curve **without rotation**; **asymmetric / absent abdominal (umbilical) reflex**
Arnold-Chiari	Tonsillar herniation blocking CSF
Tethered cord	Often silent until adult
Dysraphism	Midline skin signs
Cord tumour	Astrocytoma, dermoid, epidermoid, teratoma

Not associated atlantoaxial rotatory instability (EXCEPT answer)

EOS scoliosis < 10 yrs: infantile, juvenile, congenital, neurogenic, syndromic (Marfan, Down)

3 Presentation

History when noticed, progression, perinatal history · **no bowel/bladder control by 3–4 yrs** → neuro cause

Screening **7° on scoliometer** (Adams) ≈ 20° Cobb

Exam café-au-lait, LLD (sitting forward bend eliminates), shoulder/waist asymmetry, **cavovarus**, hairy patch, dimples, nevi, rib hump · **abdominal reflexes** (stroke each quadrant → umbilicus moves toward it), clonus, Hoffmann, Babinski, gait

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Syringomyelia on MRI

4 Diagnosis & Classification

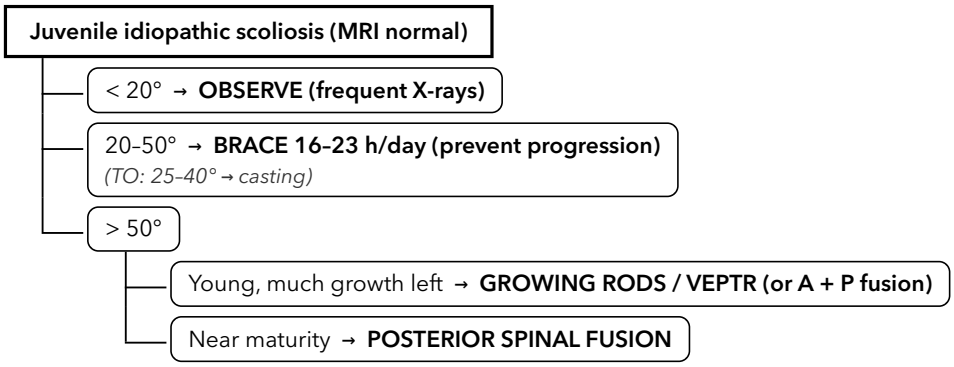
Key radiology findings

X-ray upright PA + lateral · **Cobb > 10°** = scoliosis (error 3–5°) · bending films → which curves to fuse

Diagnostic criteria – MRI (whole spine)

MRI indicated	Not an indication
Age < 10–11 with curve > 20° (even neuro-normal; e.g. 25° right thoracic in 7-yr)	Right thoracic curve per se (in older child)
Left thoracic curve	Thoracic hypokyphosis
Asymmetric / absent abdominal reflexes	Lenke C lumbar modifier
Male	No family history
Absence of thoracic apical lordosis (hypokyphosis lost) – strong predictor	Symmetric abdominal reflexes
Abnormal neuro (hyperreflexia, sensory loss) · **rapid progression**	–
Absent pedicle, congenital bar	–

5 Management



Bracing relative CI = **thoracic hypokyphosis** · goal is prevention, not correction

45° curve in 7-yr (TO) options: growing rods, **stapling**, **casting** · (✗ Wrong options: **posterior instrumented fusion** (crankshaft + ↓ thoracic growth)) · TO thresholds: **25–40° cast**, **> 40° growth-friendly surgery**

Fusion **A + P in young** (prevent crankshaft) · **PSF near maturity** (gold standard thoracic + double major) · ASF for TL / lumbar with normal sagittal profile

Growing rods

Lengthening every **~6 months → > 2 procedures**
✗ Wrong options: VBT + PSF = 2 stages · single-stage PSF · A/P 1–2 stages · halo + PSF

Law of diminishing returns each lengthening needs **↑ force** and gains **↓ length** (force doubles by 5th; < 8 mm per lengthening)

Comparison	Finding
Traditional rods vs VEPTR	**TGR: ↑ curve correction + truncal height**, fewer complications (VEPTR subcutaneous → ↑ wound problems)
Magnetic (MCGR) vs traditional	MCGR → **fewer surgeries**
Final fusion after growth-friendly vs primary fusion	**↑ reoperation + complications**
Younger age / longer duration	**↑ infection + wound complications**
Pulmonary	Growing rods ↑ PFTs + diaphragm motion

6 Complications

Crankshaft posterior fusion alone in young child

X vs Y

	Juvenile	Infantile	Adolescent
Age	4–10	0–3	10–18
Sex	F > M	M > F	F
Curve	Right thoracic	Left thoracic	Right thoracic
Neural axis anomaly	**18–25%**	~22% (> 20°)	Low
Natural history	**Progresses (70% treated)**	Mostly resolves	Variable

How it's asked

"MRI rules out all EXCEPT?"	**→ atlantoaxial rotatory instability**
"Screening MRI indications EXCEPT?"	**→ right thoracic curve** ✗ Wrong options: male, age < 11, absent apical lordosis, neuro signs
"8-yr, left curve, normal neuro – next?"	**→ MRI cervical-thoracic-lumbar**
"Progressive curve + absent abdominal reflex?"	**→ spinal cord abnormality → whole-spine MRI**

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"7-yr, MRI recommended because?"	**→ 25° right thoracic curve**
"7-yr, 45° JIS – EXCEPT?"	**→ posterior instrumented fusion** (TO)
"Growing rods with repeated lengthenings?"	**→ force ↑, length gained ↓**
"TGR vs VEPTR?"	**→ TGR more correction + truncal height**
"7-yr, 75° + 45° curves – option needing > 2 surgeries?"	**→ traditional growing rods**

KLIPPEL-FEIL SYNDROME

– Spine · 32 / 69 –

1 Basics

Epidemiology

Definition	congenital fusion of **≥ 2 cervical vertebrae** from **failure of segmentation (or formation) of cervical somites at 3–8 weeks**
Incidence	classically **1 in 40,000** · recent: 1 in 172 to 1 in 83 (asymptomatic) · **60–70% female**
Commonest fusion levels	**C2-C3 (71%)** > C5-C6 (68%) > C6-C7 (68%) > C3-C4 (29%) (TO) ✗ Wrong options: C3-4

Genes & inheritance

Pattern	Gene
Sporadic	**Commonest**; isolated fusion
Autosomal dominant	**GDF6 (chr 8)**, **GDF3 (chr 12)**; SGM1 (chr 8)
Autosomal recessive	**MEOX1 (chr 17)** (encodes MOX1, regulates vertebral separation)
Other	Notch, Pax genes; **fetal alcohol** (50% have cervical fusions)

Associated conditions

Association	Rate
Congenital scoliosis	**53.5–60%** (commonest orthopaedic association)
Sprengel deformity	30% (Orthobullets) · 33% / 50% / 20% (TO) · 6.4–16.7% (Orthobullets exam) – read the options
Renal (aplasia)	**33% → renal US**
Deafness	**30%**
Congenital heart (VSD)	15–30% (5–30%) → **echo**
Atlantoaxial instability	**~50%**
Adjacent-level degeneration	**100%**
CNS anomalies	19% (brainstem, Chiari, tethered cord, diastematomyelia)
Others	Basilar invagination, cervical stenosis, **synkinesis (mirror movements)**, cleft palate, absent ribs

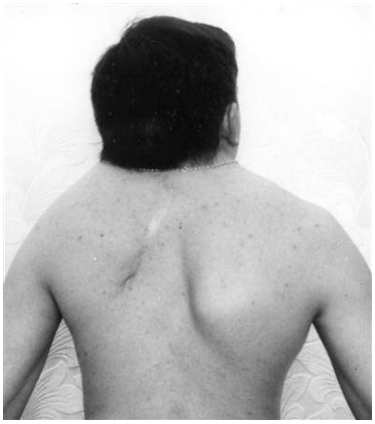
Wildervanck (cervico-oculo-acoustic)	**KFS + ossicular deafness + Duane syndrome** (abducens palsy with retraction bulbi); X-linked, female (Clarke KF4)

3 Presentation

Classic triad (< 50%)	**low posterior hairline**, **short (webbed) neck**, **limited cervical ROM**

– – Active space – –

Most consistent finding (TO)	**limitation of neck motion** ✗ Wrong options: short neck · low hairline · high scapula
Not a feature / not in triad (TO)	**bilateral SCM contractures** · **head tilt (torticollis)** may coexist but not part of triad
Symptoms	neck stiffness, neck/back pain (onset ~16 yrs, worse ~28 yrs), radiculopathy/myelopathy from hypermobile adjacent segments · ↑ cervical myelopathy rate



Klippel-Feil: low hairline, short neck

4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP, lateral, open-mouth; **whole spine**	Fusions (craniocervical / subaxial); **basilar invagination** (dens above **McRae's line**); **AAI = ADI > 5 mm**; stenosis < 13 mm; disc calcifications (resolve in 6 months)
CT	Bony anatomy pre-op
MRI	Cord anomalies; smaller cords in larger canals
Renal US + echo	Renal aplasia, VSD

Classifications

KFS (original)	Fusion	Inheritance
I	Most / all cervical	Sporadic
II	1–2 vertebrae	AD
III	+ thoracic / lumbar	AR

Samartzis (prognostic)	Pattern	Symptoms
I (25%)	Single-level fusion	Axial
II (50%)	Multiple **non-contiguous**	Myelopathy / radiculopathy
III (25%)	Multiple **contiguous**	Myelopathy / radiculopathy

Clarke	KF1 (AR, only class with C1 fusion) · KF2 (AD, C2-3 dominant, SGM1) · KF3 (isolated, e.g. C5-6) · KF4 (X-linked, Wildervanck)

High-risk fusion patterns (TO)

Pattern	Problem
1 – C1-2 fusion + occipitalised atlas	Motion concentrated at atlantoaxial → odontoid hypermobile, posterior dislocation
2 – Long fusion + abnormal occipitocervical junction	Forces through abnormal dens / C1 ring → instability
3 – Single open interspace between two fused blocks	Hypermobile segment → instability + DJD
Least symptomatic	**Multiple short fusions separated by > 1 mobile interspace** (stress shared)

Above vs below C3 fusions **above C3** → more symptoms, avoid contact sport · **below C3** → usually asymptomatic, normal life span

5 Management

Contact sports participation

Clearance	Indication
Can play contact sport	**Asymptomatic, 1–2 level fusion below C3** (e.g. C4-5), Samartzis II at/below C3 with full ROM, no instability/spondylosis
No contact sport (absolute)	**Fusion involving C2** (commonest presentation), **long / multilevel fusions**, any ↓ cervical motion, instability, spondylosis, occipitocervical anomaly, **after KFS surgery**
Relative	Samartzis II with prior transient quadriplegia

Surgery (decompression + fusion) **basilar invagination**, chronic pain, **myelopathy**, **atlantoaxial instability**, symptomatic adjacent disease, **progressive scoliosis** (82% of KFS with congenital scoliosis need surgery)

6 Complications

Cervical SCI with contact sports (small cords); central cord after minor trauma

X vs Y

	Klippel-Feil	Congenital muscular torticollis
Lesion	Bony fusion	SCM fibrosis
Neck	Short, low hairline, ↓ ROM	Tilt + rotation, SCM mass
Associations	Renal, deafness, VSD, Sprengel, scoliosis	DDH, metatarsus adductus

How it's asked

"Adolescent, short neck, renal aplasia, C4-5 fusion – football?" → **can play** (TO)

"10-yr, cervical fusion above C3, renal aplasia?" → **Klippel-Feil, no contact sports**

– – Active space – –

"14-yr, C2-4 fusion, lacrosse?"	→ **refrain from contact sports** ✗ Wrong options: after MRI · after PT · after fusion
"NOT a feature of KFS?"	→ **bilateral SCM contractures** (TO)
"Fusion pattern least likely symptomatic?"	→ **multiple short fusions separated by mobile joints** (TO)
"Most consistent finding?"	→ **limitation of neck motion** (TO)
"Most commonly fused level?"	→ **C2-C3** (TO)
"Not part of triad?"	→ **head tilt** (TO)

– – Active space – –

LOW BACK PAIN

– Spine · 33 / 69 –

1 Basics

Epidemiology & natural history

Lifetime	**50–80%** (70–85%) · ~$100 billion/yr · 2nd only to respiratory infection for office visits · leading cause of disability worldwide
Work	back injury ~2% of workforce/yr · of those off work: **60–70% back by 6 wks**, **80–90% by 12 wks**; slow after 12 wks
Course	**90% resolve within 1 yr** · chronic in 5–10% · recurrence 20–72%
Commonest cause	**muscle strain (musculogenic)**; commonest degenerative: stenosis, disc herniation, discogenic pain
Risk factors	obesity, smoking, sex, lifting, vibration, prolonged sitting, **job dissatisfaction**

Red vs yellow flags (TO)

Red flags (serious pathology)	Yellow flags (chronic disability risk)
Age < 20 or > 50	Belief pain is severely disabling
Significant trauma	**Fear-avoidance** behaviour
Constant, **night pain**, worsening	Low mood
Cancer, steroids, **IVDU**	Expects passive treatment
Unexplained fever / weight loss	Poor job satisfaction
Neurologic deficit · bowel/bladder change (CES)	**Lack of family support** (✗ Wrong options: red flag)

3 Presentation

Pain type	Features
Musculogenic	Commonest; activity-related stiffness, difficulty bending
Discogenic	Controversial; discogram
Mechanical	Facet degeneration / instability; worse lifting, standing
Sacroiliac	SI joint
Radicular	Unilateral dermatomal leg pain
Referred	Buttocks, posterior thighs, groin (L5-S1)
Neurogenic claudication	Buttock/leg pain with standing – fairly specific for **stenosis**
Myelopathy	Cord lesion (~L1 or above)
CES / conus	Bilateral leg pain, weakness, saddle anaesthesia, bowel/bladder

– – Active space – –

Non-organic (Waddell) signs

Significant if ≥ 3 of 5

Waddell sign
Superficial / non-anatomic tenderness
Simulation – pain with **axial compression** or **simulated rotation**
Distraction – **negative SLR when distracted** (seated)
Regional non-dermatomal sensory / motor disturbance
Overreaction

Organic sign — **positive SLR with distraction** · **neuroanatomic** deficit is NOT Waddell (TO)
✗ Wrong options: axial-load pain · diffuse tenderness · non-dermatomal numbness · simulated rotation

4 Diagnosis & Classification

Key radiology findings

X-ray — only if **> 1 month** failed treatment or **red flags**
✗ Wrong options: at first visit for 2-day / 2-week atraumatic pain

MRI — sensitive + specific but **many asymptomatic abnormalities**

Asymptomatic age	HNP	Bulge	Degeneration
20–39	21%	56%	34%
40–59	22%	50%	59%
60–79	**36%**	**79%**	**93%**

High-intensity zone (annular tear) — 59% symptomatic vs 25% asymptomatic (Carragee); discography provokes pain ~70% regardless

Compression fracture acuity — **MRI** (oedema ∝ recency)
✗ Wrong options: CT · flex-ext · indium scan

Differential — strain, HNP, DS, stenosis, radiculopathy, **AAA**, SI infection, AS, coccydynia, sacral insufficiency fracture

5 Management

Acute atraumatic LBP, no red flags

- REASSURANCE + LIMITED ANALGESIA + EARLY ROM / activity, follow-up ~4 wks
  (no X-ray, MRI or ESI at first visit)
- > 1 month failed / red flags → **IMAGING**

Chronic work-related pain – strongest negative predictor — **high pre-treatment VAS**
✗ Wrong options: race · smoking · occupation · gender

Chronic non-specific pain, non-surgical (smoker, depression, narcotics)	**cognitive intervention + exercise + smoking cessation** ✗ Wrong options: discography · EMG · myelogram

Coccydynia (TO)

Cause	commonest **direct trauma**; childbirth, surgery, epidural
Acute (< 2 months)	rest + **NSAIDs 8 wks**, stool softener, **U-shaped cushion**
Chronic (> 2 months)	steroid + LA injection ± manipulation under GA → after 2 failed monthly injections → **coccygectomy (60–80% success)** (TO answer **80%**)

Baastrup disease ("kissing spines") (TO)

Pathology	**crowding / contact of adjacent spinous processes** with excessive lordosis → interspinous ligament degeneration, **adventitious bursa**, sclerosis, flattening · commonest **L4-L5** ✗ Wrong options: spina bifida · cervical instability/myelopathy
Clinical	midline pain radiating up/down, **worse in extension, better in flexion**, point tenderness · rarely claudication (epidural cyst)
Imaging	lateral X-ray contact + sclerosis; MRI interspinous bursitis (< 10%)
Treatment	NSAIDs, steroid + LA injection, bursa excision / osteotomy

X vs Y

	Organic	Non-organic (Waddell)
SLR distracted	**Positive**	Negative
Deficit	Dermatomal	Regional
Tenderness	Localised	Superficial / diffuse

How it's asked

"2-day atraumatic LBP — all appropriate EXCEPT?"	**→ lumbar radiographs**
"Missed work for acute LBP — return within 6 wks?"	**→ 60–70%**
"Severe atraumatic LBP, no red flags — management?"	**→ reassurance, limited analgesics, early ROM**
"Return to work within 12 wks?"	**→ 80–90%**
"Radiographs NOT indicated initially?"	**→ 28-yr with 2 wks isolated LBP**
"Waddell signs EXCEPT?"	**→ neuroanatomic motor/sensory compromise** (TO)
"Finding indicating organic cause?"	**→ positive SLR with distraction**
"Most likely non-organic LBP?"	**→ superficial tenderness, non-dermatomal numbness/weakness, negative SLR, unrelated disc degeneration**

– – Active space – –

"Strongest negative predictor of non-op success?"	→ **high pre-treatment VAS**
"4-yr LBP, smoker, depression, narcotics, annular tear on MRI?"	→ **cognitive intervention, exercise, smoking cessation**
"Acuity of compression fracture?"	→ **MRI**
"Red flags EXCEPT?"	→ **lack of family support** (TO – yellow flag)
"Coccygectomy success after failed non-op?"	→ **80%** (TO)
"Baastrup disease?"	→ **crowding of spinous processes** (TO)

– – Active space – –

LOWER EXTREMITY SPINE EXAM

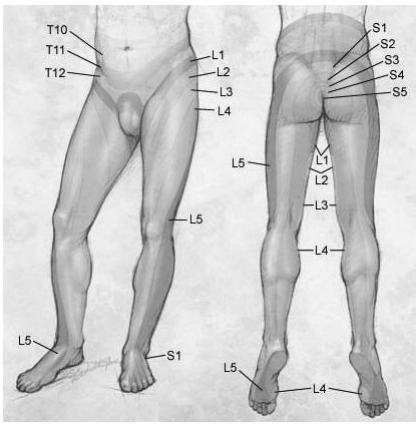
– Spine · 34 / 69 –

1 Basics

Root map

Overlap most muscles have > 1 root (e.g. **ankle dorsiflexion = L4 + L5**)

Root	Motion	Muscle (nerve)	Sensory	Reflex
L1	–	–	Iliac crest, groin	**Cremasteric (L1-2)**
L2	Hip flexion + adduction	Iliopsoas (femoral/plexus), adductors (obturator)	Anterior + inner thigh	Cremasteric
L3	Knee extension (+ L4)	Quadriceps (femoral)	Anterior/ medial thigh, medial knee	–
L4	**Ankle dorsiflexion** (+ L5)	**Tibialis anterior** (DPN)	Lateral thigh, anterior knee, **medial leg**	**Patellar**
L5	**Foot inversion**, **toe dorsiflexion**, hip extension, **hip abduction**	Tib posterior (tibial), **EHL/EDL** (DPN), hamstrings, glut max (inferior gluteal), **glut medius (superior gluteal)**	**Lateral leg + dorsum of foot**	–
S1	**Plantar flexion**, **eversion**	Gastrocsoleus (tibial), peroneals (SPN)	Posterior leg (lateral foot)	**Achilles**
S2	Toe plantar flexion	FHL, FDL (tibial)	Plantar foot	–
S3–4	Bowel + bladder	Bladder	Perianal	–



Lower limb dermatomes

- - Active space - -

3 Presentation

Symptom clues

Symptom	Think
Weight loss, axial pain, **night pain**	Tumour / infection
Mechanical pain	Instability, discogenic
Radicular pain	HNP
Referred pain	PUD, cholecystitis, renal stone, PID, pancreatitis
Psychogenic	Secondary gain

Inspection scars, café-au-lait, lumbar hairy patch

Grading (ASIA)

Motor	
0	Paralysis
1	Palpable / visible contraction
2	Full ROM, gravity eliminated
3	Full ROM against gravity
4	Against some resistance
5	Normal resistance
NT	Not testable

Sensory 0 absent · 1 impaired · 2 normal · NT

4 Diagnosis & Classification

Special tests

Test	Fact
Straight leg raise	Supine passive elevation **30–70°** (positive by ~40°) → pain **below the knee** = **L4–S1 (L5-S1)** root tension; separate from hamstring tightness
Crossed SLR	Raising the well leg reproduces pain in the affected leg (more specific)
Femoral nerve stretch	**Prone, knee flexed + hip extended** → anterior thigh pain = **L2-3, L3-4** protrusion
Babinski, ankle clonus	**UMN** lesion
Bulbocavernosus reflex	Squeeze glans/clitoris or **tug Foley** → anal contraction; tests for **spinal shock**
Lhermitte / Hoffmann / Spurling	Cervical tests (✗ Wrong options: lumbar)

Clinical findings → lesion

Finding	Cause
Medial knee paraesthesia	**Saphenous** (femoral) irritation – psoas abscess, surfer's compression
Trendelenburg gait	**L5 root** → gluteus medius (superior gluteal) – **L5/S1 far-lateral HNP** (OB) / **paracentral L4/5 HNP** (Orthobullets)
Lateral thigh numbness	**Meralgia paraesthetica (LFCN)** – positioning
Foot drop	**Common peroneal palsy** / sciatic compression (lateral knee pressure, hip dislocation)
Quadriceps weakness	**Femoral nerve palsy** (hyperflexed Pavlik) / L3-L4 roots

HNP → root (Trendelenburg logic)

Disc	Root	Weak
L3/4 far lateral	L3	Hip flexion
L3/4 central / L4/5 far lateral	L4	Knee extension, dorsiflexion
L4/5 paracentral or L5/S1 far lateral	**L5**	**Hip abduction (Trendelenburg)**, EHL
L5/S1 paracentral	S1	Plantar flexion

UMN + LMN together (TO)

Absent ankle jerks + upgoing plantars — **B12 deficiency** (subacute combined degeneration), **motor neuron disease**, **Friedreich ataxia** (also conus lesions, cervical myelopathy + lumbar stenosis)
✗ Wrong options: **multiple sclerosis** (UMN only, never LMN)

Waddell signs (≥ 3 = significant)

Sign	Description
Tenderness	Superficial (light touch) or deep non-anatomic
Simulation	Pain with light **axial skull compression** or **pelvic twist**
Distraction	**No pain with distracted SLR**
Regional	Non-anatomic motor / sensory findings
Overreaction	Any time

How it's asked

"Passive leg elevation 30–70° → pain below knee?"	→ **straight leg raise test**
"Trendelenburg gait from lumbar disc?"	→ **L5/S1 far-lateral herniation (L5 root)**
"NOT a cause of absent ankle jerks + upgoing plantars?"	→ **multiple sclerosis** (TO)
"Bulbocavernosus reflex tests for?"	→ **spinal shock**
"Femoral stretch test positive in?"	→ **L2-3 / L3-4 disc**

– – Active space – –

LUMBAR DISC HERNIATION

– Spine · 35 / 69 –

1 Basics

Epidemiology

Incidence	peak **4th–5th decades** · lifetime prevalence 10% · only ~5% symptomatic · **M:F 3:1**
Commonest level	**L5-S1** (Orthobullets) · **L4-L5** (TO) – read the options · 95% at L4-5 or L5-S1
Foraminal herniations	commonest at **L3-L4** (OB)

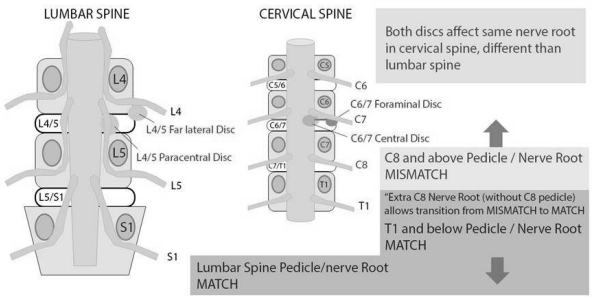
Pathoanatomy

Mechanism	recurrent **torsional strain** → outer annulus tears → nucleus herniates · **PLL weakest posterolaterally** → paracentral site
Pain fibres	**sinuvertebral nerve** → posterior annulus (vertebrogenic back pain precedes herniation)
Who herniates	young, **well-hydrated** discs (child: **ring apophysis Salter-Harris II**) · desiccated old discs less likely
Sciatica	compression + inflammation: **TNF-α, MMP, NO, PGE_2, IL-6**

Disc part	Collagen	Property
Annulus fibrosus	**Type I**, high collagen / low proteoglycan	Tensile strength, 15–25 lamellae
Nucleus pulposus	**Type II**, low collagen / high proteoglycan, 88% water	Compressibility; avascular (endplate diffusion)

Root anatomy

Lumbar	**vertical** root: L5 runs under L5 pedicle (match) → **paracentral and foraminal disc at same level hit different roots**
Cervical	horizontal root: C6 under C5 pedicle (mismatch) → central + foraminal hit same root



Vertical lumbar vs horizontal cervical roots

3 Presentation

History sudden pain after lifting · vibration + sitting with bending/rotation · relief **supine with hips + knees flexed**

Radicular pain **worse sitting**, cough, Valsalva, sneeze · better standing · walking doesn't worsen

Posture **leans away** from the radiculopathy side (opens foramen) · **sciatic scoliosis** (list)

Cauda equina 1–10% (central disc): bilateral leg pain, weakness, saddle anaesthesia, bowel/bladder → **emergent discectomy within 48 h**

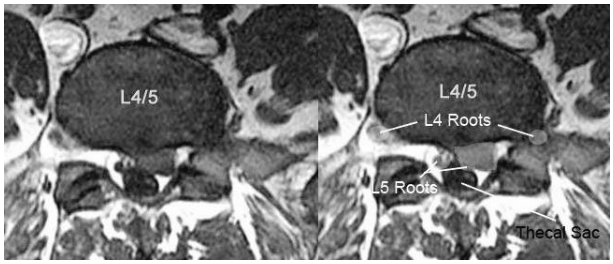
Root localisation

Root	Motor	Sensory	Reflex
L3	**Hip adduction**, knee extension	**Anteromedial thigh**, near **medial femoral epicondyle**, groin	–
L4	**Ankle DF (L4 > L5)**, quadriceps (**single-leg sit-to-stand**) · **can't heel walk** (TO)	Lateral thigh → across knee → **medial leg/malleolus**	↓ **Patellar**
L5	**EHL** (great toe extension), **foot inversion (tib post)**, **hip abduction (glut medius → Trendelenburg)**	Anterolateral leg, **dorsum of foot, first web space**	–
S1	**Plantar flexion** (10 single-leg calf raises)	Posterior calf, **lateral foot**	↓ **Achilles**

Tibialis anterior L4 or L5 dominant (~50:50); L3-4 paracentral → **traversing L4** → weak tib anterior ± quads (OB)

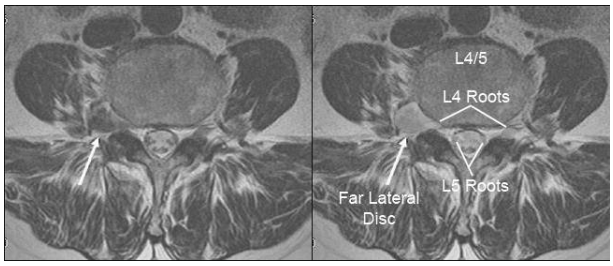
Paracentral vs far lateral

Location	Share	Root hit	Example
Central	–	Back pain; **CES**	Surgical emergency
Paracentral (posterolateral)	**90–95%**	**Traversing (lower)**	**L4-5 → L5** · L5-S1 → S1 · L3-4 → L4
Foraminal / far lateral	**5–10%**	**Exiting (upper)** + **DRG** → more severe pain	**L4-5 → L4** (TO) · **L5-S1 → L5** · L3-4 → L3
Axillary	–	Exiting + traversing	–



Paracentral disc → traversing root

– – Active space – –



Far lateral disc → exiting root

Trendelenburg gait	L5 → gluteus medius → **far lateral L5-S1** (TO) or **posterolateral L4-5** (TO answer "L4L5 herniated disc with positive Trendelenburg test")
L4-5 extraforaminal disc	L4 → **heel walk not possible** (TO) · pain worse standing + extension, eased by flexion
First dorsal web-space numbness	L5 → **L4-5 disc prolapse** (TO)
Weak great-toe dorsiflexion — causes	L5 root, peroneal palsy, sciatic lesion ✗ Wrong options: **Morton's neuroma** (TO)

L5 radiculopathy vs peroneal palsy

	L5 radiculopathy	Common peroneal palsy
Ankle inversion (tib post, L5 + tibial n.)	**Weak**	**Intact**
DF, toe extension, eversion	Weak	Weak
Hip abduction	Weak	Intact
Cough / Valsalva / SLR	Worse	No effect

4 Diagnosis & Classification

Provocative tests

Test	Fact
Straight leg raise	Reproduces leg pain at **30–70°** hip flexion · **L4, L5, S1** tension · **most predictive finding for a good surgical candidate** · false positive in hip disease (TO)
Laségue	Orthobullets: = SLR · **TO: two-step – SLR, then flex the knee → sciatic pain disappears (hip pain doesn't)** · "Laségue = SLR" is the **incorrect statement** (TO) – read the options
Bragard (reinforcement)	SLR to pain, lower slightly, **dorsiflex ankle** → pain returns · makes SLR **specific for nerve compression** (TO answer over crossed SLR)
Crossed (well-leg) SLR	**Less sensitive, more specific** · large **central** disc indenting thecal sac (TO)
Bowstring (McNab)	SLR to pain, flex knee, **press tibial nerve in popliteal fossa** → pain
Femoral stretch (Wassermann)	**Prone, knee flexed, hip extended** → anterior thigh pain · **L2, L3**
Kernig	Neck flexion + hip flexion + leg extension
Naffziger	Jugular compression → cough → pain
Milgram	Supine straight-leg elevation held 30 s
✗ Wrong options: apprehension · Murphy	Shoulder/patella instability · gallbladder (TO)

Key radiology findings

Study	Use / finding
X-ray AP + lateral	Usually normal · loss of lordosis (spasm), disc-height loss, sciatic scoliosis · counts transitional vertebrae (localisation) · poor sensitivity
Flexion-extension	**Instability → adds fusion**
MRI without gadolinium	**Modality of choice** · if pain **> 1 month** failing non-op or **red flags** · many asymptomatic findings → correlate · distinguishes **synovial facet cyst**
MRI with gadolinium	Revision: **scar enhances**, **recurrent disc doesn't**
CT myelogram	Can't have MRI (pacemaker) · 93% accurate · **misses foraminal/extraforaminal** discs

Red flags → MRI (TO)

Red flags — **unexplained fever**, IVDU, **age < 20**, **significant trauma**, cancer history, CES

✗ Wrong options: **lack of family support** (yellow flag) (TO)

– – Active space – –

Red flags	*Yellow flags*
• Age <20 years or >50 years • History of significant trauma • The pain is constant, at night and getting worse • History of cancer, steroid use, IV drug use • Unexplained fever and/or weight loss • Neurological deficit detectable	• A belief that back pain is severely disabling • Fear-avoidance behavior (avoiding activity due to anticipation of pain • Tendency to low mood • Expectation of help from passive treatment(s) • Poor job satisfaction • Lack of family support

Red and yellow flags of backache

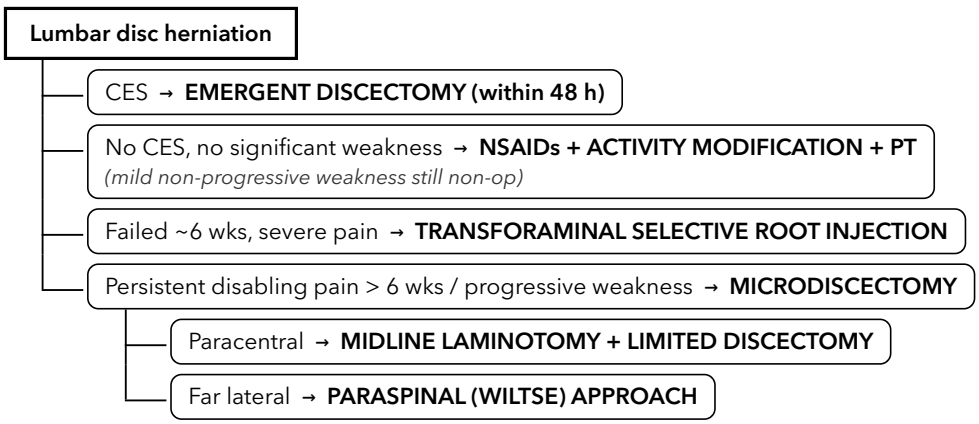
Classifications

Morphology	Definition
Protrusion	Eccentric bulge, **annulus intact**
Extrusion	Through annulus, **still continuous** with disc space
Sequestered (free)	Through annulus, **no continuity**; migrates up/down

Containment	Definition
Contained	Beneath PLL
Uncontained	Dorsal to PLL

Timing	Duration
Acute	**< 3–6 months** (surgical outcome depends on chronicity)
Chronic	> 6 months

5 Management



Natural history & non-operative care

Natural history	**90% improve** within 3 months without surgery · less likely if still symptomatic at 6 wks · TO stem: **70% resolve with NSAIDs + PT**
Resorption	herniation **shrinks over time** · **sequestered fragments resorb most** · by **macrophage phagocytosis** + enzymatic degradation
Non-op predictors	good: higher education, no radiculopathy · poor: **BMI > 30**, symptoms > 6 months before treatment

Medications	NSAIDs, muscle relaxants · oral steroid → modest function gain, no pain gain · narcotics 2–3 days only (worse surgical outcomes)
PT	from ~3 wks · **extension exercises** (✓ acute posterior-thigh radicular pain) · traction, careful manipulation
Early (2-day, 2-week) radicular pain, no deficit	**anti-inflammatories + activity modification** ✗ Wrong options: MRI · surgery · injection
Selective nerve root injection	2nd line after ~6 wks · **transforaminal best for far lateral** · durable relief **~50%** (vs ~90% surgery) · **best with extruded discs** · steroid adds nothing over lidocaine
Picking the injection level	target the **compressed root's foramen**: lateral calf → **L5 transforaminal** · anterior thigh with L3-4 foraminal disc → **L3-L4 transforaminal (L3 root)** · L5-S1 foramen block → **dorsal foot + great toe** ✗ Wrong options: lateral foot = S1 foramen · shin = L4
Facet joint injection	relief on **two occasions** → indication for **facet ablation**; medial branch block predicts better

Operative criteria (SPORT)

Indication	Detail
Persistent disabling radicular pain	**> 6 wks**, failed non-op ± injection · **refractory radicular pain = clearest indication** · better if done **within 2 months**
Progressive / significant weakness	–
Cauda equina	Emergent
No	Back pain alone · no instability → no fusion (✗ Wrong options: ALIF · PLIF)

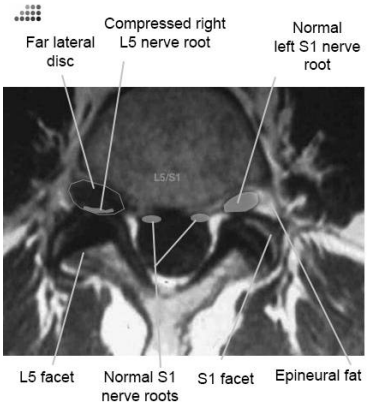
Outcome	Finding
Surgery vs non-op	**Better pain + function at 4 yrs** · **early relief sustained to 2 yrs** · equal disability at 5 yrs · long term equivalent
As-treated analysis	**Significant benefit of surgery** (pain, function, satisfaction)
Return to work	Same
Workers' compensation	**Less symptom relief, less QoL gain** at 5 yrs

Better surgical result	Worse surgical result
Leg pain dominant, **positive SLR**, weakness matching MRI	**Workers' comp**
Married, **age > 41**, **no joint problems**	Smoking
Progressively worsening symptoms	Chronic headaches, depression
Paracentral / foraminal, **caudal levels** (L5-S1 > L2-3)	Central / extraforaminal
Severe leg pain, short duration, younger, active	–

– – Active space – –

Surgical approach

Pathology	Operation
Paracentral	**Midline approach** (between spinous process and **multifidus**) → laminotomy + **limited discectomy** (e.g. L4 laminotomy, partial flavum resection, medial facetectomy)
Lateral (not far lateral) protrusion, e.g. L1-2	**Midline microdiscectomy** (OB)
Far lateral (e.g. L4-5 → L4, medial malleolus)	**Paraspinal muscle-splitting approach of Wiltse** (multifidus–longissimus) · outside-in, **preserves facet** · tubular/crank retractor
Recurrent paracentral	Revision via **midline**



Far lateral L5-S1 disc

Fragment excision vs subtotal discectomy — fragment excision: **less long-term back pain**, **more reherniation at 2 yrs** · limited beats complete discectomy

Technique — microscope = loupes · endoscopic → lower infection · all MIS techniques similar

Rehab — **early return to high-intensity exercise (~4 wks)** · Orthobullets 4–6 wks
✗ Wrong options: bed rest · avoid exercise 6–8 wks

Endoscopic anatomy (TO)

Kambin's triangle — answer not keyed by source · options: exiting root / superior endplate / traversing root vs alternatives — the commonly taught boundaries are **exiting root (hypotenuse), superior endplate of the lower vertebra (base), traversing root / dural sac (height)**

Recurrent herniation & postoperative course

Definition — recurrent sciatica at the operated level after a **6-month pain-free interval** (ipsi- or contralateral)

Rate — **5–15%** · **SPORT reoperation 15% at 8 yrs, with or without durotomy** · protective: discrete herniation, **annular defect < 6 mm**

Treatment non-op first → **revision discectomy** · **outcomes = primary** (equal leg pain + function) · fusion only if **instability**
✗ Wrong options: PLIF · ALIF · disc replacement · interspinous spacer

– – Active space – –

Post-op MRI	Diagnosis	Action
3 wks, recurrent leg pain + radicular signs, no enhancement of disc	**Recurrent herniation**	**Revision laminotomy + discectomy**
8 wks, back pain only, no neuro, **Modic** endplate change	Spondylosis	**Physical therapy**
Disc fluid + enhancement	Discitis	Biopsy + IV antibiotics

Chronic pain & central sensitisation

Pain diagram **screening tool** for pain sensitisation, psychosocial load, pain-resource use · correlates **positively** with anxiety/depression · not sensitive or specific for surgical benefit

Central sensitisation **amplified CNS neural signalling → pain hypersensitivity** · multiple surgeries, solid fusion, negative infection work-up, **PI-LL 9° (normal)**
✗ Wrong options: coronal deformity · PJK · adjacent segment

Far lateral disc + DRG DRG compression → **more severe pain** than paracentral
✗ Wrong options: CBT most cost-effective · standard laminotomy · stimulator · ALIF

Chronic discogenic back pain **CBT added to non-op care improves function** (= surgery) · disc replacement no clinically significant gain · fusion **~60% satisfaction** · discography **↑ herniation** at tested level · successful treatment **improves depression**

Return to play

Athletes **more games played before injury → better return** · **older age → worse** · RTP ~80%, only ~60% active at 2 yrs · **football: less likely to return but greatest benefit from surgery**

6 Complications

Complication	Fact
Dural tear	**0–4%** (Orthobullets) · **1%** (OB) · watertight repair · **no effect on long-term outcome**
Recurrent HNP	5–15%
Wound infection	Up to 3% · epidural abscess 0.3%
Discitis	**2.3%** → IV antibiotics ± I&D
Epidural fibrosis	Radicular symptoms; **3.2× recurrent radiculopathy** after revision
Chronic LBP	Central sensitisation; **Modic changes**
Vascular catastrophe	Through **anterior annulus** → aorta / IVC
Instability	Over-resection of lamina / **pars** → fusion

– – Active space – –

Post-op neuro deficit timing (TO)

Timing	Cause
Immediate	**Battered root** (retraction), screw / decompression injury
Delayed, day 3 after TLIF	**Haematoma** (✓) or gel foam → **immediate re-exploration + evacuation** · prevent with haemostasis + drain

Fusion approach complications

Approach / implant	At risk
ALIF transperitoneal	**Retrograde ejaculation** (4–10×) – **superior hypogastric plexus at L5-S1** → internal sphincter fails
ALIF retroperitoneal	**Ureter**
Lateral / XLIF / OLIF	**Lumbar plexus**
Long S1 pedicle screw	**L5 root** on the **anterior sacral ala**

After posterior lumbar fusion – motion most preserved — **axial rotation** (mostly thoracic) (TO)

X vs Y

	Paracentral	Far lateral
Frequency	90–95%	5–10%
Root	Traversing	Exiting + DRG
Pain	Typical	**More severe**
CT myelogram	Shows	**Misses**
Approach	Midline	**Wiltse paraspinal**
Injection	Interlaminar / transforaminal	**Transforaminal**

	Scar (epidural fibrosis)	Recurrent disc
Gadolinium MRI	**Enhances**	**Doesn't enhance**
Surgery	Poor results	Revision = primary

How it's asked

"Far lateral disc at L4-L5 compresses?"	→ **L4** (TO)
"Far lateral disc at L5-S1 causes?"	→ **Trendelenburg gait** (TO)
"Large extraforaminal L4-5 disc – finding?"	→ **heel walk not possible** (TO)
"Cannot cause weak great-toe dorsiflexion?"	→ **Morton's neuroma** (TO)

"Leg pain on walking, dorsal foot numbness, posterolateral disc?"	→ **L4-5 disc with positive Trendelenburg** (TO)
"Reduced sensation dorsal first web space?"	→ **L4-5 disc prolapse** (TO)
"Right Trendelenburg gait + leg pain?"	→ **ipsilateral far lateral L5-S1 disc** (TO)
"L4-5 posterolateral disc — exam finding?"	→ **weak foot inversion**
"L4-5 paracentral disc — all EXCEPT?"	→ **decreased patellar reflex**
"↓ Achilles reflex + positive SLR?"	→ **L5-S1 paracentral disc**
"Quad weakness, medial malleolus numbness, ↓ patellar reflex?"	→ **L4-5 far lateral disc**
"Can't do single-leg sit-to-stand, anterior leg pain?"	→ **L4-5 foraminal disc**
"Groin + anteromedial thigh pain — exam?"	→ **weak hip adduction**
"Anteromedial distal thigh pain near knee?"	→ **far lateral L3-4 disc** (L3)
"L3 root involvement — where's the pain?"	→ **near medial femoral epicondyle**
"Isolated EHL weakness + dorsal foot numbness?"	→ **L4-5 paracentral disc**
"L5-S1 MRI — weak muscle function?"	→ **hip abduction**
"14-yr tennis player, L5-S1 disc — finding?"	→ **weak plantarflexion, lateral foot numbness**
"Peroneal palsy vs L5 radiculopathy?"	→ **intact ankle inversion** in peroneal palsy
"Size of disc herniation over time?"	→ **decreases without surgery** (sequestered most)
"Severe lateral calf pain 2 wks?"	→ **L5 transforaminal selective root injection**
"L5 root block helps which pain?"	→ **dorsal foot into the great toe**
"Foraminal L3-4 disc, 6 wks anterior thigh pain?"	→ **L3-L4 transforaminal injection**
"Facet injection true statement?"	→ **relief on two occasions → ablation indicated**
"SPORT: surgery vs non-op long term?"	→ **better pain + function at 4 yrs**
"Workers' comp after discectomy?"	→ **less relief + less QoL improvement**
"Predictors of surgical benefit?"	→ **age > 41, no joint problems, married**
"Clearest surgical indication at 6 wks?"	→ **refractory radicular pain**
"Far lateral disc — approach?"	→ **paraspinal (Wiltse) muscle-splitting approach**
"Paracentral disc — approach?"	→ **midline between spinous process and multifidus**
"Revision vs primary discectomy outcomes?"	→ **equal pain + function**

– – Active space – –

"Post-discectomy activity?"	→ **early return to high-intensity exercise**
"Reoperation rate after microdiscectomy?"	→ **15% at 8 yrs, with or without durotomy**
"Post-op 8 wks back pain, Modic changes?"	→ **physical therapy**
"Pain diagram use?"	→ **screen for sensitisation + psychosocial load**
"Chronic pain after solid fusion, normal imaging?"	→ **central sensitisation**
"Far lateral disc – why so painful?"	→ **DRG compression**
"Discogenic back pain – true?"	→ **CBT improves function**
"Athlete return-to-play predictor?"	→ **more games played before injury**
"Retrograde ejaculation after L5-S1 fusion?"	→ **transperitoneal ALIF**
"Long S1 screw endangers?"	→ **L5 root**
"Clinical test for sciatic nerve compression?"	→ **Bragard ("Braggard") reinforcement** (TO)
"Incorrect statement about radicular leg pain?"	→ **"Laségue = SLR"** (TO)
"Motion most preserved after posterior lumbar fusion?"	→ **axial rotation** (TO)
"Day-3 leg weakness after TLIF?"	→ **haematoma** (TO)
"Red flags all EXCEPT?"	→ **lack of family support** (TO)

LUMBAR SPINAL STENOSIS

– Spine · 36 / 69 –

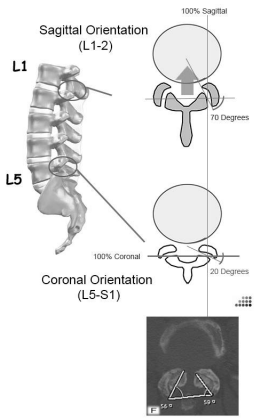
1 Basics

Epidemiology

Incidence	**commonest reason for lumbar surgery > 65 yrs** · seen in 20–25% · M:F 1.5:1 · average age 65
Level	**L4-5 (91%)**
Risk factors	Caucasian, ↑ BMI, congenital anomalies (20%: short pedicles + laminae)
Strongest factor for symptomatic degenerative disc disease	**genetic predisposition** (TO) ✗ Wrong options: heavy lifting · vibration · tobacco

Pathoanatomy

Cascade	nucleus loses cells, water, proteoglycan → disc height ↓, annulus bulges → **anterior column absorbs less stress → force transfers to posterior elements** → facet hypertrophy, osteophytes, **ligamentum flavum buckling / hypertrophy**
Compressing structures	bony: **facet osteophytes**, uncinate (posterior body) spur, spondylolisthesis · soft: bulging disc, **ligamentum flavum**, synovial facet cyst
Commonest compression source	**superior articular process** (facet hypertrophy in the lateral recess) (TO) ✗ Wrong options: disc · lamina · inferior articular process
Posture (TO)	canal space **maximal in flexion** (and axial distraction) · ↓ with **extension + axial loading** → side-lying (flexed) better than flat · uphill easier · **shopping-cart sign**
Facets	become **more coronal** moving inferiorly



Facet orientation down the lumbar spine

– – Active space – –

Ligamentum flavum

Origin	**ventral surface of the cranial lamina, 60–70% of the way up (closer to the superior edge)** · inserts on the **superior edge of the caudal lamina** (OB) · TO: origin **lower half of anterior surface of lamina above**, insertion **posterior surface + upper margin of lamina below** – read the options
Composition	**80% elastin, 20% type I collagen** (✗ Wrong options: type 2) · **posterior column** (✗ Wrong options: middle column) · not the primary flexion restraint (= **supraspinous ligament**)

3 Presentation

Neurogenic claudication	back, **buttock**, thigh pain/heaviness · **worse with extension** (walking, standing, **downhill/down stairs**, lying supine) · **relieved by flexion** (sitting, shopping cart, cycling, fetal position)
Neuro exam	often **normal** (examined seated) · SLR usually negative · no UMN signs (clonus, Babinski = cord) · no step-wise progression (= cervical myelopathy) · Valsalva doesn't worsen (unlike HNP)
Kemp sign	unilateral radicular pain from **foraminal** stenosis worse with **back extension**
Bladder	recurrent UTIs up to 10% (autonomic sphincter dysfunction) · CES rare
L2-3 / L3-4 stenosis	pain in **anterior thigh + anteromedial leg** ✗ Wrong options: plantar/lateral foot = S1 · anterolateral leg/dorsum = L5 · inguinal = L1

Neurogenic vs vascular claudication

	Neurogenic	Vascular
Postural change	**Yes**	No
Walking upright	Symptoms	Symptoms
Standing still	**Symptoms persist**	**Relieves**
Sitting	Relieves	Relieves
Stairs	**Up easier** (flexed); down worse	Down easier; up worse
Stationary bike (flexed)	**Relieves**	**Causes**
Leaning backwards	**Worse** (✓ favours neurogenic)	No effect
Pulses	Normal	**Abnormal / asymmetric**
Other	–	**Night pain relieved by hanging leg**, trophic changes, calcified iliacs

Suspected vascular → ankle-brachial index · ABI 0.4–0.8 → claudication
✗ Wrong options: SNRB · EMG/NCS · hip X-ray · CT myelogram

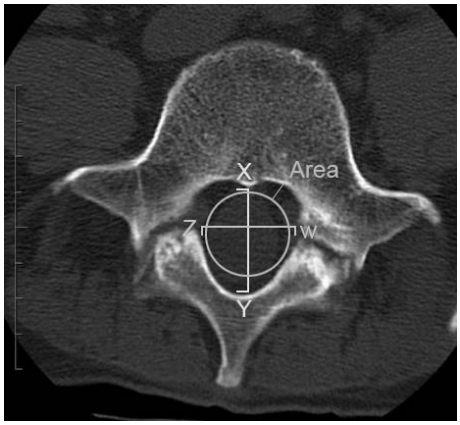
Hip-spine syndrome	hip OA + stenosis → **diagnostic lidocaine + steroid injection** (hip or epidural) to find the primary generator · severe stenosis → decompress anyway (risk of motor weakness) ✗ Wrong options: THA · decompression first · EMG

Key radiology findings

Study	Finding
Standing AP + lateral	Degenerative changes, **degenerative spondylolisthesis**, scoliosis · poor clinical correlation
Flexion-extension	**Segmental instability** → decides fusion
MRI	**Modality of choice** · thecal sac **< 100 mm²**, lost perineural fat, facet + flavum hypertrophy · positive in asymptomatic people
CT myelogram	Can't have MRI; bony detail · superior to MRI for decompression planning (OB) · can't separate vascular from neurogenic
Plain myelogram	Dynamic cut-off in extension (invasive)

Diagnostic criteria – critical central stenosis (axial CT)

Measure	Threshold
AP canal diameter	**< 10 mm**
Cross-sectional area	**< 100 mm²**
No	Transverse < 20 mm, combined < 30 mm, area 130 mm², AP < 20/50 mm



AP diameter X–Y on axial CT

Anatomic classification

Zone	Cause	Root (at L4-5)
Central	Flavum hypertrophy + bulging disc → thecal sac	**Lower / non-specific (L5)**
Lateral recess (subarticular)	Facet arthropathy, **superior articular facet overgrowth**	**Descending / traversing (L5)**
Foraminal (between pedicle borders)	Disc-height loss, foraminal disc/osteophyte, scoliosis · **superior facet overhang**	**Exiting (L4)**
Extraforaminal (lateral to pedicle)	**Far lateral disc**	**Exiting (L4)**

– – Active space – –

Etiologic classification

Type	Examples
Acquired	**Degenerative (commonest)**, post-surgical, post-traumatic, AS, Paget, acromegaly, fluorosis
Congenital	Short pedicles, medial facets · idiopathic or developmental (**achondroplasia**)

5 Management

Lumbar spinal stenosis

- First line → **NSAIDs, PT, WEIGHT LOSS, BRACE, EPIDURAL STEROID**
- Failed 3–6 months / progressive deficit → **WIDE DECOMPRESSION**
  - Stable (normal flexion-extension) → **DECOMPRESSION ALONE**
  - Instability / spondylolisthesis / > 50% facets removed → **DECOMPRESSION + INSTRUMENTED FUSION**

Natural history & SPORT

Non-op prognosis	**most get some relief within 3 months** · > 50% non-op satisfactory at 10 yrs · **delay of surgery doesn't worsen result** · complete myelographic block responds poorly · catastrophic neuro progression unlikely
SPORT (as-treated)	surgery → **better pain + function (+ disability, satisfaction) at 2 and 4 yrs** · same 4-yr mortality · benefit persists at 8 yrs (some convergence)
Reoperation	**~13% at 4 yrs**, ~half restenosis at same level ✗ Wrong options: 25%
Commonest surgical complication (SPORT)	**dural tear** ✗ Wrong options: infection
ESI	may avoid surgery (Orthobullets) · efficacy vs placebo questioned (OB) · temporary relief only → surgery

Operative criteria

Procedure	Indication
Decompression alone	Persistent pain **3–6 months** failing non-op · **progressive deficit** (weakness, bowel/bladder) · **no instability**
Decompression + instrumented fusion	**Segmental instability** (isthmic / degenerative spondylolisthesis, degenerative scoliosis) · **surgical instability** (complete laminectomy + **> 50% facets**)
✗ Wrong option: fusion	Stable spine (even stable grade I listhesis elsewhere) — fusion adds nothing (Försth: same 2- and 5-yr outcomes, more time, blood, cost)
✗ Wrong option: microdiscectomy	Flavum / facet stenosis (any approach)
✗ Wrong option: fusion alone	Doesn't decompress
✗ Wrong options: facet injection · manual therapy	Don't treat neurologic symptoms

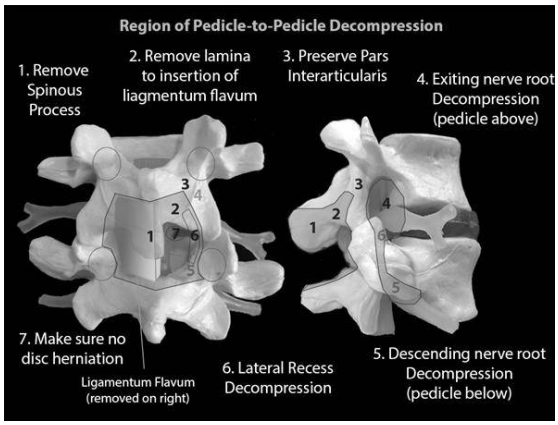
Iatrogenic instability (TO)

Destabilising (✓ needs fusion)	Not destabilising
Bilateral inferior facet (inferior articular process) resection	Near-total discectomy (> 50% nucleus)
> 50% facet removal	**Laminotomy + spinous process removal** (+ interspinous ligament)
–	**Medial facetectomy ≤ 1/4 (20%)**
–	Unilateral hemilaminectomy

Rule of thumb (TO) unilateral decompression: up to **half** of a facet · bilateral: **one third** each side

Surgical technique

Wide pedicle-to-pedicle decompression (L4-5) inferior half L4 spinous process · L4 lamina to flavum insertion · flavum · **medial facetectomy + lateral recess undercutting** · explore L4-5 + L5-S1 foramina, palpate L4 + L5 pedicles



Pedicle-to-pedicle decompression

Level matching single-level → **single-level laminectomy** · multilevel → **multilevel laminectomy** · stenosis at L3-4 + stable L4-5 listhesis → **L3-4 laminectomy alone**
✗ Wrong options: L4-5 laminotomy only

Midline-sparing **unilateral laminectomy undercutting the spinous process** preserves spinous process + inter/supraspinous ligaments

Stenosis + spondylolisthesis / instability **bilateral laminectomies + single-level (L4-5) instrumented fusion**
✗ Wrong options: L4-S1

Fusion instrumentation controversial · PLIF/TLIF accepted, not superior · **adjacent segment degeneration > 30% at 10 yrs**

Outcomes commonest cause of failure **recurrence above/below** · **comorbidities = strongest predictor of outcome**

Prognostic factors

Strongest predictor (stenosis surgery) **comorbid medical conditions**
✗ Wrong options: smoking · multilevel · income

– – Active space – –

Degenerative disc disease surgery	**workers' compensation** most predictive ✗ Wrong options: duration · severity · levels · arthroplasty
Chronic preop opioids	**prolonged hospital stay**, ↑ costs, harder analgesia, ↓ PROs ✗ Wrong options: more SSI

Fusion assessment & bone health

Most reliable fusion assessment	**CT** (X-ray only 60–80% accurate)
Osteoporosis + fusion	**teriparatide** improves fusion rates ✗ Wrong options: denosumab · alendronate · clodronate · calcitonin

Related concepts

Adult deformity HRQL	**sagittal balance** = best predictor ✗ Wrong options: coronal · curve correction · shoulder
SLR	pain below knee at **30–70°**
Far lateral disc (L3-4 → L3, quad weakness)	**lateral paraspinal (Wiltse) approach + partial discectomy** ✗ Wrong options: midline · fusion

6 Complications

Complication	Rate / fact
Wound infection	**10%** (fusion) · discectomy 0.7% → **1.4% with microscope** · deep → **I&D + cultures + antibiotics**
Pneumonia / renal failure	5% each
Persistent neuro deficit	4%
Transient neuro deficit	**36%**
UTI / transfusion / confusion	34% / 27% / 27%
Risk ↑ with	Age, blood loss, levels fused

Persistent leg pain after decompression

Timing	Cause
No improvement from day 1	**Residual foraminal stenosis** (inadequate decompression)
Recurrence after pain-free interval	Recurrent herniation, instability, restenosis
Slow axial pain	Infection
Immediate saddle anaesthesia	CES (rare)
Decompression alone in degenerative listhesis	Instability

Dural tear

Repaired watertight	**outcome not affected** · drains safe · commoner in revision · rate 2–17%

Persistent CSF leak positional headache + nausea weeks later, normal WBC/ESR → **MRI** → **reoperation = gold standard** (fibrin glue, subarachnoid drain, blood patch less invasive) · lumbar drain only after failed repairs

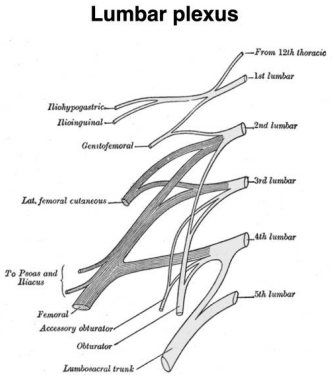
Surgical site infection

Risk factors **HbA1c > 7.5%** (✓), op time **> 3 h**, posterior approach, **BMI > 35** (> 30 obese), age > 70, active smoking, albumin low

Morphine paste on dura **↑ SSI** (epidemic) · ✓ protective: **intrawound vancomycin**, **clipping**, **alcohol prep** · chlorhexidine wash not proven

Lateral transpsoas fusion

Structure	Injury result
Genitofemoral nerve (L1-2) – on psoas, mid-axis **L3-4**; at risk at **L4-5**	**Groin / medial thigh numbness + burning** ("liquid trickling") – **commonest lateral-approach nerve injury**
Sympathetic chain (anterolateral spine)	**Ipsilateral leg anhidrosis**, warmth (post-sympathectomy)
Femoral / obturator	Prolonged retraction at L4-5
Ilioinguinal	Base of penis
Iliohypogastric	Suprapubic skin
Superior hypogastric plexus	Retrograde ejaculation



Lumbar plexus on psoas

X vs Y

	Stenosis	Disc herniation
Age	65	40s
Worse	**Extension**, walking, standing	**Sitting**, flexion, cough
SLR	Negative	Positive
Treatment	Laminectomy	Microdiscectomy

– – Active space – –

How it's asked

"Buttock pain, MRI stenosis — expected history?"	→ improvement on a stationary bicycle
"Pain climbing stairs, relieved standing still, MI history?"	→ vascular claudication
"Stenosis — also likely to report?"	→ more pain walking down stairs
"Pain relieved by standing, hangs leg at night, weak pulses — next?"	→ ankle-brachial index
"Calf pain not helped by flexion — test?"	→ ABI
"Favours neurogenic claudication?"	→ worse leaning backwards
"Smoker, neurogenic not vascular?"	→ ascending stairs easier than descending
"Neurogenic claudication — exam?"	→ no neurologic deficit
"Critical stenosis on CT?"	→ AP diameter < 10 mm / area < 100 mm^2
"Ligamentum flavum — true?"	→ originates on ventral L4 lamina closer to superior edge
"Surgery vs non-op at 4 yrs?"	→ better pain + function (+ satisfaction)
"Laminectomy alone at L4-5 leads to?"	→ better pain, function, disability at 2 and 4 yrs
"Non-op prognosis?"	→ most get some relief in 3 months
"Failed non-op, counselling — true?"	→ decompression beats non-op care
"Iatrogenic instability after stenosis surgery?"	→ bilateral inferior facet removal (TO)
"Decompression needing L4-5 fusion?"	→ bilateral L4 inferior articular process resection
"Stenosis, stable flexion-extension, failed PT + ESI?"	→ laminectomy + bilateral lateral recess decompression + foraminotomies
"L4 spinous process + 80% lamina + 20% medial facetectomy, no fusion — for?"	→ 65-yr single-level stenosis
"Bilateral leg pain, stable, failed PT?"	→ posterior lumbar decompression L4-5
"Multilevel stenosis, short ESI relief?"	→ multilevel laminectomy
"Stable L4-5 stenosis — option preserving posterior ligaments?"	→ unilateral approach laminectomy
"Stenosis + instability on standing film?"	→ bilateral laminectomies + L4-L5 instrumented fusion
"L3-4 stenosis + stable grade I L4-5 listhesis?"	→ L3-4 decompressive laminectomy
"L2-3 + L3-4 decompression — dermatome?"	→ anterior thigh + anteromedial leg

"Cyclist, pain walking, stable films?"	→ **laminectomy + medial facetectomies + foraminotomies**
"Strongest prognostic factor for stenosis surgery?"	→ **comorbid medical conditions**
"DDD surgery – most predictive factor?"	→ **workers' compensation**
"Chronic preop opioids?"	→ **prolonged hospital stay**
"No relief of leg pain after laminectomy?"	→ **residual foraminal stenosis**
"Repaired dural tear – effect?"	→ **clinical outcome not affected**
"Positional headache 1 month after durotomy?"	→ **CSF leak**
"Wound dehiscence + epidural abscess post-discectomy?"	→ **I&D, cultures, antibiotics**
"Increases SSI after decompression?"	→ **morphine paste on dura**
"Post-fusion wound infection – risk factor?"	→ **HbA1c 7.8**
"Structure A on cadaver (ALIF/transpsoas) injured?"	→ **ipsilateral leg anhidrosis** (sympathetic chain)
"Groin + inner thigh burning after XLIF L4-5?"	→ **genitofemoral nerve**
"Transpsoas L3-4 nerve C injury?"	→ **medial thigh numbness + burning**
"Hip OA + L2-3 stenosis – next?"	→ **diagnostic hip injection**
"Most reliable assessment of fusion?"	→ **CT**
"Drug improving fusion in osteoporosis?"	→ **teriparatide**
"Adult deformity revision – key HRQL goal?"	→ **sagittal balance**
"SLR tension range?"	→ **30–70°**
"Far lateral L3-4 disc, quad weakness, failed non-op?"	→ **lateral paraspinal approach + partial discectomy**
"Canal space maximum in?"	→ **flexion** (TO)
"Commonest compression source in degenerative stenosis?"	→ **superior articular process** (TO)
"Strongest factor for symptomatic DDD?"	→ **genetic predisposition** (TO)

– – Active space – –

LUMBAR SPINE ANATOMY

– Spine · 37 / 69 –

1 Basics

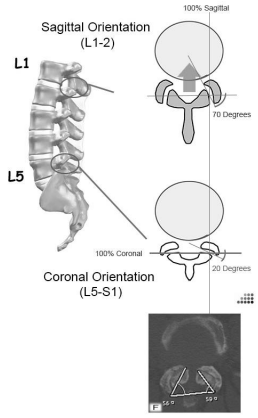
Alignment

Lumbar lordosis — average **60°** (range 20–80°) · apex **L3** · mostly from the **disc spaces**

Osteology

Vertebral bodies — **largest** in the axial spine

Part	Fact
Pedicles	Project posteriorly from posterolateral body corners
Laminae	Posteromedial from pedicles, join midline
Mammillary processes	**Separate ossification centres**; project posteriorly from the **superior articular facet** · pedicle screw landmark
Pars interarticularis	Bone between superior + inferior facets · **site of spondylolysis**
Facet (zygapophyseal) joint	From pedicle–lamina junction · **more coronal moving distally**



Facet orientation becomes coronal distally

Ligamentum flavum

Attachments (TO) — origin **lower half of the anterior surface of the lamina above** · insertion **posterior surface + upper margin of the lamina below** · OB: origin ventral cranial lamina 60–70% of the way up (closer to superior edge), insertion superior edge of caudal lamina – read the options

MRI — **low signal on T2** · thickened flavum → stenosis
✗ Wrong options: synovial cyst and epidural CSF are high signal

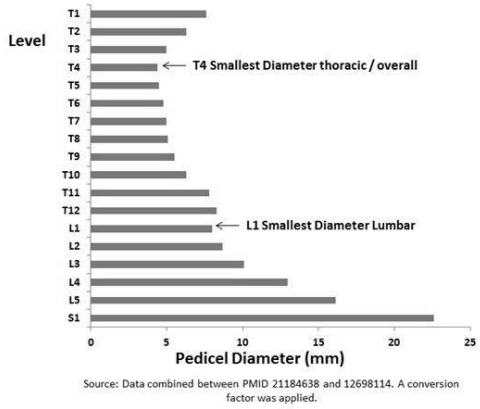
Pedicle anatomy

Landmarks — superior-inferior midpoint = **midpoint of transverse process** · medial-lateral midpoint = **lateral border of pars**

Pedicle screw entry (TO) — **mammillary process, lateral to pars** · lumbar: intersection of line through lateral facet and line bisecting transverse process, ~1 mm below the facet joint · laterality: **mammillary > intersection > pars** start points

Danger (TO) — lumbar root lies in the **upper third** of the foramen, just **inferior to the medial pedicle wall → medial or inferior breach more dangerous** than lateral/superior

Measure	Fact
Medial angulation	Increases distally: **L1 12° → L5 30° → S1 39°** · largest horizontal angle at L5 (TO)
Smallest lumbar diameter	**L1** (OB, Orthobullets)
Smallest overall	**T4** (Orthobullets) · **T5** horizontal (TO) – read the options
Widest	**L5** horizontal; **T11** sagittal · S1 ~19 mm
Sagittal angle	Caudal at L5, cephalad L3–T1 (TO)
Thoracic pedicle	62–79% cancellous · **lateral wall thinner than medial**



Pedicle diameter by level

Cortical : cancellous ratio

Highest → lowest
Thoracic pedicles
Lumbar pedicles
Vertebral bodies
Sacrum (lowest)

Cervical pedicles (TO)

Size — **C3 smallest**, C2 + C7 largest · C3–C5 need < 4.5 mm screws

Angulation — medial at all levels: **most C5**, least C2 + C7 · upward at C2-3, parallel C4-5, downward C6-7

Vertebral artery — at risk **C3–C6** (lateral cortex thinnest; medial cortex ~2× thicker) · less at risk C2 + C7-T1 (outside foramen at C7)

– – Active space – –

Neural structures

Root rule	exits under the **same-numbered pedicle** · central/paracentral → traversing root · far lateral → exiting root
Cervical vs lumbar	cervical C6 under C5 pedicle (mismatch, extra C8 root) · lumbar L5 under L5 (match) · vertical lumbar roots → paracentral + foraminal discs hit **different** roots
Cauda equina	begins ~**L1**

Pain innervation

Disc	**sinuvertebral nerve → outer annulus only** (nociception + proprioception) · discogenic pain: axial, worse sitting/bending, no radicular signs ✗ Wrong options: nucleus · DRG direct
Dura	intradural communicating nerve (central herniation pain)
Dorsal ramus branches	**medial → facet capsule + multifidus** · intermediate → longissimus, spinalis · lateral → iliocostalis + skin
Facet dual supply	each facet from the **medial branch of the level above + same level** · **L4-L5 facet block = L3 + L4 medial branches** ✗ Wrong options: ventral rami · lateral/intermediate branches
Ventral rami	anteromedial trunk

Blood supply

Segmental arteries	dorsal branches → dura + posterior elements · tie off for corpectomy

4 Diagnosis & Classification

Lumbopelvic parameters

Parameter	Definition	Formula
Pelvic incidence (PI)	Line S1 endplate centre → femoral head centre vs **perpendicular to S1 endplate**	**PI = PT + SS** · **fixed, independent of pelvic position**
Pelvic tilt (PT)	Same line vs **vertical**	PT = PI − SS
Sacral slope (SS)	S1 endplate vs **horizontal**	SS = PI − PT

PI	correlates with spondylolisthesis severity (**Meyerding grade**)

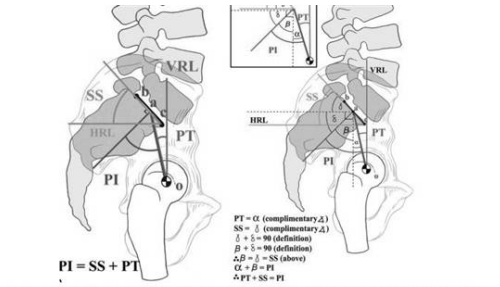


Figure 4. A, the association between PI, SS, and PT. PI, which describes the shape of the pelvic unit, is linked to the orientation (SS and PT) of the pelvis in space since it can be demonstrated that PI = SS + PT. Depending on the shape and orientation of the sacropelvic foundation, the lumbosacral junction will experience a different combination of normal and shear forces. **B,** the "geometric proof" demonstrates the mathematical association of PI, PT, and SS, where a change in one parameter affects the other measurements and the overall alignment of the sacropelvic foundation.

PI = PT + SS

Image-guided interventions

Procedure	Use
Selective nerve root injection	Unilateral radicular symptoms · diagnostic + therapeutic · **transforaminal (outside-in)**
Facet injection	Confirms facet as pain generator · may be therapeutic
Epidural injection	**Lumbar spinal stenosis**
Discography	Controversial · **concordant** vs discordant pain · shows annular tears

Needle 22–25 G, CT or fluoroscopy

5 Management

Surgical approaches

Approach	Facts
Posterior midline	PLIF / TLIF
Wiltse paraspinal	Far lateral pathology
Retroperitoneal (anterolateral)	**Aortic bifurcation L4-5 · superior hypogastric plexus on L5 body → retrograde ejaculation**
Lateral transpsoas	**Left side** (aorta tolerates retraction) · ideal **L1-2, L2-3, L3-4** · **L4-5** = highest plexus risk (hip flexion + knee extension weakness) · T12-L1 needs rib + diaphragm takedown

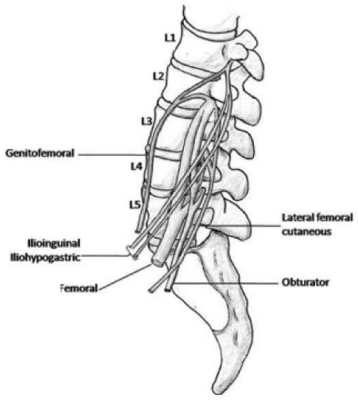
– – Active space – –

Transpsoas safe zone

Cranial → caudal — **safe zone shrinks because the lumbar plexus moves ventral** · triggered EMG helps
✗ Wrong options: vessels move; great vessels roughly constant

Nerve injured	Result
Ilioinguinal + iliohypogastric	**Delayed (~2 wks) abdominal wall paresis** — bulge mimics hernia (CT excludes) · groin paraesthesia · resolves with observation
Genitofemoral	Genital numbness, **lost cremasteric reflex**
Femoral	Knee extension weakness, anterior thigh numbness
LFCN	Lateral thigh (aggressive retroperitoneal fat dissection)
Obturator	Medial thigh numbness, **adduction weakness**

Risk factors — multilevel, long retraction, excessive bed break, hips not flexed



Nerves at risk in transpsoas approach

Retroperitoneal relations (TO)

Anterior to L3 body — answer not keyed by source (options aorta, ureter, right iliac artery, IVC) · Orthobullets: aorta bifurcates at **L4-5**, so the iliac arteries lie below L3

X vs Y

	Cervical root	Lumbar root
Course	Horizontal	Vertical
Pedicle	C6 under C5 (mismatch)	L5 under L5 (match)
Central vs foraminal disc	**Same root**	**Different roots**

How it's asked

"Entry point for pedicle screw?"	→ **mammillary process, lateral to pars** (TO)
"T12–L4 fusion — smallest screw diameter?"	→ **L1**
"Highest cortical : cancellous ratio?"	→ **thoracic pedicles**

"Pelvic parameter independent of pelvic position?"	→ **PT + SS (= pelvic incidence)**
"Transpsoas safe zone cranial → caudal?"	→ **decreases — lumbar plexus moves ventral**
"Abdominal bulge 2 wks after L3-5 lateral fusion?"	→ **ilioinguinal / iliohypogastric injury**
"Ligamentum flavum origin and insertion?"	→ **lower half anterior surface of lamina above → posterior surface + upper margin of lamina below** (TO)
"Arrow on T2 axial — thick low-signal band?"	→ **ligamentum flavum**
"Discogenic pain pathway?"	→ **sinuvertebral nerve to the annulus fibrosus**
"Block for L4-L5 facet RFA?"	→ **medial branches of L3 + L4 dorsal rami**

– – Active space – –

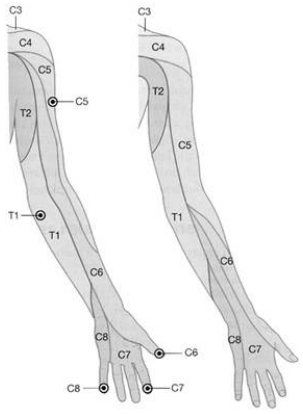
NECK & UPPER EXTREMITY SPINE EXAM

– Spine · 38 / 69 –

1 Basics

Root map

Root	Motion	Muscle	Sensory	Reflex
C4	Scapular stabilisation (winging)	Upper serratus (variable)	Upper shoulder, over clavicle	–
C5	**Shoulder abduction**, elbow flexion (palm up)	**Deltoid**, **biceps**	**Lateral arm** below deltoid	**Biceps**
C6	Elbow flexion (thumb up), **wrist extension**	**Brachioradialis**, **ECRL**	**Thumb + radial hand/forearm**	**Brachioradialis**
C7	**Elbow extension**, **wrist flexion**	**Triceps**, **FCR**	**Fingers 2, 3, 4**	**Triceps**
C8	Finger flexion, **grip**, thumb extension	**FDS**	Finger 5	–
T1	Finger abduction	Interossei	Medial elbow	–



Upper limb dermatomes

Motor testing

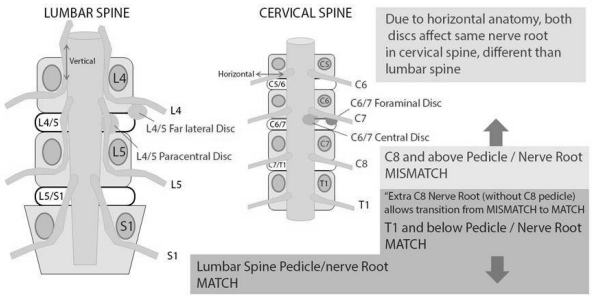
Motion	Muscle	Nerve	Root
Scapular stabilisation	Serratus	Long thoracic	C4
Shoulder abduction	Deltoid	Axillary	C5
Shoulder IR	Subscapularis	Subscapular	C5
Shoulder ER	Infraspinatus	Suprascapular	C5
Elbow flexion (palm up)	Biceps, brachialis	Musculocutaneous	C5
Elbow flexion (thumb up)	Brachioradialis	Radial	C6
Wrist extension	ECRL	Radial	C6
Supination	Supinator	PIN	C6
Elbow extension	Triceps	Radial	C7
Wrist flexion	FCR, PL	Median	C7
Pronation	PT, PQ	Median	C7
MCP + PIP flexion	FDS	Median	C8
DIP flexion	FDP	Ulnar + AIN	C8
Thumb extension	EPL	PIN	C8
Finger abduction	Interossei	Ulnar	T1

Cervical root anatomy

Mismatch — **C7 root exits below C6 pedicle** (C6 under C5) · extra C8 root (no C8 pedicle)

Horizontal root — **central and foraminal disc at same level hit the same root** (C6-7 disc → **C7**)

Commonest degenerative levels — **C4-5 (C5)** and **C5-6 (C6)**



Cervical vs lumbar root–pedicle relation

3 Presentation

Inspection — sagittal/coronal alignment (kyphosis), scars (ulnar transposition, carpal tunnel), **café-au-lait** (NF), atrophy (deltoid + biceps in palsy)

Normal cervical ROM — flexion **50°** · extension **60°** · rotation **80°** · lateral bend **45°**

Motor / sensory grading — ASIA 0–5 for one muscle per root C5–T1 · sensory: always **pain (pin) + light touch**; vibration, temperature, 2-point for focused exam

– – Active space – –

Gait

Gait	Cause
Antalgic	Hip/knee pathology, severe radicular pain
Trendelenburg	Painful hip arthritis, **gluteus medius weakness**
Wide-based shuffling	**Myelopathy** / neurologic
Steppage / lateral swing	**Foot drop**

4 Diagnosis & Classification

Provocative tests

Test	Technique	Accuracy
Spurling	Rotate toward affected side + extend + **axial load**, 30 s → ipsilateral arm pain	**Specific, not sensitive** – acute **radiculopathy**
Hoffmann	Flick distal phalanx of long finger into extension → **thumb IP flexes**	**Sensitive, not specific** – **myelopathy**
Lhermitte	Neck flexion/extension → electric shock down spine/ limbs	**Specific, not sensitive** – cord compression

X vs Y

Root	Distinguishing finding
C5	Weak shoulder abduction + ER
C6	**Absent brachioradialis reflex**, **thumb numbness**
C7	**Weak wrist flexion + elbow extension**, middle-finger numbness, ↓ triceps
C8	**↓ Grip**

How it's asked

"Left foraminal C6-7 disc – exam?"

→ **weak wrist flexion** (C7)
✗ Wrong options: absent brachioradialis = C6 · thumb numbness = C6 · shoulder abduction/ ER = C5 · grip = C8

NEUROMUSCULAR SCOLIOSIS

– Spine · 39 / 69 –

1 Basics

Definition spinal curvature from disorders of **brain, spinal cord or muscle**

Neuromuscular vs idiopathic curves

Neuromuscular curves	
Progression	**More rapid**; **may progress after maturity**
Pelvis	**Pelvic obliquity**
Length	**Longer**, more vertebrae (may reach cervical) – long C-shaped
Surgery	**↑ Pulmonary complications**



Long C-shaped neuromuscular curve with pelvic obliquity

Paralytic scoliosis after childhood SCI (TO)

Cervical complete SCI before age 10 paralytic scoliosis in **virtually 100%** (TO answer: 6-yr with C4-5 fracture-dislocation → **100%**) · risk depends on level + age at injury

5 Management

Outcome **spinal fusion is the only treatment with proven benefit on deformity** · carers report excellent QoL improvement

– – Active space – –

Classification & treatment

Category	Disease	Non-op	Operative
Upper motor neuron	**Cerebral palsy**	Boston underarm brace until puberty (10–12), wheelchair modification	Group I → **PSF** · Group II → PSF ± ASF + **fusion to pelvis (Luque-Galveston)** · indications: **curve > 50°**, worsening **pelvic obliquity with sitting imbalance**
	Rett syndrome	Brace C-shaped curves	PSF – thoracic curve affecting sitting/ balance, C-curves failing brace
Muscle weakness	**Spinal muscular atrophy**	Boston brace until puberty	**PSF to pelvis** (wheelchair sitting) · release **hip contractures first** · may transiently lose UE function
	Muscular dystrophy	**Bracing contraindicated**	PSF ± pelvis (controversial) · **curve > 20°** or progressive · operate early (**20–30°**) **before pulmonary function declines**
Paralytic	**Spina bifida, SCI**	**Bracing contraindicated**	**ASF + PSF** with pelvic fixation (**anterior fusion needed – deficient posterior elements**)
	Polio	Boston brace until puberty	PSF

6 Complications

Wound complication risk factors

Risk factor	Not a risk factor
Serum albumin < 3.5 g/dL	**Hemiplegia**
WBC / lymphocytes < 1,500 cells/µL	Number of levels fused
VP shunt	Blood loss, transfusion
Severe spastic quadriplegia, non-ambulatory, seizures	Preoperative curve magnitude
–	Age, length of surgery

Infection rate **4–14%** (> idiopathic)

How it's asked

"Wound complication risk factors in CP – EXCEPT?"	→ **hemiplegia**
"Known risk factor for wound infection after NM fusion?"	→ **albumin < 3.5 g/dL** ✗ Wrong options: levels · blood loss · curve size
"Paralytic scoliosis after complete C4-5 SCI at 6 yrs?"	→ **100%** (TO)

OCCIPITOCERVICAL INSTABILITY

– Spine · 40 / 69 –

1 Basics

Epidemiology

Traumatic	15–30% of cervical injuries at the occipitocervical junction · in **19% of fatal** cervical injuries
Acquired	commonest in **Down syndrome** · usually asymptomatic, found on screening (surgery, Special Olympics)
Synonyms	atlanto-occipital dissociation (AOD), occipitocervical dislocation

Pathophysiology

Traumatic	high-energy **translation or distraction** · head usually displaces **anteriorly**
Acquired	bony dysplasia, ligament laxity · RA, Down syndrome
Associations	atlantoaxial instability, neuro deficit, **vertebral / carotid artery injury**, Down syndrome

Anatomy

Occipital condyles	lateral foramen magnum · near vertebral arteries, medulla, **CN IX–XII**
Joints (6)	anterior + posterior median atlanto-odontoid, paired atlanto-occipital, paired atlantoaxial

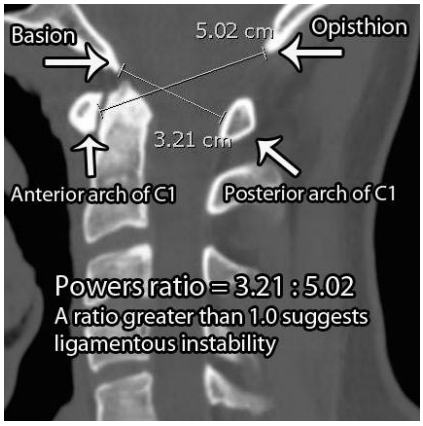
Ligament	Attachment	Role
Transverse (TAL)	Behind dens → C1 medial tubercles (horizontal limb of cruciate)	**Primary atlantoaxial stabiliser** · "strongest ligament of the spine" (TO)
Alar (paired)	Dens → occipital condyles (+ C1 lateral mass)	Strong · OC stability · **limits C1 rotation on C2** (TO)
Apical	Dens tip → foramen magnum	**Weak**
Tectorial membrane	Posterior C2 body → anterior foramen magnum	**Cephalad continuation of the PLL** · atlanto-occipital stability
Anterior AO membrane	Foramen magnum → C1 anterior arch	Continuation of ALL (TO)
Posterior AO membrane	Foramen magnum → C1 posterior arch	–

C1-2 joint (TO)	near-horizontal facets → **60% of cervical rotation**

4 Diagnosis & Classification

Key radiology findings

Measure	Normal / abnormal
X-ray (AP, lateral, odontoid)	**Low sensitivity (57%)**
Powers ratio = BC / OA (basion–posterior arch ÷ anterior arch–opisthion)	**~1 normal** · **> 1 → anterior dislocation** · < 1 → posterior AOD, odontoid or atlas ring fracture
Harris rule of 12	**BDI or BAI > 12 mm** → dissociation
Condyle–C1 interval (CCI)	**Most sensitive + specific** · normal **< 2.5 mm**
CT	**Gold standard for bone** · midsagittal reconstruction
CT angiogram	Vertebral artery injury; anatomy before fusion
MRI	Ligament injury with preserved alignment, occult injury, deficits · AOD with normal CT measurements → **MRI**



Powers ratio

Traynelis classification (direction)

Type	Displacement
I	**Anterior** occiput
II	**Longitudinal** (distraction)
III	**Posterior** occiput

Harborview classification (instability)

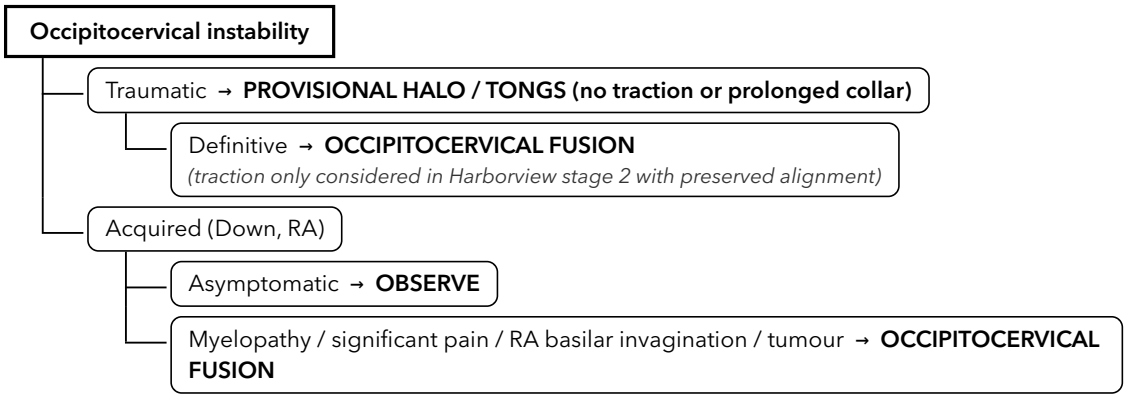
Stage	Injury	Stability
I	Minimal / undisplaced unilateral ligament injury	Stable
II	Minimally displaced, significant MRI soft-tissue injury (traction test)	Stable or unstable
III	**Gross malalignment (BAI or BDI > 2 mm beyond normal)**	**Unstable**

Atlantoaxial instability criteria (TO)

Measure	Threshold
ADI (flexion lateral)	**> 3 mm adult**, **> 5 mm child** → TAL rupture
PADI / SAC	Normal 17–29 mm · **< 14 mm → cord compression**
Rule of Spence (open mouth)	C1 lateral mass overhang **> 6.9 mm** → TAL disruption

Causes of AAI (TO) trauma (hyperflexion), **Morquio**, **Down**, skeletal dysplasia, **os odontoideum**, **Grisel** (infection), **RA**, NF

5 Management



Traumatic OC dissociation (intubated polytrauma) **instrumented occipitocervical fusion**
✗ Wrong options: Philadelphia collar · halo 12 wks as definitive · ACDF

Occipitocervical fusion technique

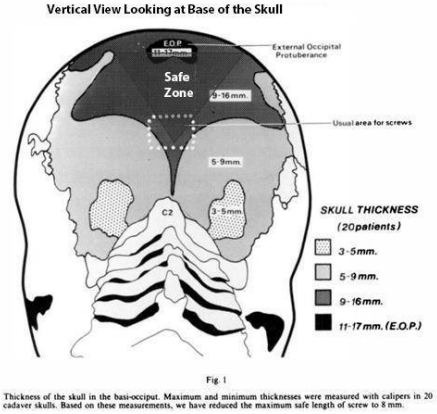
Set-up prone, **Mayfield**, set **O-C2 angle** on lateral fluoroscopy before draping · Occiput–**C3** instrumentation

C1 dissection stay **< 1 cm lateral to midline** above the C1 posterior arch → protects **vertebral artery**

Occipital screws **unicortical** (dural **venous sinuses** under the external occipital protuberance) · safe zone: **2 cm lateral + 1 cm inferior to EOP** along superior nuchal line (Orthobullets) · triangle **2 cm lateral, 2 cm inferior to EOP** (OB) – read the options

Thickest occiput **5 cm lateral to EOP** → too lateral for screws; **Mayfield posterior pin** site

– – Active space – –



Occipital screw safe zone

C1 anterior cortex breach — **internal carotid artery** (< 3 mm from C1) · also bicortical C1 or C1-2 transarticular screws
✗ Wrong options: vertebral · basilar · spinal arteries

C2 — pars, pedicle, transarticular or translaminar · **C3** lateral mass aimed cephalad + lateral · complications: **nonunion**, bleeding (internal carotid, vertebral artery)

Atlantoaxial subluxation – Fielding & Hawkins (TO)

Type	Injury	Treatment
I	Rotatory, **TAL intact**	Collar 6–8 wks
II	Anterior **3–5 mm**, one lateral mass pivot, TAL deficient	Collar / surgery
III	Anterior **> 5 mm**	**C1-2 fusion** (unstable)
IV	**Posterior** C1 on C2 (odontoid erosion)	**C1-2 fusion** (unstable)

Surgery (TO) — adult: **ADI > 5 mm**, dynamic instability · child: deficit, **persistent ADI > 4 mm**, deformity > 3 months, recurrence despite 6 wks immobilisation

AARS (TO) — **"cock robin"** (20° tilt one way, 20° rotation other, 10° flexion) · Grisel syndrome · CT shows rotation · child, trivial fall, transient quadriparesis, recovered → **congenital atlantoaxial instability**
✗ Wrong options: bifid atlas · ossified flavum · syrinx

How it's asked

"Down syndrome OC fusion — site for 8 mm unicortical screw?"	**→ point B, within 2 cm lateral/inferior of the EOP**
"C1 drill through anterior cortex endangers?"	**→ internal carotid artery**
"Tectorial membrane on MRI?"	**→ posterior C2 body → anterior foramen magnum (PLL continuation)**
"Intubated MVC, OC dissociation — definitive treatment?"	**→ occipital-cervical fusion with instrumentation**
"10-yr, trivial fall, transient quadriparesis?"	**→ congenital atlantoaxial instability** (TO)

ODONTOID FRACTURE

– Spine · 41 / 69 –

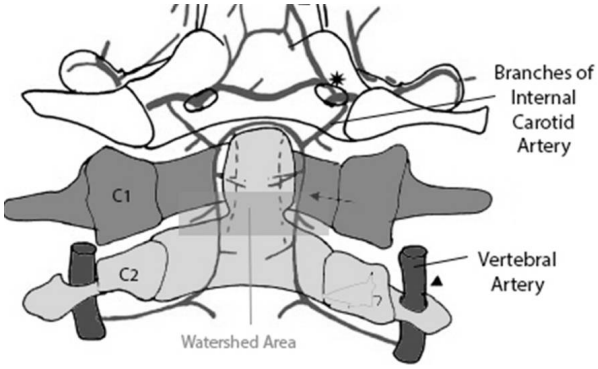
1 Basics

Epidemiology

Incidence	**commonest axis fracture** · 10–15% of cervical fractures · **commonest cervical fracture in the elderly (> 65)**
Bimodal	young: high-energy head trauma (hyperflexion/extension) · elderly: **simple falls**, often missed, ↑ morbidity + mortality · children: rare, **almost always at the basilar synchondrosis**
Displacement	anterior (hyperflexion) → TAL failure + AAI · posterior (hyperextension) → C1 anterior arch strikes dens

Anatomy

Ossification	**5 centres**: body, 2 neural arches, odontoid, **secondary (tip) centre** · **subdental (basilar) synchondrosis fuses ~6 yrs** (5–7) · secondary centre **appears ~3, fuses ~12 yrs** → fused synchondrosis + unfused tip = **8–10 yrs**
Ligaments	**transverse = primary atlantoaxial stabiliser** (limits anterior C1 translation) · **alar + apical limit rotation**
Blood supply	**watershed at the waist**: apex from **internal carotid** branches · base from **vertebral arteries → type II nonunion**



Odontoid watershed blood supply

Cervical kinematics

Joint	Flex/ext	Rotation	Lateral bend
Occiput–C1	50	4	8
C1-2	**10**	**50**	**0**
C3–C7	50	50	60
Total	**110**	**100**	**68**

C1-2 fusion	**~50% loss of rotation**

– – Active space – –

Os odontoideum

Cause	likely **old trauma** (not failed fusion) · looks like a type II fracture · **observe** if no deficit/instability · **C1-2 fusion** if deficit or instability

3 Presentation

Symptoms	neck pain worse with **rotation** · **dysphagia** (retropharyngeal haematoma)
Neuro deficit	**very rare** (large canal at C1-2)

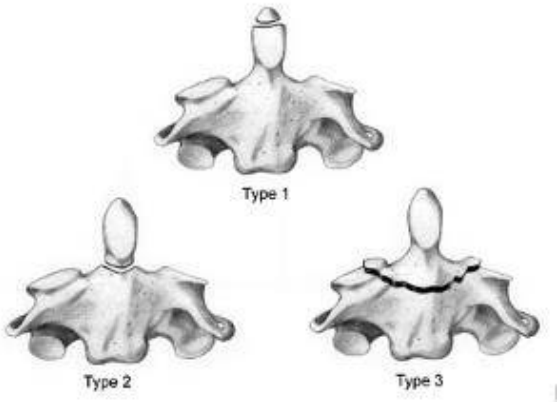
4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP, lateral, **open-mouth**	Fracture best seen on **open-mouth view**
Flexion-extension	Type I (rule out **OC instability**) + os odontoideum · instability: **ADI > 10 mm**, **SAC < 13 mm**
CT	**Study of choice** – pattern + stability
CT angiogram	**Vertebral artery course before posterior screws** (aberrant VA up to 30%)
MRI	Only if neuro symptoms

Anderson & D'Alonzo

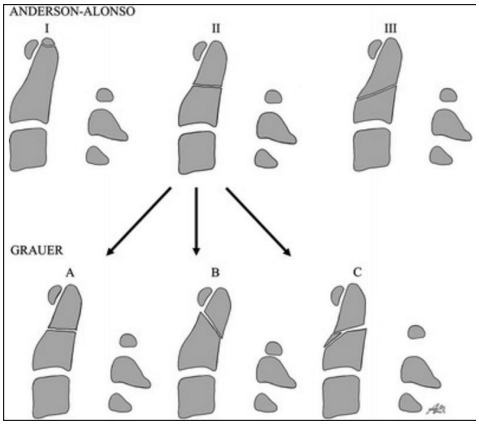
Type	Fracture	Note
I	**Oblique tip avulsion – alar ligament**	Rare; rule out **atlanto-occipital instability** (flexion-extension)
II	**Waist**	**Highest nonunion** (watershed) · 35% of C2 fractures
III	Into **cancellous C2 body** ± C1-2 joint	Good blood supply + large cancellous surface → heals



Anderson & D'Alonzo types

Grauer (type II)

Subtype	Pattern	Treatment
IIA	Undisplaced / minimal, no comminution	**External immobilisation**
IIB	Displaced, line **anterosuperior → posteroinferior** ("anterior oblique")	**Anterior odontoid screw** (good bone)
IIC	**Anteroinferior → posterosuperior** or **comminuted**	**Posterior C1-2 stabilisation**



Grauer classification

Nonunion risk factors (type II)

Factor	Threshold
Posterior displacement	**> 2 mm (strongest predictor)** (Orthobullets) · **> 5 mm** (TO, OB) – read the options
Displacement magnitude	**≥ 5 mm (> 50% nonunion)** (Orthobullets) · **> 6 mm = best predictor** (OB)
Age	**> 40** (Orthobullets) · **> 50** (OB) · > 65 (TO) · not age < 40
Delay	**> 4 days** (✗ Wrong options: 36–48 h)
Angulation	**> 10°** (≥ 11° OB) · not 7°
Gap	**> 1 mm**
Others	**Smoking**, comminution, diabetes, osteoporosis
No	Gender

Overall nonunion 33% (up to 88%) · elderly non-op **60–71%**

5 Management

Type / patient	Treatment (Orthobullets)
Type I	**Collar** (soft collar acceptable, OB)
Type II < 40 (no risk factors)	**Halo vest** (or hard collar)
Type II 40–80	**Surgery**
Type II > 80	**Collar**
Type III	**Collar 6–12 wks** (✗ Wrong options: halo advantage)

– – Active space – –

Type II odontoid

- Young, no nonunion risk → **HALO VEST (or hard collar)**
  (halo if displacement < 6 mm, reduced within 1 wk, age < 60 → > 90% union)
- Risk factors, reducible, oblique IIB, good bone, normal habitus → **ANTERIOR ODONTOID SCREW**
- Risk factors otherwise / elderly surgical candidate → **POSTERIOR C1-2 FUSION**
- Elderly, comorbid / > 80 → **RIGID COLLAR (fibrous union acceptable)**

Elderly type II

Halo	**highest morbidity + mortality** (aspiration, pneumonia, dysphagia, pin problems) → **contraindicated** in elderly, especially dementia, COPD
Posterior C1-2 fusion (surgical candidate)	↑ union + ↓ **mortality vs non-op** · ↑ SF-36 · less dysphagia than halo · **higher fusion than anterior screw**
Not a surgical candidate	**rigid collar 6–8 wks** · **fibrous union likely** and usually stable enough
> 80 yrs	**high morbidity + mortality regardless of treatment** · collar recommended
Nonop vs op in elderly	nonop → ↑ **nonunion** ✗ Wrong options: VA occlusion · myelopathy · radiculopathy
Anterior screw in elderly	avoid with **barrel chest (COPD)**, short neck, osteoporosis (cut-out), **dysphagia**, displaced fractures, fractures **> 4 wks old** · higher failure than posterior fusion

Anterior odontoid screw

Indications	type II with nonunion risk **AND** acceptable/reduced alignment + **fracture line perpendicular to screw (anterior oblique, IIB)** + habitus allowing trajectory
Pros	**preserves C1-2 rotation**, fewer infections than posterior · single screw adequate
Cons	**higher failure than posterior C1-2 fusion** · not for transverse, posterior oblique, type I/III



Anterior oblique fracture line suits odontoid screw

Posterior C1-2 fusion

Indications	type II with nonunion risk (50–80) · **type II/III nonunion** · os odontoideum with deficit/instability ✗ Wrong options: PT · collar · halo · anterior screw
C1 lateral mass screws	**10° medial, 22° cephalad** · don't breach C1 anterior cortex (**internal carotid**)
C2	**pars (commonest)**, pedicle, laminar · Gallie/Brooks wiring needs halo (rare)
Transarticular C1-2 screws	**contraindicated with aberrant vertebral artery**
Transoral odontoidectomy	severe posterior dens displacement + cord compression + deficit · rarely; C1 laminectomy usually enough · combine with posterior fusion

Halo

Pin-site infection	superficial → **tighten pins + oral antibiotics**

Paediatric synchondrosis fracture

Features	**commonest C-spine fracture < 7 yrs** · through basilar synchondrosis · neuro deficit rare · nonunion rare
Treatment	**closed reduction (extension)** + **Minerva cast or halo** 6–8 wks ✗ Wrong options: soft collar · screw · fusion initially

6 Complications

Nonunion	type II · symptomatic → **posterior C1-2 fusion**
Mortality	> 80 poor with any treatment, worst with halo

X vs Y

	Type II	Type III
Site	Waist (watershed)	C2 body
Union	Poor	Good
Default	Age/risk-based	Collar

How it's asked

"Major nonunion risk in type 2?"	→ **posterior displacement > 5 mm** (TO)
"Nonunion risks EXCEPT?"	→ **age < 40**
"Greatest nonunion risk on images?"	→ **type II with posterior displacement**
"Best predictor of nonunion non-op?"	→ **magnitude of displacement**
"72-yr declines surgery — risk with non-op?"	→ **fracture nonunion**

– – Active space – –

"40-yr ladder fall, type II — consider?"	→ high nonunion from vascular watershed
"78-yr smoker, emphysema, delayed type II?"	→ high nonunion risk → C1-C2 posterior fusion
"Elderly type II — highest morbidity/mortality treatment?"	→ halo vest
"72-yr COPD, undisplaced type III?"	→ rigid cervical collar
"85-yr comorbid, displaced type II?"	→ cervical collar
"80-yr COPD + dementia, undisplaced type II?"	→ rigid orthosis 6–8 wks
"79-yr posterior C1-2 fusion — true?"	→ lower mortality than non-op
"75-yr non-ambulatory comorbid, < 50% displaced?"	→ rigid collar
"65-yr diabetic smoker COPD, type II?"	→ reduction + posterior C1-C2 fusion
"84-yr demented, undisplaced type II — true?"	→ fibrous union likely in hard orthosis
"71-yr posterior displaced horizontal type II?"	→ posterior C1-2 instrumented fusion
"81-yr, 5 mm posterior displacement — true?"	→ morbidity + mortality high regardless
"Young diabetic smoker, IIB, anterior screw — associated with?"	→ high union, few complications
"Aberrant vertebral artery — contraindicated?"	→ C1-C2 transarticular screws
"Symptomatic type II nonunion at 7 months?"	→ posterior C1-C2 fusion
"42-yr smoker, 5-day delay, favourable obliquity?"	→ anterior odontoid screw
"Single anterior screw suits?"	→ type II anterior oblique pattern
"37-yr type III?"	→ external orthosis 6–8 wks
"Rigid orthosis acceptable EXCEPT?"	→ Jefferson fracture with TAL disruption
"28-yr, 2 mm displacement, 7° angulation, 36 h?"	→ halo or hard orthosis (low nonunion risk)
"Fused synchondrosis, unfused tip centre — age?"	→ 8–10 yrs
"Paediatric odontoid fracture?"	→ almost always at the basilar synchondrosis
"5-yr unstable synchondrosis fracture?"	→ closed reduction + Minerva cast or halo

OSSIFICATION OF THE POSTERIOR LONGITUDINAL LIGAMENT

– Spine · 42 / 69 –

1 Basics

Epidemiology

Population **East Asian (Japanese)** commonest · Asian 2–3% (up to 4% of cervical patients) · Caucasian 0.2–0.7% (TO) · not European ancestry

Sex / age **men ~2× women** (not prognostic) · peak 6th decade

Level **C4-C6** (C4-5, C5-6) · **> 95% cervical** · then thoracic, lumbar

Risk factors & labs

Risk factor	Not a risk factor
Hyperinsulinaemia (even without diabetes) · **diabetes**	HLA-B27 (= AS)
Obesity / BMI (correlate with extent)	Traumatic brain injury (= HO)
Fluoride intoxication (fluorosis) – urinary fluoride ∝ severity (TO)	Female sex
Hyperparathyroidism, **hypophosphataemic rickets**, abnormal Ca metabolism	–
Growth hormone imbalance, recurrent minor trauma, infection (TO)	–
High-salt low-meat diet, poor Ca absorption, mechanical PLL stress	–
Genetic linkage	–

Pathology vascular infiltration → hypertrophy → **endochondral ossification** of PLL → lamellar bone

Associations (TO) **DISH**, **ossified ligamentum flavum**, **ankylosing spondylitis**, fluorosis

3 Presentation

Symptoms often asymptomatic · **myelopathy** (hand clumsiness, gait instability, Hoffmann, hyperreflexia) · radiculopathy · **cord injury from minor trauma** (central canal stenosis)

MRI clue compression at **mid-vertebral body levels**, not only discs

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Role
Lateral X-ray	**Diagnosis** + **sagittal alignment (lordosis)** · classification
CT	**Study of choice for extent + thickness** of ossification · screw planning, vertebral artery
MRI	**Study of choice for cord compression**
CT myelogram	MRI contraindicated / complex bony stenosis

Pre-op work-up (TO) **lateral + dynamic X-rays AND CT** (both) · laminectomy only if lordosis maintained



OPLL on sagittal CT

Classification (lateral X-ray) (TO)

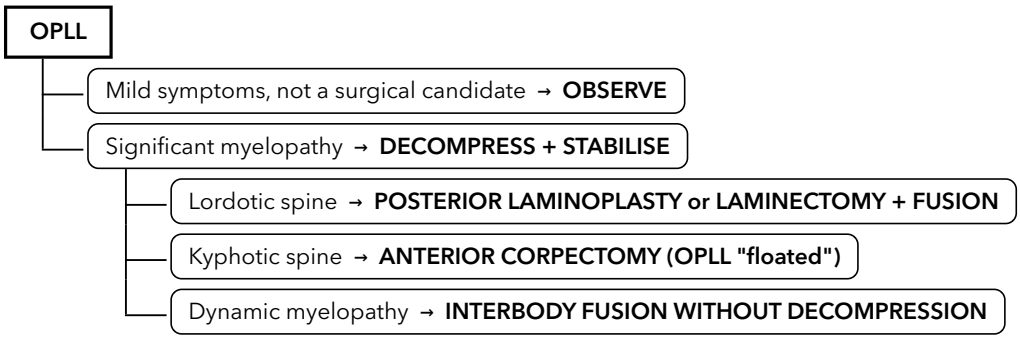
Type	Pattern
1 **Continuous**	Long lesion over several bodies
2 **Segmental**	Behind each body, interrupted at discs
3 **Mixed**	Continuous + segmental
4 **Localised**	Over disc space, 1–2 segments

Myelopathy risk **mixed + continuous** progress most · **occupancy ratio (OPLL ÷ canal AP) 30–60%** predicts myelopathy (TO)



OPLL types and occupancy ratio

5 Management



Posterior (laminoplasty / laminectomy + fusion)	**needs lordosis** (cord drifts back) · **safer, preferred** (avoids peeling OPLL off dura) · fuse after laminectomy (prevents **post-laminectomy kyphosis**) · risk **OPLL growth**
Anterior corpectomy	**kyphotic** spine · **"float" the OPLL** in the corpectomy site instead of excising → avoids **dural tear / CSF leak**

6 Complications

Post-laminoplasty C5 palsy	**commonest root palsy (~6%)** · **deltoid / lateral arm (C5 dermatome)** · short C5 root at apex of lordosis, cord drifts back
Recurrence / growth	less with complete resection or floating

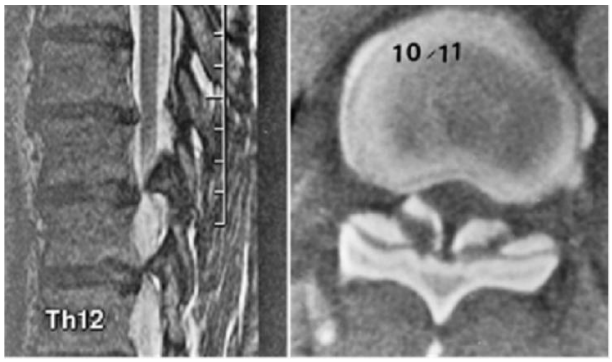
X vs Y

	OPLL	DISH	AS	Fluorosis
Ossification	**Posterior** longitudinal ligament	**Anterior**, flowing, ≥ 4 levels	**Marginal syndesmophytes**	**Diffuse** — PLL, flavum, interosseous membranes
Marker	Hyperinsulinaemia	Diabetes	**HLA-B27**	Endemic fluoride (Punjab)
Anterior osteophytes	Small, non-syndesmotic	Large	–	–

Ossified ligamentum flavum (TO)

Features	rare cause of **thoracic myelopathy**, East Asians · **posterior** compression → **spastic gait, LL hyperreflexia, normal UL reflexes**
Imaging	**T2 sagittal MRI** · **whole-spine screen mandatory** (tandem lesions) · fine CT for planning
Treatment	**decompression only** (non-op fails) · better prognosis than other thoracic myelopathy · prognosis ← preop severity + time to surgery

– – Active space – –



Thoracic ossified ligamentum flavum

How it's asked

"65-yr from Punjab, myelopathy, MRI sagittal – diagnosis?"	→ **OPLL** (TO)
"65-yr from Punjab, myelopathy, CT neck – diagnosis?"	→ **fluorosis** (TO) – read the image and options
"Mixed OPLL with myelopathy – pre-op tests?"	→ **lateral + dynamic X-rays and CT** (TO)
"Known risk factor for OPLL?"	→ **hyperinsulinaemia**
"Commonest OPLL level?"	→ **C4-6**
"42-yr myelopathy, cord compressed at mid-body levels, thick ossified PLL?"	→ **OPLL**
"Image suggestive of OPLL?"	→ **ossified band behind the vertebral bodies on CT**
"Commonest root palsy after laminoplasty?"	→ **C5**
"Spastic gait, LL hyperreflexia, normal UL reflexes?"	→ **thoracic ossified ligamentum flavum** (TO)

OSTEOPOROTIC VERTEBRAL COMPRESSION FRACTURE

– Spine · 43 / 69 –

1 Basics

Epidemiology

Commonest fragility fracture	✓ VCF · 700,000/yr US · 70,000 admissions · $15 billion ✗ Wrong options: distal radius · femoral neck · intertrochanteric · proximal humerus
Prevalence	**25% > 70 yrs** · **50% > 80 yrs** (up to 40% of women > 80, OB)
Strongest predictor of further vertebral fractures	**history of 2 VCFs** (postmenopausal women)

Osteoporosis

Definition	**normal bone quality, decreased quantity** · thin cortices, ↓ trabecular continuity ✗ Wrong options: abnormal quality = osteomalacia
WHO	**T-score < −2.5** · after a fragility fracture → **treat regardless of T-score**
Peak lumbar BMD	**females 33–40** · **males 19–33** (variable)
Screening	all women **≥ 65** + younger women with osteoporotic fractures

Associations & mortality

Pulmonary	kyphosis → **each VCF ↓ FVC up to 9%** → **↑ death from pulmonary causes** (pneumonia) · ↓ chest/abdominal volume → **early satiety**
Comorbid	associated with hypertension, smoking, calcium/vitamin D use · frailty drives both fracture + death

Mortality	Value
1 yr	**~15%** (7–15%) – **less than hip** (hip ~25%)
2 yr	**~20% – equivalent to hip**
Ranking (1 yr)	**Hip > VCF > distal radius** (distal radius: no ↑ mortality)
Medicare cohort	~2× controls · **men > women** · **younger age at fracture → greater risk** · HR 1.8 · survival 54% / 31% / 11% at 3 / 5 / 7 yrs

Further fracture risk	**20%** after first VCF (OB) · **~40% within 2 yrs without osteoporosis treatment** (OB) · **5× more likely** (OB) – read the options
Progressive kyphosis	**load transfers to the superior adjacent vertebra** · deformity progresses even when pain-free · worse gait, satiety, lungs

– – Active space – –

3 Presentation

Pain	only **25%** seek attention · localised; may wrap around ribs dermatomally · tenderness on deep spinous palpation
Neuro	**< 1%** (SCI very rare; lumbar foraminal root deficits) · no chronic pain in majority

4 Diagnosis & Classification

Key radiology findings

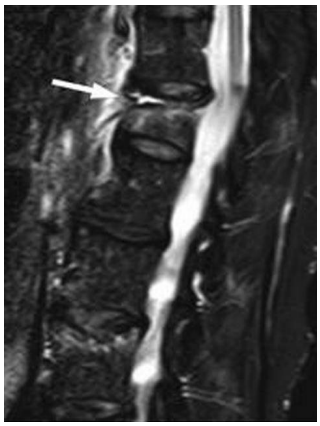
Study	Use
X-ray whole spine	Concomitant fractures **20%** · height loss **≥ 20% or ≥ 4 mm**
CT	Inadequate films, LE deficit, posterior wall
MRI	**Acuity (STIR/T2 oedema = acute)**, PLC, cord compression, **benign vs metastatic** · ✓ next study after X-ray
Bone scan (3-phase)	Acuity if MRI contraindicated
No	Flexion-extension, tagged WBC scan



L4 compression fracture

Benign vs malignant on MRI

Osteoporotic (benign)	Metastatic (malignant)
Fluid sign (fluid cleft)	**Complete marrow replacement** (diffuse homogeneous low T1)
Band-like low T1 / high T2 signal	**Convex posterior border**
Retropulsed fragment	**Pedicle / posterior column involvement**
–	Paraspinal or epidural mass



MR fluid sign

Suspect metastasis	**above T5**, atypical films, constitutional symptoms, **young without a fall**

Labs

Work-up	CBC, BMP · **ESR** (infection) · **SPEP + UPEP** (myeloma)

5 Management

Osteoporotic VCF, neuro intact

- Acute → **ANALGESIA + EARLY MOBILISATION ± TLSO**
  (calcitonin 4 wks if < 5 days; no bed rest)
- Severe pain > 6 wks / Kümmell → **KYPHOPLASTY**
- Progressive deficit / PLL injury / unstable → **DECOMPRESSION + LONG FIXATION**
  (multiple fixation points ± anterior support)

Non-operative

First line	**oral analgesia (opioid acceptable) + gradual activity, walking unrestricted** ± TLSO for comfort · even **> 30° kyphosis or > 50% height loss if PLL intact** ✗ Wrong options: strict bed rest → DVT, PE, deconditioning, bone loss
AAOS 2010 guideline	only **moderate** recommendation: **calcitonin 4 wks if presenting within 5 days** · only **strong** recommendation: **against vertebroplasty** · kyphoplasty = limited/inconclusive option · brace, bed rest + opiates, exercise, ibandronate/strontium, L2 nerve block = inconclusive
TLSO	RCTs: same pain, disability, radiographs with or without brace · not strongly recommended
Calcitonin	**directly inhibits osteoclasts** (cell-surface receptor)
Medical	**bisphosphonates** prevent further fractures (atypical femur fracture risk) · **teriparatide ↑ BMD without atypical femur fracture risk** · exercise ↑ BMD

Vertebral augmentation

Indication	**severe pain after 6 wks** of non-op (narcotic-dependent, sedentary, constipated) · not first line · not for **healed VCF with kyphosis** (TO)

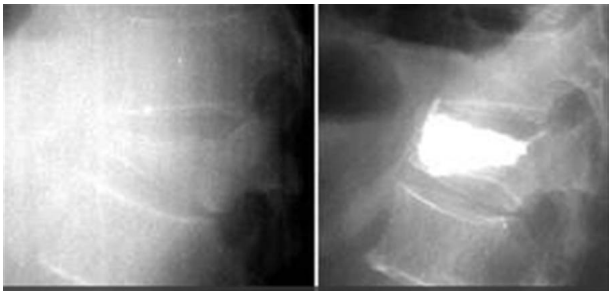
– – Active space – –

Evidence **double-blind sham-controlled RCTs: vertebroplasty no benefit at any time point** · surgery meta-analysis (OB): **greater pain improvement**

	Vertebroplasty	Kyphoplasty
Cavity	None	**Balloon cavity**
Cement viscosity	**Less viscous**	More viscous
Injection pressure	**Higher**	**Lower** (TO: "kyphoplasty higher pressure" = false)
Extravasation	**Higher**	Lower
Height restoration	No	Goal; partial reduction
Cost / equipment	Less	**More**
Pain relief	**Comparable** (TO)	Comparable
Future VCF risk	–	Yes (TO)

TO trap vertebroplasty cement: **high viscosity + more radiopaque than arthroplasty cement** (true statement in TO stem)

Mechanism **eliminates micromotion**



Kyphoplasty balloon

Kümmell disease

Definition **delayed post-traumatic vertebral body osteonecrosis** · pain-free interval (months–years) → new pain, collapse, **intravertebral cleft** (pathognomonic) · thoracolumbar junction

Treatment **kyphoplasty**
✗ Wrong options: continued non-op · vertebroplasty (leak, no correction) · fusion without deficit



Intravertebral cleft of Kümmell disease

Operative stabilisation

Indications	**progressive neuro deficit**, **PLL injury / unstable** (very rare) · not stable 30% height loss, 15° kyphosis · retropulsion makes cement risky
Technique	**long constructs**, multiple fixation points, consider anterior support

6 Complications

Cement extravasation	into canal → **neuro injury** · vertebroplasty > kyphoplasty · check posterior cortex defects
Kümmell disease	avascular necrosis + nonunion

How it's asked

"71-yr L2 VCF, 30% height loss – management?"	→ **analgesia, TLSO, progressive activity**
"Ambulatory with oxycodone – first line?"	→ **opioids + unrestricted walking**
"78-yr T11 20% height loss, walker?"	→ **analgesics + progressive rehabilitation**
"AAOS-supported treatment within 5 days?"	→ **activity modification + calcitonin**
"Vertebroplasty vs kyphoplasty – false?"	→ **kyphoplasty uses higher cement pressure** (TO)
"Cementing techniques – EXCEPT?"	→ **healed VCF with kyphosis can be cemented** (TO)
"Kyphoplasty – EXCEPT?"	→ **cement under high pressure** (TO)
"Narcotic-dependent 6 wks after T11 VCF?"	→ **percutaneous vertebral augmentation**
"Vertebroplasty in sham-controlled RCTs?"	→ **no benefit at any time point**
"Old VCF with AVN + collapse, failed conservative care?"	→ **kyphoplasty**
"Surgery vs non-op meta-analysis?"	→ **greater improvement in pain**
"Old T12 VCF – compared with peers?"	→ **greater risk of death from pulmonary conditions**
"Acute T12 VCF – true?"	→ **2-yr mortality ≈ hip fracture**
"Mortality risk after VCF?"	→ **greater in men than women**
"Fragility fracture mortality, decreasing?"	→ **hip > VCF > distal radius**
"T-score −3.0 – commonest fragility fracture?"	→ **vertebral compression fracture**
"VCFs associated with?"	→ **20% further fracture risk**
"Progressive kyphosis after VCF due to?"	→ **load transfer to the superior adjacent vertebra**
"Spinal deformity of progressive osteoporotic collapse?"	→ **deformity progression**

– – Active space – –

"Untreated, risk of another fragility fracture in 2 yrs?"	**→ ~40%**
"MRI fluid sign?"	**→ osteoporotic VCF**
"75-yr L1 VCF 8 wks — next imaging?"	**→ lumbar MRI**
"Pain-free interval then pain 10 months after VCF?"	**→ post-traumatic osteonecrosis (Kümmell)**
"Breast cancer history — MRI finding favouring osteoporotic?"	**→ band of high T2 signal**
"Osteoporosis — true?"	**→ decreased bone quantity**

PATHOLOGIC SCOLIOSIS

– Spine · 44 / 69 –

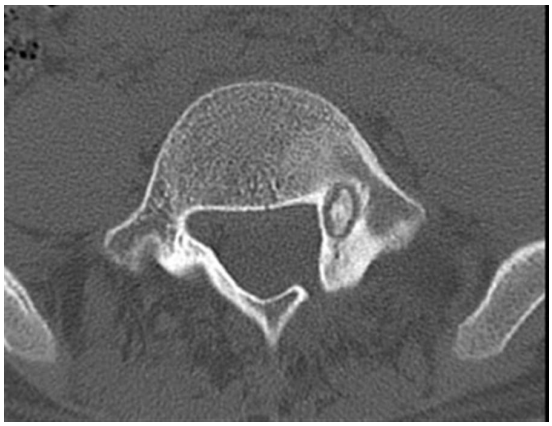
1 Basics

Definition scoliosis from a **benign bone tumour**, usually in the **posterior elements** · any age; OO + osteoblastoma typically **2nd decade**

Mechanism painful **paraspinal muscle spasm**

	Osteoid osteoma	Osteoblastoma
Size	**< 2 cm** nidus	**> 2 cm**
Site	**Apex of the concavity** of the curve · posterior elements > body	Posterior elements
Curve	**Rigid** (✗ Wrong options: flexible)	–
Pain	Severe, **night**, **relieved by NSAIDs**	Less severe, **NSAIDs help less**
Neuro	–	**May cause deficits**
Histology	**Same as peripheral OO** (✗ Wrong options: more malignant)	Less ossified; ± secondary ABC

NSAID mechanism nidus makes **PGE_2 + PGI_2**; **COX-2** high in nidal osteoblasts



Osteoid osteoma at the concave apex

Syndromic scoliosis (TO)

Syndrome	Scoliosis incidence
Marfan (FBN1, fibrillin-1)	**Two-thirds – highest**
Rett (**MECP2**)	Up to half
Prader-Willi, Down, Loeys-Dietz	Fewer

3 Presentation

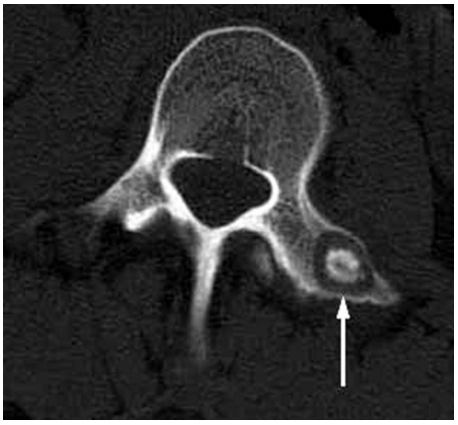
Symptoms **painful scoliosis**, back pain **worst at night**, NSAID relief · curve usually **mild**

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Finding
Standing AP + lateral	Level + severity · cortical thickening with **radiolucent nidus**
Fine-cut CT	**Best for outlining the lesion + planning** · low-attenuation nidus with central mineralisation, perinidal sclerosis
MRI	Neurovascular proximity · may show only oedema, **miss the nidus**
Bone scan / SPECT	**Markedly ↑ uptake**



Nidus on axial CT

Differential of painful posterior-element lesions

Lesion	Clue
Osteoid osteoma	Small sclerotic nidus, painful scoliosis
Osteoblastoma	Larger, less ossified, ± ABC
ABC	Purely lytic, no sclerotic nidus
Eosinophilic granuloma	**Vertebral body**, purely lytic, **vertebra plana**
Metastasis	Mixed blastic-lytic, no nidus

5 Management

Pathologic scoliosis

- Minimal curve → **NSAIDs + OBSERVE (up to 36 months to resolve)**
  (osteoblastoma rarely responds to NSAIDs)
- Painful, progressive scoliosis → **EN BLOC RESECTION**
  (RFA case by case – usually too near neural elements)

Prognosis curve resolves after resection if done **within 15–18 months of curve onset** in a child **< 11 yrs**

How it's asked

– – Active space – –

"Spinal osteoid osteoma — true?"	→ **occurs on the concave side of the curve** ✗ Wrong options: flexible curve · never RFA · always body · more malignant
"13-yr painful scoliosis, hot bone scan, sclerotic nidus in posterior element?"	→ **osteoid osteoma**
"Highest scoliosis incidence?"	→ **Marfan syndrome** (TO)

– – Active space – –

PEDIATRIC CERVICAL TRAUMA OVERVIEW

– Spine · 45 / 69 –

1 Basics

Epidemiology

Share	uncommon, but **60% of paediatric spinal injuries** · **33% present with neuro deficit** · other organs in ~40%
Level	**< 8 yrs: 87% at C3 or above** · > 8 yrs: lower cervical (adult pattern)
Mechanism	**MVA commonest** · falls (toddlers, school age) · sports (adolescents)

Why under-8s are vulnerable

Anatomy	**large head : trunk**, **horizontal facets**, **ligamentous laxity**, weak muscles · column stretches **5 cm** without rupture
SCI	commoner + more lethal < 8 · recovery better > 8
Mortality	**C1 injury 17%** · C4 ~4% · higher at C3 and above

3 Presentation

Exam	full exam (associated injuries) · suspect C-spine with **head + facial trauma** · document sensation incl. **sacral sparing**, motor, reflexes · **repeat — 20% with fractures examine normal** initially

4 Diagnosis & Classification

Key radiology findings

Mandatory X-rays	**AP, open-mouth odontoid, cross-table lateral** · oblique (facets) optional · **flexion-extension only under physician supervision**

Normal paediatric lateral (diagnostic criteria)

Parameter	Normal
Prevertebral soft tissue	**< 2/3 of adjacent body width**
Retropharyngeal space	**< 6 mm at C2** · at C6: **< 22 mm** (Orthobullets) · **< 14 mm in children, < 22 mm adults** (TO) – read the options
Retrotracheal space	**< 14 mm**
ADI	**< 5 mm child**, **< 3 mm adolescent**
Pseudosubluxation C2-3 / C3-4	**< 4 mm** if **posterior laminar line contiguous**
Contour lines	Smooth anterior + posterior body, spinolaminar, spinous tips · parallel facets
C3 wedging	Normal in **7%**
Loss of lordosis	Normal in **14%**



Normal paediatric lateral C-spine

CT upper cervical fractures, **atlantoaxial rotatory subluxation**, canal compromise · not as screening study

MRI **obtunded / closed head injury**, canal compromise

5 Management

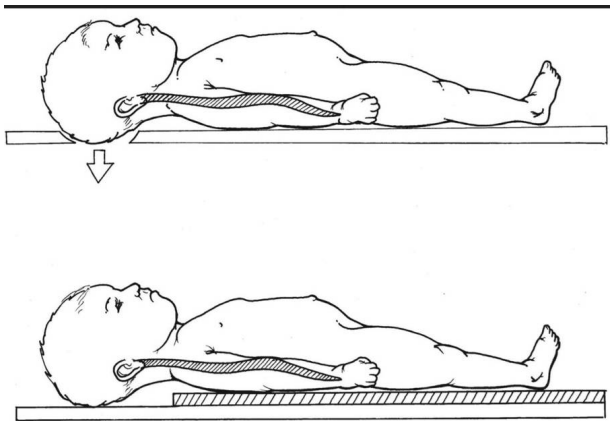
Transport immobilisation

Board **standard adult backboard → cervical flexion (kyphosis)** – avoid · use **occipital recess (head cut-out)** or **thoracic elevation (torso pad)** for < 8 yrs

Alignment in-line, **external auditory meatus in line with the shoulders** (relative extension)
✗ Wrong options: EAM anterior/posterior · adult board + sandbag under head

Collars commercial collars often misfit → sandbags · paediatric collars acceptable

– – Active space – –



Backboard with occipital recess vs standard

Treatment by injury

Treatment	Indications
Observation	**C2-3 pseudosubluxation**
Collar	Stable odontoid fracture, AAI, **acute AARD**, stable subaxial injury
Halo (thin calvaria — many low-torque pins)	Unstable odontoid, OC instability, AAI, **subacute AARD**, **Jefferson**, unstable subaxial
Surgery	Unstable spine + SCI, AAI, **chronic AARD**

Car seat & airbag safety

Stage	Recommendation
Rear-facing, rear centre	Until **≥ 1 yr and ≥ 20 lb**
Forward-facing seat	Until **≥ 4 yrs or ≥ 40 lb**
Booster	Until **60–80 lb and 8 yrs**
Seat belt, rear middle	8–12 yrs
Front seat	From **13 yrs · airbag OFF if < 80 lb**

How it's asked

"8-yr unconscious after fall — acute stabilisation?"	**→ in-line, external auditory meatus in line with shoulders**
"Avoid in young children after trauma?"	**→ standard backboard**
"Consistent car-safety recommendation?"	**→ 75 lb 13-yr in front seat — airbag OFF**

PEDIATRIC SPINAL CORD INJURY

– Spine · 46 / 69 –

1 Basics

Epidemiology

Incidence	**2–5% of all SCI** · **83% cervical** · M:F 1.5:1
Age pattern	**< 8 yrs: cervical** (**large head-to-body ratio**) · **> 8 yrs: thoracolumbar** · upper cervical (C1-4) > lower (C5-7)
Mechanism	**MVA / RTA commonest** · falls, sport, **child abuse** · sledding + ATV → thoracolumbar
Associations	**head injury > 30%**, intrathoracic, intra-abdominal

Injury patterns

Region	Patterns
Cervical	OC / atlantoaxial dislocation, **atlantoaxial rotatory fixation**, odontoid (synchondrosis), subaxial ligamentous, **SCIWORA**
Thoracolumbar	**Compression (commonest)**, burst, flexion-distraction, fracture-dislocation

Anatomy & biomechanics

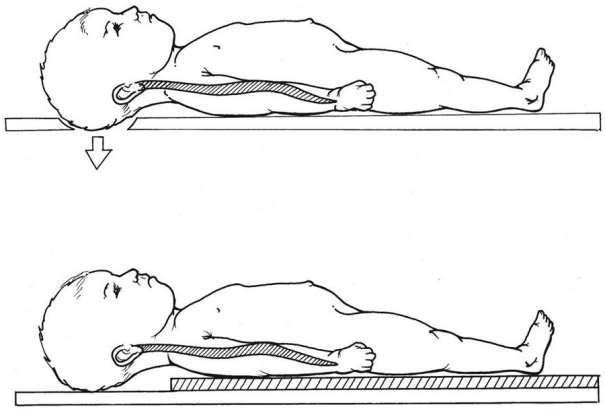
Cord	ends at **L3 at birth → L1-2** · adult size by **10 yrs**
Flexibility	ligamentous laxity, immature supports, **shallow horizontal facets**, nucleus with ↑ water + ↓ collagen cross-links · column stretches **5 cm** without rupture · forces pass through weak **physeal cartilage**

3 Presentation

Primary survey (paediatric ATLS)

C	**Broselow tape** estimates weight
D	log-roll + immobilisation on **paediatric board** or adult board with **torso pad / head cut-out** (avoid C-spine flexion) · **Paediatric GCS**

– – Active space – –



Board with occipital recess

Secondary survey myotomes + dermatomes, **rectal + genital exam**, bulbocavernosus reflex

Paediatric GCS

Score	Motor	Verbal	Eyes
6	Normal spontaneous movement	–	–
5	Withdraws to touch	Smiles, orients to sound, follows, interacts	–
4	Withdraws to pain	Cries but consolable, confused	Spontaneous
3	Abnormal flexion	Inconsistently consolable, moaning	To voice / touch
2	Extension	Inconsolable, agitated	To pain
1	None (flaccid)	None	None

4 Diagnosis & Classification

Key radiology findings

Study	Use
X-ray AP + cross-table lateral of whole spine	**Multilevel injury common** · ± swimmer's, open-mouth, flexion-extension
CT	Polytrauma, high energy, altered mental state, head/face injury · **radiation risk** · **not a screening tool**
MRI	**Deficit with normal X-rays (SCIWORA)** · soft tissue, cord · no radiation

Retropharyngeal space (TO) **< 6 mm at C2** (adults + children) · at C6 **< 14 mm children**, **< 22 mm adults**

SCIWORA (TO)

Definition **spinal cord injury without radiographic abnormality** · deficit with **normal X-ray + CT** · often recovers (TO stem: quadriplegia, normal imaging, complete recovery in 3 months – answer not keyed by source; SCIWORA is the option matching the definition) · MRI is the study

5 Management

– – Active space – –

Treatment	Indications
Analgesia + activity	**Apophyseal, spinous process, transverse process** fractures
Collar 8–12 wks	Stable odontoid, AAI, acute AARD, stable subaxial
TLSO 8–12 wks	**Compression < 50% height loss**, **burst < 50% retropulsion, no deficit**, **purely osseous flexion-distraction (Chance)**
Halo	Unstable cervical (**thin calvaria → skull penetration**)
Cervical fusion	OC, atlantoaxial, subaxial instability (OC fusion, transarticular screws, pedicle screws)
Thoracolumbar fixation	**Unstable burst, cord compression, irreducible fracture-dislocation, ligamentous flexion-distraction** · pedicle screws 1–2 levels above + below

6 Complications

List complete deficit, **progressive spinal deformity**, poor wound healing, CES

Post-injury scoliosis cervical complete SCI **before age 10** → paralytic scoliosis in **virtually 100%** · follow all to maturity

Prognosis most paediatric SCI **incomplete** · SCI commoner + more lethal < 8 · recovery better > 8

How it's asked

"9-yr complete C5-6 SCI – scoliosis risk?"	→ **100%**
"Paediatric SCI – EXCEPT?"	→ **"cervical injury in < 8 from small head-to-body ratio"** (it's large) (TO)
"Child quadriplegic, normal X-ray + CT, recovers in 3 months?"	→ **SCIWORA** (TO)

– – Active space – –

PEDIATRIC SPONDYLOLYSIS & SPONDYLOLISTHESIS

– Spine · 47 / 69 –

1 Basics

Epidemiology

Prevalence	**4–6%** of population (defects **not present at birth**) · **6–7% of adolescent athletes** · up to **47% of adolescent LBP** · up to **47% in gymnasts, weightlifters, football linemen, divers**
Ethnicity	↑ **Native Americans** · possible **autosomal dominant** inheritance
Level	**L5 pars (85–95%)**, L5 on S1 · then L4, L3 (TO)
Laterality (TO)	**80% bilateral**, 20% unilateral · unilateral defect stresses the contralateral pars
Risk factors	**repetitive hyperextension** (+ rotation) · ↑ **pelvic incidence**, sacral slope, sacral table index · ↓ sacral table angle

Continuum

Stage	Definition
Pars stress reaction	Sclerosis, no complete break
Spondylolysis	Complete pars fracture
Isthmic spondylolisthesis	Forward slip through the pars defect
Spondyloptosis	**100%** translation

Morita stages (TO) early (hairline) → progressive (wider, fragments) → **terminal (sclerotic, pseudarthrosis)** · healing predictors: **unilateral**, earlier stage, **level other than L5**

Progression

Lysis → listhesis **~15%** (all bilateral defects) · most pars defects **never need surgery**

Progression risk ↑
Higher grade at presentation (> Meyerding 2, > 50%)
Dysplastic (Wiltse I) slips
Slip angle > 45–50° – best predictor of progression (OB), instability, pseudarthrosis, pain
Young age at presentation

Severity of current slip correlates most with **pelvic incidence** (PI = PT + SS; constant, ~57° adult; high PI → ↑ lumbosacral shear)
✗ Wrong options: SS · PT · slip grade

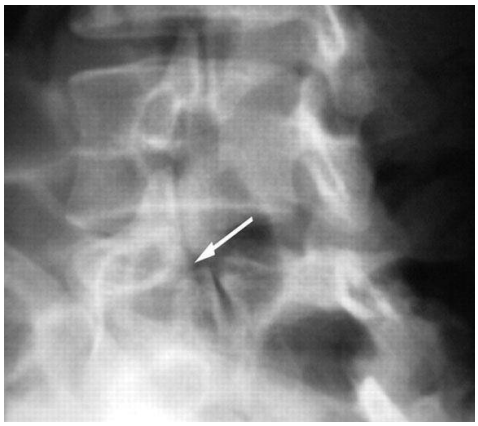
3 Presentation

Classic	active adolescent, **activity-related LBP**, relieved by rest · many asymptomatic · **grade ≠ symptoms**
Leg	**hamstring tightness (commonest)**, buttock pain, L5 radicular pain
Commonest exam finding	**pain with lumbar extension**, especially **single-leg-stance extension** ✗ Wrong options: pain on flexion · SLR · lost Achilles reflex
High-grade / dysplastic	**heart-shaped buttocks**, flat lumbar lordosis, palpable **step-off**, ↑ **popliteal angle**
Gait (Phalen-Dickson)	**short stride, hips + knees flexed (crouched)**
Listhetic crisis	severe pain worse with extension, neuro deficit, hamstring spasm · CES rare

4 Diagnosis & Classification

Key radiology findings

Study	Fact
AP + lateral X-ray	**Best initial**; lateral shows **80%** of defects · obliques add 15% (OB) but **no extra diagnostic value + more radiation** (Orthobullets, OB) · TO: AP, lateral + **bilateral obliques**
Oblique	**Scotty dog collar** (neck = pars, nose = TP, eye = pedicle, ear = SAP, front leg = IAP)
SPECT	**Most sensitive** after normal X-rays (OB) · "gold standard" (TO) · now rarely used (radiation)
MRI	**As sensitive + specific as SPECT, no radiation → next test** after negative X-rays · STIR oedema = acute · neuro deficit, stenosis
CT (thin cut)	**Best to delineate the defect** · age + healing
Bone scan	Most sensitive screen (lesion may be cold)
Flexion-extension	Instability (**> 3 mm** translation abnormal, TO)



Scotty dog collar

– – Active space – –

Wiltse-Newman

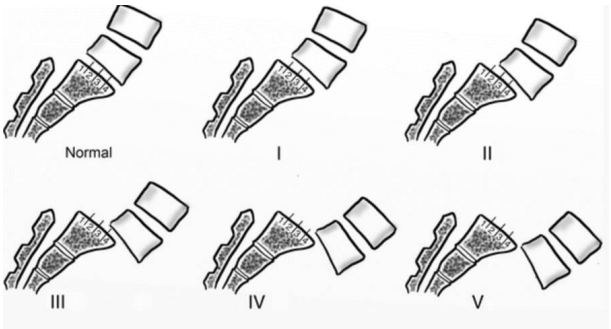
Type	Description
I Dysplastic	Congenital lumbosacral facets/sacrum, **pars intact** · **more neuro symptoms, more progression**
II-A Isthmic	**Pars fatigue (stress) fracture** – lytic
II-B Isthmic	**Pars elongation** (healed microfractures)
II-C Isthmic	**Acute pars fracture**
III	Degenerative (commonest overall)
IV	Traumatic
V	Neoplastic / pathologic
VI	Iatrogenic (TO)

Marchetti-Bartolozzi developmental (Wiltse I + II) vs acquired (traumatic, post-surgical, pathologic, degenerative)

Meyerding

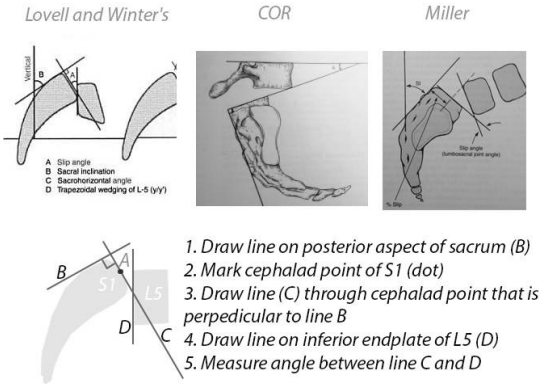
Grade	Slip
I	< 25%
II	25–50%
III	50–75%
IV	75–100%
V	**Spondyloptosis** (> 100%)

Low grade I–II · **high grade** III–V



Meyerding grades

Slip angle lumbosacral kyphosis



Slip angle

5 Management

Spondylolysis / spondylolisthesis

- Asymptomatic (any grade) → **OBSERVE, NO ACTIVITY RESTRICTION**
- Symptomatic lysis / low grade → **ACTIVITY MODIFICATION + PT 6 MONTHS**
  - Acute stress reaction or failed PT → **BRACE (TLSO / LSO) 6–12 WKS**
- Failed non-op
  - L1–L4 defect, multiple defects → **PARS REPAIR**
  - L5 lysis / low grade (I–II) → **L5-S1 POSTEROLATERAL FUSION**
  - High grade (III–V) → **L4-S1 POSTEROLATERAL FUSION ± reduction**

Non-operative

PT	6 months: **hamstring stretching, pelvic tilts, abdominal strengthening** · most improve
Bracing	**acute pars stress reaction** (brace > activity restriction alone; prevents defect formation) · failed PT · chronic L5 lysis failing 3 months PT (no STIR signal) → **rigid LSO 6–12 wks** · **antilordotic (hypolordotic) brace** ✗ Wrong options: hyperextension brace
TO stepwise protocol	avoid hyperextension → **non-rigid corset** → after 2–4 wks PT (core) → still painful at 4 wks → **TLSO / Boston antilordotic brace** · healing 3–6 months · return to play from **8 wks** if pain-free
TO answers	SPECT-proven bilateral stress fracture → **activity modification + rest** first · **2-month-old bilateral pars defect → antilordotic brace**
Sport	stopping sport ≥ 3 months improves outcome · return controversial

– – Active space – –

Operative criteria

Procedure	Indication
Direct pars repair	**L1–L4 isthmic defect** failing non-op, **multiple defects** · preserves motion · screw, tension wiring, screw + hook · not with **disc degeneration** (MRI first), facet arthropathy, listhesis
L5-S1 posterolateral fusion ± ALIF	**L5 spondylolysis** failing non-op · **low grade (I–II)** failing non-op, progressive, deficit or **dysplastic** · in situ, ± instrumentation, TLSO
L4-S1 posterolateral fusion ± reduction	**High grade (III–V)** · L5-S1 alone fails for > 50% slip
Gaines L5 vertebrectomy	**Spondyloptosis** (grade V)
No	Isolated ASF, pars repair with listhesis

TO surgical indications	pain > **6–12 months** of conservative care, neuro deficit, worsening symptomatic listhesis, **grade ≥ III**, progression > 1 grade, instability > 3 mm · fusion avoided in pro athletes (adjacent segment disease: **36%** reoperation at 10 yrs after posterior lumbar fusion)
TO acute grade 3 in a gymnast, no deficit	**observation** initially (then elective fusion) – controversial
Return to sport after fusion	non-contact 3–6 months, contact 6–12 months · ALIF may help · 90% success

Reduction

Pros	restores sagittal alignment, ↓ lumbosacral kyphosis · TO: partial slip-angle reduction favoured for high grade
Cons	complications **8–30%** · **L5 root commonest injury** (**weak EHL / great-toe extension**, hip abduction, dorsal foot numbness) · sexual dysfunction · catastrophic deficit → neuromonitoring
360° fusion + decompression	more in-hospital complications

6 Complications

List	**L5 root injury (commonest neuro)**, pseudarthrosis, slip progression, hardware failure

X vs Y

	Dysplastic (I)	Isthmic (II)
Pars	Intact (elongated)	Defect
Neuro	**More** (intact arch traps cauda)	Less
Progression	**Higher**	Lower

How it's asked

"Most sensitive test when AP + lateral normal?"	→ **SPECT**
"13-yr, 6 months back pain, SPECT + CT?"	→ **spondylolysis**
"Lineman with insidious LBP — exam?"	→ **pain with lumbar extension in single-leg stance**
"Commonest exam finding in chronic spondylolysis?"	→ **pain with lumbar extension**
"14-yr, extension pain, negative X-rays — next?"	→ **MRI**
"15-yr lineman, extension pain — true?"	→ **best initial studies are AP + lateral alone**
"Gymnast, MRI finding fitting rest + PT?"	→ **pars fracture (Figure A)**
"Gymnast, SPECT bilateral pars stress fracture — next?"	→ **activity modification + rest** (TO)
"2-month-old bilateral pars defect?"	→ **antilordotic brace** (TO)
"13-yr gymnast, acute pain, SPECT hot L5 pars?"	→ **molded lumbosacral orthosis**
"Asymptomatic low-grade isthmic slip in soccer player?"	→ **observe, no activity restriction**
"Chronic L5 lysis, failed 3 months PT, no STIR signal?"	→ **LSO brace**
"7-yr gymnast with L5 pars defect — true?"	→ **likely never needs surgery**
"Spondylolysis, extension pain, hamstring tightness — initial?"	→ **activity modification ± brace + PT**
"Grade I isthmic slip in 13-yr gymnast?"	→ **short activity modification + structured PT**
"Parameter correlating with severity?"	→ **pelvic incidence**
"Lines measuring the angle best correlating with severity?"	→ **pelvic incidence lines (C + D)**
"Grade III — predicts progression?"	→ **slip angle**
"Factor associated with progression?"	→ **> Meyerding 2 at presentation**
"Gymnast acute grade 3 L5-S1, no deficit?"	→ **observation** (TO)
"12-yr, 80% slip, failed non-op?"	→ **posterior instrumented fusion L4-S1**
"Reduction of high-grade L5-S1 slip — complication?"	→ **weak great-toe extension (L5)**
"14-yr, grade 4, crouched gait — surgery?"	→ **posterior fusion L4–S1 ± instrumentation**
"Gait pattern in spondylolisthesis?"	→ **short stride, hips + knees flexed**
"Same lesion at L3 — surgical difference?"	→ **pars interarticularis repair indicated**

– – Active space – –

PSEUDOSUBLUXATION OF THE CERVICAL SPINE

– Spine · 48 / 69 –

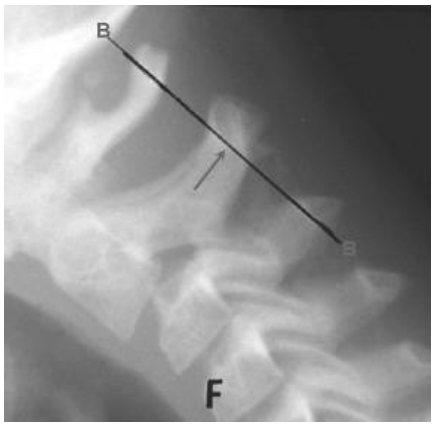
1 Basics

Definition	**physiologic** anterior translation (up to **4 mm / 40%**) in young children
Incidence	**~20%** of children admitted with polytrauma · **no association with gender**, trauma, intubation or injury severity ✗ Wrong options: commoner in girls
Age	**< 8 yrs**
Level	**C2 on C3 commonest** · C3 on C4 second
Cause	**horizontal facet joints** → become vertical with age

4 Diagnosis & Classification

Key radiology findings

Views	lateral + **flexion-extension**
Swischuk's line	spinolaminar line **C1 → C3** · C2 spinolaminar point within **1.5 mm** = pseudosubluxation



Swischuk's line

Pseudosubluxation vs true subluxation

Pseudosubluxation	True traumatic subluxation
Reduces on extension	Persists
Spinolaminar line **within 1.5 mm** of C2	> 1.5 mm
No significant trauma history/exam	Trauma (e.g. **Hangman's fracture**)
No anterior soft-tissue swelling	Prevertebral swelling



True subluxation — Hangman's fracture

5 Management

Observation no ↑ morbidity or mortality

How it's asked

"Pseudosubluxation — EXCEPT?"	**→ more common in female children** (TO)
"5-yr, no trauma, C2-3 step reduces on extension?"	**→ pseudosubluxation of the cervical spine**

– – Active space – –

RHEUMATOID CERVICAL SPONDYLITIS

– Spine · 49 / 69 –

1 Basics

Epidemiology

Prevalence	cervical involvement in **90% of RA** (often missed) · cervical spine = **2nd commonest RA site** after peripheral joints (TO) · **C1-C2 (synovial joint) = commonest segment** (TO)
Risk factors (TO)	older age at onset, active synovitis, **↑ CRP**, rapidly erosive peripheral disease, early joint subluxation
DMARDs	have **↓ cervical surgery** rates

Three patterns

Pattern	Frequency (Orthobullets)	Frequency (TO)
Atlantoaxial subluxation (commonest)	**50–80%**	**19–70%**
Basilar invagination / cranial settling	40%	38%
Subaxial subluxation	20%	7–29%

Others (TO)	inflammatory discitis, **atraumatic odontoid erosion / fracture**

Genes & lab markers

Anti-CCP	**most sensitive + specific RA marker** (✓ for C2 pannus) ✗ Wrong options: CRP/WBC (non-specific) · HLA-B27 (AS) · M-spike (myeloma)

3 Presentation

Symptoms	**neck pain** (craniocervical, 69%), stiffness · **occipital headache** (**lesser / greater occipital nerve, C2**), mastoid pain · **myelopathy** (hand clumsiness, gait instability, hyperreflexia) · vertebrobasilar: tinnitus, vertigo, visual change, dysphagia (TO)
C1-2 facet arthrosis / O-C1 ankylosis	**axial neck pain + ↓ rotation** with occipital headache ✗ Wrong options: Spurling · Hoffmann · Lhermitte

Ranawat classification (neurologic)

Class	Findings
I	Pain, no deficit
II	**Subjective** weakness, hyperreflexia, dysaesthesia
IIIA	**Objective** weakness + UMN signs, **ambulatory**
IIIB	Objective weakness + UMN, **non-ambulatory → do NOT operate** (poor results)

4 Diagnosis & Classification

Key radiology findings

Flexion-extension X-rays — **always before elective surgery** (anaesthesia)

CT — bony anatomy, **vertebral artery** course before transarticular screws

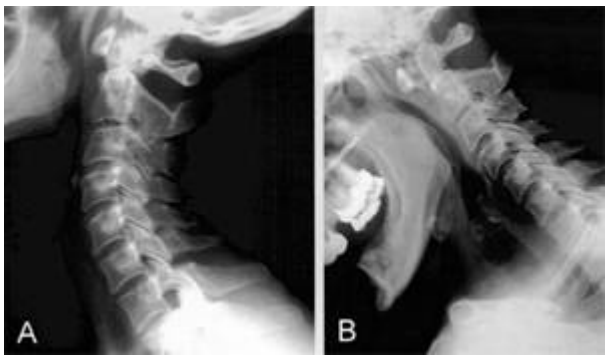
MRI — **cord compression + myelomalacia** (study of choice) · **cervicomedullary angle**

Atlantoaxial subluxation – diagnostic criteria

Measure	Threshold
AADI normal	**< 3 mm** adult (< 5 mm child)
Instability	**> 3.5 mm** change flexion↔extension (Orthobullets) · AADI > 3 mm not fixed (TO) – **not alone an indication for surgery**
> 7 mm motion	Alar ligament disruption
AADI surgery	**> 10 mm** (Orthobullets, OB) · 6–10 mm or > 8 mm cited (TO)
PADI / SAC surgery	**< 14 mm** (Orthobullets, OB) · **< 13–14 mm** (TO) · PADI < 14 mm **97% sensitive** for deficit
PADI predicting recovery	**> 13 mm** (Orthobullets, OB, TO) · **> 14 mm** (OB) – read the options

Best predictor of neuro deficit — **PADI** · cord 10 mm + 1 mm dura + 1 mm CSF each side → PADI ≥ 14 mm (TO)
✗ Wrong options: AADI

Mechanism — **pannus** between dens and C1 ring → destroys **transverse ligament + dens** · usually **anterior** C1 on C2



Dynamic atlantoaxial subluxation

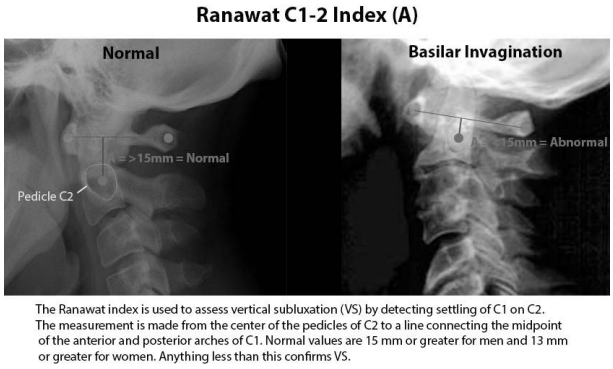
– – Active space – –

Basilar invagination – lines

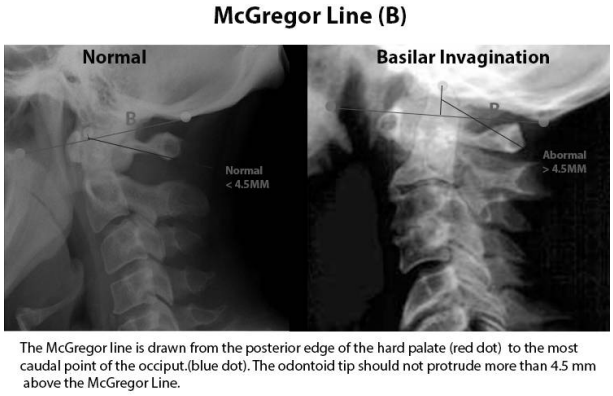
Line	Landmarks	Abnormal
Ranawat C1-2 index	**Centre of C2 pedicle → C1 arch line**	Normal **17 mm men / 15 mm women** · **< 13 mm = impaction** (Orthobullets) · **< 15 mm in men** (TO) · **most reproducible**
McGregor	**Posterior hard palate → most caudal occiput**	Dens tip **> 4.5 mm above** · **routine screening line** (TO)
Chamberlain	Dorsal hard palate → **posterior foramen magnum (opisthion)**	Dens **> 5 mm above** · hard to see
McRae	**Foramen magnum** opening (basion → opisthion)	Dens below = no impaction (asymptomatic)
Fischgold-Metzger	**AP view** – digastric grooves	Normally 10.7 mm above dens tip (TO)

Best combination for cranial settling (TO) **Clark station + Ranawat + Redlund-Johnell**

MRI **cervicomedullary angle < 135°** → impending deficit → surgery



Ranawat C1-2 index



McGregor's line

Subaxial subluxation

Features multilevel, with upper instability · ↑ with **steroids, males, seropositive, nodules, severe RA** · pannus + facet/Luschka laxity

Cord compression slip **> 4 mm or > 20%** · **cervical height index (body height ÷ width) < 2.0** → ~100% sensitive + specific for deficit

5 Management

Goal prevent progression; **may not reverse existing deficit** · pharmacotherapy first

Operative criteria

Pattern	Indication	Operation
AAS	**AADI > 10 mm** or **PADI < 14 mm** (even without deficit), **progressive myelopathy** · not 3.5 mm change alone	**Posterior C1-2 fusion** (reducible; **transarticular screws** avoid halo; CT for VA)
AAS + **basilar invagination** / irreducible AAS	–	**Occiput-C2 fusion ± C1 posterior arch resection**
Basilar invagination	**Progressive migration > 5 mm**, deficit, **CMA < 135°**	**Occiput-C2 fusion** · transoral / retropharyngeal odontoid resection for **brainstem compromise**
Subaxial	**> 4 mm / > 20% slip + intractable pain + neuro**	Posterior fusion + wiring

Pannus **resolves after posterior fusion** (stability) → **posterior fusion ± decompression** for periodontoid pannus myelopathy · **odontoidectomy** only for residual anterior compression after fusion
✗ Wrong options: transoral resection first (CSF leak, wound, more morbidity)

TO basilar impression surgery by **symptoms, not degree** · anterior impingement → **occiput-C1-2 fusion in extension** ± later anterior odontoid excision · posterior impingement → suboccipital craniectomy + C1 (± C2) laminectomy

6 Complications

Failure to improve **Ranawat IIIB**

Pseudarthrosis **10–20%** · ↓ by **extending to occiput**

Adjacent level degeneration

How it's asked

"Commonest RA cervical complication?"	→ **atlantoaxial instability** (TO)
"Commonest instability in RA?"	→ **atlanto-axial** (TO)
"RA cervical – FALSE?"	→ **"PADI more than 13–14 mm is a surgical indication"** (it's less than) (TO)
"Most useful measure predicting deficit?"	→ **PADI** (TO)
"RA cervical involvement occurs as?"	→ **C1-2 instability** (TO)
"Most frequently involved segment?"	→ **C1-C2** (TO)
"Spastic quadriplegia – good prognostic parameter?"	→ **PADI > 13 mm** (TO)

– – Active space – –

"RA myelopathy + AAS on flexion-extension?"	**→ posterior C1-C2 instrumented fusion**
"Predicts neural recovery after decompression?"	**→ PADI > 13 mm**
"Surgical indications EXCEPT?"	**→ 3.5 mm ADI change on flexion-extension**
"Predicts full recovery with stabilisation?"	**→ PADI > 14 mm**
"RA periodontoid pannus myelopathy – treatment?"	**→ posterior fusion ± decompression**
"Hard palate → most caudal occiput?"	**→ McGregor line** (TO)
"RA, occipital headaches, C1-2 arthrosis – exam?"	**→ axial neck pain + ↓ rotation**
"Radiographic surgical indication?"	**→ cervicomedullary angle 120°**
"C2 pannus – lab?"	**→ positive anti-CCP**

SACROILIAC JOINT DYSFUNCTION

– Spine · 50 / 69 –

1 Basics

Epidemiology

Incidence overlooked; **15–30% of outpatient LBP**

Risk factors **previous lumbar fusion, especially > 3 levels** (≈ adjacent segment disease) · **pregnancy + vaginal delivery** (3rd-trimester oestrogen/relaxin → hypermobility) · pelvic trauma · **iliac crest graft harvest**
✗ Wrong options: age > 65 · male sex · THA · prior SI injection

Causes

Mechanism	Examples
Idiopathic (commonest)	**Repetitive trauma** → capsule/ligament tension, shear, hypo/hypermobility
Arthritis	**~100% of spondyloarthropathies** (AS, Reiter's) · HLA-B27
Infection	Haematogenous, **unilateral** · S. aureus, Pseudomonas, Cryptococcus, TB · IVDU, endocarditis, immunosuppression
Metabolic	CPPD, gout, ochronosis, hyperparathyroidism, renal osteodystrophy, acromegaly
Tumour	Metastasis commonest (pelvis 40% of bony mets, 2nd to spine) · primary rare: GCT, PVNS, chondrosarcoma
Extra-articular	Enthesopathy (posterior ligaments), **insufficiency fractures**, post-traumatic (**lateral compression** pelvic injuries)

Anatomy & biomechanics

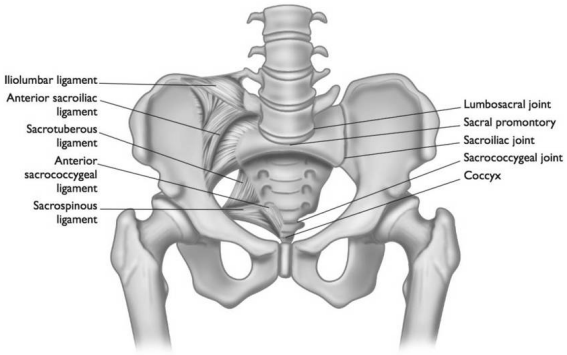
Joint **largest axial joint**, ~17.5 cm² · called synovial though **superior 75% non-synovial** · flat until puberty → ridges by 30 → erodes by 50 · ankylosis common in men by 50

Ligaments thin anterior capsule · **posterior interosseous ligament** (posterior capsule rudimentary) · sacrotuberous (→ ischial tuberosity), sacrospinous (→ ischial spine)

Innervation anterior **L2–S2 ventral rami** + sacral plexus · posterior **L4–S4 dorsal rami** (RFA targets **L5–S3 lateral branches**)

Biomechanics **triplanar shock absorber** · "keystone in arch" · takes 6× medial load of lumbar spine but fails at 1/20 of its axial load · motion **< 4° rotation, 1.6 mm translation** · **nutation (nodding) = commonest motion** (backward rotation of ilium on sacrum)

– – Active space – –



SI joint anatomy

3 Presentation

Pain	**just inferior to the PSIS** · only 4% above L5 · may radiate past knee · **relieved by a tight belt**
Tenderness	**sacral sulcus (most)** > **PSIS (2nd)**
Neuro	intact (pain-inhibited weakness only)

Provocative tests

Rule	no single test 100% · **> 3 positive = highly suggestive** ✗ Wrong options: 2

Test	Technique
FABER (Patrick)	Flexion, abduction, ER → SI pain (groin pain = hip/iliopsoas)
Fortin finger	Patient points **twice** to area **inferomedial to PSIS**
Gaenslen	**Affected hip extended off the table**, contralateral hip + knee flexed → shear
SI compression	Lateral decubitus, medial force on iliac crest
Anterior sacral thrust	Prone, anterior force on sacrum
SI distraction	Supine, posterior force on ASIS
SLR	Usually negative; may hurt **> 60°** (SI motion)



Gaenslen's test

4 Diagnosis & Classification

Diagnostic criteria – SI injection

Result	Meaning
> 75% pain relief after a single injection	**Confirms diagnosis**
> 50% relief after **two** injections	Diagnostic
Injection response	Does **not** predict pain relief after fusion

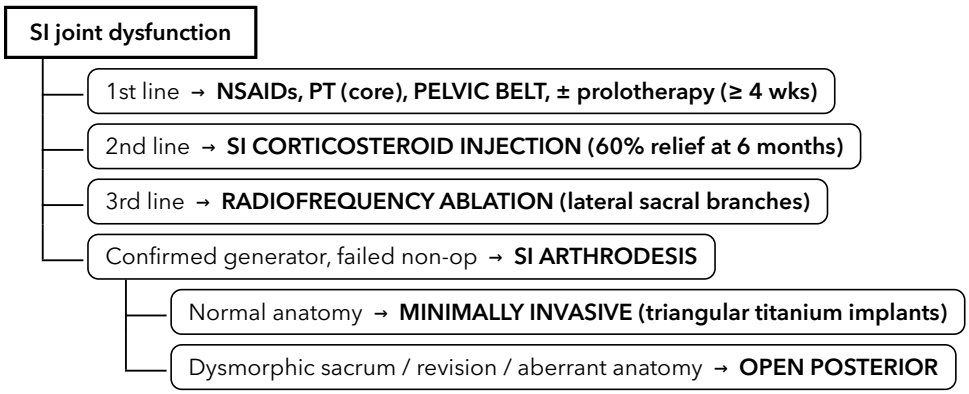
Guidance fluoroscopy/US (blind injection reaches joint only **22%**) · max **3 in 6 months / 4 per year**

Key radiology findings

Study	Fact
X-ray: pelvis AP, inlet, outlet, obliques, **SI views**, lumbar flexion-extension	Narrowing, sclerosis, cysts, osteophytes, ankylosis · **25% of asymptomatic > 50 abnormal**
Flamingo views	Single-leg standing → **pelvic instability**
CT	Surgical planning · poorer than injection diagnostically
MRI	Exclude tumour, infection
Bone scan	**Specificity 90%**, sensitivity 12%, PPV 86%, NPV 72%

Differential (top 5) lumbar stenosis, DDD, hip OA, hip labral tear, disc herniation

5 Management



Pelvic belt 4–8 inch belt around **greater trochanters** (medial compression) · best **after pregnancy**

Prolotherapy phenol/glucose at ligament bases → scarring · best for **ligamentous laxity**

Injection **lower success after lumbar fusion** · not first line

RFA limited (can't denervate anterior joint)

MIS vs open arthrodesis MIS: shorter stay, smaller incision, less blood loss, **earlier full weight bearing**, fewer SSIs (theory) · "fusion" by **ongrowth onto implants** · open: posterior approach, protected WB **12 wks**

– – Active space – –

6 Complications

Complication	Risk
SSI	Immunocompromise, smoking, diabetes
Wound	**Open** (dependent wound)
Nerve injury (L5, S1, S2)	**MIS**, **sacral dysmorphism**
Pseudarthrosis	**Up to 5%** → open revision

QoL worse than COPD + mild heart failure; ≈ hip/knee OA

How it's asked

"Buttock pain after spine fusion, >75% relief with injection – risk factor?"	**→ spine fusion**
"Post-fusion SI pain, positive provocative tests – confirm with?"	**→ diagnostic SI joint injection**
"SI dysfunction after 3-level fusion – true?"	**→ injection less successful after lumbar fusion**
"SI joint pathology – exam finding?"	**→ pain extending hip off table with contralateral hip/knee flexed (Gaenslen)**

SACROILIITIS

– Spine · 51 / 69 –

1 Basics

Definition	**inflammation** of the SI joint → LBP + posterior pelvic pain
Demographics	teens to middle age · **M > F**
Causes	chronic inflammation (**AS, Reiter's**), infection, trauma, overuse, **pregnancy** · may end in fibrosis + ossification
Septic	**IVDU**, haematogenous · gram-positive cocci (Staph ± Strep)

Genes & lab markers

Marker	Fact
HLA-B27	Ankylosing spondylitis · **1–2% of all people** carry it
RF	Usually **negative** in AS
ESR / CRP	**Usually ↑**
WBC	Usually normal; ↑ in infection
Blood cultures	Positive in **~50%** of septic cases

Reiter's	**oligoarthritis + conjunctivitis + urethritis**

3 Presentation

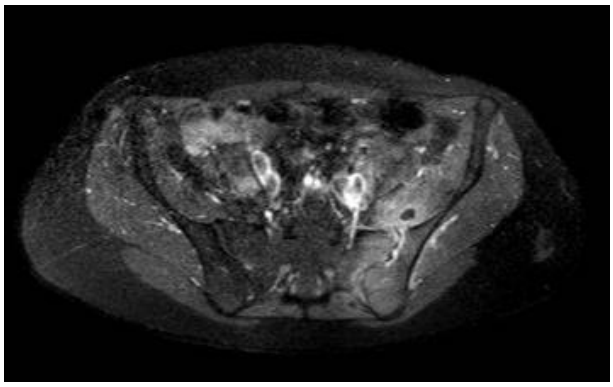
Symptoms	pain with prolonged standing, **difficulty climbing stairs**, generalised LBP, ipsilateral hip weakness, **morning stiffness**
Exam	**FABER → posterior (SI) pain** with otherwise painless hip ROM · AS: thoracolumbar flexion deformity
Septic	fever, leukocytosis, limp; may mimic hip sepsis (hip US/exam normal) · L5 root irritation possible

4 Diagnosis & Classification

Key radiology findings

Study	Finding
X-ray	**Non-specific erosions**, sclerosis, calcification
MRI with gadolinium	**Study of choice** – T2 **fluid / inflammation** in SI joint, **abscess**
Nuclear medicine	Historical

– – Active space – –



Septic sacroiliitis on axial T2 MRI

5 Management

Cause	Treatment
Infection	**IV antibiotics** (until symptoms + CRP settle) → oral · **surgery only if antibiotics fail or large abscess**
IVDU septic shock (GPC clusters + chains)	**Vancomycin + clindamycin** (✗ Wrong options: 1st-gen cephalosporin alone · immediate I&D)
Trauma / overuse	Rest, activity modification, NSAIDs, steroid injection
Pregnancy	**Observe** – resolves after delivery
Spondyloarthropathy	Aggressive PT, NSAIDs, **TNF inhibitors** / biologics

Septic sacroiliitis

- No abscess → **EMPIRIC BROAD-SPECTRUM IV ANTIBIOTICS, narrow on cultures**
- Failed antibiotics / large abscess → **SURGICAL DEBRIDEMENT**

Prognosis most resolve

X vs Y

	Septic sacroiliitis	Septic hip	Transient synovitis
Hip ROM	**Painless**	Painful	Painful
FABER	**Posterior pain**	Groin pain	–
Hip US	Normal	Effusion	Effusion
Treatment	**IV antibiotics**	I&D	NSAIDs

How it's asked

"IVDU, septic shock, SI fluid without abscess, GPC clusters + chains?"	**→ vancomycin + clindamycin**
"12-yr febrile, painful FABER, normal hip, SI fluid on MRI?"	**→ broad-spectrum IV antibiotics + monitor**

SCHEUERMANN'S KYPHOSIS

– Spine · 52 / 69 –

1 Basics

Epidemiology

Incidence	0.4–8.3% · **commonest structural kyphosis in adolescents** · 2nd commonest adolescent deformity after idiopathic scoliosis (TO)
Age / sex	onset **10–12 yrs** (diagnosed 12–17) · **M:F 2:1 to 7:1**
Location	**thoracic** (commonest) · thoracolumbar/lumbar less common

Genes & pathology

Inheritance	**autosomal dominant**
Theory (most accepted)	**developmental error in collagen aggregation → abnormal endplate** (defective endochondral ossification) · others: ring apophysis osteonecrosis, disc herniation, relative osteoporosis, biomechanical growth arrest
Histology (TO)	abnormal endplate cartilage, irregular mineralisation, **↓ collagen : proteoglycan ratio**
HLA-B27	association stated in one OB explanation

Associations

Association	Rate
Lumbar hyperlordosis	Compensatory
Lumbar spondylolysis	**33%** (Orthobullets) · **50–54%** (TO) – read the options
Scoliosis	**33%**
Dural cysts	–
Tight hamstrings, iliopsoas, anterior shoulder	–
Pulmonary compromise	Curves **> 100°**

3 Presentation

Symptoms	cosmesis, dull thoracic/lumbar pain (worse with activity, better with rest) · neuro deficit **rare**
Exam	**rigid kyphosis, sharper angulation on forward bending** · doesn't correct prone/extension · compensatory cervical/lumbar lordosis
Normal thoracic kyphosis	**20–45°** (Orthobullets) · **20–50°** T5-T12 (OB) – 38° in a 13-yr = normal · any thoracolumbar kyphosis abnormal

– – Active space – –



Clinical Scheuermann kyphosis

4 Diagnosis & Classification

Diagnostic criteria (Sorensen)

Criterion
Anterior wedging > 5° of 3 consecutive vertebrae (at apex)
Irregular endplates / apophyseal lines
Schmorl's nodes (disc into endplate)
Disc space narrowing
TO: + thoracic kyphosis **> 50°**

✗ Wrong options: **Traps** – posterior wedging, lumbar hypolordosis, preserved disc height, **marginal syndesmophytes (AS)**



Anterior wedging over three vertebrae

Key radiology findings

Study	Use
AP + lateral standing	Wedging, endplates, Schmorl's, scoliosis · **C7 plumb line** (sagittal balance)
Hyperextension lateral (supine over bolster)	**Separates postural kyphosis** · Scheuermann's is **inflexible**
MRI	Pre-op (controversial) – disc herniation, epidural cyst, cord, stenosis · **any neuro deficit**
Lumbar films	Spondylolysis if LBP

Types

	Thoracic (typical, type 1)	Thoracolumbar / lumbar (atypical, type 2)
Curve	T1/2–T12/L1, **apex T6–T8**	T4/5–L2/3, apex TL junction
Frequency	Commonest	Less common
Radiograph	Wedging	**Endplate irregularity, Schmorl's, narrow discs – less wedging**
Course	**Better prognosis**	**More pain, more progressive**
Surgical threshold	70–75°	**25–30°** (OB)

5 Management

Scheuermann's kyphosis

- < 60° (TO: < 50°), mild symptoms → **OBSERVE + PT (extension, hamstring stretch)**
- 60–80° with growth left → **EXTENSION BRACE (Jewitt / Milwaukee)**
- > 75°, neuro deficit, cord compression, severe adult pain → **POSTERIOR SPINAL FUSION ± PONTE / SPO**

Bracing

Thresholds **60–80° with growth remaining** (Orthobullets) · **50–74°** skeletally immature (OB) · **> 50° with ≥ 1 yr growth + 40–50% flexibility** (TO) – read the options

Brace **extension orthosis (Jewitt, high chest pad)** · **Milwaukee** (TO) full time 12–18 months then part time · TLSO if apex **T7–T9 or lower** · Boston for < 70°

Outcome stops progression; best if **< 65°** and **> 15° correction in brace** · compliance poor
✗ Wrong options: Charleston night brace · electrical stimulation

TO answer 12-yr, 55°, pain, no deficit → **Milwaukee brace**

– – Active space – –

Operative criteria

Indication	Source
Kyphosis > 75°	Orthobullets, OB
> 70–75° + pain not relieved + unacceptable cosmesis – all	TO
> 60° + persistent pain	OB
Neuro deficit, cord compression	All
Severe pain in adults	Orthobullets
Not if painless < 75°	OB

Technique — **posterior-only PSF, dual rods**, pedicle screws ± hooks · **cantilever** or **compression across posterior anchors** · **Ponte (posterior shortening)** for stiff curves: remove spinous processes, flavum, facets · anterior release rarely now · **MEP + SSEP monitoring**
✗ Wrong options: intracanal hooks at apex

Osteotomy choice — **Smith-Petersen** for **long, sweeping kyphosis** (~10°/level, less in rigid Scheuermann)
✗ Wrong options: PSO (AS, 30–40°/level) · VCR (sharp angular kyphosis)

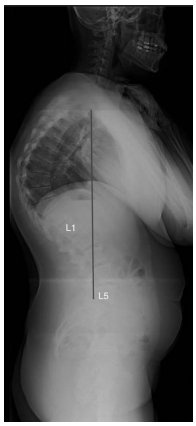
Goal — final kyphosis **40–50°** · **correct ≤ 50%** of original

Fusion levels

Proximal — whole kyphotic Cobb

Distal (LIV) — **sagittal stable vertebra (SSV)**: most proximal body **bisected by / ≥ 50% anterior to the posterior sacral vertical line** · old **first lordotic disc** method → more DJK

Too long (e.g. L5 when SSV is L3) — protects against DJK but **excessively stiff** → early disc degeneration



Posterior sacral vertical line selecting LIV

Outcomes — 60–90% pain improvement (not correlated with correction) · residual > 75° → worse function

6 Complications

Complication	Fact
Neurologic	**0.6–0.8%** (> AIS surgery) · cord stretch → ensure posterior shortening · reverse correction on signal change
Distal junctional kyphosis	**20–30%** · avoid: fuse to **SSV**, **don't overcorrect > 50%**
Proximal junctional kyphosis	**Overcorrection + negative sagittal balance** (less common)
Others	Pseudarthrosis, hardware failure, loss of correction, **SMA syndrome**

Junctional kyphosis Cobb **> 10°** between fused and unfused segments

Prognosis at least some progression in **80%** · adult pain rarely limits activity (mean 71°) · > 75° → severe thoracic pain

X vs Y

	Scheuermann's	Postural kyphosis
Rigidity	**Rigid**	Corrects on hyperextension
Wedging	> 5° × 3	None
Forward bend	**Sharp angulation**	Smooth

How it's asked

"Not true regarding Scheuermann's?"	→ **"cannot be corrected by postural control"** (TO key – read carefully; Orthobullets calls it rigid)
"13-yr, 38° thoracic kyphosis?"	→ **normal range**
"16-yr chronic back pain, wedged vertebrae – expected finding?"	→ **Schmorl's nodes**
"15-yr athlete, mid-back pain – underlying condition?"	→ **anterior wedging > 5° across three consecutive vertebrae**
"12-yr, 55° kyphosis with pain, no deficit?"	→ **Milwaukee brace** (TO)
"Risser 2, kyphosis corrects to 50°?"	→ **extension-type spinal orthosis**
"Surgical indication?"	→ **all – > 70° curve, pain + rigidity, unacceptable cosmesis** (TO)
"Fixed 80° kyphosis, no correction on hyperextension?"	→ **Smith-Petersen osteotomies + posterior arthrodesis**
"Fusion stopped at L5 beyond SSV (L3)?"	→ **protects against DJK but excessively stiff**
"14-yr Risser 4, rigid 90° kyphosis?"	→ **posterior spinal fusion + instrumentation**

– – Active space – –

SPINAL CORD ANATOMY

– Spine · 53 / 69 –

1 Basics

Cord overview

Length	~45 cm adult · foramen magnum (continuous with medulla) → **lower border of L1** (TO)
Conus medullaris	**L3 at birth → L1 at maturity** · lower L1 in women, L1 pedicle in men (OB)
Filum terminale	**modification of pia mater** (TO) · conus → adheres to dura → inserts on **first coccygeal segment** (TO)
Thecal sac	dura containing CSF, roots, cauda equina
Meninges	dura (outer) → arachnoid → pia (inner) · **CSF in subarachnoid space** (arachnoid–pia)

Embryology

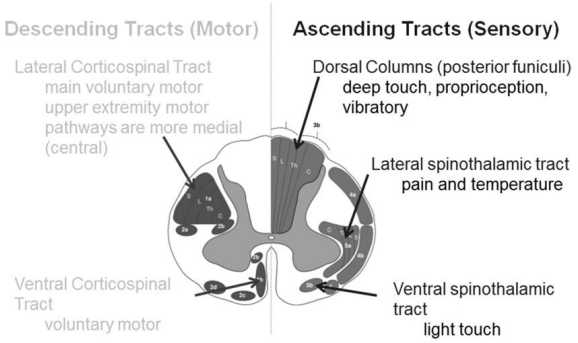
Structure	Becomes
Neural tube	Spinal cord · failed closure cranially → anencephaly · distally → spina bifida occulta, meningocele, myelomeningocele
Neural crest (dorsal to tube)	PNS, **pia mater**, spinal ganglia, sympathetic trunk
Notochord (ventral to tube)	Vertebral bodies, **nucleus pulposus** (annulus from sclerotome)

Autonomic

Sympathetic chain	**22 ganglia** (3 cervical, 11 thoracic, 4 lumbar, 4 sacral) · cervical: superior, **middle**, stellate · **middle ganglion at C6**, medial edge of **longus colli** → injury → **Horner's**
Hypogastric plexus	S2–S4 parasympathetic + lumbar sympathetic splanchnics

Functional tracts

Tract	Function
Dorsal columns	**Deep touch, proprioception, vibration**
Lateral spinothalamic	**Pain + temperature** · cordotomy site
Ventral spinothalamic	Light touch
Lateral corticospinal	Main voluntary motor · **UE fibres medial (central) → central cord: arms > legs**
Ventral corticospinal	Voluntary motor



Ascending tracts

CSF

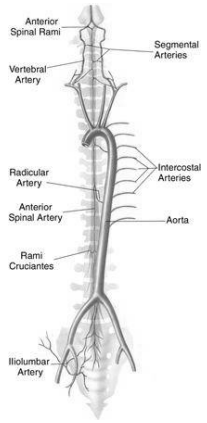
Production	**choroid plexus** (3rd, 4th, lateral ventricles) · ultrafiltrate of plasma · **~500 mL/day** · total **~150 mL** → turned over **3–4×/day** · mechanical + immunological protection

Blood supply

Vessel	Supply
Anterior spinal artery (single)	**Anterior 2/3** incl. lateral + ventral corticospinal tracts — **predominant supply**
Posterior spinal arteries (paired)	Posterior 1/3 — **dorsal columns**
Artery of Adamkiewicz	**Largest anterior segmental (radicular) artery** · from **left posterior intercostal** · **lower 2/3 of cord** via ASA

Adamkiewicz level	**left T8–L1 in 75%** (Orthobullets) · **left T9–L1** (OB, TO) · **T9–T11 in 80%** (TO, Dommisse) · range T5–L4, either side · "anywhere between T8 and T12" (TO true statement) — read the options
Adamkiewicz injury	**paraplegia NOT common** (TO EXCEPT) · thoracic watershed ischaemia with **sacral sparing** · ASA infarct → **anterior cord syndrome** (motor loss, preserved sensation/proprioception) · e.g. after wide **left anterolateral** corpectomy approach
Critical vascular zone (TO)	**T4–T9**: narrowest canal + poorest supply → paraplegia risk
Dommisse principles (TO)	3 longitudinal channels (1 anterior, 2 posterolateral) · largest at **cervical + lumbar enlargements** (grey matter demand) · 2–17 anterior and 6–25 posterior medullary feeders · **vertebral arteries supply 80% of neck radicular arteries** · reversible flow like circle of Willis

– – Active space – –



Artery of Adamkiewicz

Nerve root anatomy

Region	Exit
Cervical	**Above** same-numbered pedicle (C5 above C5) · horizontal · **extra C8 root**
Thoracic	**Below** same-numbered pedicle (T4 between T4-5)
Lumbar	**Below** pedicle (L5 below L5) · descend vertically
Intrathecal lumbar	**Exiting roots ventral**; **motor fibres ventral** to sensory

Why thoracic/lumbar roots lie below their vertebrae (TO)
7 cervical vertebrae, 8 cervical nerves · 31 pairs total

Vertebral artery (TO)
through transverse foramina **C1–C6**
✗ Wrong options: C7 (foramen present but artery usually outside)

Saddle anaesthesia, no other LL deficit (TO)
lesion **below S2** (S3: inner thighs, buttocks, perianal)

Neurogenic bladder (TO)

Lesion	Bladder
Above pons (micturition centre in pons, **locus coeruleus**)	**Loss of voluntary inhibition**, mild detrusor overactivity, **no DSD**, normal volume + pressure, no hypertrophy
Above S2 (suprasacral cord)	**Detrusor-sphincter dyssynergia**, high pressures → functional outlet obstruction
Below S2 / pelvic nerves	**Detrusor areflexia**, no DSD, **high volume, low pressure**

Neuroanatomy – brain & cranial nerves (TO)

Topic	Fact
Trigeminal-associated ganglia	**Ciliary (V1), otic + pterygopalatine (V2 per TO), submandibular (V3) – all four** · trigeminal ganglion is sensory
Brodmann	**4 primary motor** · 6 premotor/SMA · 3,1,2 somatosensory · **17 visual** · 41,42 auditory · **44 Broca** · 8 frontal eye field · **Wernicke = 39 + 40**
Internal acoustic meatus	Transverse crest (horizontal) + **Bill's bar** (vertical, deeper) · **"7 up, Coke down"** – facial + nervus intermedius anterosuperior, **cochlear anteroinferior** (✗ Wrong options: superior), vestibular nerves posterior
Vernet (jugular foramen)	**IX, X, XI** – taste posterior 1/3, palate/pharynx/larynx, trapezius + SCM · not **tongue deviation (XII, hypoglossal canal)**
Cavernous sinus	**Inside: VI (abducens) + ICA + sympathetic plexus** · lateral wall: III, IV, V1, V2 · "OH TOM CAT"
Annulus of Zinn	Through: III (sup + inf), **VI**, **nasociliary**, optic nerve + ophthalmic artery · **outside: frontal, lacrimal, trochlear**, superior ophthalmic vein
Facial nerve branches	**1st = greater superficial petrosal** (geniculate) → stapedius → chorda tympani → posterior auricular, digastric, stylohyoid → 5 terminal
Facial muscles	Orbicularis oris/oculi, buccinator · not **lateral pterygoid (V3)**
Facial nerve blood	**External carotid** branches
Olfaction	Unmyelinated, cribriform plate, entorhinal cortex · **only sense not relaying in thalamus** (✗ Wrong options: VPL)
Left optic tract	**Right homonymous hemianopia** (temporal right + nasal left)
Posterior pituitary	**Neuroectoderm** (3rd ventricle floor) · anterior = Rathke's pouch
Asterion	Lambdoid + occipitomastoid + parietomastoid sutures
Kocher's point	**1 cm anterior to coronal suture**, 2–3 cm from midline (mid-pupillary) – EVD into frontal horn
Vein of Labbé	**Sylvian fissure (superficial middle cerebral vein) → transverse sinus**
Pineal infratentorial supracerebellar approach	Sacrifice **superior vermian + precentral cerebellar veins** · not basal vein of Rosenthal

X vs Y

	Cervical	Thoracic / lumbar
Root vs pedicle	Above	Below
Reason	7 vertebrae, 8 nerves	–

How it's asked

"Spinal anatomy – EXCEPT?"	→ **vertebral artery through C1–C7 foramina** (it's C1–C6) (TO)
"Artery of Adamkiewicz – EXCEPT?"	→ **paraplegia after its injury is common** (TO)

– – Active space – –

"Critical vascular zone?"	→ **T4–T9** (TO)
"Predominant cord blood supply?"	→ **anterior spinal artery (A)**
"Motor loss with preserved sensation after left anterolateral corpectomy?"	→ **left-sided artery T9–L1 (Adamkiewicz)**
"Filum extends to?"	→ **first coccygeal segment** (TO)
"Saddle anaesthesia only?"	→ **below S2** (TO)
"Why thoracic/lumbar nerves exit below their vertebrae?"	→ **7 cervical vertebrae, 8 cervical nerves** (TO)
"Nerve between T4-5?"	→ **T4** (TO)
"Correct root–foramen grouping?"	→ **cervical roots above, lumbar below their pedicle (Group D)**
"Conus level in adults?"	→ **L1**
"Ganglia associated with trigeminal?"	→ **all of the above** (TO)
"Veins sacrificed in infratentorial supracerebellar approach?"	→ **superior vermian + precentral cerebellar** (TO)
"Brodmann 4?"	→ **primary motor cortex** (TO)
"IAM – NOT true?"	→ **cochlear nerve in superior-medial compartment** (TO)
"Vernet syndrome – FALSE?"	→ **tongue deviation** (TO)
"Wernicke's area?"	→ **Brodmann 39 + 40** (TO)
"Direct content of cavernous sinus?"	→ **6th cranial nerve** (TO)
"Posterior pituitary develops from?"	→ **neuroectoderm** (TO)
"Junction of lambdoid, occipitomastoid, parietomastoid?"	→ **asterion** (TO)
"Not through annulus of Zinn?"	→ **frontal nerve** (TO)
"Cranial nerve supplied by ECA?"	→ **VII** (TO)
"Olfaction – false?"	→ **relays to VPL of thalamus** (TO)
"Facial nerve supplies all EXCEPT?"	→ **lateral pterygoid** (TO)
"First branch of facial nerve?"	→ **greater superficial petrosal nerve** (TO)
"Left optic tract lesion?"	→ **temporal right + nasal left field loss** (TO)
"Kocher's point?"	→ **1 cm anterior to coronal suture** (TO)
"Vein of Labbé connects?"	→ **Sylvian fissure to transverse sinus** (TO)
"Urgency, detrusor overactivity, no DSD, normal pressures?"	→ **lesion above the pons** (TO)

SPINAL CORD INJURIES

– Spine · 54 / 69 –

1 Basics

Epidemiology

Incidence 11,000/yr US · **incomplete tetraplegia 34% (central cord commonest)** · complete paraplegia 25% · complete tetraplegia 22% · incomplete paraplegia 17%

Distribution bimodal: young high-energy · elderly minor trauma + spondylotic stenosis · **50% cervical** · **MVA 50%**, falls, GSW · **3–25% occur after the initial trauma** (poor immobilisation/transport)

Pathophysiology

Primary injury direct, **irreversible**

Secondary injury ↓ perfusion, **lipid peroxidation**, free radicals/cytokines, apoptosis → target of methylprednisolone

Associated head injury, **non-contiguous spine fractures**, **vertebral artery injury** (atlas fractures, facet dislocations; usually asymptomatic unilateral; **MRA** least invasive; **stent only if basilar insufficiency symptoms**)

3 Presentation

Spinal vs neurogenic vs hypovolaemic shock

	Spinal shock	Neurogenic shock	Hypovolaemic
BP	↓	↓	↓
Pulse	**Bradycardia**	**Bradycardia**	**Tachycardia**
Bulbocavernosus reflex	**Absent**	Variable / **can be intact**	Variable
Motor	**Flaccid areflexic**	Variable	Variable
Mechanism	Neurons **hyperpolarised**, unresponsive	**Loss of sympathetic tone** (injury above **T6**), ↓ SVR, pooling, unopposed vagus	↓ preload
Skin / urine	–	**Warm extremities**, urine output preserved (✗ Wrong options: **absent urine output**)	Cold

Bulbocavernosus reflex anal contraction on squeezing glans/clitoris or **tugging the Foley** · **its return = end of spinal shock** · conus/cauda equina injury → **may never return**
✗ Wrong options: "returns in all cauda equina injuries" (TO)

Spinal shock ruled out BCR present **or** 48 h elapsed

– – Active space – –

Spinal shock phases

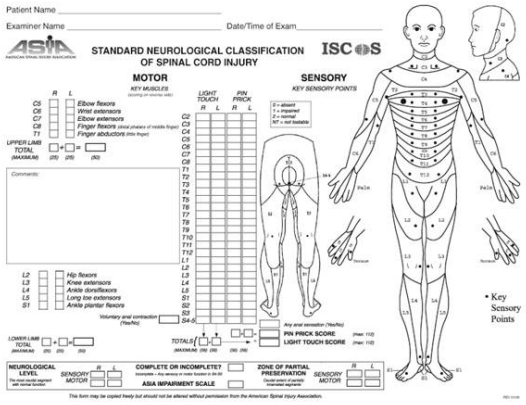
Phase	Orthobullets	TO (Ditunno)
1	**0–48 h** areflexia	**0–24 h** areflexia; **delayed plantar** first, then BCR, abdominal, cremasteric
2	1–2 days: **polysynaptic (BCR) return**, monosynaptic (patellar) absent	Day 1–3: cutaneous reflexes, DTRs absent
3	1–4 wks hyperreflexia	Day 4–1 month: DTRs return, Babinski
4	1–12 months spasticity	1–12 months: hyperreflexia, **autonomic dysreflexia**

First reflex to return — **delayed plantar** (TO) · classical teaching **bulbocavernosus** — read the options · order: delayed plantar → BCR → cremasteric → ankle jerk → Babinski → knee jerk

4 Diagnosis & Classification

ASIA impairment scale

Grade	Motor	Sensory
A Complete	None	None (incl. **S4-5**)
B Incomplete	**None** (none > 3 levels below NLI)	**Preserved** (incl. sacral)
C Incomplete	Preserved; **> half key muscles < grade 3**	Preserved
D Incomplete	Preserved; **≥ half key muscles ≥ grade 3**	Preserved
E	Normal	Normal



ASIA classification

Diagnostic criteria – classifying an injury

Step 1 – neurologic level — **lowest segment with normal motor + sensory both sides** (e.g. intact wrist extension, absent triceps → **C6**)

Step 2 – complete? — **no voluntary anal contraction + 0/5 distal motor + no perianal sensation + BCR intact** → complete

Step 3 – grade — any **perianal sensation = at least B** (sacral sparing → incomplete)

BCR absent → cannot assign ASIA grade (spinal shock) · prognosis by **repeated examinations over 48–72 h**

✗ Wrong options: MRI · SSEP · EMG

Descriptive — **tetraplegia** (cervical: arms, trunk, legs, pelvis) · **paraplegia** (arms spared)

ASIA B at C6 (OB) — deltoid 5/5, long-finger + medial forearm sensation may be present, intrinsics 0/5 · no **triceps power** (2/5 OB; 3/5 in the unkeyed TO variant)

Incomplete cord syndromes

Syndrome	Mechanism	Deficit	Prognosis
Central cord (commonest)	**Hyperextension in elderly spondylotic neck** (cord pinched between body + buckled flavum)	**UE > LE**, distal > proximal (UE fibres medial), sacral sparing	Good
Brown-Séquard	Hemisection, **penetrating**	**Ipsilateral motor + proprioception/vibration**, **contralateral pain + temp** (2–3 levels below)	**Best**
Anterior cord	Hyperflexion / compression, **anterior spinal artery** (e.g. **aortic dissection**)	Motor + pain/temp lost, **proprioception preserved**	**Worst** (~90% no recovery)
Posterior cord	Rare (tabes dorsalis atraumatic mimic)	Proprioception lost	–
Conus / cauda equina	–	–	Conus recovers better than proximal lesions

Rear-end collision → quadriparesis (TO) — **extension–distraction** (whiplash mechanism)

5 Management

Field & initial care

Immobilisation — rigid collar + spine board (transport only) · **decubiti after 30–60 min on board** · log roll · athletes: **leave helmet + shoulder pads on** until controlled simultaneous removal

Airway — injury **above C5** → likely intubation

Exam — remove collar, inspect face/scalp, rotational head posture (unilateral facet dislocation), **posterior midline tenderness** (absent in alert patient → low probability of injury), log roll, neuro exam · **seat-belt sign → flexion-distraction**

Prevention — DVT prophylaxis (LMWH, rotating bed, pneumatic compression) · avoid hypotension · pulmonary toilet · skin

Neurogenic shock treatment

TO — IV fluids first → **vasopressors/inotropes** if persists · phenylephrine (pure α1 → reflex bradycardia) · bradycardia: **atropine**, glycopyrrolate (before suction), isoproterenol, theophylline/aminophylline

– – Active space – –

OB	hypotension unresponsive to initial fluids → **pressors** · Swan-Ganz-guided fluids ✗ Wrong options: more rapid fluid → overload, CHF

Methylprednisolone

Current	**not recommended**: no clear benefit, **↑ complications** (infection, GI bleed, ventilator days, LOS) · no effect on cord oedema · OB variant: "**controversial, may be given in appropriate circumstances**" — read the options
Historical NASCIS III	non-penetrating SCI **within 8 h** · **30 mg/kg bolus** + **5.4 mg/kg/h** for **23 h (< 3 h)** or **47 h (3–8 h)** · CI: **GSW, pregnancy, < 13 yrs, > 8 h**, peripheral nerve injury
Presenting at 10 h	**don't start methylprednisolone**
GM-1	controversial; no long-term benefit · **hypothermia** — not recommended (coagulopathy, sepsis, pneumonia, arrhythmia)

Reduction & surgery

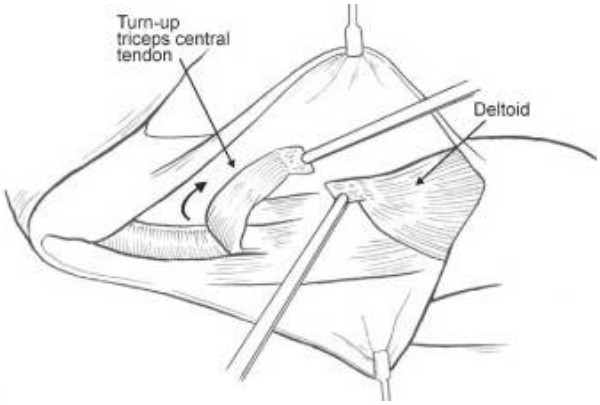
Closed reduction	**awake, alert** patient with deficit from fracture-dislocation (commonest: **bilateral facet dislocation**) · add 10 lb every 10 min with exams · **abort**: overdistraction, **neuro worsening (→ remove traction)**, failure
TO (stable C4-5 fracture-dislocation with weakness)	**MRI first** (cord, compression, disc) ✗ Wrong options: immediate traction
Decompression + stabilisation	**most incomplete and complete SCI (except GSW)** · **early (< 24 h) improves recovery even in complete injury** · may recover 1–2 root levels · stabilise for rehab
GSW	**most → brace / treat extraspinal injury** (stable lateral masses, no CSF leak) · surgery for **progressive deficit with canal bullet**, **cauda equina**, **bullet in thecal sac (lead poisoning)**
C6 SCI + spondylotic C6 fracture	**early decompression + stabilisation alone** ✗ Wrong options: steroids · hypothermia

Functional level

Level	Function
C1-3	Ventilator; electric chair head/chin control
C3-4	Ventilator → may wean; head/chin (**C4 puffer**)
C5	Deltoid, biceps · **electric wheelchair with hand control** · can't feed (no wrist extension/ supination)
C6	Wrist extension + supination → feeds self · **manual wheelchair + sliding-board transfers** · drives with hand controls
C7	Triceps · **manual wheelchair + independent transfers**
C8-T1	Hand dexterity · fully independent transfers
T2-6	Normal UE, wheelchair
T7-12	Abdominal control; bracing walk possible
L1-5	Variable; aids
S1-5	Walk with minimal assistance

Posterior deltoid → triceps transfer

complete C5/C6 with **5/5 deltoid + biceps, 0/5 triceps**
✗ Wrong options: axillary nerve injury · weak deltoid



Posterior deltoid-to-triceps transfer

FES

stimulates **skeletal muscle**

TENS

presynaptic inhibition in the dorsal horn (small myelinated afferents block C-fibre input) + endorphins

– – Active space – –

6 Complications

Complication	Fact
Pressure injury	Prevention: **frequent repositioning**, off board early · NPUAP 2016: "**injury**", **Arabic stages** (TO keys: "Grade II pressure injury" FNB 2022 · "Grade 2 pressure injury" – read the options)
VTE	Immediate prophylaxis
Urosepsis	**Common cause of death**
Sinus bradycardia	**Commonest arrhythmia** acutely
Orthostatic / supine hypotension, arrhythmia	Lost sympathetic control (**T6 or above**) · not **spasticity** (not sympathetic)
Autonomic dysreflexia	**Above T5-6** · headache, agitation, **hypertension** (SBP ↑ ≥ 20%), bradycardia, sweating · **unchecked visceral stimulus** → check **Foley**, disimpact, X-ray for **occult fracture** (swollen thigh)
Major depression	**~11% (5–20%)** · suicidal ideation
Shoulder	Long-term paraplegia → **4× rotator cuff tears**
Post-traumatic instability	Missed adjacent ligamentous slip below construct → **revise + extend fusion**

Prognosis

Complete	**80% gain 1 root level**, 20% gain 2 · **1% full recovery** · **ASIA A: least recovery, highest lifetime cost** (ASIA B 46.6% marked recovery)
Incomplete	more sparing + faster early recovery = better · plateau rarely resumes
MRI predicting complete injury	**intramedullary cord haematoma** ✗ Wrong options: oedema · ALL disruption · prevertebral oedema
SCIWORA	commoner < 10 yrs · **severity of initial deficit = best predictor**

Dural tears (TO)

Commonest site	**lateral**, then root sheath, axilla · **suture whenever possible** · sealants (Tachosil, fibrin glue) if not · small unsuturable → tight fascia + skin ✗ Wrong options: posterior

SIADH vs cerebral salt wasting (TO)

	SIADH	Cerebral salt wasting
Volume	**Euvolaemic / hypervolaemic**	**Hypovolaemic**
Serum osm	< 275	↓
Urine osm	> 100	–
Urine Na	**> 40 mEq/L**	High
Criteria exclude	Diuretics, thyroid/adrenal disease (✗ Wrong options: **renal insufficiency is not a criterion**)	–
Supplementary	Uric acid < 4, corrected by fluid restriction, ↑ ADH	–
Treatment	Na < 125 acute/severe → **3% saline** + furosemide · correct **≤ 8 mEq/L per 24 h** (osmotic demyelination) · chronic → **fluid restriction**, demeclocycline, conivaptan	Saline

Whiplash (TO)

Mechanism rear impact → lower cervical **hyperextension** + upper flexion (S-shape) · commonest MVA injury (up to 83%) · "WAD"

Poor outcome predictors **rapid-onset pain**, **severe early pain**, hospital admission, **radiation to arm**, **headache** · **least predictive: pain isolated to neck** (TO key)

Management MRI not initially (false positives) · **early mobilisation** > soft collar

How it's asked

"Spinal shock – EXCEPT?"	→ **BCR returns in all cauda equina injuries** (TO)
"Cervical fracture-dislocation, no motor/sensation/anal tone, BCR intact?"	→ **complete SCI** (TO)
"Spinal shock – HR, BP, BCR?"	→ **bradycardia, hypotension, absent BCR** (TO)
"First reflex to return after spinal shock?"	→ **delayed plantar** (TO)
"ASIA B at C6 – NOT seen?"	→ **triceps power** (TO unkeyed; OB: **2/5 triceps**)
"Thoracic trauma, bradycardia, no catheter-tug response – ASIA?"	→ **cannot be determined** (TO unkeyed; OB same scenario keyed)
"Wrist extension 5/5, triceps 0/5, no sacral function, BCR intact?"	→ **complete C6 (ASIA A)**
"Grade 3 in most muscles, sensation intact, BCR intact?"	→ **ASIA D**
"No motor, diminished perianal + LE sensation?"	→ **ASIA B**
"ASIA B motor definition?"	→ **no motor > 3 levels below NLI**
"Grade 2 power in 80% of LE key muscles, sacral sensation intact?"	→ **ASIA C**

– – Active space – –

"Complete injury — early decompression?"	→ **improves chance of neurologic recovery**
"Biceps 4/5 left, deltoid 5/5, nothing below, no sacral function?"	→ **complete C5**
"No recovery at 48 h with BCR back?"	→ **complete cord syndrome**
"BCR absent — best prognostic tool?"	→ **repeat physical examinations**
"Diver, HR 57, BP 92/70, absent BCR?"	→ **unable to determine ASIA**
"Neurogenic shock — NOT seen?"	→ **absent urine output** (TO)
"Hypotension + relative bradycardia, BCR present?"	→ **neurogenic shock**
"Expected vitals in neurogenic shock?"	→ **HR 55, BP 65/40**
"Hypotension unresponsive to fluids after cervical SCI?"	→ **pressors**
"Methylprednisolone 10 h after injury?"	→ **don't start**
"Methylprednisolone in SCI — evidence?"	→ **not supported; more complications, no clear benefit**
"C6 fracture SCI — treatment?"	→ **early decompression + stabilisation**
"ASIA A C5-6 facet dislocation — steroids?"	→ **controversial, may be given in appropriate circumstances**
"Best prognosis syndrome?"	→ **Brown-Séquard** (TO)
"Central cord mechanism?"	→ **hyperextension in elderly with hypertrophied facets + flavum** (TO)
"Worst prognosis syndrome?"	→ **anterior cord** (TO)
"Rear-end crash, quadriparesis?"	→ **extension–distraction** (TO)
"82-yr hyperextension, UE weakness, LE spared?"	→ **central cord syndrome**
"Left neck GSW, Brown-Séquard?"	→ **left limb weakness, right pain/temp loss**
"LE weakness + pain/temp loss after aortic dissection, proprioception kept?"	→ **anterior cord syndrome**
"Brown-Séquard deficits?"	→ **ipsilateral motor, contralateral pain + temp**
"Right C7 pain/temp loss, left C5 motor + vibration loss?"	→ **left hemisection at C5**
"Complete C5 function?"	→ **electric wheelchair with hand control**
"Deltoid-to-triceps transfer candidate?"	→ **C6 ASIA A, 5/5 biceps**
"C6 SCI maximal function?"	→ **manual wheelchair + sliding-board transfers**
"SCIWORA — best predictor?"	→ **severity of initial neurologic injury**
"MRI predicting complete SCI?"	→ **6-mm cord haematoma**

"Lowest recovery + highest cost?"	→ **ASIA A**
"Quadriplegic sacral sore — description?"	→ **Grade II / Grade 2 pressure injury** (TO — keys differ)
"T6 lesion — NOT sympathetic loss?"	→ **spasticity**
"FES acts on?"	→ **skeletal muscle**
"MDD in SCI?"	→ **5–20%**
"Paraplegic since 14 — shoulder MRI?"	→ **rotator cuff tear**
"Slip below thoracolumbar construct on sitting films?"	→ **revise + extend posterior fusion**
"T3 paraplegic, dysreflexia, swollen thigh?"	→ **occult fracture**
"ASIA C → B during traction?"	→ **remove traction**
"Low-velocity GSW, stable, no CSF leak?"	→ **treat extraspinal injuries + orthosis**
"TENS mechanism?"	→ **presynaptic inhibition in the dorsal horn**
"C4-5 fracture-dislocation with weakness, stable — next?"	→ **MRI** (TO)
"Not an SIADH criterion?"	→ **renal insufficiency** (TO)
"SIADH vs CSW?"	→ **euvolaemia** (TO)
"Dural tears — true?"	→ **suture whenever possible** (TO)
"Least predictive of poor whiplash outcome?"	→ **pain isolated to neck** (TO)

– – Active space – –

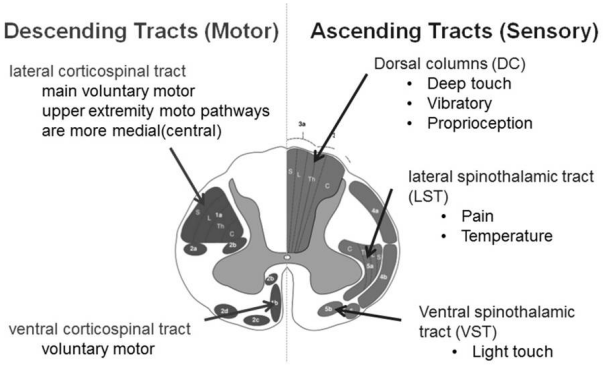
SPINAL CORD MONITORING

– Spine · 55 / 69 –

1 Basics

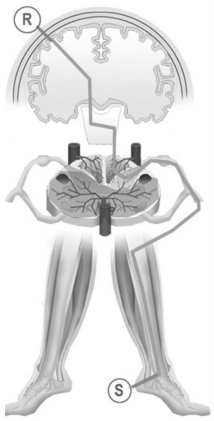
Modalities

Modality	Monitors	Stimulate → record	Accuracy
SSEP	**Dorsal columns** (afferent; gracilis LE, cuneatus UE)	**Posterior tibial** (ankle) / peroneal; **ulnar/median → transcranial recording of somatosensory cortex** (or neck)	**25% sensitive, 100% specific**
tcMEP	**Lateral + ventral corticospinal tracts** (efferent; anterior 2/3, ASA territory)	**Transcranial motor cortex →** muscle (gastroc, soleus, EHL)	**100% sensitive, 100% specific · most sensitive early indicator**
Spontaneous (mechanical) EMG	Specific **nerve roots**	Surgical manipulation → muscle · **burst = contact (insignificant) · sustained train = significant traction/injury**	Over-sensitive
Triggered (electrical) EMG	**Pedicle breach** (medial/inferior wall → exiting root)	Stimulate screw → muscle · bone insulates	–
Stagnara wake-up test	Gross motor (move feet)	Intraoperative waking (amnestic)	Not continuous; delays



Monitored pathways and blood supply

	SSEP	MEP
Anaesthetic effect	**Unaffected / reliable**	**Highly affected** (TO trap: "MEPs not affected" = false)
Anterior cord ischaemia	**Unreliable** (paralysis with normal SSEP reported)	**Detects**
Existing motor deficit	–	**Still measurable** (TO)



Anterior infarct missed by SSEP

4 Diagnosis & Classification

Alert criteria

Modality	Significant change
SSEP	**≥ 50% ↓ amplitude and/or ≥ 10% ↑ latency** (or loss) · not 10% amplitude / 50% latency
MEP	**> 50% ↓ amplitude** · **> 100 V ↑ threshold** (early injury)
Triggered EMG	**< 8 mA → breach** (Orthobullets, OB) · Stecker: < 10 mA inspect, **< 5 mA likely malpositioned** · **> 8 mA (e.g. 11, 12, 14 mA) = intact pedicle**

Lumbar (L4-5) surgery (TO) **triggered EMG**
✗ Wrong options: SSEP · MEP · D-waves (cord monitors)

Patient with existing motor deficit (TO) **transcranial MEPs**

5 Management

Responding to signal loss

Significant signal change

- Check technical → **LEADS, EQUIPMENT, interference (irrigation, cautery, fluoroscopy, warming blanket)**
- Check anaesthesia → **NO PARALYTIC / INHALATIONAL bolus, steady state**
- Optimise physiology → **RAISE MAP (80–90 TO · > 90 OB), Hb > 8, O2, reverse hypocapnia + hypothermia**
- Surgical → **REVERSE LAST MANOEUVRE (distraction/correction), REMOVE IMPLANT, widen decompression**
  (signals may take ≥ 10 min to return; close 3-column osteotomy)
- Still absent → **STAGNARA WAKE-UP TEST**

TO EXCEPT ✗ Wrong options: **lower BP to reduce blood loss** (raising MAP recovers signals in ~50%)

– – Active space – –

Corpectomy cage placed → 75% SSEP ↓ + MEP loss	**remove the cage** and monitor
Deformity correction → sustained 80% SSEP ↓, technical + anaesthetic causes excluded	**reverse correction ± wake-up test** ✗ Wrong options: steroids · remove all instrumentation · continue
DMD child, diffuse 80% MEP ↓, MAP 55 on propofol	**restore MAP > 90 and retest** ✗ Wrong options: remove one screw (diffuse change) · switch to halogenated gas
Most concerning in AIS correction	**65% drop in LE MEPs** ✗ Wrong options: 30% SSEP ↓ · 5% latency ↑ · 14 mA screw
Most concerning in PSO	**> 100 V MEP threshold rise**

Reversible causes

Commonest non-anaesthetic reversible cause	**patient positioning** (nerve compression, plexus traction, neck alignment) ✗ Wrong options: screw · ischaemia · retractor
Unilateral cortical SSEP loss only during ACDF (subcortical + Erb's normal)	**cerebral ischaemia from carotid retraction** (incomplete circle of Willis)
Hypothermia SSEP drop (TO)	**flood the field with warm saline** (fastest – exposed pathway) ✗ Wrong options: warmer air, room, blankets

Physiological effects on evoked potentials (TO)

Variable	Effect
Cerebral blood flow	Normal to 20 · deteriorate 18–13 · **absent < 10–12 mL/100 g/min**
Hypotension	↑ latency
↑ ICP	↑ latency, ↓ amplitude
PaO_2	Deteriorate **< 40 mmHg**
Haematocrit	Latency ↑ at 10–15%; amplitude ↓ < 10%
$PaCO_2$	Extreme hypocapnia **< 25 mmHg**
Hypothermia	**↑ latency**; no recording < 18 °C
Inhalational agents	Depress: **N_2O > isoflurane > halothane**
IV agents	**Propofol** no spinal effect (cortical depression recovers fast) · opioids, ketamine none · benzodiazepines mild

EMG myotomes (TO)

Root	Muscle
C5	Deltoid, biceps
C6	Biceps, triceps, brachioradialis, PT, FCR
C7	Triceps, PT, FCR, forearm extensors
C8 / T1	Hand intrinsics (APB, FDI, ADM), FCU
L2–L4	Iliopsoas, adductors, quadriceps
L5	**Tibialis anterior, peroneus longus**
S1-2	**Gastrocnemius, abductor hallucis**
S2–S5	Anal, urethral sphincter
CN VII / XI / XII	Facial muscles / SCM + trapezius / tongue

How it's asked

"Neuromonitoring – EXCEPT?"	→ **"tcMEP unaffected by anaesthetics vs SSEP"** (TO)
"Monitoring in a patient with existing motor deficit?"	→ **transcranial MEPs** (TO)
"AIS fusion monitoring – true?"	→ **SSEPs monitor the dorsal columns**
"SSEPs – true?"	→ **unreliable for anterior cord pathways**
"Best detector of evolving motor tract injury in cervical surgery?"	→ **tcMEP**
"Most sensitive early indicator of cord injury?"	→ **transcranial MEPs**
"SSEP alert criteria?"	→ **50% ↓ amplitude, 10% ↑ latency**
"Most concerning change during AIS correction?"	→ **65% LE MEP drop**
"Cage placed, SSEP + MEP loss?"	→ **remove strut cage, monitor**
"80% SSEP drop during correction, causes excluded?"	→ **reverse correction ± wake-up test**
"DMD, diffuse MEP drop, MAP 55?"	→ **restore MAP > 90, retest**
"PSO – most concerning change?"	→ **> 100 V MEP threshold increase**
"Isolated right cortical SSEP loss during ACDF?"	→ **cerebral ischaemia from retraction**
"Commonest non-anaesthetic reversible cause?"	→ **patient positioning**
"Signal loss – steps EXCEPT?"	→ **lower BP to reduce blood loss** (TO)
"SEP drop from hypothermia – fastest fix?"	→ **warm saline into the field** (TO)
"L3 screw stimulates at 11 mA?"	→ **intact pedicle**
"L4-5 surgery – which monitoring?"	→ **triggered EMG** (TO)

– – Active space – –

SPINAL CORD TUMORS

– Spine · 56 / 69 –

1 Basics

Epidemiology

Primary CNS tumours	mostly intracranial · only **2–4% spinal** (850–1,700/yr US)
Compartments	**intradural extramedullary** (60–70%) · **intradural intramedullary** (20–30% adults, **50% children**) · extradural



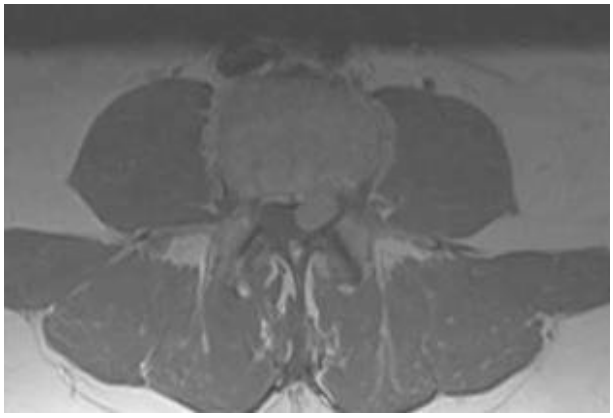
Spinal tumour compartments

3 Presentation

Compartment	Symptoms
Intradural extramedullary	Cord compression, **radicular pain**, deep **back pain worst at night**, cauda equina-like
Intramedullary	Local/radicular pain, motor (65%), sensory (60%), **sphincter dysfunction (38%) – often first symptom**

Intradural extramedullary

	Schwannoma	Meningioma
Frequency	**Commonest intradural tumour** (with meningioma)	**25%** of primary spinal tumours
Demographics	**40s–50s, M = F**	**Women 80%**, 5th–6th decades
Site	**Dorsal nerve root** · cervical 31%, cauda equina 24%, thoracic 22%	**Thoracic**
Risk	**NF2**	**NF2**, prior radiation
MRI	T1 iso, **T2 hyperintense**	**Dura-based**, T1 iso/hypo, T2 slightly hyper, homogeneous enhancement
Histology	**Antoni A (hypercellular) + B (hypocellular)**, **S-100+**	Lobulated, **meningothelial whorls**
Treatment	Resection ± RT if malignant	Resection (curative if complete); RT for recurrence



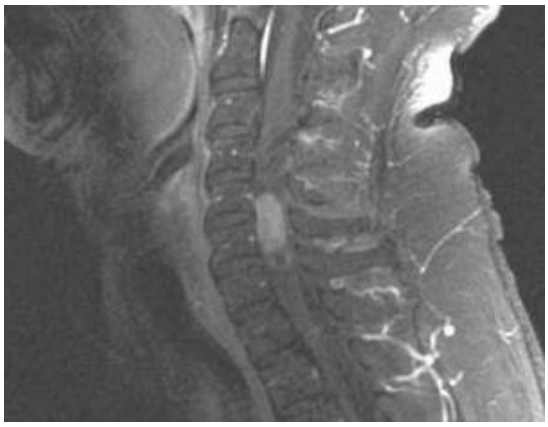
Schwannoma from dorsal root

Intramedullary

	Ependymoma	Astrocytoma
Frequency	**Commonest intramedullary (adults)** (Orthobullets)	**Commonest in children (80–90%)** · **commonest overall** (TO)
Age	Adults	Peak **3rd decade**
Site	**Cervical** commonest spinal site (TO) · myxopapillary in **filum terminale**	**Cervicothoracic** junction (children)
Types	Myxopapillary 40–50%, cellular 42% (benign) · anaplastic	30% benign
MRI	T2 hyper, heterogeneous · myxopapillary: **scalloping, foraminal enlargement, scoliosis, sacral seeding** · cellular: **polar cysts**, haemorrhage	**Fusiform, irregular margins**, infiltrative
Histology	**Rosettes + pseudorosettes**	Hypercellular, mitoses, **eosinophilic granules**
Treatment	**Gross total resection** – good prognosis (clean plane)	GTR difficult → **RT** ± chemo

– – Active space – –

TO	paediatric ependymoma intracranial (**4th ventricle**); adult spinal



Ependymoma polar cyst

Extradural & metastatic

Metastasis	extramedullary = advancing disease · **drop metastases** via CSF · intramedullary rare (< 1%) from **lung > breast**, cervical + conus · **intramedullary mets → life expectancy < 3 months** · RT/chemo
GBM drop metastasis (TO)	**intradural extramedullary** · 1–2% of GBM · CSF spread (debulking ↑ seeding) · lag ~8.5 months · steroids for rapid cord compression
Lymphoma	image whole neuraxis · T2 ill-defined hyperintense, marked homogeneous enhancement, little cord enlargement · **cervical** · **methotrexate**

5 Management

Metastatic cord compression

Frameworks	**SINS** (instability) + **NOMS** (neurologic, oncologic/radiosensitivity, mechanical, systemic) · **Tokuhashi 9–11 → 6–12 months**
Renal cell metastasis with anterior compression, myelopathy	**pre-op embolisation → corpectomy + instrumented fusion → post-op RT** ✗ Wrong options: posterior decompression alone · surgery without embolisation · pre-op RT
Unkeyed TO item	structure spared and cut last during intramedullary tumour removal (denticulate ligament / midline myelotomy / functional root / DRG) — answer not provided by source

Brain tumour facts (TO)

Fact	Answer
Commonest brain tumour	**Metastases** (> 50%)
Commonest primary (adult CNS)	**GBM**
Commonest primary non-glial	**Meningioma**
Commonest paediatric CNS tumour	**Astrocytoma** (~50%), then medulloblastoma
Commonest paediatric brain malignancy / vermian tumour	**Medulloblastoma**
Brainstem glioma presentation	**Cranial neuropathies first** (then weakness/ataxia; headache, vomiting late) · 77% < 20 yrs
Petroclival meningioma	Feeders from **ICA dural branches (meningohypophyseal trunk)** · PVA particles not more complications · liquid agents (NBCA, Onyx) infrequent
Arachnoid cyst	Commonest **Sylvian fissure** (middle fossa) · M:F 4:1, left · EMA+, CEA– · middle fossa cysts bleed (bridging veins) · suprasellar → hydrocephalus, precocious puberty, **bobble-head doll** · incidental adult → 1 repeat scan at 6–8 months

How it's asked

"Renal cell met with cord compression, 6–12 months expected?"	→ **embolisation, corpectomy + fusion, post-op RT**
"Commonest site for spinal ependymoma?"	→ **cervical** (TO)
"Commonest intramedullary tumour?"	→ **astrocytoma** (TO — overall; ependymoma in adults)
"GBM drop metastases?"	→ **intradural extramedullary** (TO)
"Petroclival meningioma — true?"	→ **feeders from internal carotid artery** (TO)
"Brainstem glioma first presents with?"	→ **cranial neuropathies** (TO)
"Commonest arachnoid cyst location?"	→ **Sylvian fissure** (TO)

– – Active space – –

SPINAL EPIDURAL ABSCESS

– Spine · 57 / 69 –

1 Basics

Epidemiology

Definition	pus / granulation tissue between **dura and epidural fat**
Patients	usually **> 60 yrs** · 2–10 per 10,000 admissions (TO) · location usually **dorsal thoracolumbar** · cervical SEA commoner in **IVDU injecting arms/ neck** (TO)
Risk factors	**IVDU**, immunodeficiency, HIV, malignancy, immunosuppressants, **recent spinal procedure**, diabetes, endocarditis

Pathogenesis

Routes	**haematogenous ~50%** (commonest non-iatrogenic) · contiguous discitis ~33% · direct inoculation
Nidus (TO)	**skin** commonest ✗ Wrong options: respiratory
Organisms	**Staph aureus 50–65%** · gram-negatives (E. coli 18%) · **Pseudomonas in IVDU**
Association	**spondylodiscitis** (SEA in ~18% of spondylodiscitis)
Neurologic	deficit in **33%** · permanent paralysis **4–22%** (compression or cord infarction) · risk highest cervical, lowest lumbar (TO)

3 Presentation

Symptoms	**severe insidious pain (87%)** · **fever only ~50%** · sicker than osteomyelitis · **rapid neuro progression (days)** (TO) · meningism, radicular pain · sequence: back ache → root pain → weakness → paralysis
Exam	radiculopathy or myelopathy (clonus, Hoffmann → look above)

Labs

Marker	Fact
WBC	Mean 22,000; raised in only **~42%**
ESR	Raised **> 90%**
CRP	Raised **90%**

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray	Usually normal
CT	Poor sensitivity
CT myelogram	90% sensitive, invasive
MRI + gadolinium	**Study of choice** · **ring-enhancing lesion = pathognomonic**, T2 bright · pus vs CSF · extent, osteomyelitis, compression



Cervical epidural abscess on MRI

Whole-spine MRI (skip lesions)

Indication
Concomitant infection outside the spine
Delayed presentation > 7 days (strongest predictor among listed options ; ✗ Wrong options: IVDU · diabetes · CRP · fever)
ESR > 95 mm/h
Neuro signs not explained by the lesion found (e.g. Hoffmann with lumbar SEA; clonus with L2-3 discitis)

5 Management

Spinal epidural abscess

- Neuro intact, small abscess, minimal compression, can be observed → **ADMIT, IV ANTIBIOTICS ± BRACE**
  (also non-surgical candidates)
- Deficit / cord compression / failure risk / persistent infection / instability → **SURGICAL DECOMPRESSION ± FUSION**
  - Posterior abscess, no discitis → **LAMINECTOMY (avoid wide facetectomy)**
  - Anterior abscess / spondylodiscitis → **ANTERIOR DEBRIDEMENT + STRUT ± POSTERIOR FIXATION**

– – Active space – –

Failure of medical treatment

Predictor	Note
Neurologic deficit	**Strongest predictor**
Diabetes	–
CRP > 115 mg/L	–
WBC > 12–12.5 k	–
Positive blood cultures	–
Age > 65	–
MRSA	–
No	Smoking, IVDU

> 40% of medically managed patients fail (pain, deficit)

Neuro-intact 53-yr, MSSA bacteraemia — **admit, brace, IV cefazolin 6 wks** · serial exams
✗ Wrong options: discharge on oral linezolid · laminectomy

Most likely to need surgery — **67-yr diabetic, leg weakness, MRSA bacteraemia** (all risk factors)

Operative

Indications — **neuro deficit**, cord compression on imaging, failure risk factors (> 65, diabetes, new deficit, MRSA), persistent infection, deformity/instability

Antibiotics after surgery — **2–4 wks if no bone involvement · 6 wks if bony involvement**

Cervical anterior abscess + deficit (IVDU) — **anterior decompression + fusion**
✗ Wrong options: posterior · steroids · antibiotics alone

Cervical spondylodiscitis C5-7 + kyphosis + deficit + sepsis — **anterior debridement, corpectomy, fusion + posterior instrumentation** (instrument despite infection)

Lumbar 3-level destruction + abscess after endocarditis — **I&D, corpectomy, fusion**

Antibiotic timing — hold until intraoperative cultures if stable (OB 213125)

Prognosis

Best predictor — **pre-operative neuro deficit** · mortality ~5% · early diagnosis key · only 18% with frank abscess / 23% with paralysis fully recover

Intracranial infections (TO)

Brain abscess — **congenital cyanotic heart disease**, endocarditis, lung, sinusitis (**frontal/ethmoid → frontal lobe**), otitis, trauma · overall **Streptococcus**; trauma S. aureus; transplant fungi; post-neurosurgery S. epidermidis/aureus · stages: early cerebritis 1–3 d → late cerebritis 4–9 d → early capsule 10–13 d → late capsule > 14 d · IV antibiotics 6–8 wks then oral · medical alone if **< 3 cm, cerebritis stage, < 2 wks** symptoms

Neurocysticercosis	**Taenia solium** larvae (pork / contaminated vegetables) · **commonest CNS parasite**, commonest acquired epilepsy in low-income countries · C. cellulosae: 5–15 mm transparent cyst with **scolex** · stages: vesicular → colloidal → granular nodular → calcific · **seizures** commonest · **EITB serology** ~100% specific · **MRI choice** (scolex pathognomonic) · treatment: **albendazole** + steroids (start before) + antiepileptics ± surgery/shunt

Pneumorrachis (TO)

Definition	**air in the spinal canal** (traumatic, iatrogenic, spontaneous) · mostly cervical · from pneumothorax/pneumomediastinum tracking via retropharynx
Gas from necrotising fasciitis / emphysematous osteomyelitis	hypointense epidural lesion · CT best (can't separate air from gas) · usually resolves conservatively

How it's asked

"53-yr neuro-intact SEA, MSSA?"	→ **admission, bracing, IV cefazolin 6 wks**
"Most likely to need surgery?"	→ **67-yr diabetic, leg weakness, MRSA bacteraemia**
"Strongest predictor of failure in IVDU with weakness?"	→ **lower extremity weakness**
"Lumbar SEA + positive Hoffmann – next?"	→ **MRI cervical + thoracic spine**
"IVDU, diffuse weakness, anterior cervical abscess?"	→ **anterior decompression + fusion**
"Post-endocarditis, 3-level lumbar destruction, weakness?"	→ **I&D, corpectomy, fusion**
"C5-7 spondylodiscitis, kyphosis, deficit, sepsis?"	→ **anterior debridement, corpectomy, fusion + posterior instrumentation**
"Cervical anterior abscess, 1 month fevers, weakness?"	→ **anterior decompression + fusion**
"SEA – EXCEPT?"	→ **commonest nidus is respiratory infection** (TO – it's skin)
"Post-endocarditis back pain, clonus – imaging?"	→ **thoracic ring-enhancing epidural collection (Figure B)**
"L2-3 discitis, neuro signs above – next?"	→ **MRI of the entire spine**
"Factor predicting skip lesions?"	→ **duration of symptoms (≥ 7 days)**
"Large fronto-parietal lesion + congenital heart disease?"	→ **brain abscess** (TO)
"Multiple white transparent cysts?"	→ **neurocysticercosis** (TO)
"Necrotising fasciitis, hypointense epidural lesion?"	→ **pneumorrachis** (TO)

– – Active space – –

SPINAL TUBERCULOSIS

– Spine · 58 / 69 –

1 Basics

Epidemiology

Pott's disease	M. tuberculosis spondylitis · ↑ with immunocompromise (**HIV, CD4 50–200**)
Frequency	15% of TB extrapulmonary · **spine = commonest extrapulmonary site** (thoracic) · 5% of all TB
Level (TO)	**lower thoracic** > lumbar > upper thoracic > cervical > sacral · children: **anterior lower thoracic** (OB)

Pathoanatomy

Seeding	haematogenous to end-arterioles in the **anterior vertebral body next to the endplate (metaphysis)** · **commonest pattern: anterior** (TO key) · spreads **under the ALL** → multilevel, **skip lesions (15%)**, **paraspinal abscess (50%)** – large, anterior (» pyogenic)
Disc	**spared early** (vs pyogenic disc-first) → may mimic tumour
Aneurysmal phenomenon (TO)	**long-standing abscess** → concave anterior scalloping of bodies · thoracic abscess = **bird-nest shadow**
End result in spine (TO)	**bony ankylosis** (weight-bearing + cancellous ends) vs fibrous in limb joints

Kyphosis

Adults	static after healing · mean 15° non-op; > 60° in 5%
Children	**progresses in 40%** after healing (growth) · **anterior + posterior element involvement = best predictor of progression** ✗ Wrong options: age · level count · bracing timing; thoracic-only does better than thoracolumbar

Rajasekaran progression	Course
Type I	↑ until growth stops → **surgery**
Type II	Decreasing progression with growth
Type III	Minimal change

3 Presentation

Onset	insidious · malaise, **night sweats**, weight loss · back pain often late
Earliest (TO)	symptom **back pain** · sign **paraspinal muscle spasm** · first neuro sign **ankle clonus** (motor before sensory – anterior disease near motor tracts), then extensor plantar

Neuro deficit — **10–47%** · abscess, granulation, debris, caseation, instability, **ossified flavum** near severe kyphosis · upper thoracic commonest (narrow canal) · Pott's paraplegia 10–30% (TO)

Pott's paraplegia (TO)

	Early onset	Late onset
Timing	**Active disease, < 2 yrs**	**> 2 yrs**, healed disease
Cause	Oedema, granulation, abscess, caseation, ischaemia	**Deformity** (internal gibbus), debris, sequestra, stenosis
Prognosis	**Better**	Worse
Treatment	ATT ± decompression	Deformity correction + stabilisation

Commonest cause of late-onset paraplegia — **deformity** (TO)

Prognosis by cause (Orthobullets) — abscess / bony sequestra better than **meningomyelitis**

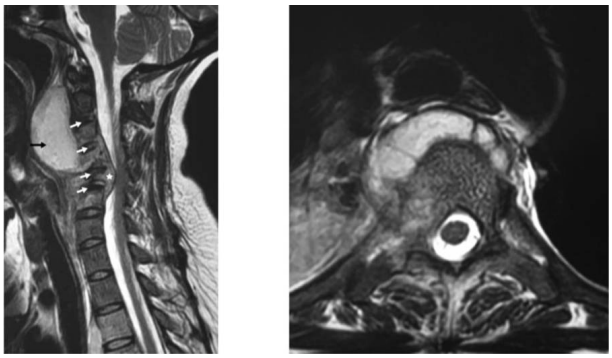
Modified Tuli stage	Findings
1	Subtle — clonus, brisk reflexes, Babinski
2	UMN deficit + spasticity, **ambulatory**
3	**Bedridden**, spastic, lateral column sensory loss
4	Bedridden, severe sensory loss (posterior + lateral), sores
5	Stage 4 + **bladder/bowel**, flexor spasms, flaccidity

4 Diagnosis & Classification

Key radiology findings

Study	Fact
CXR	Abnormal in **66%** — always obtain
X-ray early	Anterior body involvement, **disc preserved**
X-ray late	Collapse, disc destruction, kyphosis
MRI + gadolinium	**Preferred** · T1 low, T2 high · **septated smooth-walled para/prevertebral abscess with subligamentous spread** · multilevel 16–70% · endplate disruption (sens 100%, spec 81%), paravertebral shadow (97/85) · cord oedema, myelomalacia, syrinx
CT	Lesions < 1.5 cm; fragmentary, osteolytic, subperiosteal, sclerotic destruction · poor for epidural extent
Tc + gallium scan	Highest sensitivity for infection

– – Active space – –

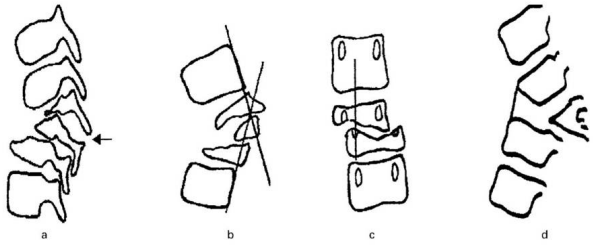


Subligamentous TB abscess on MRI

Disc-confined lumbar abscess pyogenic, not TB (TO)

Spine-at-risk signs (Rajasekaran)

✓ Sign
Facet subluxation / separation at apex
Retropulsion (posterior displacement)
Lateral translation
Toppling
✗ Wrong options: kyphosis angle (> 50°, > 55°) · abscess · neuro worsening



Spine-at-risk signs

Labs & tissue

Test	Fact
CBC	Relative lymphocytosis, low Hb
ESR	Usually ↑ (normal in 25%)
PPD	**+ in ~80%**
CT-guided biopsy	**Most specific test (TO)** · **AFB** stain (smear + 52%), culture + 83% (**up to 10 wks**), **PCR** 95% sensitive
No	**Empirical antibiotics without biopsy** (TO)

Differential Actinomyces, Nocardia, **Brucella**, fungi (Coccidioides, Blastomyces, Cryptococcus, Aspergillus), syphilis

Pyogenic vs TB (TO)

Treat as pyogenic	Treat as TB
Acute, septic, severe pain	Subacute / chronic
Lumbar, disc-predominant, no big abscess	**Skip / multiple lesions**
Older, immunocompromised	**Paraspinal abscess**
–	**Thoracic / thoracolumbar**

5 Management

Medical

Regimen — **HRZE 2 months → HR 9–18 months** (Orthobullets) · **HRZE 2 months → HRE ≥ 10 months, total 12–16 months, daily** (TO, India) · with surgery: continue H, R, Z

HIV + cervical Pott's with Ziehl-Neelsen + — **isoniazid, rifampin, pyrazinamide, streptomycin**

Orthosis — pain + deformity prevention · **steroids only for arachnoiditis/meningitis** (TO)

Operative criteria

Indication	Detail
Neuro deficit	Worsening deficit, acute severe paraplegia, **not responding to ≥ 3 wks ATT** (TO)
Instability	Panvertebral disease ± subluxation · **Ganga score > 3**, **≥ 2 spine-at-risk signs** (TO)
Kyphosis	**> 60° adults** (Orthobullets) · **thoracic > 40°, lumbar > 30°** (TO) · **progressive in a child** · ≥ 3 vertebrae with loss of ≥ 1.5 bodies (thoracic) · child ≤ 7 with ≥ 3 bodies + ≥ 2 at-risk signs
Late-onset paraplegia	Kyphosis correction
Others	**Caseation** blocking drugs, failure after 3–6 months, uncertain diagnosis, > 4 levels (TO), large abscess

Benefits — less kyphosis, faster healing, fewer sinuses, better neuro recovery · implant colonisation rarer than pyogenic · autograft or allograft struts fine

– – Active space – –

Approach

Situation	Approach
Classic	**Anterior decompression / corpectomy + strut ± posterior instrumentation ± posterior shortening** (single-stage transpedicular, two-stage, extrapleural anterolateral)
TO modern	**Posterior / transpedicular** preferred (lung already involved; diseased body poor for screws) · **posterior abscess (e.g. D6-7) → posterior laminectomy** · lumbar: anterior abscess drainable via laminectomy (dura retractable)
TO progressive deficit, T7-9, no paravertebral collection	**Posterior decompression + stabilisation**
TO worsening paraplegia + bladder after 3 wks ATT	**Decompression + evaluate for MDR-TB** (✗ Wrong options: add streptomycin · steroids · repeat MRI)
Cervical kyphosis	Halo traction → anterior decompression, graft, plate
Lumbar kyphosis	**Pedicle subtraction osteotomy**
Healed thoracic/TL kyphosis	**Internal kyphectomy** (cord moved anteriorly)
Transthoracic	**Left side** (aorta easier than IVC)
Atypical TB (granuloma)	Laminectomy (extradural/subdural) · myelotomy (intramedullary)

Rajasekaran (TO) active disease: posterior stabilisation then anterior debridement + graft · healed rigid deformity: anterior first then posterior

Drug-resistant TB (TO)

Rif-resistant on Xpert next **second-line LPA / DST for fluoroquinolone resistance** (MDR vs pre-XDR) · XDR work-up: SL-LPA + **gene sequencing** (Bdq/Lzd)

WHO 2022 group	Drugs
A (all three)	**Levofloxacin/moxifloxacin, bedaquiline, linezolid**
B	Clofazimine, cycloserine/terizidone
C	Ethambutol, delamanid, pyrazinamide, amikacin/streptomycin, ethionamide

Regimens shorter all-oral **9–12 months** (FQ-sensitive) · longer **18–20 months** · **BPaLM** 6 months · injectables last resort · never add a single drug to a failing regimen

XDR (2021) MDR + FQ resistance + bedaquiline or linezolid resistance

Monitoring **bedaquiline → QTc (ECG) · linezolid → myelosuppression (CBC)**

6 Complications

Deformity highest after **anterior graft alone** (graft slips/breaks, ≥ 2 levels) · lowest with **anterior + posterior fusion**

Others retropharyngeal abscess (dysphagia, hoarseness), TB arteritis/pseudoaneurysm, costopelvic impingement, **sinus formation**, Pott's paraplegia

Atypical spinal TB	compressive myelopathy **without visible deformity** · intraspinal granuloma, neural arch disease, concentric collapse, sclerotic bridging
Spina ventosa (TO)	**tubercular dactylitis**: phalanges of the hand, destruction + new bone, **spindle-shaped finger swelling**

How it's asked

"Aneurysmal phenomenon in dorsal Pott's?"	→ **long-standing abscess** (TO)
"MRI NOT suggestive of TB?"	→ **abscess confined to lumbar disc space** (TO)
"Commonest TB spine pattern?"	→ **anterior** (TO)
"End result of TB spine?"	→ **bony ankylosis** (TO)
"9-yr gibbus – true?"	→ **vertebral body involved, disc spared**
"Commonest site in children?"	→ **anterior lower thoracic**
"Early finding in Pott's?"	→ **preserved disc space**
"Late-onset paraplegia cause?"	→ **deformity** (TO)
"T7-9 lesion, progressive deficit, no collection?"	→ **posterior decompression + stabilisation** (TO)
"Worsening paraplegia + bladder after 3 wks ATT?"	→ **decompression + evaluate for MDR TB** (TO)
"D6-7 posterior abscess failing ATT – approach?"	→ **posterior laminectomy** (TO)
"HIV, cervical Pott's, ZN positive – drugs?"	→ **INH, rifampin, PZA, streptomycin**
"Predicts kyphosis progression after healing in child?"	→ **anterior + posterior element involvement**
"Xpert: rifampicin resistance – next?"	→ **second-line LPA / DST for fluoroquinolones** (TO)
"Group A drugs?"	→ **bedaquiline, linezolid, fluoroquinolone** (TO)
"Monitoring on bedaquiline + linezolid?"	→ **ECG QTc + weekly CBC** (TO)
"Tool for FQ + Group A resistance?"	→ **SL-LPA + targeted gene sequencing** (TO)
"Spine-at-risk EXCEPT?"	→ **kyphosis > 55° / > 50°** (TO)
"Spine-at-risk sign?"	→ **facet subluxation** (TO)
"First neurological sign?"	→ **ankle clonus** (TO)
"Earliest sign of Pott's?"	→ **paraspinal muscle spasm** (TO)
"Most specific test for spinal infection?"	→ **biopsy** (TO)
"Spina ventosa?"	→ **tubercular dactylitis** (TO)

– – Active space – –

SPINE BIOMECHANICS

– Spine · 59 / 69 –

1 Basics

Functional spinal unit

FSU cephalad + caudad vertebrae, disc, facet joints · physiologic motion + neural protection · intradiscal pressure varies with position

Stability no abnormal strain or excess motion under physiologic load · maintained by FSU, muscle tension, abdominal/thoracic pressure, rib cage

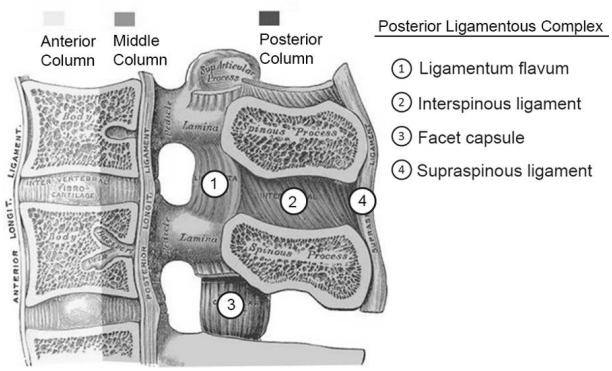
Activity	Intradiscal Pressure (kPa)	Compressive Force (N)	% Weight Above L3-L4
Standing	270	380	100%+
Twisting	480	670	180%
Lateral Bending	620	870	230%
Flexion	710	990	260%
Extension	720	1,010	270%
Flexion + 40 N in Each Hand	1,620	2,270	600%

Intradiscal pressure by posture

Denis three columns

Column	Structures
Anterior	**ALL**, **anterior 2/3** of body + annulus
Middle	**PLL**, **posterior 1/3** of body + annulus
Posterior	Pedicles, facets, **ligamentum flavum**, spinous process, **PLC**

Instability middle column injury (**widened interpedicular distance**, loss of posterior body height) · PLC disruption + anterior and/or middle column injury · model only moderately reliable



Denis three columns

Ligaments

Ten ligaments restrict FSU motion, absorb energy · all **type I collagen** except **ligamentum flavum (mostly elastin)** · viscoelastic, non-linear

Posterior ligamentous complex

Components	**supraspinous**, **interspinous**, **ligamentum flavum**, **facet capsules** ✗ Wrong options: facet joints · PLL
Ruptured if	bony Chance fracture, interspinous widening, progressive kyphosis on non-op care, facet diastasis · ambiguous: MRI signal between spinous processes
TLICS	PLC intact 0 / indeterminate / ruptured 3 · burst without deficit + intact PLC = 2 → **non-op (< 4)** · + ruptured PLC = 5 → **surgery (> 4)**

4 Diagnosis & Classification

Sagittal balance

Segment	Normal
Cervical lordosis	**20–40°**
Thoracic kyphosis	Average **35°** (20–50°)
Lumbar lordosis	Average **60°** (20–80°) · **75% at L4–S1**, **47% at L5-S1**

Gravity line	centre of C2 → anterior T7 → mid T12-L1 disc → posterior to L3 → posterosuperior S1 · radiograph: **C7 plumb line** to posterosuperior S1
Negative balance	axis **behind** sacrum (lumbar hyperlordosis) · **positive** – axis **in front** (hip flexion contracture, **flat back**)
Best radiographic predictor of good outcome after adult deformity surgery	**sagittal balance correction** ✗ Wrong options: coronal · rotation · listhesis · PI
Pelvic incidence	fixed angle; **correlates with lumbar lordosis, thoracic kyphosis, sagittal alignment** (PT + SS change with pelvic rotation)



Negative sagittal balance

– – Active space – –

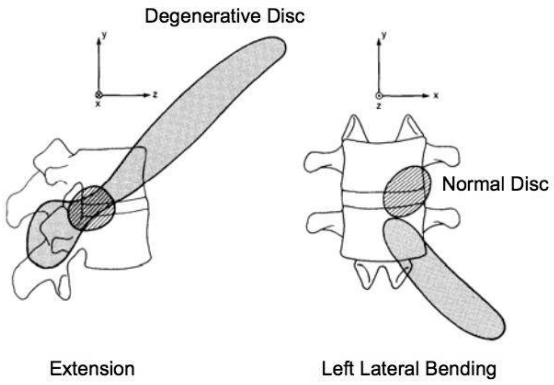
Facet orientation & motion

Region	Coronal / sagittal angle	Motion
Cervical (C3-7)	**0° coronal, 45° sagittal**	Flex-ext, lateral bend, rotation
Thoracic	**20° coronal, 55° sagittal**	Rotation; little flex-ext (ribs) · protects heart/lungs
Lumbar	**50° coronal, 90° sagittal**	**Flex-ext**, minimal rotation

Instantaneous axis of rotation

Definition **theoretical axis about which an FSU rotates at an instant** · not fixed; moves with bending/rotation · **3 axes, 6 degrees of freedom**

Pathology **degeneration, fracture, ligament injury shift the IAR** outside the FSU (enlarged)
✗ Wrong options: fixed · 12 DOF · from AP/lateral films



IAR in degenerative disc

Not IAR CORA (deformity correction) · vertebral incidence (L5) · plumb line (C7)

Pedicle anatomy

Region	Fact
C2, C7	Pedicle screws viable
C3–C6	Small pedicles → **lateral mass screws**
Thoracic wall	**Medial wall 2× thicker than lateral**
T4	**Narrowest diameter + shortest (14 mm)** (T1 20 mm, T10 20 mm)
T7	Narrow on concave side in AIS
T12 vs L1	T12 larger than L1
Transverse angle	**10° mid-thoracic → 30° L5**
Sagittal angle	**15–17° cephalad** thoracic · neutral lumbar · caudal at L5
Lumbar	**L1 smallest lumbar diameter** · medial angulation L1 12°, L5 30°, S1 39° · S1 ~19 mm
Landmarks	Transverse process midpoint (cranio-caudal) · lateral pars border (medio-lateral)

5 Management

Instrumentation (TO)

System	Approach
Harrington rods	Posterior
Luque rods (sublaminar wires)	Posterior
Steffee plate + pedicle screws	Posterior
Dwyer, Zielke	**Anterior** (scoliosis)
Pedicle screw	Posterior, spans **all three columns** (Boucher 1959)

Sacropelvic fixation (TO)

	Direct iliac screw	S2 alar-iliac (S2AI)
Entry	Posterior iliac crest near PSIS → ASIS	Between S1 and S2 foramina (Kebaish 2007)
Prominence	**Prominent** → pain, wounds, removal	**15 mm deeper** → seldom removed
Alignment	Needs **connector**	**In line** with lumbar screws
Muscle	Iliac devascularisation	Less
SI joint	Less violation	**More SI joint violation** (main drawback)

How it's asked

"PLC components?"	→ **supraspinous, interspinous, flavum + facet capsules**
"Theoretical axis of FSU rotation at an instant?"	→ **instantaneous axis of rotation**
"IAR – true?"	→ **altered by degeneration, fracture, ligament injury**
"Best predictor of good outcome after adult deformity surgery?"	→ **sagittal balance correction**
"Angle correlating with lordosis, kyphosis, sagittal alignment?"	→ **pelvic incidence (A)**
"Not a posterior instrumentation?"	→ **Dwyer-Zielke** (TO)
"S2AI vs iliac screws – incorrect?"	→ **"less SI joint violation"** (it's more) (TO)

– – Active space – –

SPINE SURGICAL SITE INFECTIONS

– Spine · 60 / 69 –

1 Basics

Epidemiology

SSI commonest early post-op hospital-acquired infection · cost ~$200,000/patient · ↑ morbidity, reoperation, stay, mortality

Procedure	Rate
Overall spine	**0.7–16%** (orthopaedic spine 2.0%)
Microdiscectomy	**0.7%** with prophylaxis · **1.4% with microscope**
Elective instrumented fusion	**2.8–6%**
Trauma	**Up to 10%** (multilevel, delay > 160 h, complete deficit, cognitive impairment)
Posterior vs anterior	**Posterior higher** · combined A/P not higher than posterior alone

Average time to infection 14 months

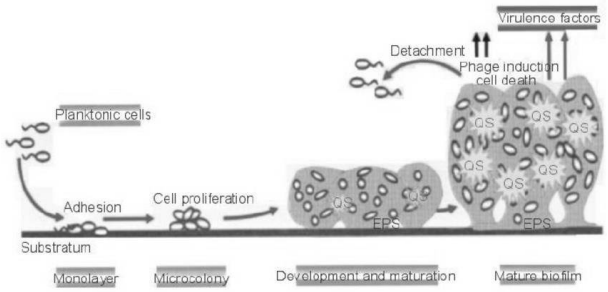
Risk factors

Category	Factors
Medical	**> 70 yrs**, ASA, **diabetes**, CVD, cancer, steroids, prior lumbar surgery, COPD, immunocompromise, prior infection, **pre-op stay > 1 wk**, prior radiation
Lifestyle	**Obesity**, **smoking**, alcohol, **malnutrition (15×)**
Intra-op	Transfusion, **instrumentation**, staged surgery, **levels fused**, OR traffic, **> 3 h**, **blood loss > 1 L**

Pathogenesis

Implants seroma + inflammation · **glycocalyx biofilm** shields bacteria from host + antibiotics · metallosis granulomas

Organism	Context
S. aureus (73%)	Commonest · **MRSA 5–18%**
S. epidermidis	Increasing
E. coli, Enterococcus	**Incontinence / faecal soiling**
Cutibacterium acnes	Low virulence, **late hardware infection**, immunocompromised
Gram-negative rods	Trauma (ISS > 18), **neuromuscular scoliosis**
Polymicrobial	Post-op wound contamination (neuromuscular)



Biofilm on implant

3 Presentation

Commonest presentation	**wound drainage** · pain increasing instead of settling · night sweats
Fever	commonest general sign · **> 39 °C** worrying for deep infection
Sepsis / organ failure	emergent debridement
Wound	superficial: erythema, discharge · **deep may look clean** · clear persistent drainage = seroma; purulent = infection

4 Diagnosis & Classification

Classifications

Anatomic	
Superficial	Skin + subcutaneous, fascia intact
Deep	Fascia/muscle – **won't settle with antibiotics alone**

Timing	
Early	**< 3 wks**
Late	**> 4 wks**
Latent	Years

Mechanism	
Direct inoculation	> 10^5 organisms; within 30 days
Early outside-in	Drains, seroma, soiling
Late haematogenous	Dental work, UTI, bacteraemia

– – Active space – –

Thalgott	Host	Infection
A / B / C	Normal / local-systemic disease (incl. smoking) / immunocompromised	–
1	–	Single organism
2	–	Deep, polymicrobial (~3 debridements)
3	–	Deep, polymicrobial + **myonecrosis** (poor)

Labs

Marker	Behaviour
CRP	Peaks **day 2–3**, normal by **5–21 days (~2 wks)** · **most sensitive marker** · **persistent rise or second peak = infection**
ESR	Peaks day 4–11, **elevated up to 6 wks** · rising after day 4 suggests infection
WBC	Unreliable
Malnutrition	**Albumin < 3.5 g/dL**, transferrin < 150
HbA1c	Pre-op diabetics **< 7.0**

Resolution **CRP normalised + ESR improving**

TO post-discectomy severe pain + spasm, no fever, normal TLC/ESR, **high CRP → infective discitis**

Key radiology findings

Study	Fact
X-ray	Early normal · late: disc-height loss, endplate erosion, **lucency around hardware**
CT	Fusion status, implant position, peri-screw lysis, pseudarthrosis
MRI + gadolinium	**Most useful** · **rim-enhancing large collection = pathognomonic** · epidural collection, bony destruction, progressive marrow change · post-op change mimics infection
WBC-labelled / bone scan	No MRI (pacemaker)



Rim-enhancing post-op collection

Cultures **intra-operative tissue = gold standard** (ideally before antibiotics) · superficial swabs unreliable · culture removed hardware in latent cases

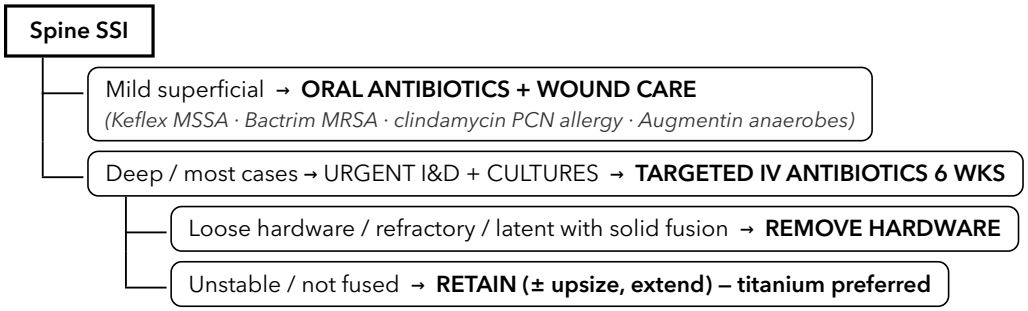
Differential adjacent segment disease, inadequate decompression, seroma, haematoma

Failed back surgery syndrome (TO)

Timing	Commonest cause
Immediate / early	**Incorrect diagnosis**
Few weeks (e.g. 3 wks)	**Spinal infection**
Months	Recurrent herniation, **epidural fibrosis**, arachnoiditis
Years	Degeneration, instability, stenosis, adjacent segment disease

Overall commonest poor patient selection · patient factors: workers' comp, smoking, obesity, stress

5 Management



Surgery indications most cases, antibiotic failure, deformity, deficit, MRI progression

Debridement old incision, layered, pulse lavage, remove necrotic muscle + **loose graft** + loose hardware · multiple debridements · **NPWT**, flaps, SPY perfusion, non-braided tension-free closure, beads, drains

Antibiotics **6 wks IV** (PICC), follow ESR/CRP · **serial MRI not needed** · prolonged IV antibiotics may allow fusion before hardware removal

6 Complications

List wound breakdown, pseudarthrosis, neuro deficit, sepsis, organ failure, death, worse outcomes

– – Active space – –

Prevention

Phase	Measures
Pre-op	Weight loss, **stop smoking**, alcohol abstinence, **glucose control**, treat UTI, **povidone-iodine nasal swab for MRSA carriers**
Prep	**Clippers** (✗ Wrong options: razor), alcohol prep
Antibiotics	**Within 1 h of incision**, **redose if > 4 h** · **cefazolin 2 g** (3 g if > 100 kg / BMI > 35) · PCN allergy/MRSA risk: **vancomycin** (start > 1 h before) or clindamycin · **stop at 24 h**
Intra-op	↓ time, blood loss, invasiveness · ↓ traffic · **dilute betadine 0.3% × 2 min** · **release retractors every 2 h** · debride necrotic muscle · haemostasis
Vancomycin powder	**↓ SSI** · if infection occurs → **↑ gram-negative** + culture-negative collections (✗ Wrong options: VRE · nonunion · ↑ cost)
Drains	Evacuate haematoma; NASS: not routine after single-level surgery

How it's asked

"Post-discectomy pain + spasm, only CRP high?"	→ **infective discitis** (TO)
"Most sensitive marker of post-op inflammation?"	→ **CRP**
"Resolution of wound infection?"	→ **CRP normalised, ESR improving**
"Topical vancomycin powder leads to?"	→ **more gram-negative SSI if infection occurs**
"Commonest cause of FBSS at 3 weeks?"	→ **spinal infection** (TO)

SUBAXIAL CERVICAL VERTEBRAL BODY FRACTURES

– Spine · 61 / 69 –

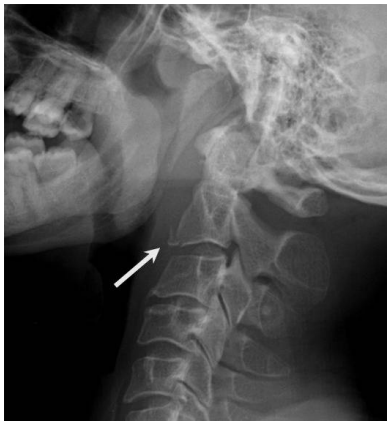
1 Basics

Fracture types

Type	Features	Stability / treatment
Compression	Anterior body fails, **posterior cortex intact**, no retropulsion · often PLC injury	Mild + intact PLC → **collar 6–12 wks**
Burst	Through **posterior cortex**, **retropulsion** · ± PLC injury · complete/incomplete SCI	**Unstable → surgery**
Flexion teardrop	Anterior column fails in flexion-compression, **posterior fragment retropulsed / retrolisthesis**, posterior column fails in tension · large anterior lip = **quadrangular fracture**	**Unstable → surgery** · SCI
Extension teardrop avulsion	Small fleck off anterior endplate (ALL/annulus), usually **C2**, no malalignment	**Stable → collar**, no SCI



Flexion teardrop fracture



Extension teardrop avulsion

– – Active space – –

Flexion teardrop (TO)

Features triangle off **anteroinferior** body · commonest **C5** · **posterior displacement of the upper column** · **frequently with sagittal body + laminar fractures** · **anterior cord syndrome** characteristic
✗ Wrong options: "unusual"

Quadrangular fracture **flexion-compression** · anterior third fragment, posterior subluxation, kyphosis, interspinous widening, facet subluxation

Mechanism

Mechanism	Usual injury
Hyperextension	Central cord
Flexion-distraction	**Facet dislocation**
Rotation	Soft tissue / high cervical
Pure axial load	**Burst**
Rear-end collision	**Extension–distraction** (TO)

Commonest subaxial injury pattern (TO) **distractive flexion**, commonest **C6–C7** · stages: 1 PLC failure → 2 unilateral facet dislocation → 3 bilateral → 4 floating vertebra

4 Diagnosis & Classification

Classifications

Allen-Ferguson (research, mechanism)
Flexion-compression · vertical compression · flexion-distraction · extension-compression · extension-distraction · lateral flexion

Clinical descriptive compression, burst, flexion-distraction, facet dislocation (uni/bilateral), facet fracture

SLIC (TO example)

Component	Example score
Morphology (translation/rotation)	4
Discoligamentous complex disrupted	2
Neurology (incomplete cord 3 + continuous compression 1)	4
Total	**10 → surgery**

Key radiology findings

MRI **PLC status** · TO: before reduction of fracture-dislocation · MRI within **48 h** most sensitive for oedema/haemorrhage · obtain if CT shows prevertebral soft tissue **> 5 mm at C3**, disc-space widening > 1–2 mm, interspinous widening

5 Management

Operative criteria

Procedure	Indication
Collar 6–12 wks	**Mild compression** (intact PLC, no kyphosis), **extension teardrop avulsion**
Halo	Stable pattern
Anterior corpectomy + strut + plate	Compression with **≥ 11° angulation or 25% height loss** · **unstable burst/ teardrop with cord compression** · minimal posterior injury
Posterior decompression + fusion	**Significant posterior element injury**, no anterior decompression needed

Timing — **early decompression < 24 h improves neuro outcome** · delay → pulmonary complications, ulcers, DVT
✗ Wrong options: steroids · delay to avoid "second hit" · collar

TO conflict — awake, alert C4-5 jumped facets within 30 min → **hard collar + MRI** first (TO) · ASIA C C4-5 fracture-dislocation with MRI done, SLIC 10 → **immediate reduction + stabilisation** (TO)

Anterior approach (TO)

Plane — **carotid sheath (lateral) vs trachea + oesophagus (medial)** · no internervous plane

Side — **left preferred**: right RLN loops under subclavian, less protected · left RLN under aortic arch (ligamentum arteriosum) in tracheo-oesophageal groove

Commonest complication — **vocal cord palsy (RLN) → hoarseness**
✗ Wrong options: superior laryngeal

C5 palsy (TO)

Definition — deltoid ± biceps weakness ≥ 1 grade after decompression without myelopathy worsening · **mostly posterior surgery**, also anterior (similar incidence)

Features — half with C5 pain/sensory change · **92% unilateral** · within **1 wk** · **good prognosis** · theories: segmental cord ischaemia-reperfusion vs **root tethering** (favoured)

Prevention / treatment — prophylactic C5 foraminotomy, limit posterior cord shift + excess lordosis · physiotherapy, steroids

Cervical disc replacement (TO)

Pros — preserves motion + IAR, avoids pseudarthrosis + adjacent segment disease · equal neuro outcomes, **fewer reoperations** than ACDF

Ideal — mature patient, 1–2 level radiculopathy ± myelopathy C3–C7 from soft disc / foraminal stenosis

Avoid — disc height **< 3 mm**, major spondylosis, axial neck pain alone, **facet arthrosis**, **≥ 3 levels**, instability (> 3 mm translation, > 11° rotation), osteoporosis, obesity, **kyphosis**

– – Active space – –

How it's asked

"Flexion teardrop — EXCEPT?"	→ **sagittal body + laminar fractures are unusual** (TO)
"Quadrangular fracture mechanism?"	→ **flexion-compression**
"Large anteroinferior fragment + retrolisthesis?"	→ **unstable flexion compression injury**
"Awake C4-5 jumped facets at 30 min?"	→ **hard collar + MRI** (TO)
"ASIA C C4-5 fracture-dislocation, MRI done?"	→ **immediate surgery: reduction + stabilisation** (TO)
"Diving C6 burst, ASIA C — best recovery?"	→ **urgent decompression < 24 h**
"Bilateral deltoid weakness day 2 after C5-6 surgery?"	→ **C5 palsy** (TO)
"Anterior cervical approach — side?"	→ **left (right RLN more at risk)** (TO)
"Hoarseness + unilateral cord palsy after ACDF?"	→ **recurrent laryngeal nerve palsy** (TO)
"Anterior cervical approach plane?"	→ **carotid sheath laterally, trachea/oesophagus medially** (TO)
"Rear-end crash, quadriparesis?"	→ **extension–distraction** (TO)
"Commonest subaxial injury type?"	→ **distractive flexion** (TO)
"Image after ACD with articulating device?"	→ **disc replacement** (TO)

SYNOVIAL FACET CYST

– Spine · 62 / 69 –

1 Basics

Epidemiology

Incidence	rare · lumbar
Level	**L4-5 60–89% (most mobile segment)** · L3-4 ~14% · L5-S1 ~12% · cervical rare: C5-6, C7-T1 ✗ Wrong options: "L5-S1 commonest (sagittal facets", TO)

Pathophysiology

Cause	**facet microinstability** → synovium extrudes through capsule · myxoid degeneration, fibroblast proliferation, ↑ hyaluronic acid · trauma controversial
Associations (TO)	facet OA (40.5%), **degenerative spondylolisthesis** (43.4%), disc degeneration (13.2%)
Contents	ganglion, flavum cysts; serous, mucoid, haemorrhagic

Histology

Cyst	Lining
Synovial cyst	**Synovial (cuboidal) cells over vascular granulation stroma** · clear/xanthochromic fluid
Ganglion	No synovial lining, gelatinous myxoid content
✗ Traps	Concentric fibrocollagen lamellae = annulus · gelatinous matrix with oval chondrocytes = nucleus · nidus + halo = osteoid osteoma · storiform atypia = UPS

3 Presentation

Symptoms	**mechanical back pain**, **radicular leg pain** (lateral recess → **traversing root**, e.g. L5: EHL + hip abductor weakness), neurogenic claudication · **sudden deterioration = cyst haemorrhage** · CES rare

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

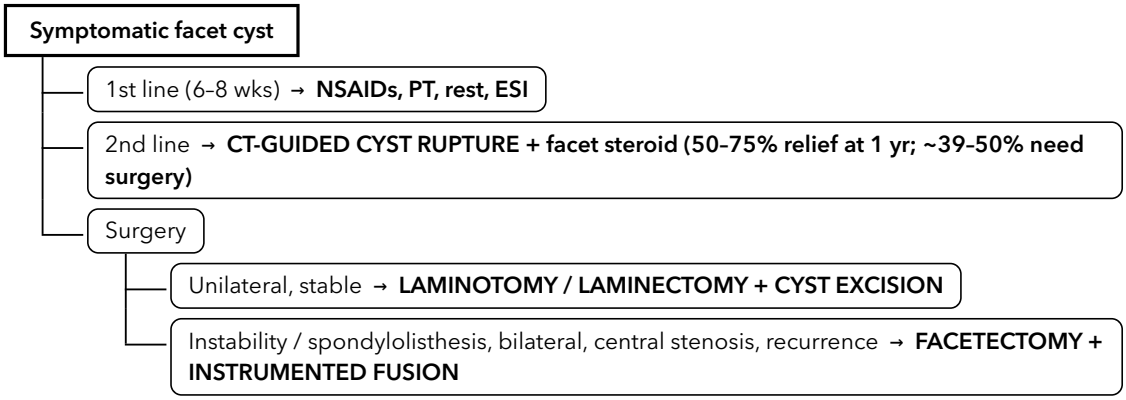
Study	Finding
X-ray AP, lateral, **flexion-extension**	Usually normal · **segmental instability**
MRI	**Axial + sagittal T2** · **T2 hyperintense centre, hypointense rim** · T1 hypointense core · **rim enhancement** · contiguous with degenerative facet, narrowing lateral recess
Positional MRI	Supine 89% → **standing 97%** sensitivity (cysts enlarge standing)
T2 signal → aspiration	**High T2 (synovial fluid) → better CT rupture success** · low T2 (gelatinous/ calcified) → worse



Facet synovial cyst on axial T2

Differential ligamentum flavum, **Tarlov cyst**, pseudomeningocele, HNP

5 Management



Operative criteria

Procedure	Indication / outcome
Decompression + cyst excision	Persistent symptoms, **unilateral** · **TO: surgery of choice without instability** · 80–90% success · risk: **high recurrent back pain + cyst within 2 yrs**, iatrogenic listhesis
Facetectomy + instrumented fusion	**Recurrence after laminectomy**, bilateral symptoms, central stenosis, **instability / degenerative spondylolisthesis** · some: **first-line surgery** · **lowest recurrent back pain + cyst recurrence** · 80–90% resolution
✗ Wrong option: ALIF	Doesn't remove cyst or decompress lateral recess
✗ Wrong option: uninstrumented fusion + facetectomy	↑ pseudarthrosis
✗ Wrong option: aspiration	High failure, short-term only

Technique unilateral laminotomy + medial facetectomy (burr) · develop dura–cyst plane, Allis traction, Epstein curette/Woodson · pedicle screw at **superior TP border × mid superior articular process**

6 Complications

Recurrence high with excision alone → fusion

Iatrogenic spondylolisthesis laminectomy without fusion → instrumented fusion

Dural tear revision, **cyst adherent to dura (50–55%) → partial excision** (88% success, ↓ durotomy)

How it's asked

"Facet cyst – wrong statement?"	→ **commonest at L5-S1** (it's L4-5) (TO)
"Hip abductor + EHL weakness, MRI lesion off facet?"	→ **facet synovial cyst**
"Histology of this lesion?"	→ **synovial cells over vascular granulation stroma**
"Arrow on T2 at a degenerative facet, bright cyst in lateral recess?"	→ **lumbar synovial cyst**
"Recurrent synovial cyst after laminectomy?"	→ **revision laminectomy + instrumented fusion**
"Lowest recurrent back pain at 2 yrs?"	→ **decompression + facetectomy + instrumented fusion**
"Unstable L4-5 listhesis + cyst, EHL weakness – definitive?"	→ **laminectomy + cyst excision + instrumented fusion**
"Grade 1 listhesis + cyst – best long-term?"	→ **L4-5 posterior instrumented fusion, laminectomy, cyst excision**

– – Active space – –

SYRINX & SYRINGOMYELIA

– Spine · 63 / 69 –

1 Basics

Definitions

Syringomyelia	fluid cavity in the **cord** that expands → deficits · **syringobulbia** – in the brainstem
Epidemiology	8.4 per 100,000 · **M > F** · 3rd–4th decades; insidious from adolescence to 45

Causes (partial CSF obstruction)

Cause	Fact
Chiari malformation	**> 50%** · tonsils act as **pistons** at foramen magnum → pressure wave each heartbeat · syrinx in **69% of adult, 40% of child Chiari** (TO)
Spinal cord trauma	**4–10% of SCI**, may appear years later
Cord tumour	**30%** develop syrinx (TO: tumour cysts not true syringomyelia)
Post-infectious	Arachnoiditis (meningitis, subarachnoid blood)

Primary spinal syringomyelia (TO)	pathology **entirely within the spine** (trauma, arachnoid cysts/webs, meningitis scarring) (✗ Wrong options: "secondary")
Chiari syrinx mechanism (TO)	**Oldfield: pulsatile tonsils on a closed spinal subarachnoid space drive CSF into the cord** (+ arterial pulsations, Stoodley) · (✗ Wrong options: "pressure differential from partial block" (TO EXCEPT key)) · old "communicating" 4th-ventricle theory abandoned · cause: **small posterior fossa** relative to cerebellum

Associations

Condition	Fact
Scoliosis	**25–80%** of syringomyelia · neural axis anomaly in 1.8–25% of "idiopathic" curves
Klippel-Feil	–
Charcot joint	**< 5%**, usually **shoulder**

3 Presentation

Symptoms	**occipital headache worse with Valsalva** (cough, sneeze – tussive, dural stretch), back + radicular pain · **central cord syndrome** picture
Dissociated sensory loss	**pain + temperature lost in a cape** over neck/shoulders · **light touch, vibration, position preserved**
Reflexes	**asymmetric abdominal reflexes**, asymmetric/hyperactive reflexes, asymmetric sweating, spasticity

Motor / deformity	asymmetric bulk, weakness, LE contractures, scoliosis
Syringobulbia	CN XII tongue atrophy, XI SCM/trapezius, IX–X dysphagia/dysarthria, VII facial palsy
Chiari brainstem signs (TO)	nystagmus, hoarseness, swallowing/speech problems, sleep apnoea, imbalance



Cervical syringomyelia

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray	**Lack of thoracic apical lordosis** suggests syrinx · scoliosis series
MRI	**Diagnostic** · **gadolinium** to exclude tumour
CT myelogram	Post-traumatic syrinx / arachnoid webs – shows obstruction (TO)

Diagnostic criteria – MRI in scoliosis

Indication
Atypical curve – left thoracic apex, double curve
Neurologic deficit (e.g. ↓ flank/hand pinprick, absent abdominal reflex)
Infantile or juvenile onset
Male with atypical or large curve
Thoracic kyphosis > 30°
Valsalva headache

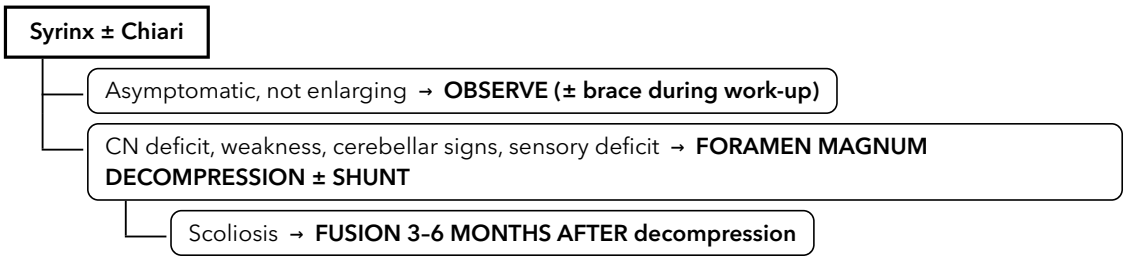
Histology	grey matter cavitation, near/continuous with central canal, **gliotic lining**
Differential	hydromyelia, glioependymal cyst, myelomalacia, cystic tumour, persistent central canal

Neuropathic shoulder

Clue	severe destruction both sides of joint, little pain, minor trauma, chronic dislocation · **MRI brain + cervical spine** → Chiari / syrinx ✗ Wrong options: RF, HLA-B27, synovial fluid, bone scan

– – Active space – –

5 Management



Chiari decompression (TO)	suboccipital craniectomy to uncover tonsils + **C1 laminectomy** + intradural inspection ± tonsil reduction + **duraplasty**
Primary spinal syringomyelia (TO)	shunting (pleura/peritoneum) has **high failure** (cavity collapses on tube) → **resect focal obstruction**
Scoliosis	decompression alone may improve curve but large curves / **> 8 yrs** still progress

6 Complications

Undiagnosed syrinx + fusion	**↑ neurological deterioration / paralysis**

How it's asked

"Arnold-Chiari – EXCEPT?"	→ **"partial CSF block pressure differential" as syrinx cause** (TO)
"Syringomyelia – EXCEPT?"	→ **"pathology wholly in spine = secondary"** (it's primary) (TO)
"62-yr painless destroyed shoulder – confirm with?"	→ **MRI brain + cervical spine**
"Chronic dislocation + destruction after minor trauma – positive test?"	→ **MRI of upper cervical spine**
"5-yr, 20° left thoracic, hand pinprick loss?"	→ **MRI of the entire spine**
"12-yr left thoracic curve + Valsalva headaches – MRI?"	→ **syrinx + tonsillar herniation (Chiari I)**

THORACIC DISC HERNIATION

– Spine · 64 / 69 –

1 Basics

Epidemiology

Frequency	**~1% of all HNP** (Orthobullets) · **0.25–1%** (TO) · rare: limited motion, short discs, rib cage
Age	**4th–6th decades** (Orthobullets, TO) · 3rd–5th (OB)
Level	**lower third** · **T11-12 commonest** (maximal mobility, thin PLL) · **75% at T8–12** · T1–5 only 1% · flexion + torsion concentrate T10–L1 (floating ribs 11, 12)
Risk factor	**Scheuermann's disease** (also ↑ neuro deficit)
Asymptomatic prevalence	abnormalities 73%, **frank herniation 37%**, cord compression 29% (Orthobullets) · **~40%** (OB) – read the options
Natural history	**most symptomatic cases resolve non-operatively**; most imaging lesions never cause symptoms; large herniations shrink

Classification

Type	Definition
Bulge	Annulus intact
Extruded	Through annulus, contained by PLL
Sequestered	Free in canal
Location	Central · posterolateral · lateral

3 Presentation

Commonest symptom	**axial back / chest pain** (dorsal backache, TO)
Radicular	**band-like** chest/abdominal pain along the intercostal nerve, **around or straight through to the anterior chest wall** · **T10 dermatome** commonest (TO) · upper thoracic + lateral discs · arm pain with T2–5
Myelopathy	**LE hyperreflexia with normal UE reflexes** (T11-12 cord), clonus, Babinski, wide-based spastic gait, paraparesis, rectal tone · **bowel/bladder 15–20%** · sexual dysfunction
Horner's	T2–5 herniation
Work-up of LE UMN signs with normal UE	**thoracic MRI** · hip OA + wide-based gait → **thoracic + lumbar MRI before THA** ✗ Wrong options: cervical/lumbar surgery

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Study	Fact
Lateral X-ray	Disc narrowing, **calcification**
MRI	**Most useful** · tumour, intradural pathology, myelomalacia · may **under-show calcified disc** · high false positives
CT / CT myelogram	Calcified component



Thoracic disc herniation on MRI

5 Management

Thoracic disc herniation

- Axial / radicular pain → **REST, PT, NSAIDs, gabapentin, injections (6–12 wks)**
- Myelopathy / progressive deficit / intolerable pain → **DISCECTOMY ± FUSION**
  - Central (calcified) → **TRANSTHORACIC (± VATS)**
  - Lateral, extruded, sequestered → **COSTOTRANSVERSECTOMY / LATERAL EXTRACAVITARY**
  - Posterior → **TRANSPEDICULAR + instrumented fusion** *(NOT LAMINECTOMY ALONE)*

Operative criteria

Indication
Myelopathy from the disc (sensory loss + clumsy gait → **discectomy**), especially progressive
Persistent intolerable pain
✗ Wrong option: most cases need surgery (TO EXCEPT)

Approaches

Approach	Fact
Posterior laminectomy	**Avoid – highest neurological deterioration** – cord can't be retracted; thoracic kyphosis drapes cord over disc
Transthoracic (transpleural, T4–T12; VATS)	**Best for central discs** · T12-L1 needs diaphragm takedown · complication **intercostal neuralgia**
Costotransversectomy	Lateral, extruded, sequestered · no pleural violation · hard for calcified discs/osteophytes
Lateral extracavitary	More bone resection + blood loss, less neuro risk than laminectomy
Transpedicular + instrumented fusion	Posterior option for myelopathy with cord signal change

Thoracic stenosis (degenerative, flavum/facet) — **laminectomy** appropriate (posterior compression) – differs from disc

FVC — transthoracic not safe with FVC 20%

6 Complications

Intercostal neuralgia — **commonest after transthoracic / endoscopic** surgery (portal compression)

X vs Y

	Thoracic disc	Thoracic stenosis
Compression	**Anterior** (disc, often calcified)	**Posterior** (flavum, facets)
Approach	Anterior / lateral / transpedicular	**Laminectomy**

How it's asked

"Commonest thoracic disc level?"	→ **T11-12** (TO)
"Large T11-12 disc with imbalance – reflexes?"	→ **normal UE, hyperreflexic LE**
"Posterolateral thoracic disc pain pattern?"	→ **around/through the chest to the anterior wall**
"Thoracic disc herniation – true?"	→ **found in 40% of asymptomatic people**
"New LE weakness, UMN signs, normal arms – next?"	→ **thoracic MRI**
"Thoracic discs typically occur?"	→ **lower third**
"Hip pain + wide-based gait + LE weakness – next?"	→ **thoracic + lumbar MRI, spine referral**
"Thoracic disc – EXCEPT?"	→ **most cases need surgery** (TO)
"2 wks dermatomal rib-cage pain, normal neuro?"	→ **PT + NSAIDs**

– – Active space – –

"Axial pain, thoracic disc — true?"	→ **usually resolves with conservative care**
"T7-8 disc, sensory loss, clumsy gait?"	→ **discectomy** (TO)
"Endoscopic transthoracic surgery — commonest complication?"	→ **intercostal neuralgia**
"29-yr thoracic myelopathy from disc?"	→ **thoracic discectomy**
"Thoracic disc + myelopathy — approach with most neuro morbidity?"	→ **posterior laminectomy**
"Thoracic myelopathy, cord signal change, normal cervical MRI?"	→ **laminectomy + transpedicular discectomy + instrumented fusion**
"Thoracic stenosis with claudication, hyperreflexia, incontinence?"	→ **laminectomy**

– – Active space – –

THORACIC SPINE ANATOMY

– Spine · 65 / 69 –

1 Basics

Alignment

Thoracic kyphosis	average **35°**, normal **20–50°** (✗ Wrong options: 40–50°) · apex **T5–T8** (T6-7) (✗ Wrong options: T4) · adolescents 30–40° (TO)
Lumbar lordosis	**majority below L3** (TO) · L4-S1 ~60% (TO) · 75% (Orthobullets, previous chapter) · **discs give 78%** (47°), bodies 12° · apex L3-L4 · range 20–60° (TO) · 20–80° (Orthobullets)
Frontal plane	straight; **> 10° = scoliosis** ✗ Wrong options: < 20° normal
Plumb line (sagittal vertical axis)	from **C7 centre → posterosuperior corner of S1** (within 2 cm) · anterior = positive · posterior = negative · ideally from dens: anterior to thoracic, posterior to lumbar
Pelvis	sacral slope < 35° + low PI → flat short lordosis · > 45° + high PI → long curved lordosis (TO)
Conus (TO)	**lower border of L2 adult**, L3 neonate (Orthobullets: L1) · filum to first coccygeal segment

Osteology

Vertebra prominens	C7 spinous process
Costal facets	bodies + transverse processes **T1–T9** · ribs → **most rigid region** of axial skeleton
Bodies	size ↑ T1 → T12 · canal dimension varies

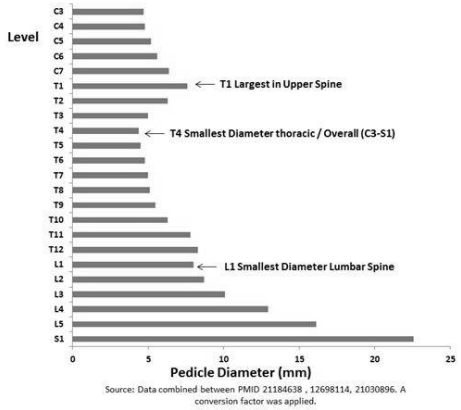
Facet orientation & motion

Region	Coronal / sagittal	Motion
Cervical	0° / 45°	All planes
Thoracic	**20° / 55°** (coronal facets), 6 DOF	Rotation; little flex-ext (ribs) · protects heart/lungs
Lumbar	50° / 90° (sagittal facets)	Flex-ext; little rotation

– – Active space – –

Pedicles

Measure	Fact
Wall	**Medial 2× thicker than lateral**
Narrowest diameter	**T4** (Orthobullets, OB) · T4–T6 dip; **T6** smallest among T1/T6/T12/L3/L5 (OB)
Largest	**T1** (upper) · **T12 (> L1)** at TL junction · T10, T11 > L1 · L5 largest lumbar · C7 largest cervical
T7	Narrow on concave side in AIS
Length	T1 20 mm → **T4 14 mm (shortest)** → T10 20 mm
Transverse angle	**T1 25° medial, T2 15°, T3–T10 ~10°** (5–7°), minimal T11-12 (OB) · 10° mid-thoracic → 30° L5 (Orthobullets)
Sagittal angle	**15–17° cephalad** thoracic · neutral lumbar · caudal L5



Thoracic pedicle diameters

Interpedicular distance (TO)

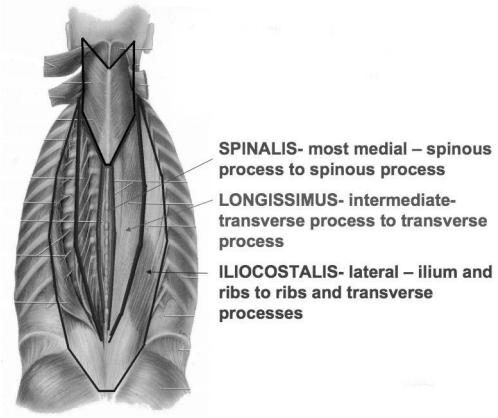
Minimum **T4–T8** · narrowed: **achondroplasia** (caudal narrowing), thanatophoric dysplasia · widened: **diastematomyelia**, burst fracture (middle column)

Erector spinae

Function trunk extension · **dorsal rami**

Muscle (medial → lateral)	Attachment
Spinalis (medial)	Spinous → spinous process
Longissimus (intermediate)	Transverse → transverse process
Iliocostalis (lateral)	Ilium + ribs → ribs + TPs

Mnemonic "I Love Spine" lateral → medial



Erector spinae layers

5 Management

Thoracic approaches

Approach	Access
Laminectomy / hemilaminectomy	Posterior (not for central disc)
Transpedicular	Posterolateral
Costotransversectomy	Lateral
Lateral extracavitary	Lateral, more ventral
Anterior intracavitary	**Thoracotomy / VATS** – central

How it's asked

"Interpedicular distance minimal at?"	→ **T4–T8** (TO)
"Smallest average transverse pedicle diameter?"	→ **T6** (of options)
"Greatest pedicle isthmic width?"	→ **T12**
"Thoracic pedicle transverse angulation?"	→ **25° T1, 15° T2, 10° T3–T10**
"Plumb line passes through?"	→ **posterosuperior margin of S1** (TO)
"Correct normal alignment statement?"	→ **majority of lumbar lordosis below L3** (TO)

– – Active space – –

THORACOLUMBAR BURST FRACTURES

– Spine · 66 / 69 –

1 Basics

Epidemiology & mechanism

Mechanism high-energy **axial load ± flexion** (falls, MCC) · anterior + middle column compression → **retropulsion**

Level **thoracolumbar junction T10–L2** (commonest **L1**, OB key **L2** among L2–S1) · > 50% of thoracic + lumbar fractures (T12–L2) · TL junction = stress riser between rigid ribbed thoracic (T1–10) and mobile lumbar spine · T10–12 floating ribs

Neuro maximal canal occlusion at impact (recoil after) · **retropulsed fragments resorb**, rarely cause late deterioration · deficit type depends on level vs conus: cord / **conus** / cauda equina

Associated injuries

Injury	Fact
Non-contiguous spine fractures	**20%** → image whole spine
Traumatic durotomy	**Laminar fracture, split spinous process** (entrapped roots)
Thoracic fracture + deficit	1/3 haemopneumothorax, great-vessel injury, diaphragm rupture
Flexion-distraction / dislocation	Bowel rupture, vessels, kidney, liver, spleen, pancreas
Long-bone fractures	Hamper rehab

Anatomy

Denis columns anterior (ALL, anterior 2/3 body) · **middle (PLL, posterior 1/3 body)** · posterior (pedicles, laminae, facets, flavum, spinous process, PLC)

Instability middle column injury (**widened interpedicular distance**, posterior body height loss) · PLC + anterior/middle column injury

PLC supraspinous, interspinous, flavum, facet capsule · **clearly ruptured**: bony Chance, interspinous widening, progressive kyphosis, facet diastasis · indeterminate: MRI signal only

Conus ends **L1-2**; **L1 fracture → conus medullaris syndrome** (bowel/bladder paralysis ± mixed UMN/LMN LE)

3 Presentation

Vitals hypotension + **bradycardia = neurogenic** · + tachycardia = haemorrhage

Exam log roll · palpable gap / **↑ interspinous distance** (posterior injury) · full neuro incl. reflexes · **absent BCR = spinal shock** (up to 72 h) · **hyperactive BCR** = disinhibition, complete cord injury

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray (whole spine; often trauma CT)	AP: **widened pedicles**, coronal deformity · lateral: retropulsion (underestimated), kyphosis · flexion-extension once stable: **spinous process diastasis = PLC injury**
CT	**Best for retropulsion (axial)**, comminution · more sensitive than X-ray
MRI	**PLC (STIR/T2 signal, sagittal)**, conus level, compression, cord oedema/ haematoma · **investigation of choice (TO)** · nearly every case unless CT/X-ray already shows instability
CT myelogram	MRI contraindicated; durotomy

MRI prognosis **cord oedema > 2 levels** or **haematoma (low T2 with oedema halo)** → poor motor recovery



Axial CT retropulsion

Denis burst types

Type	Pattern
A	**Both endplates**, retropulsion
B	**Superior endplate** – commonest (axial + flexion)
C	Inferior endplate
D	Burst-rotation (mimics fracture-dislocation)
E	Burst-lateral flexion (↑ interpedicular distance vs lateral compression)

TLICS

Category	Points
Morphology	Compression 1 · **burst 2** · rotation/translation 3 · **distraction 4**
Neurology	Intact 0 · **root 2** · **incomplete cord/conus 3** · **complete cord/conus 2** · **cauda equina 3**
PLC	Intact 0 · indeterminate 2 · **disrupted 3** (largest single contribution)
Total	**< 4 non-op · 4 either · > 4 surgery**

– – Active space – –

TO rules max 10 · score each level, highest decides · translation/rotation always with PLC (3 + 3 = 6) · posterior distraction always PLC (4 + 3 = 7) · **incomplete scores more than complete**

Not in TLICS **% canal compromise**

The Thoracolumbar Injury Classification and Severity (TLICS) Score

Injury Category	Point Value
Injury morphology	
Compression	1
Burst	2
Translation or rotation	3
Distraction	4
PLC status	
Intact	0
Injury suspected or indeterminate	2
Injured	3
Neurologic status	
Intact	0
Nerve root involvement	2
Spinal cord or conus medullaris injury	
Incomplete	3
Complete	2
Cauda equina syndrome	3

TLICS

Worked scores burst + intact PLC + intact neuro = **2 → non-op** · burst + indeterminate PLC + root = **6 → surgery** · burst + incomplete + PLC = **8 → surgery** · burst + translation + PLC + ASIA B = 9

5 Management

TL burst fracture

- Neuro intact + PLC intact (TLICS ≤ 3) → **MOBILISE AS TOLERATED ± TLSO** *(no bed rest or traction)*
- PLC injury / progressive kyphosis / polytrauma, intact → **POSTERIOR INSTRUMENTED STABILISATION**
- Deficit with compression / TLICS ≥ 5 → **DECOMPRESSION + STABILISATION**
  - Below conus, PLC injured, fracture-dislocation → **POSTERIOR (transpedicular, posterior corpectomy, cage)**
  - Anterior compression above conus (≤ L2), comminution, kyphosis > 30°, > 4–5 days → **ANTERIOR / LATERAL CORPECTOMY ± posterior**

Non-operative

Neuro intact + stable **equivalent outcomes** (pain, function, RTW, kyphosis, fragment resorption at 5 yrs; better PROs at 16–22 yrs) with **fewer complications** than surgery · surgery → **↑ complications and disability**

TLSO **no clear advantage**; use for comfort · may begin **early ambulation without brace** if pain controlled · older teaching: TLSO/body cast 3–6 months

Old thresholds kyphosis > 30°, height loss > 50% (controversial; no longer indications alone)

Operative criteria

Indication	Detail
PLC injury on MRI	**Indication even in a compression fracture** (TO) (✗ Wrong options: radiculopathy · middle column · kyphosis 15°)
Neuro deficit + compression	> single root level (TO) · complete and incomplete
Progressive deficit / **CES** after bracing	**Anterior decompression + strut → posterior instrumentation**
Progressive kyphosis, polytrauma	Stabilise
Laminar fracture + unilateral L5–S4 deficit	**Laminectomy, free entrapped roots, repair dura**, posterior fusion

TO burst L4 with deficit	**posterior decompression + stabilisation** ✗ Wrong options: laminectomy alone · bed rest · brace
Near-complete canal compromise, incomplete deficit at L1	**anterior decompression + stabilisation**
L1 burst + incomplete cord + PLC (TLICS 8)	**lateral L1 corpectomy + T12-L2 fusion** (cord at level → avoid posterior retraction)

Technique

Posterior	pedicle screws · historically 3 above / 2 below; modern **1 above / 1 below (short segment)** – not at TL junction · "**fuse short, instrument long**" · percutaneous option · **two above + two below (T11–L3)** for L1 burst with deficit (OB) · no laminectomy alone (destabilises)
Indirect decompression	**distraction + lordosis → ligamentotaxis** (PLL + annulus pull fragments) · **less effective after 4–5 days** · **monoaxial screws** give more distraction · equal to anterior + 360° in studies · overdistraction → pseudarthrosis, pain, deformity
Direct posterior	transpedicular removal **below conus (L2)** · above conus → cord retraction risk
Anterior	corpectomy to contralateral pedicle wall · strut (tricortical iliac, allograft, expandable cage) · lumbar: retroperitoneal (below L1) · **T11–L1: left lateral lumbotomy** (avoid liver) · **above T11: right thoracotomy** (avoid great vessels) · diaphragm level
Pedicle screw entry (TO)	**intersection technique** commonest free-hand (lateral facet line × TP bisector) · mammillary point more lateral
Unkeyed TO item	1 above + 1 below with index-level screw = short-segment construct (source gives no key)

– – Active space – –

6 Complications

Complication	Fact
Pain	**Commonest** · overdistraction
Dural tear / entrapped roots	Laminar fractures · fewer with anterior approach
Progressive kyphosis	Missed PLC/PLL injury, comminution
Flat back	Pain, stooped posture; post-traumatic syrinx
SSI	**Up to 10%**
Pseudarthrosis	Overdistraction
Iatrogenic neuro injury	**1%** – medial screws, cord manipulation
Approach	Ileus (transperitoneal), pleural effusion (thoracotomy), loss of correction (posterior)

Chance fracture (TO)

Mechanism **flexion-distraction** around an axis **anterior to the spine** (lap belt, "jack-knife") · **middle + posterior columns fail in distraction** · no body crush · bony Chance → hyperextension brace; ligamentous/unstable → surgery

How it's asked

"ASIA E L1 burst, intact PLC – surgery?"	**→ increased complication rates**
"Neuro intact, PLC intact – advantage of surgery?"	**→ none; equivalent outcomes**
"68-yr L3 burst, ASIA E, 20% canal occlusion?"	**→ orthosis + early mobilisation**
"Stable burst, no deficit – long-term surgery outcome?"	**→ more disability + complications**
"L1 burst, 25% canal, intact PLC?"	**→ ambulation ± TLSO**
"L1 burst, intact PLC – true?"	**→ early ambulation without brace if pain controlled**
"L2 burst TLICS 2?"	**→ external orthosis**
"Snowboarder, stable burst, non-op?"	**→ equivalent clinical outcome**
"5-yr difference surgery vs non-op?"	**→ incidence of complications**
"Rock climber L1 burst, 20° kyphosis, intact?"	**→ TLSO / body cast with mobilisation**
"Intoxicated fall, displaced L3 burst, intact?"	**→ early mobilisation ± TLSO**
"Fall from tree, L1 burst, intact?"	**→ early mobilisation ± orthosis**
"Stable L1 burst – true?"	**→ non-op: fewer complications, similar outcomes**
"L1 burst + L2 compression, intact, TLICS < 4?"	**→ immediate mobilisation in TLSO**
"Elderly fall – non-operative candidate?"	**→ T12 burst, intact PLC, intact neuro**
"Flexion-distraction picture – true?"	**→ PLC injury confirmed** (TO)

"Burst, indeterminate PLC, root deficit?"	→ **TLICS 6, surgery**
"Greatest TLICS contribution?"	→ **PLC disruption**
"Not in TLICS?"	→ **% canal compromise**
"L2 burst, intact PLC + neuro?"	→ **TLICS 2, non-op**
"L1 compression 40% — surgical indication?"	→ **PLC injury on MRI** (TO)
"L4 burst + deficit?"	→ **posterior decompression + stabilisation** (TO)
"L4 burst in TLSO, new CES day 5?"	→ **anterior decompression + strut + posterior instrumentation**
"L1 burst, multi-root weakness, fall on feet?"	→ **decompression + T11–L3 fusion with postural reduction**
"Compression fracture + MRI PLC disruption, intact?"	→ **posterior instrumented fusion**
"L1 burst, ~100% canal, sacral sparing?"	→ **anterior decompression + stabilisation**
"L2 burst, unilateral L5–S4 deficit, laminar fracture?"	→ **L2 laminectomy + T11–L4 posterior fusion**
"L1 burst + translation, ASIA B, PLC oedema?"	→ **corpectomy + cage + posterior instrumentation**
"L1 burst, incomplete cord + PLC injury?"	→ **lateral L1 corpectomy + T12-L2 fusion**
"L1 burst, mixed UMN/LMN, can't void?"	→ **conus medullaris injury**
"Chance fracture mechanism?"	→ **flexion distraction** (TO)
"Chance — columns failing?"	→ **middle + posterior in distraction** (TO)
"Commonest site of compression/burst fractures?"	→ **L2 (of options)**
"Burst fracture, intact PLC — mechanism?"	→ **axial load**
"Investigation of choice for burst fracture?"	→ **MRI** (TO)
"Commonest pedicle screw technique?"	→ **intersection technique** (TO)

– – Active space – –

THORACOLUMBAR FRACTURE-DISLOCATION

– Spine · 67 / 69 –

1 Basics

Epidemiology

Incidence	**~4% of SCI** at level 1 centres · **50–60% have SCI** · M:F 4:1
Mechanism	high energy (**MVA commonest**, falls, sport, violence) · **acceleration/deceleration → hyperflexion + rotation + shear**
Site	**thoracolumbar junction T10–L2** (kyphosis → lordosis transition; vulnerable to shear) · injures **cord, conus or cauda equina** by level
Associations	head, thoracic, abdominal injuries

3 Presentation

Pre-hospital	backboard + collar · ATLS · inspect for open injury, kyphosis · palpate step-off, crepitus · **ASIA score**, rectal exam, GCS

4 Diagnosis & Classification

Key radiology findings

Study	Fact
AP + lateral X-ray	Fracture type + **dislocation**
CT	Pattern (unilateral facet dislocation), canal, screening in blunt trauma
MRI	Cord + ligaments (**PLL**) · deficit level not matching bony level



Thoracolumbar fracture-dislocation

TLICS

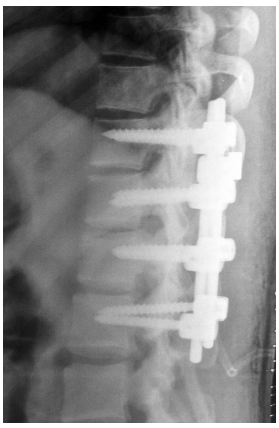
Thresholds	**≤ 3 non-op · 4 indeterminate · ≥ 5 surgery**

Translation/rotation	3 + PLC 3 = **≥ 6** → surgery
Example (OB)	T12 burst (2) + incomplete cord (3) + intact PLC (0) = **5 → posterior decompression + arthrodesis**

– – Active space – –

5 Management

Posterior open reduction + instrumented fusion	**most fracture-dislocations** · pedicle screws for distraction-reduction · **2 levels above + 2 below** · post-op CT/MRI for residual anterior compression · **early surgery > delayed or non-op**
Avoid isolated anterior procedure	ALL usually intact; anterior surgery further destabilises; anterior reduction difficult · no closed reduction + TLSO or bed rest
With deficit	**posterior fusion + instrumentation + posterolateral decompression**



Posterior reduction and instrumentation

Pedicle screws (TO)

Entry	**intersection technique** commonest (free-hand) · lumbar: line just lateral to inferior articular process × TP bisector · thoracic: mid inferior facet × mid TP insertion · **S1: inferolateral base of superior articular process** · mammillary point more lateral than intersection, more lateral than pars point
Osteoporotic bone – best pull-out	**larger thread depth** · > 60% of hold from **pedicle cortex** · fine threads proximally, coarse distally · insertion at 0–10° better · **cement augmentation** (PMMA, CaP, HA), expandable screws ✗ Wrong options: cannulation · larger core (reduces thread depth) · larger pitch (narrower pitch better)
Failure	loosening 0.6–11% · breakage 2.6–60% · pedicle screws span **all 3 columns** (Boucher 1950s, Roy-Camille 1980s)

Left retroperitoneal lumbar approach (TO)

Structure	Fact
Ureter	Adherent to posterior peritoneum, falls away from psoas
Sympathetic trunk	On the **medial** border of psoas (✗ Wrong options: lateral) → injury: **warm red ipsilateral (left) foot**, so the normal right foot seems **cold + pale**
Ilioinguinal nerve	Emerges from **lateral** psoas border → quadratus lumborum

– – Active space – –

Perched facets (TO)

Mechanism	**distractive flexion** (subaxial) · spectrum: sprain → perched → unifacet → bifacet dislocation → floating vertebra

6 Complications

Post-traumatic pain	**commonest**; ↑ with kyphosis
Deformity	scoliosis, **progressive kyphosis (missed PLL injury)**, **flat back** (pain, stooped posture, fatigue), post-traumatic syrinx
Others	neuro injury, CES, DVT, nonunion

X vs Y

	Burst	**Fracture-dislocation**
Mechanism	Axial load ± flexion	Flexion + rotation + shear
Alignment	Adjacent levels aligned	**Translation**
PLC	Variable	Disrupted
Treatment	Often non-op if intact	**Posterior reduction + fusion**

How it's asked

"T12 burst, incomplete cord, PLC intact – TLICS + treatment?"	→ **5; posterior decompression + arthrodesis**
"Fracture-dislocation with quad weakness – definitive?"	→ **posterior instrumented fusion + posterolateral decompression**
"Commonest pedicle screw technique?"	→ **intersection technique** (TO)
"Pedicle screw entry point?"	→ **mammillary process, lateral to pars** (TO)
"Osteoporotic bone – screw property?"	→ **larger thread depth** (TO)
"Left retroperitoneal approach – EXCEPT?"	→ **sympathetic trunk on the lateral psoas border** (it's medial) (TO)
"Perched facets seen in?"	→ **distractive flexion** (TO)

TRAUMATIC SPONDYLOLISTHESIS OF AXIS (HANGMAN'S FRACTURE)

– Spine · 68 / 69 –

1 Basics

Definition	**bilateral C2 pars (or pedicle) fractures** → traumatic spondylolisthesis of C2 on C3
Mechanism	**primary hyperextension** (pars fracture) + **secondary flexion** (tears PLL + disc → anterolisthesis) · MVA/falls: hyperextension-compression · judicial hanging: hyperextension-**distraction** (cord transection) ✗ Wrong options: primary hyperflexion (TO)
Associations	**30% other C-spine fracture** · vertebral artery injury (MRA if suspected)

3 Presentation

Neck pain	usually **neurologically intact** (fragments separate → canal widens) · deficit mainly **type III**

4 Diagnosis & Classification

Key radiology findings

Study	Fact
CT	**Study of choice** for pattern
Flexion-extension X-ray	Subluxation
MRA	Vertebral artery injury

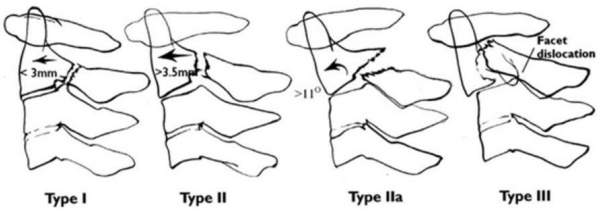


Hangman's fracture lateral

– – Active space – –

Levine-Edwards

Type	Mechanism	Features	Treatment
I	Axial compression + hyperextension	**< 3 mm** displacement, no angulation, **C2-3 disc intact**, stable	**Rigid collar 4–6 wks**
II	Hyperextension + axial load → rebound flexion	**> 3 mm**, significant angulation, **vertical** fracture line, disc + PLL torn	**< 5 mm: traction + extension → halo 6–12 wks** · **> 5 mm: surgery** or prolonged traction · usually auto-fuse C2 on C3
IIA	**Flexion-distraction**	**No displacement**, significant **angulation**, **horizontal/oblique** line	**No traction** (worsens) → **gentle axial compression + hyperextension → compression halo** 6–12 wks
III	Flexion-distraction → hyperextension	Type I + **bilateral C2-3 facet dislocation**	**Open reduction + fusion (posterior)**



Levine-Edwards types

Halo duration 6–12 wks (Orthobullets) · 8–12 wks · "2 months" (TO)

5 Management

Hangman's fracture

- Type I → **RIGID COLLAR 4–6 wks**
- Type II
  - 3–5 mm → TRACTION + EXTENSION → **HALO**
  - > 5 mm, severe angulation → **SURGERY**
- Type IIA → NO TRACTION · hyperextension + compression → **HALO**
- Type III → **OPEN REDUCTION of facets + C2-3 FUSION**

Surgical options **anterior C2-3 interbody fusion** (disc injury) · **posterior C1-3 fusion** · **bilateral C2 pars screw osteosynthesis** · anterior facet fixation for failed posterior fusion

X vs Y

	Type II	Type IIA
Displacement	> 3 mm	**None**
Angulation	Significant	**Severe**
Fracture line	Vertical	Horizontal
Reduction	**Traction** + extension	**No traction** — compression + extension

How it's asked

"Polytrauma, undisplaced C2 pars fractures, intact disc?"	→ **type I — rigid collar**
"Hangman's — EXCEPT?"	→ **"primary hyperflexion mechanism"** (TO)
"CT shows bilateral C2 pars fracture — called?"	→ **Hangman's fracture** (TO, unkeyed — options Jefferson, Hangman's, odontoid II, flexion teardrop)

– – Active space – –

VERTEBRAL ARTERY INJURY

– Spine · 69 / 69 –

1 Basics

Epidemiology

Setting	Rate
All trauma	**0.5%** · 70% of blunt VAI have a cervical fracture
Cervical spine fractures	**33–39%**
Anterior cervical surgery	0.3%
Posterior cervical surgery	**4.1–8.2% (C1-2 transarticular screws)**

Anatomy

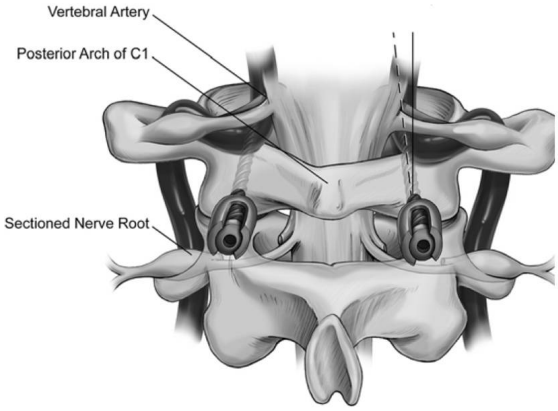
Segment	Course	Risk
V1	**Subclavian** → anterior to C7 TP → enters **C6 transverse foramen**	**Highest blunt injury risk at C6 entry**
V2	Foramina **C6 → C1**	**Most traumatic injuries**; lateral mass/pedicle screw drilling
V3	Over C1 arch → foramen magnum	C1 lateral exposure, laminectomy, caudal-lateral **transarticular screws** (2nd highest risk at C1-2)
V4	Intradural → joins opposite VA → **basilar**	–

Anomalies 2.7% in anterior surgery (foramen medial to / within 1.5 mm of uncovertebral joint) · **2.3–20% at C1-2**: **high-riding C2 foramen**, C2 pedicle erosion, thin lateral mass → CTA before transarticular screws

Safe zones

C1 posterior dissection stay **< 1.5 cm lateral to midline above the C1 arch** (Orthobullets, OB) · < 1 cm (OC fusion chapter) · retraction **> 2 cm** past midline injures VA (OB) – read the options · dissect **subperiosteally on the inferior arch**

C1 lateral mass screw **10° medial, 22° cephalad → inferior + medial to the VA** · C2 root crossed first (some sacrifice it)



C1 lateral mass screw trajectory

3 Presentation

History	C1/C2 fractures, **subaxial facet fractures/dislocations**, cervical/CVJ surgery
Vertebrobasilar insufficiency	dizziness, vertigo, nausea, **diplopia**, blindness, **ataxia**, bilateral weakness, **dysphagia** · onset variable (days–years); many asymptomatic

4 Diagnosis & Classification

Key radiology findings

Study	Fact
X-ray	Fracture patterns raise suspicion
CT angiography	**Sensitivity ~100%** · screening test
MRA	93.9% · deficits from VA/basilar ischaemia
Intra-op / post-op angiography	Collaterals before ligation, fistula/pseudoaneurysm embolisation

Diagnostic criteria – CTA screening (any one)

Indication
C1, C2 or C3 fracture · **fracture into transverse foramen** (≥ 1 mm displacement) · subluxation (VAI in 20%)
Cervical SCI · **hanging injury** · occipitocervical dissociation, basilar skull fracture
Unexplained central/lateralising deficit · acute infarct on CT · **GCS < 9** · diffuse axonal injury
Le Fort II/III or facial fracture · major thoracic injury / **1st rib fracture**

C1 floating lateral mass + visual change, nausea, dysphagia, dizziness	**CT angiogram first** ✗ Wrong options: traction → may be lethal

– – Active space – –

Biffl grading

Grade	Lesion
I	Dissection / intramural haematoma **< 25% stenosis**
II	Thrombus, intimal flap, **> 25% stenosis**
III	**Pseudoaneurysm**
IV	**Occlusion**
V	**Transection**

5 Management

Vertebral artery injury

- Traumatic → **ANTICOAGULATION / ANTIPLATELET (aspirin first line, heparin)**
  (CI: major intracranial infarct, intraspinal haematoma/AVF) · stabilise fracture as needed
- Intra-operative bleeding → **TAMPONADE (haemostatic agents, bone wax, insert the screw)**
  - Controlled → **consider DIRECT MICROVASCULAR REPAIR**
  - Uncontrolled + adequate collaterals → **ENDOVASCULAR EMBOLISATION / clip / ligation**

Tamponade risks	delayed haemorrhage, fistula
Ligation	angiogram first (collaterals; left VA usually dominant) · risks **cerebellar infarct (PICA)**, cranial nerve palsy, hemiplegia
Complex right VA injury during O-C3 fusion, tamponade failed, segmental defect, good collaterals	**embolisation** ✗ Wrong options: antifibrinolytic · direct repair · graft
Post-op	observe or immediate angiography (fistula, pseudoaneurysm, collaterals)

6 Complications

Late	AV fistula, delayed haemorrhage, **pseudoaneurysm**, thromboembolism, **stroke** (persistent vertigo), death

Cerebrovascular (TO)

Topic	Fact
SAH work-up	NCCT → LP if negative · CTA first for source · **DSA = gold standard** (finds source in 80–85%)
Aneurysm sites	Carotid system 85–95%: **ACoA (commonest, 30%)**, PCoA 25%, MCA 20% · posterior 5–15% (basilar tip commonest) · multiple in 20–30%
PCoA aneurysm	**3rd nerve palsy (ptosis, dilated pupil)** in ~9% · new CN III palsy = impending rupture
ISUIA	**> 10 mm → ~5%/yr rupture** · 7–12 mm 2.6%, 13–24 mm 14.5%, ≥ 25 mm 40% (5-yr) · anterior commoner, posterior ruptures more
AVM	No capillary bed, **no intervening parenchyma** in nidus · 2–4%/yr bleed · **small AVMs bleed more** (higher feeder pressure) · death 1%/yr · **rebleed 6–18% in yr 1, evenly spread** (✗ Wrong options: highest in first 2 wks) · treatment: embolisation, stereotactic radiosurgery, surgery — **all**
Spetzler-Martin	Size < 3 / 3–6 / > 6 cm = 1/2/3 · eloquent 1 · deep drainage 1
Cavernoma	**No intervening brain parenchyma**
Anterior choroidal occlusion	Contralateral hemiplegia, hemihypaesthesia, homonymous hemianopia · cognition spared
Carotid endarterectomy	Symptomatic **50–99%**, asymptomatic 60–99% → **endarterectomy** (less stroke than stenting, more MI) · clamp release **ECA → CCA → ICA**
Subclavian steal	**Left subclavian occluded proximal to the vertebral origin**
ICA : ECA flow	**80 : 20**
Malignant MCA infarct + haemorrhagic transformation	**Decompressive craniectomy**

Spinal vascular malformations (TO)

Type	Fact
I — dural AVF	**Commonest**, **acquired**, slow progressive myelopathy from **venous congestion + steal** (✗ Wrong options: recurrent bleeding) · treat by **ligating/embolising the draining (leptomeningeal) vein** · 15–20% other AVMs
II — glomus	**Intramedullary** nidus (congenital)
III — juvenile	Intramedullary-extramedullary, high flow, SAH risk
IV	Perimedullary (intradural extramedullary) fistula

Aggressiveness — **leptomeningeal vein** (fistulous) determines outcome
✗ Wrong options: location · symptoms

Unkeyed TO items — SAH CT appearance (classic: hyperdensity in basal cisterns + sulci) and basilar tip aneurysm treatment (coiling usual) — answers not provided by source

– – Active space – –

How it's asked

"C1 lateral mass screw lies relative to VA?"	→ **inferior + medial**
"VA injured during C1 dissection — where?"	→ **cephalad to C1 arch > 1.5 cm lateral to midline**
"Structure injured > 2 cm lateral on atlas arch?"	→ **vertebral artery** — from subclavian through transverse foramina
"Uncontrolled VA bleeding, segmental defect, good collaterals?"	→ **embolisation**
"C1 floating lateral mass + VBI symptoms — next?"	→ **CT angiogram**
"AVM without intervening parenchyma?"	→ **cavernous haemangioma** (TO)
"Clamp removal after CEA?"	→ **ECA, CCA, ICA** (TO)
"5 mm ACoA aneurysm — ISUIA true?"	→ **> 10 mm ≈ 5%/yr rupture** (TO)
"Anterior choroidal occlusion?"	→ **hemiplegia, hemihypaesthesia, hemianopia** (TO)
"Perimesencephalic SAH — gold standard?"	→ **DSA** (TO)
"AVMs — EXCEPT?"	→ **rebleed highest in first 2 weeks** (TO)
"Subclavian steal?"	→ **left subclavian occluded proximal to vertebral origin** (TO)
"Frontal AVM — options?"	→ **all (embolisation, radiosurgery, combined)** (TO)
"80% symptomatic ICA stenosis?"	→ **open endarterectomy** (TO)
"Malignant MCA infarct with bleed, hemiplegia?"	→ **decompressive craniectomy** (TO)
"Headache, neck pain, right ptosis?"	→ **PCoA aneurysm SAH** (TO)
"ICA : ECA flow?"	→ **80 : 20** (TO)
"Spinal dural AVF — EXCEPT?"	→ **recurrent bleeding causes symptoms** (TO)
"Aggressive spinal AVM factor?"	→ **leptomeningeal vein** (TO)
"Glomus AVM?"	→ **type II** (TO)